

Shatter Clustering: Merge Overlap Threshold (min_intersection) Analysis

Sweep over `min\_intersection` in `merge\_ensembles` with Default Growth Defaults

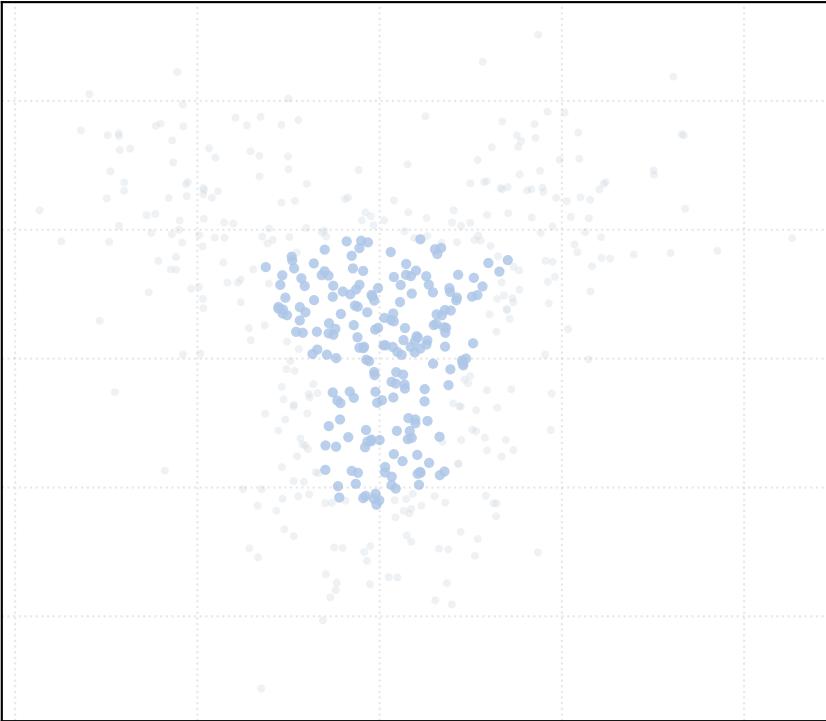
Experimental Configuration

- Total Evaluated Datasets: 30
- Tested min_intersection Values: [1, 2, 3, 5, 8, 10, 15, 20]
- Total Executions: 240 (8 runs per dataset)
- Fixed Default Parameters: alpha_MAD = 0.5, alpha_accept = 0.5, t_percentile = 0.15, r = adaptive
- Target Output PDF: benchmark_report_min_intersection.pdf

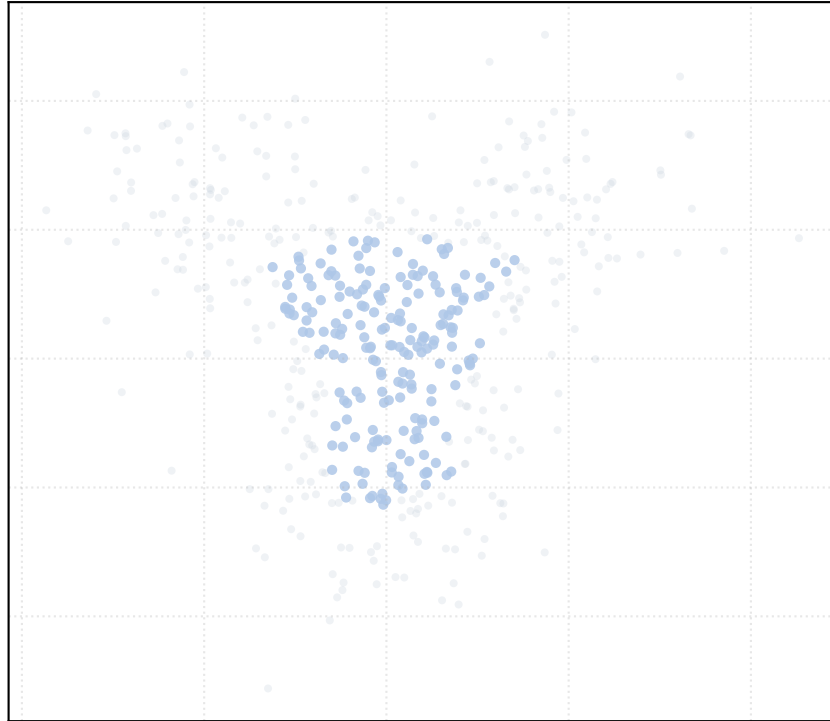
Parameter	Value / Range	Algorithmic Role
min_intersection	{1, 2, 3, 5, 8, 10, 15, 20}	Minimum overlapping vectors required to merge raw ensemble
alpha_MAD	0.50 (Default)	MAD multiplier for surface growth compatibility
alpha_accept	0.50 (Default)	Log2 fold-change acceptance cutoff
t_percentile	0.15 (Default)	Top 15% densest seeds
mean_dist_quant	0.15 (Default)	Quantile for adaptive search radius r

Dataset: bifurcation_2way_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

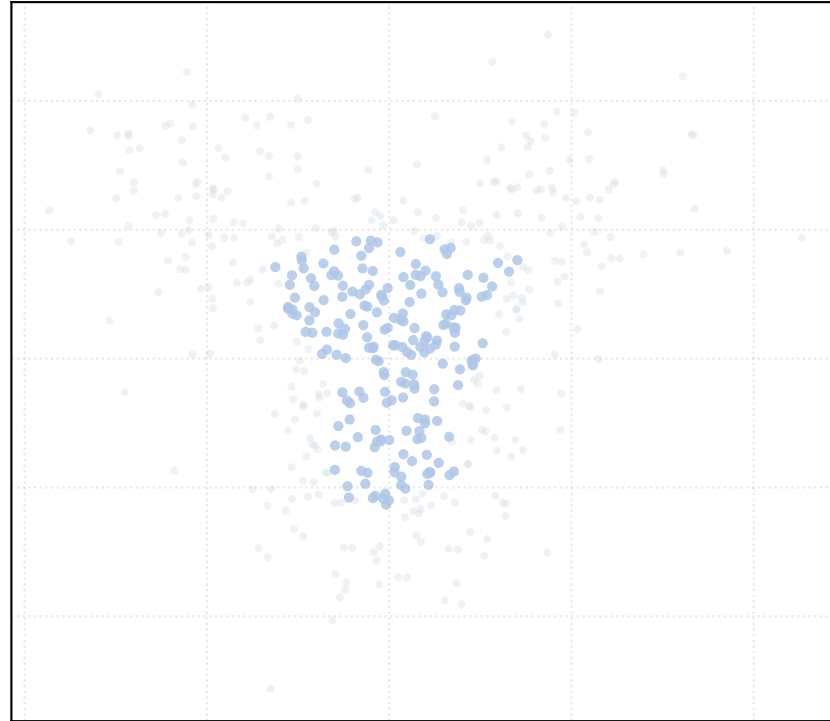
min_intersection = 1 (Default) [1 Clusters]



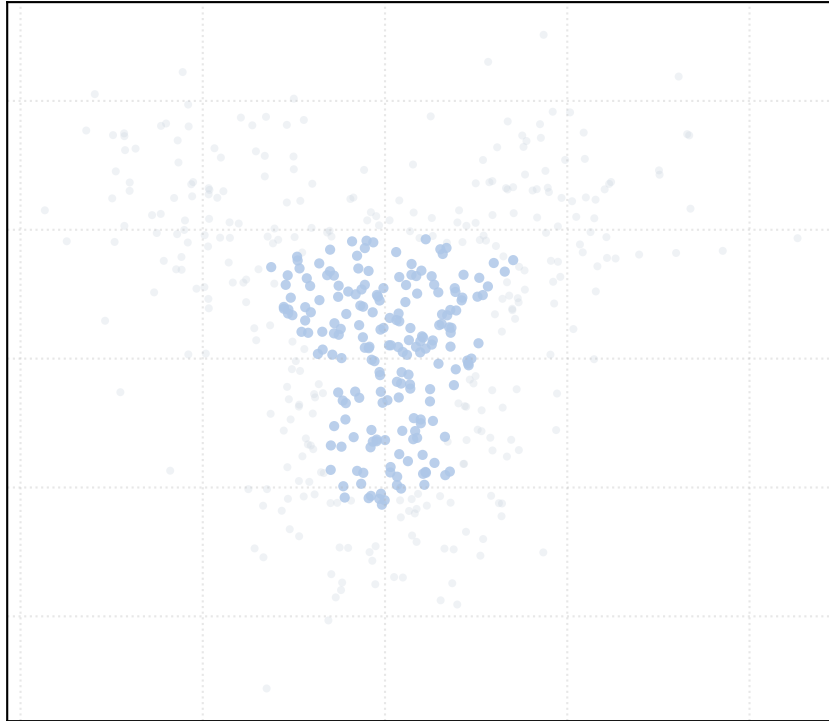
min_intersection = 2 [1 Clusters]



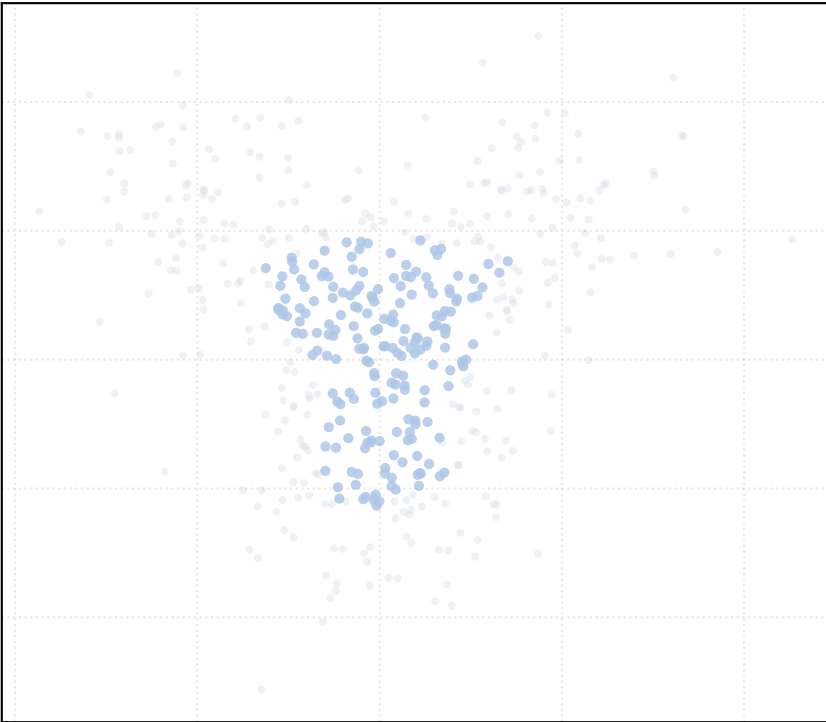
min_intersection = 3 [1 Clusters]



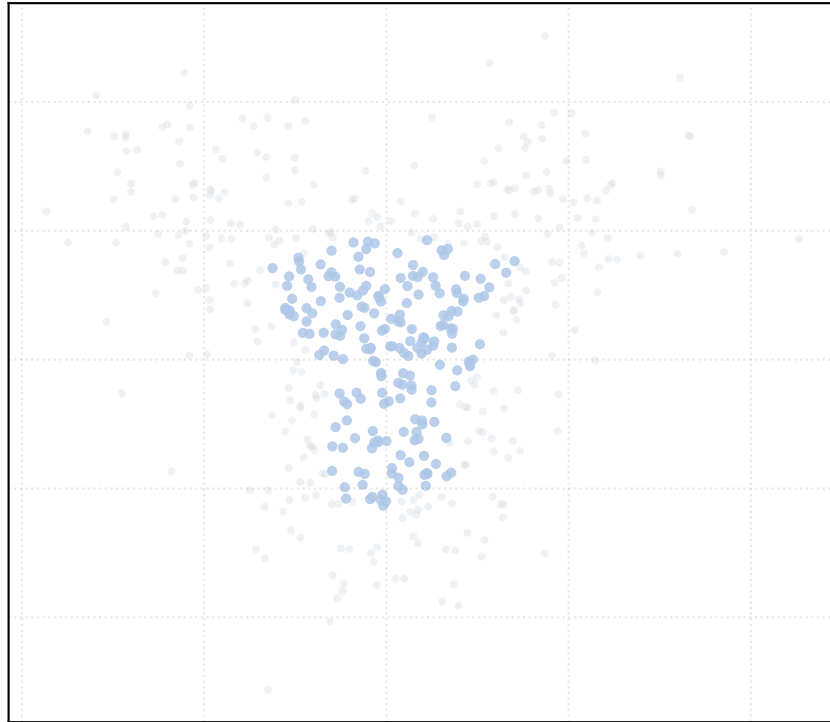
min_intersection = 5 [1 Clusters]



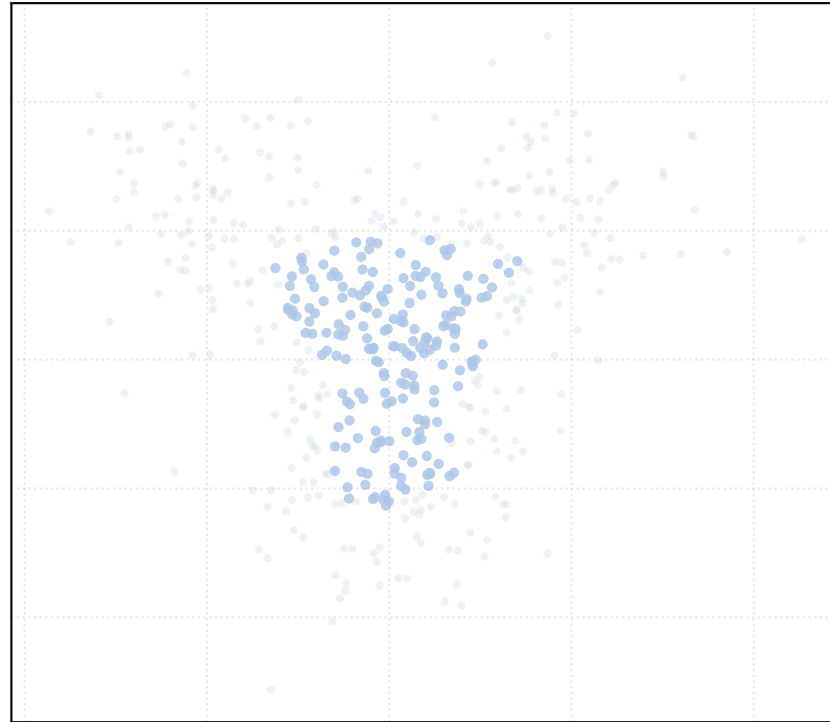
min_intersection = 8 [1 Clusters]



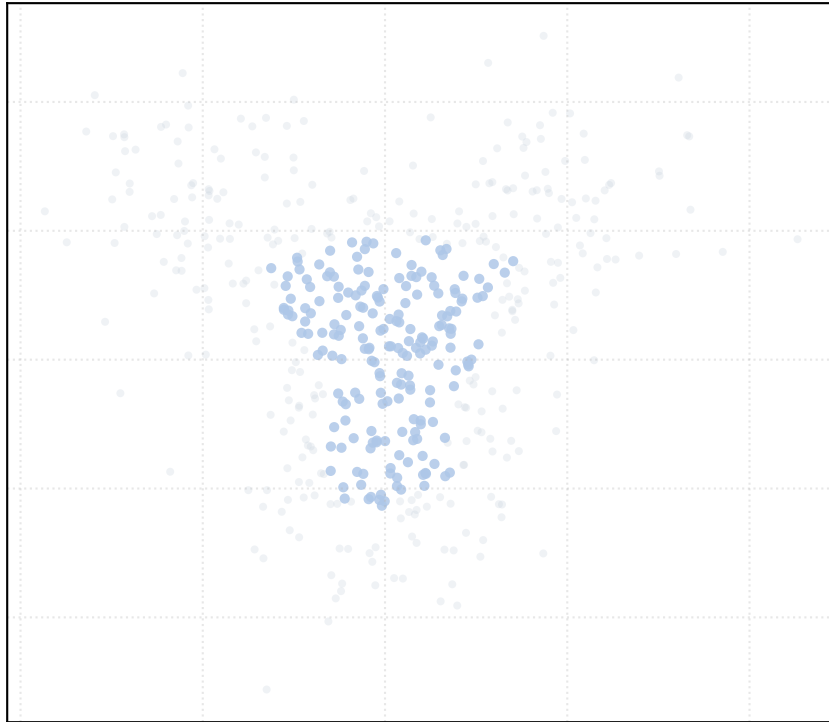
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

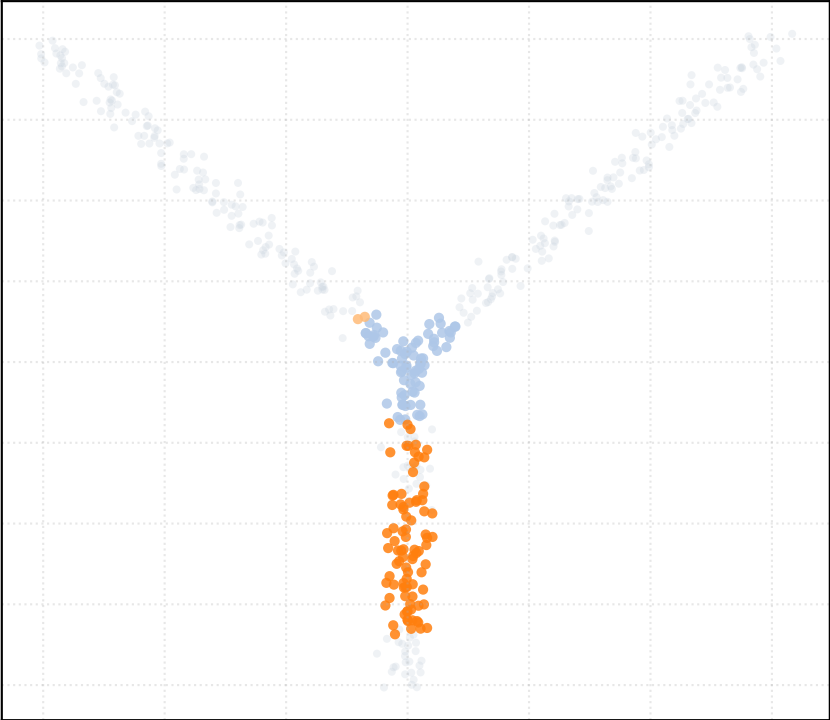


min_intersection = 20 [1 Clusters]

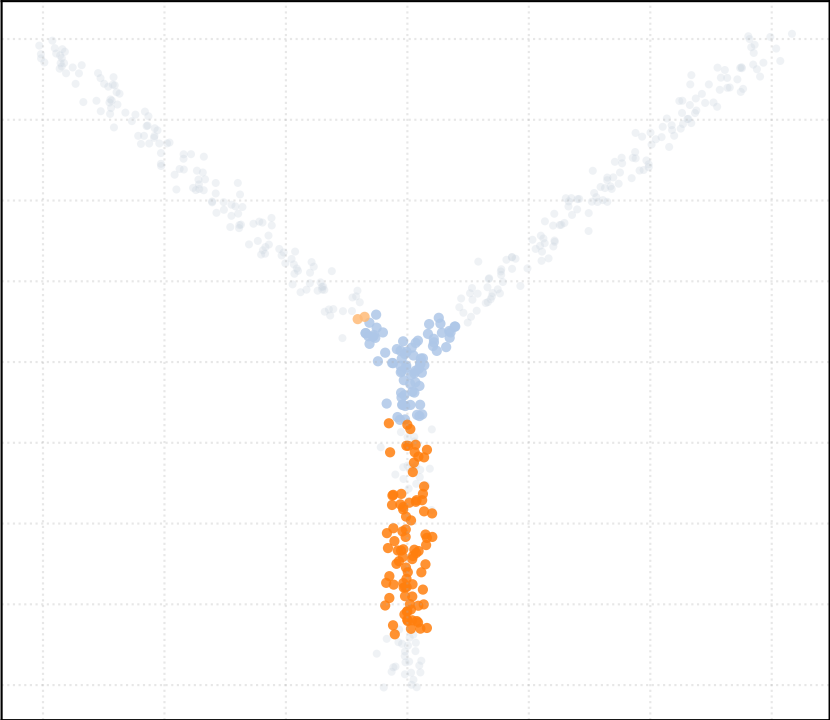


Dataset: bifurcation_2way_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

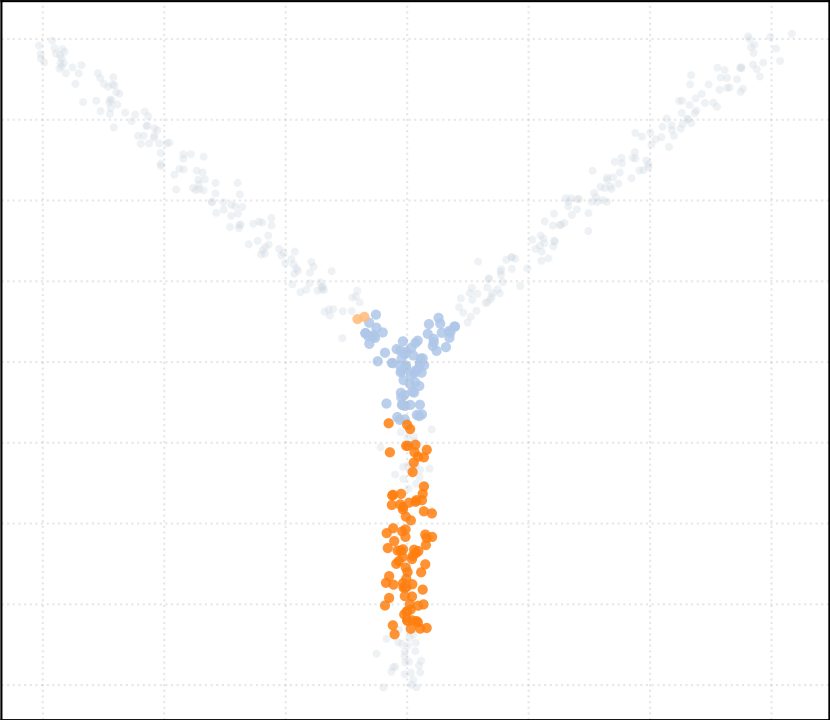
min_intersection = 1 (Default) [3 Clusters]



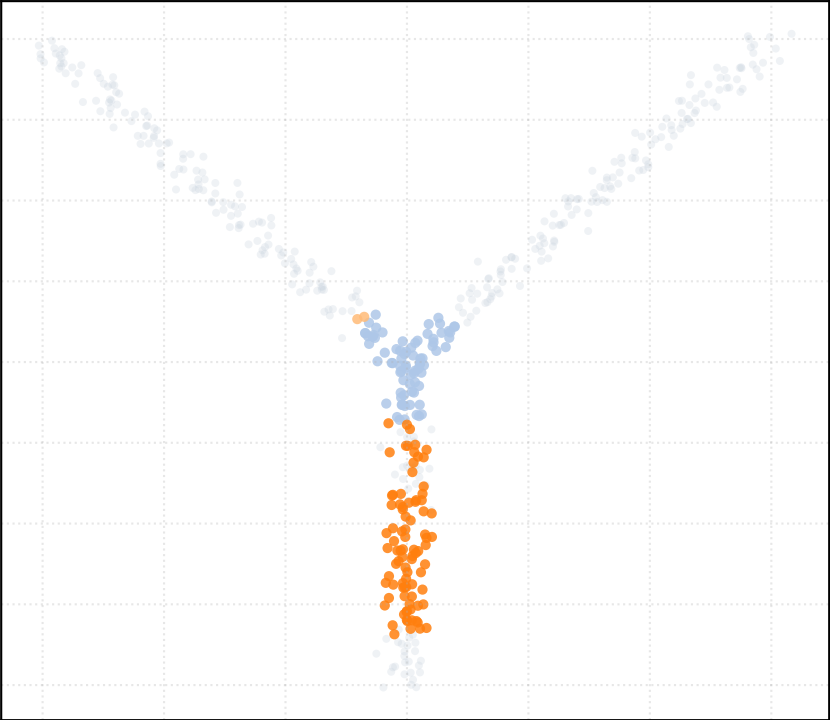
min_intersection = 2 [3 Clusters]



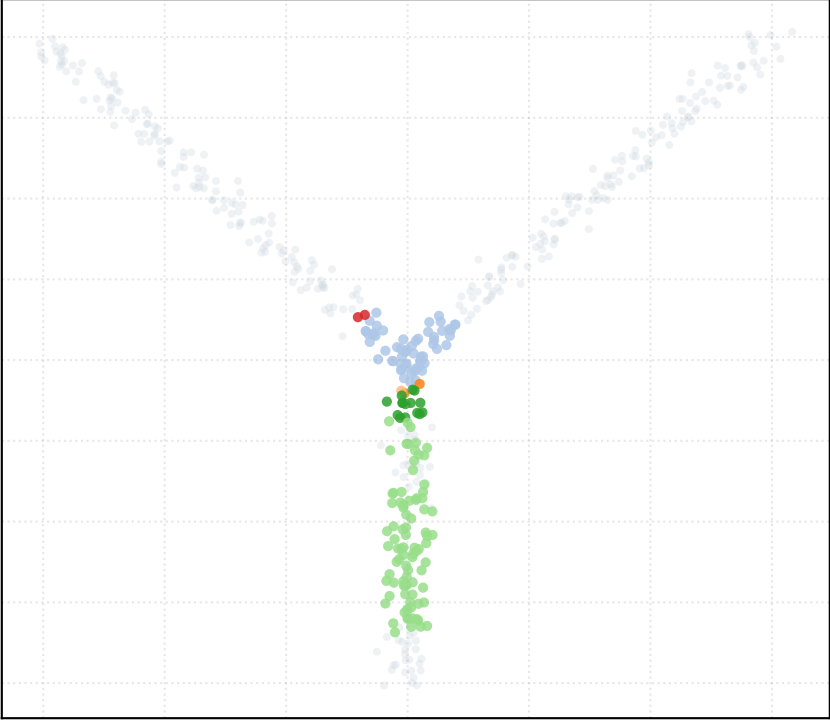
min_intersection = 3 [3 Clusters]



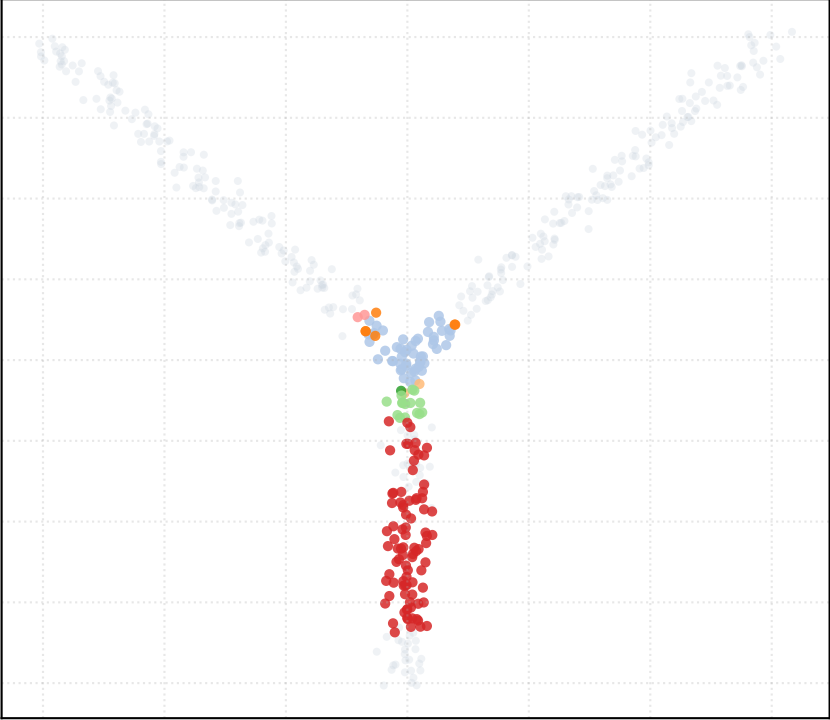
min_intersection = 5 [3 Clusters]



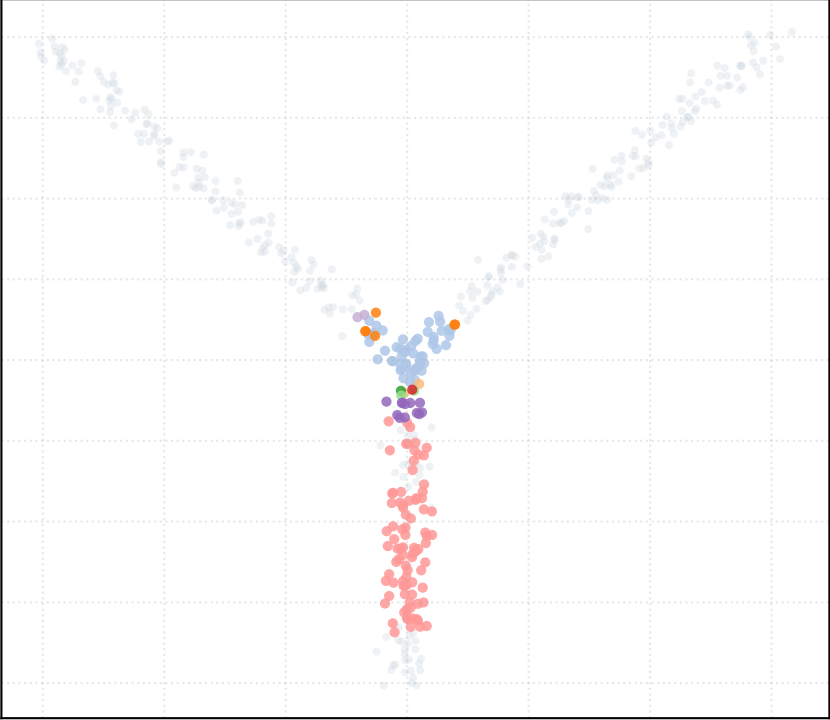
min_intersection = 8 [6 Clusters]



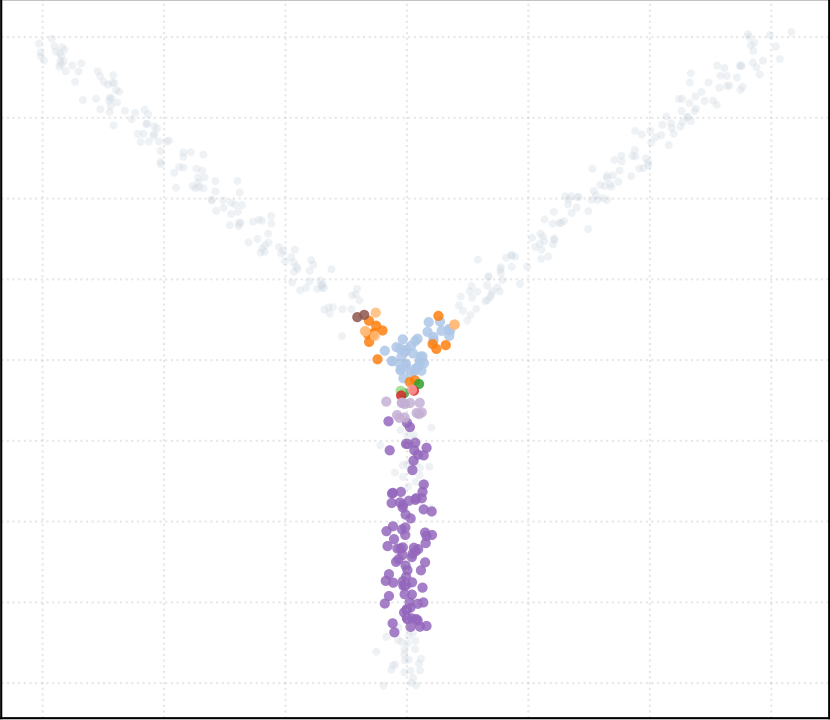
min_intersection = 10 [7 Clusters]



min_intersection = 15 [9 Clusters]

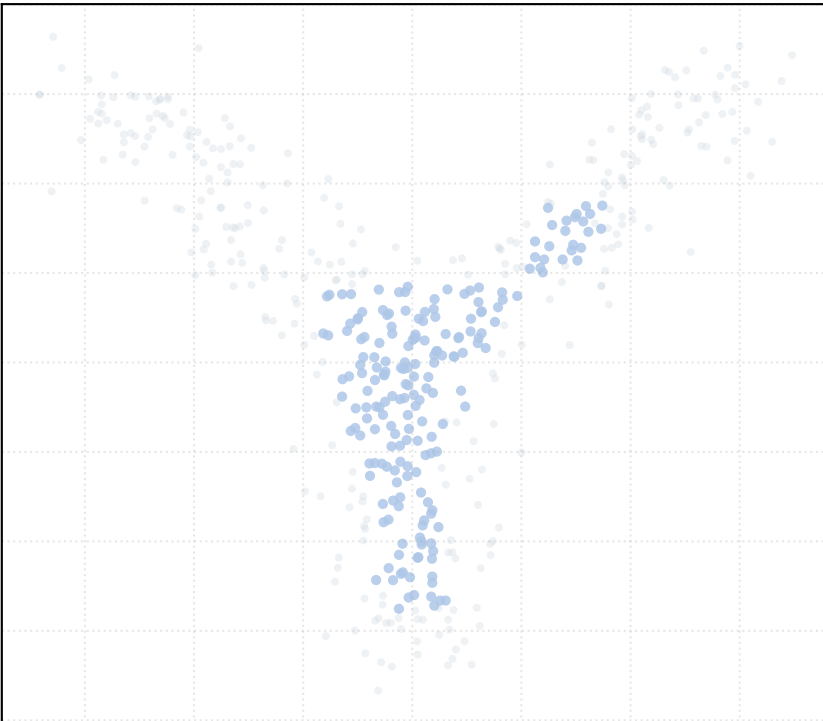


min_intersection = 20 [10 Clusters]

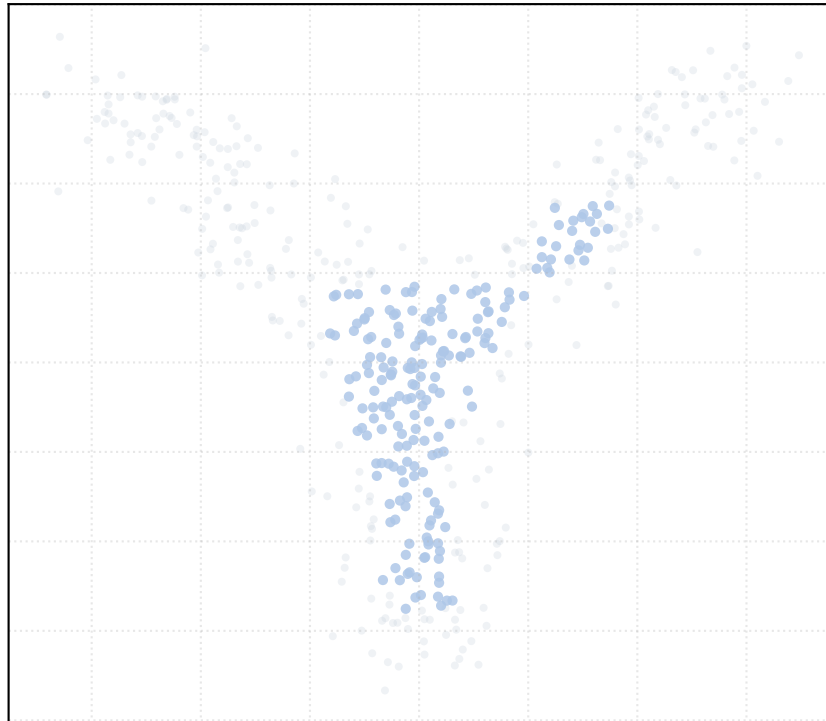


Dataset: bifurcation_2way_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

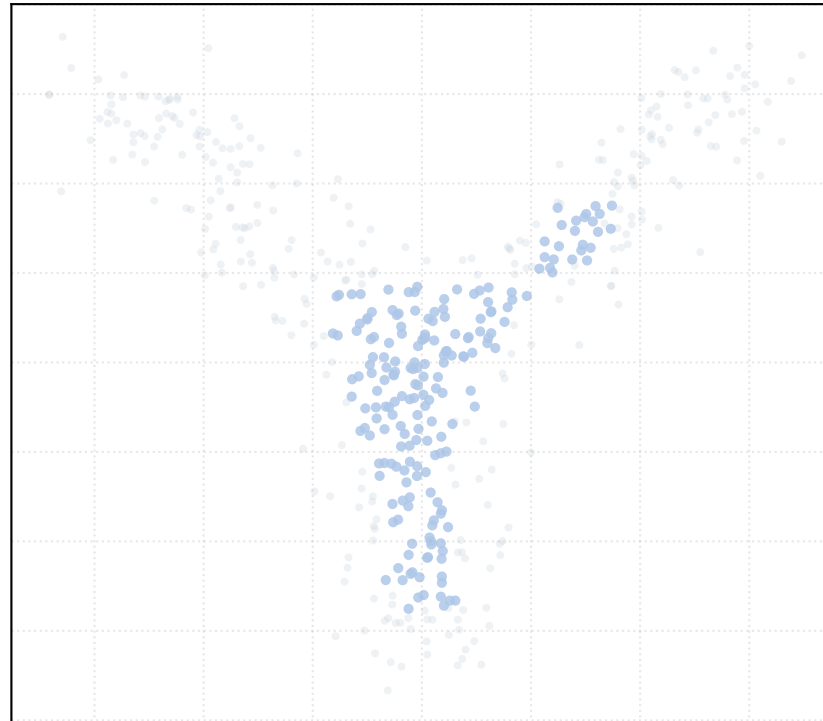
min_intersection = 1 (Default) [1 Clusters]



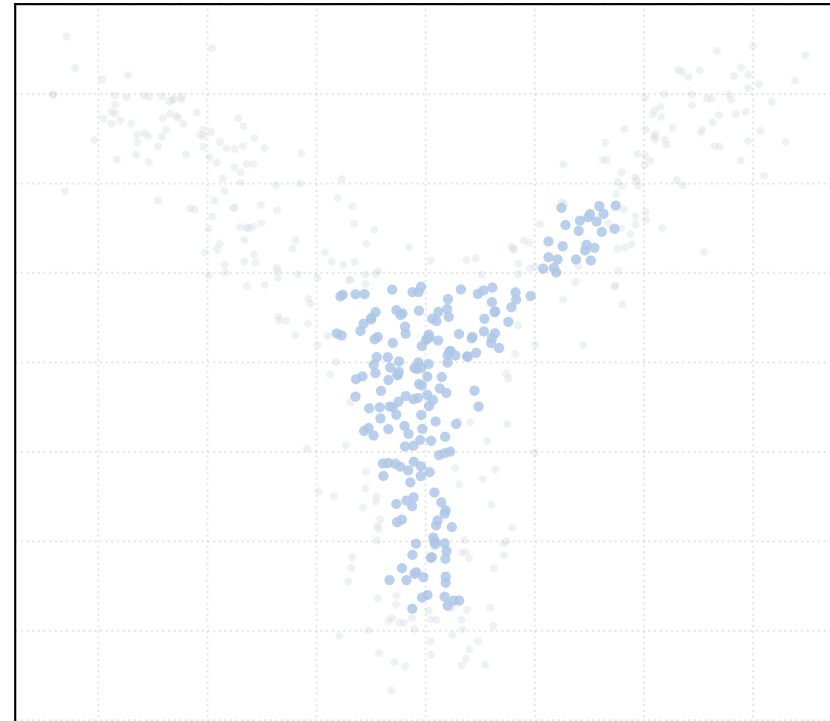
min_intersection = 2 [1 Clusters]



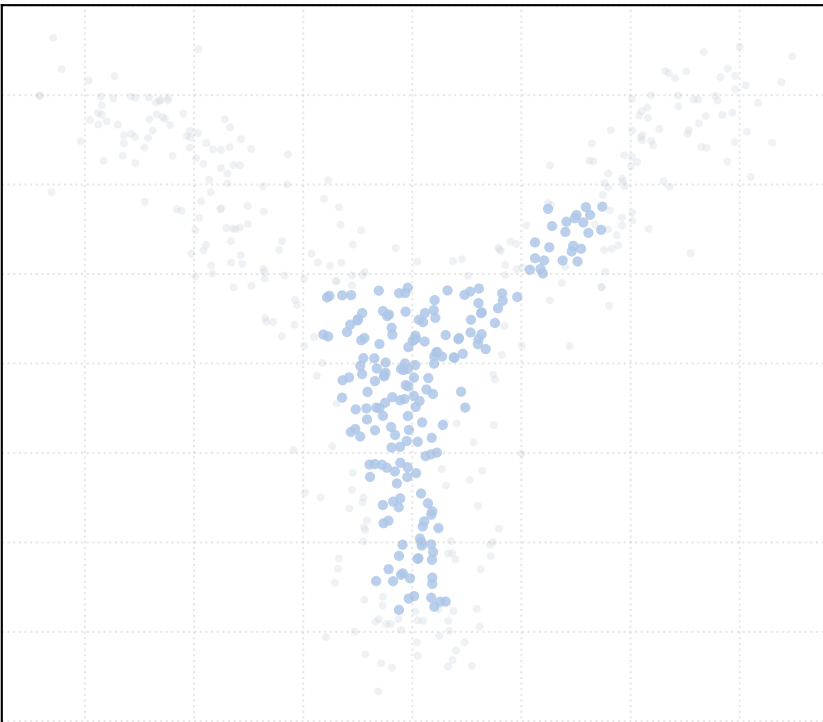
min_intersection = 3 [1 Clusters]



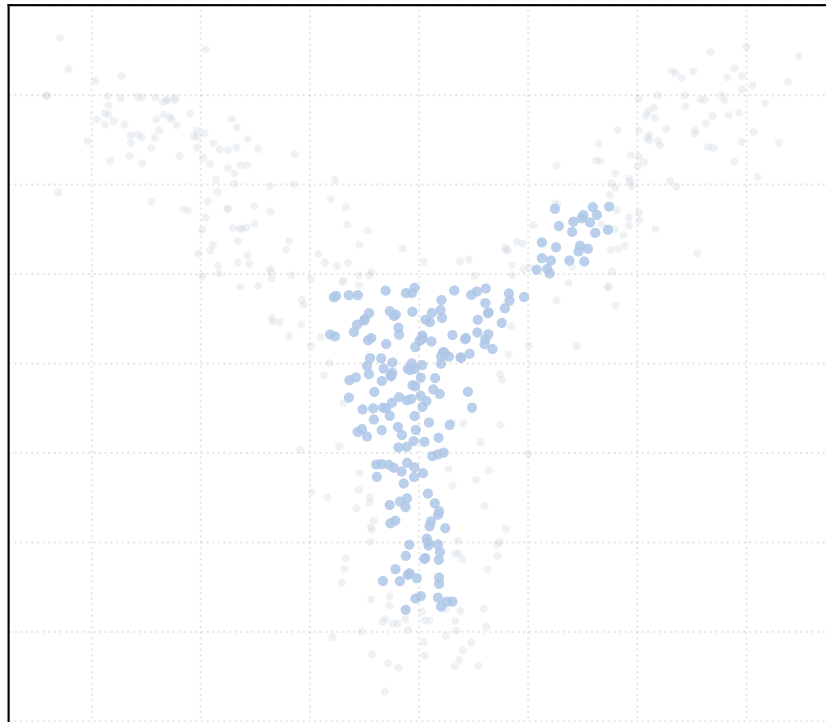
min_intersection = 5 [1 Clusters]



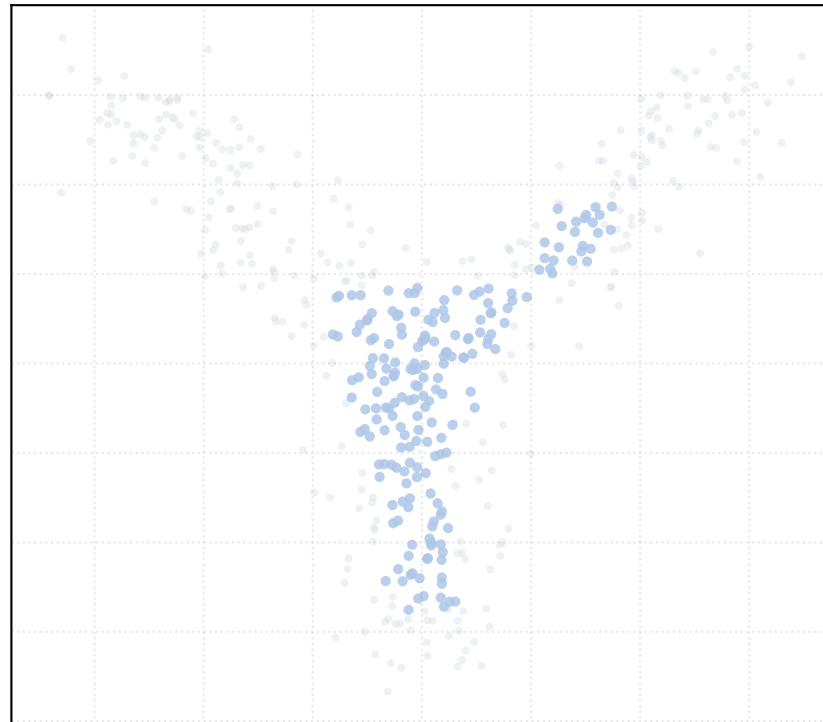
min_intersection = 8 [1 Clusters]



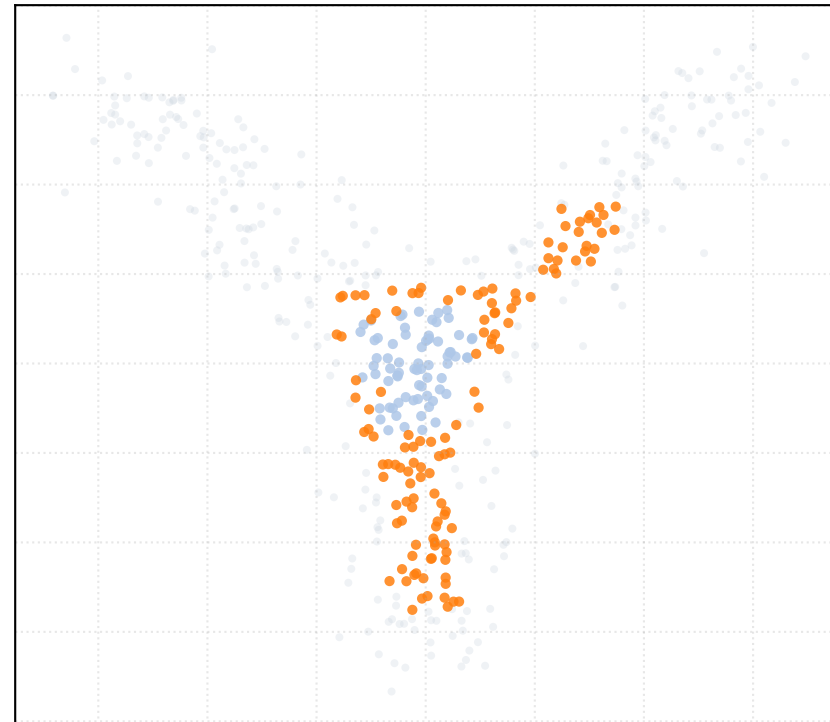
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

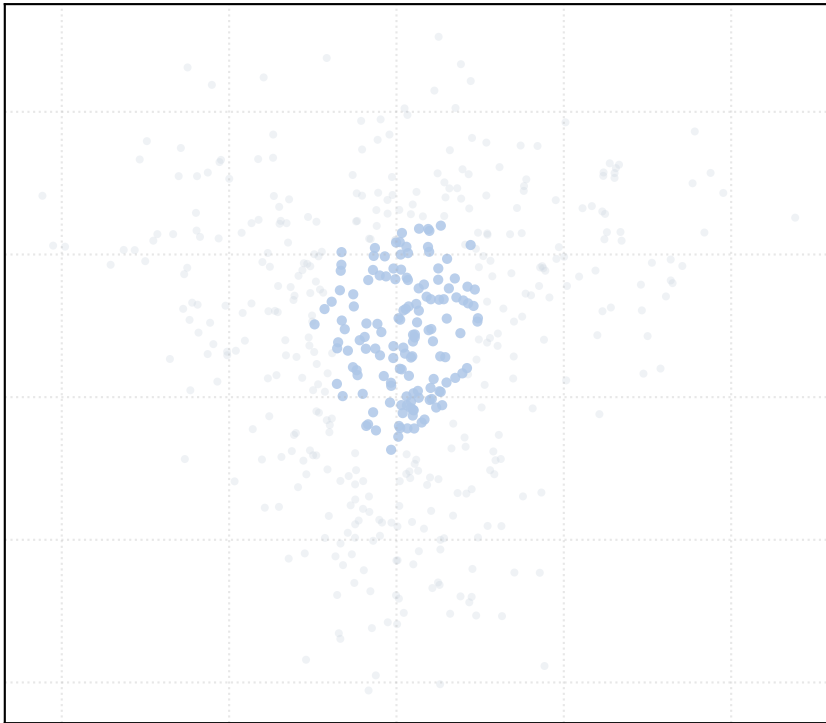


min_intersection = 20 [2 Clusters]

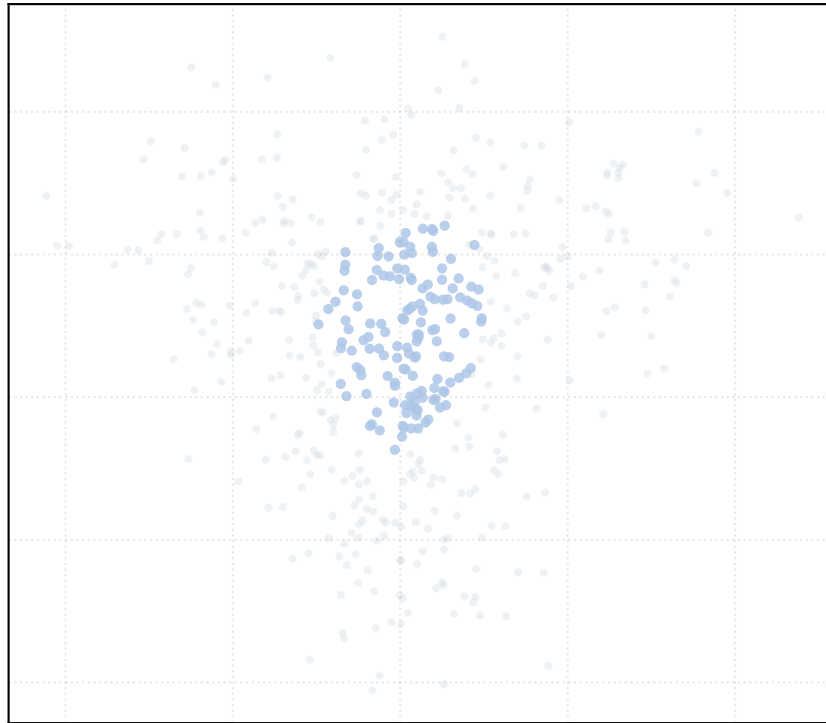


Dataset: bifurcation_3way_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

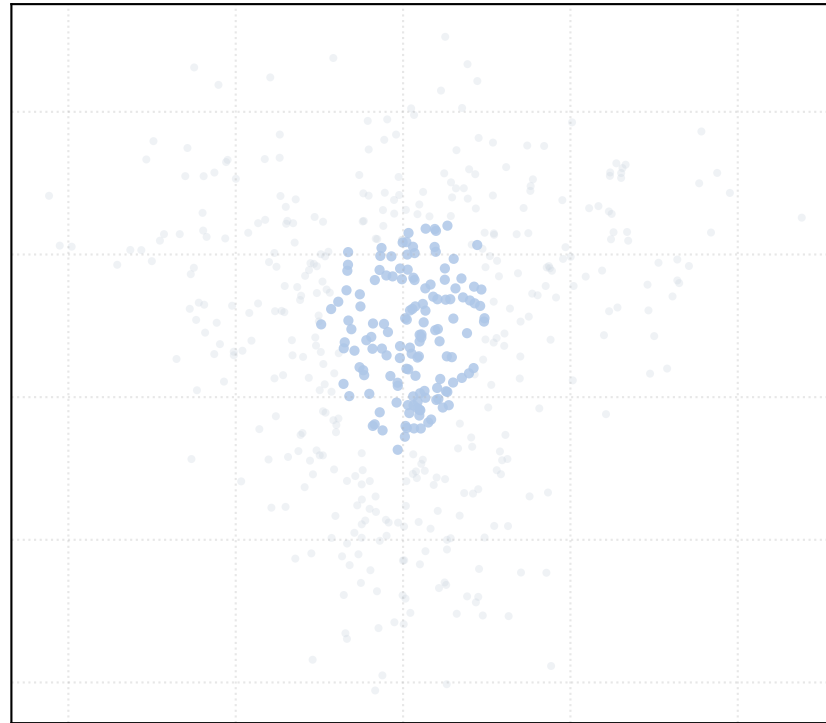
min_intersection = 1 (Default) [1 Clusters]



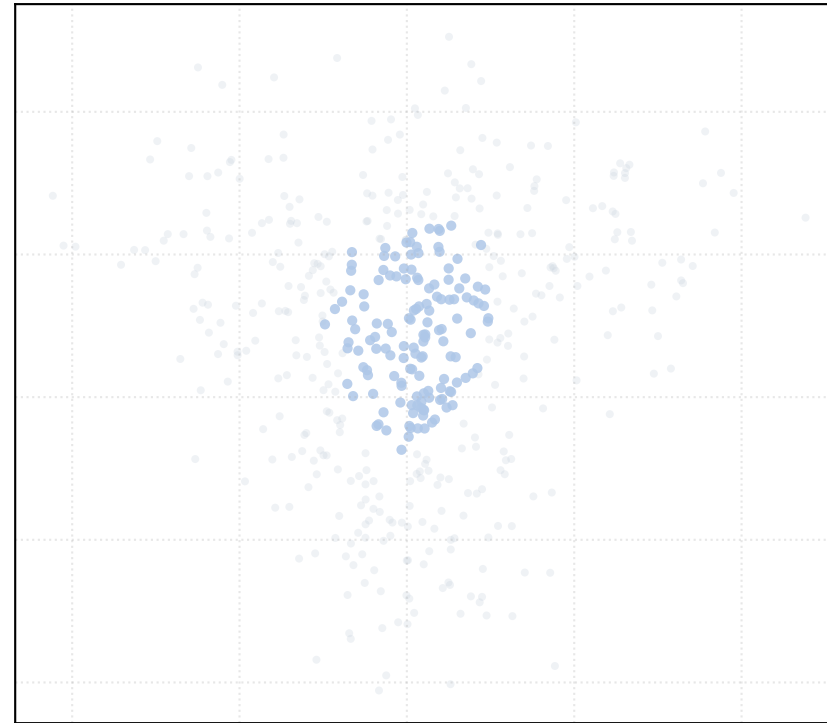
min_intersection = 2 [1 Clusters]



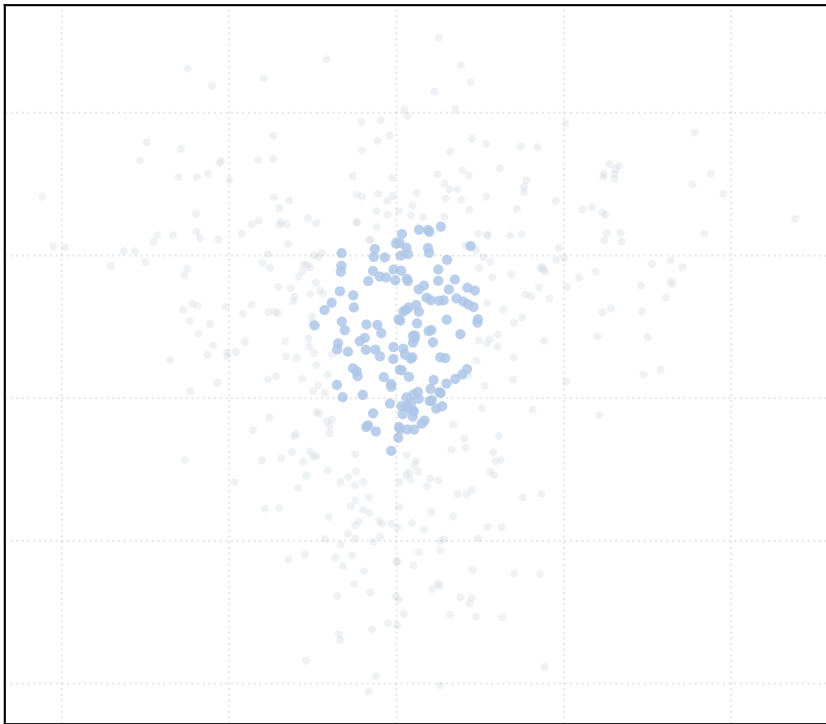
min_intersection = 3 [1 Clusters]



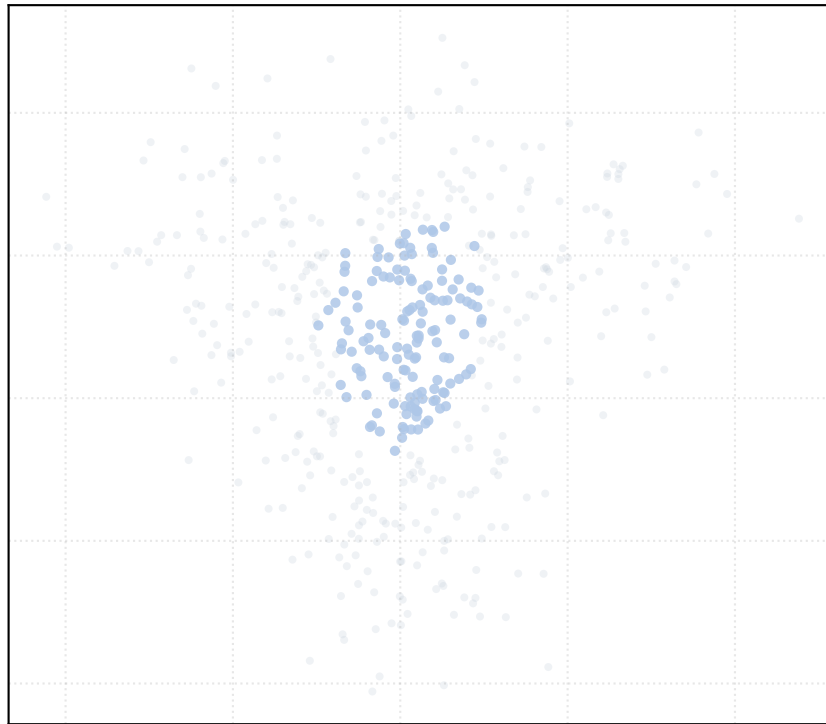
min_intersection = 5 [1 Clusters]



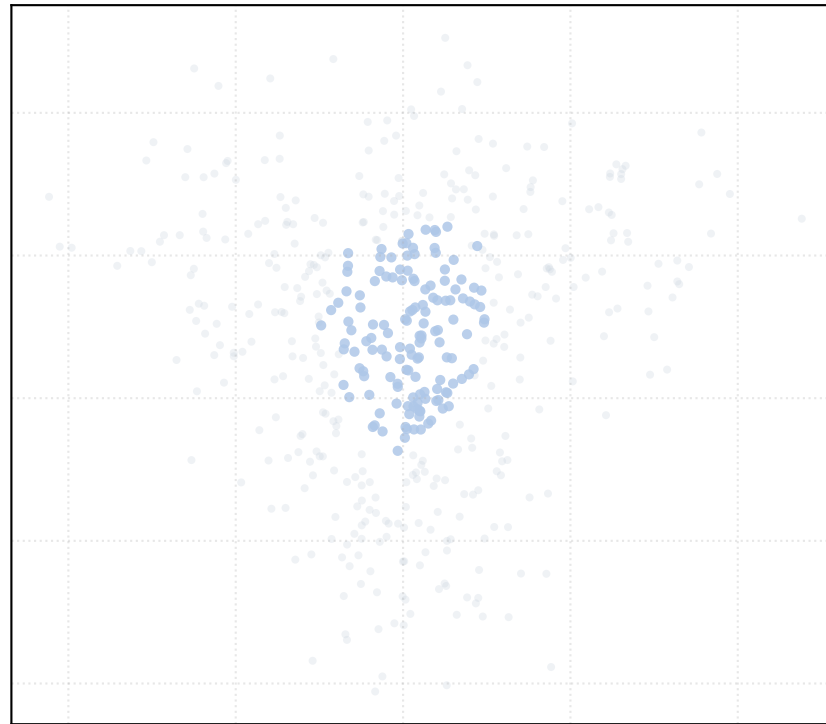
min_intersection = 8 [1 Clusters]



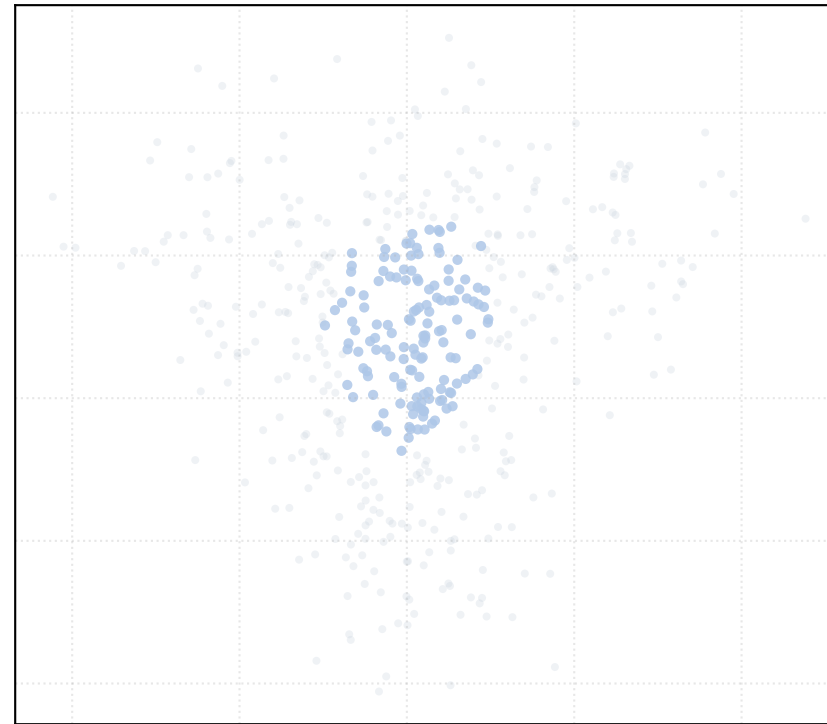
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

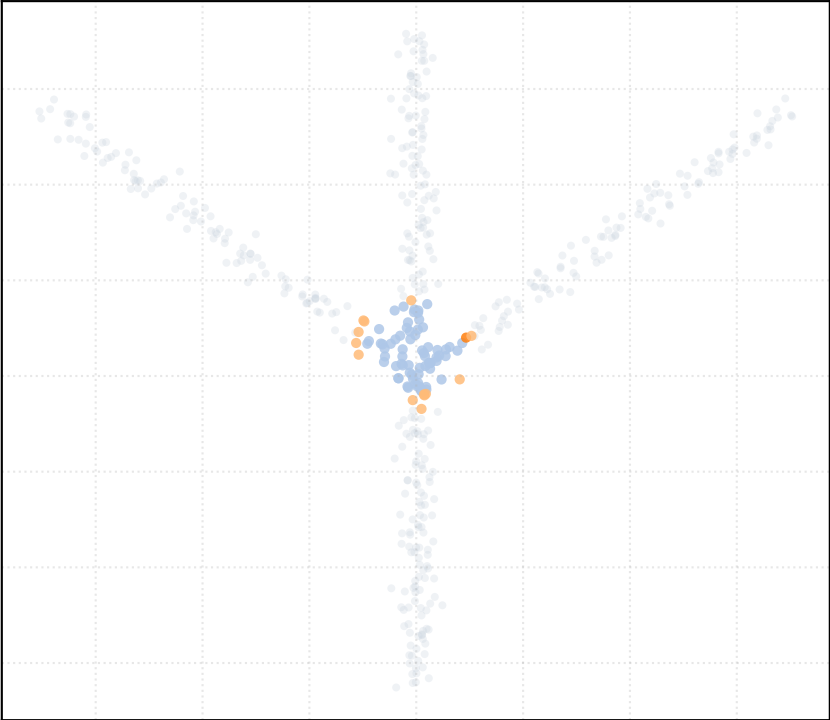


min_intersection = 20 [1 Clusters]

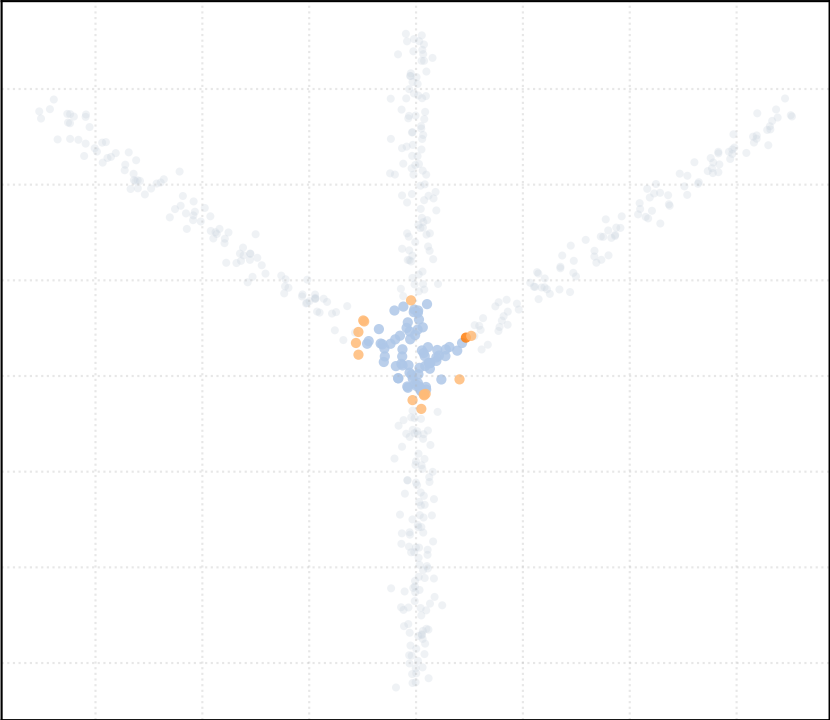


Dataset: bifurcation_3way_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

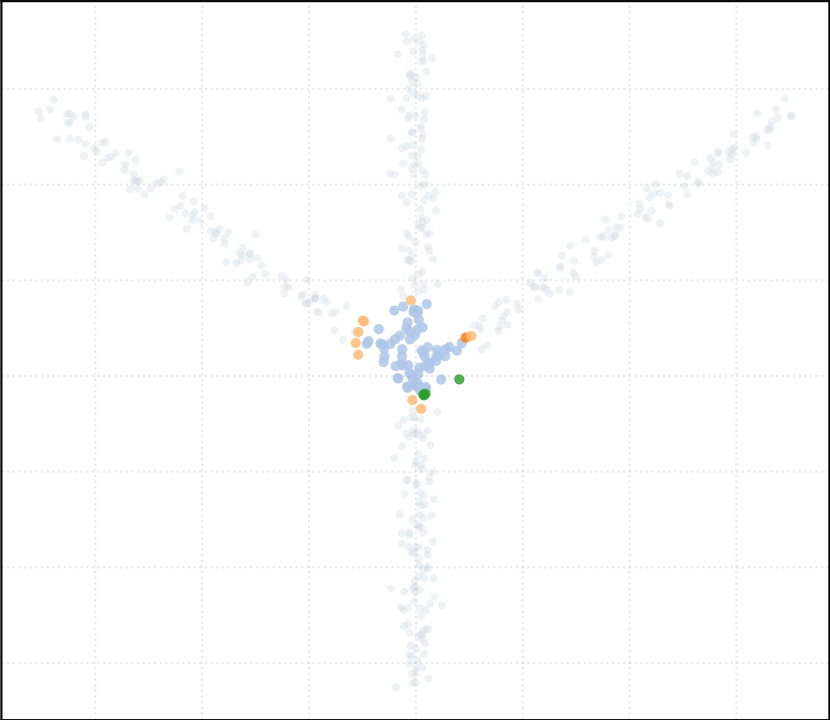
min_intersection = 1 (Default) [3 Clusters]



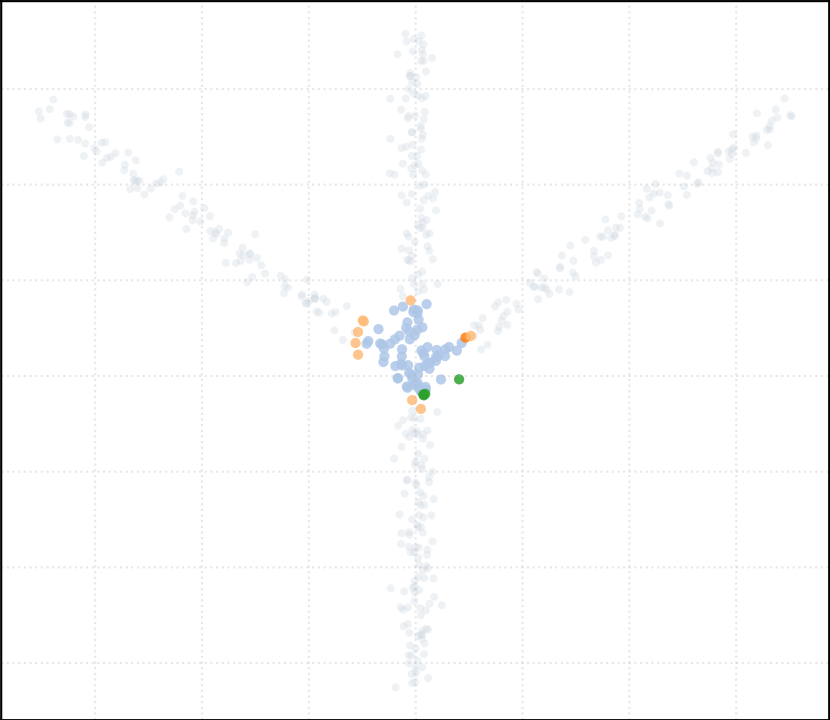
min_intersection = 2 [3 Clusters]



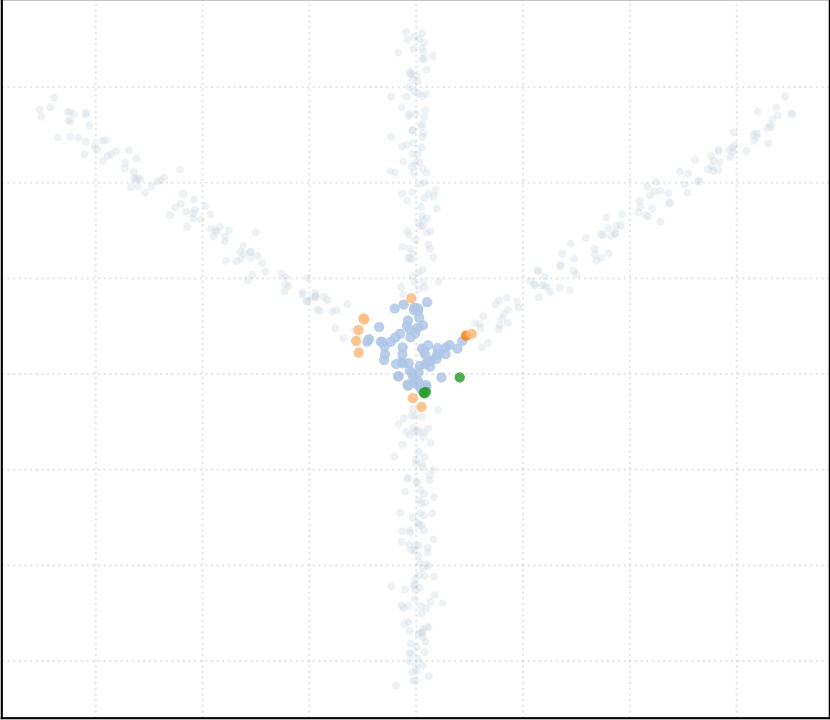
min_intersection = 3 [4 Clusters]



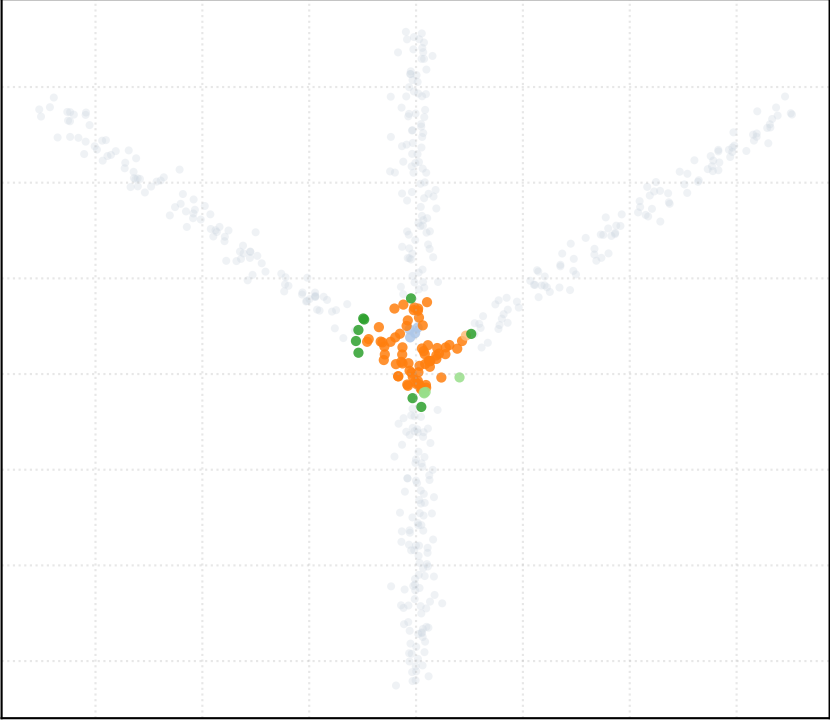
min_intersection = 5 [4 Clusters]



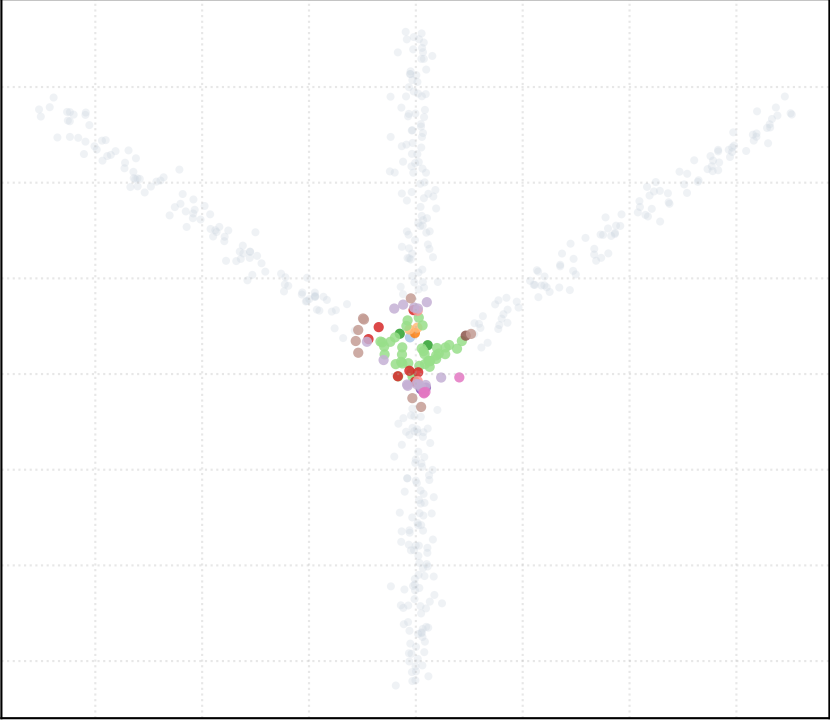
min_intersection = 8 [4 Clusters]



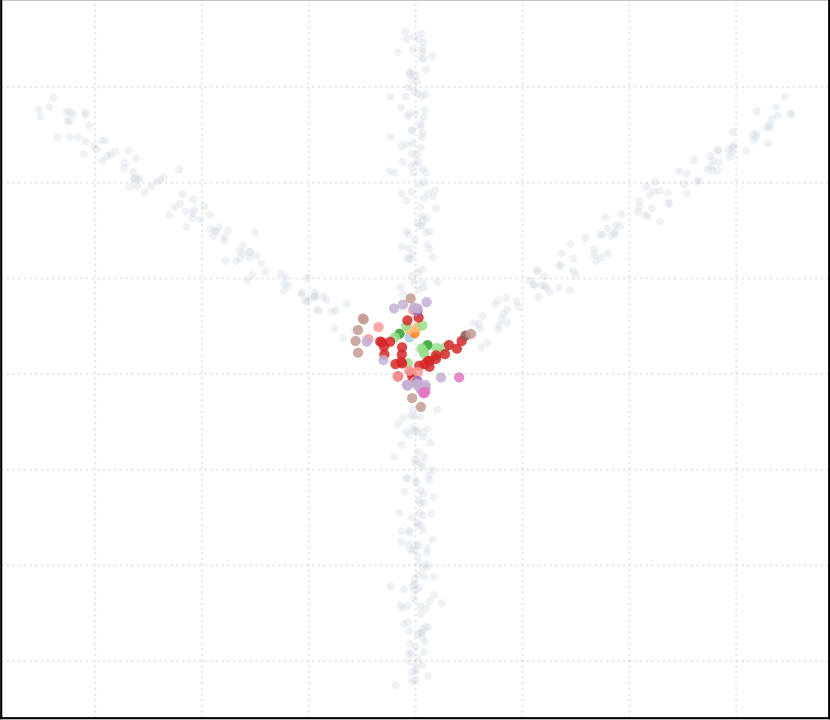
min_intersection = 10 [5 Clusters]



min_intersection = 15 [12 Clusters]

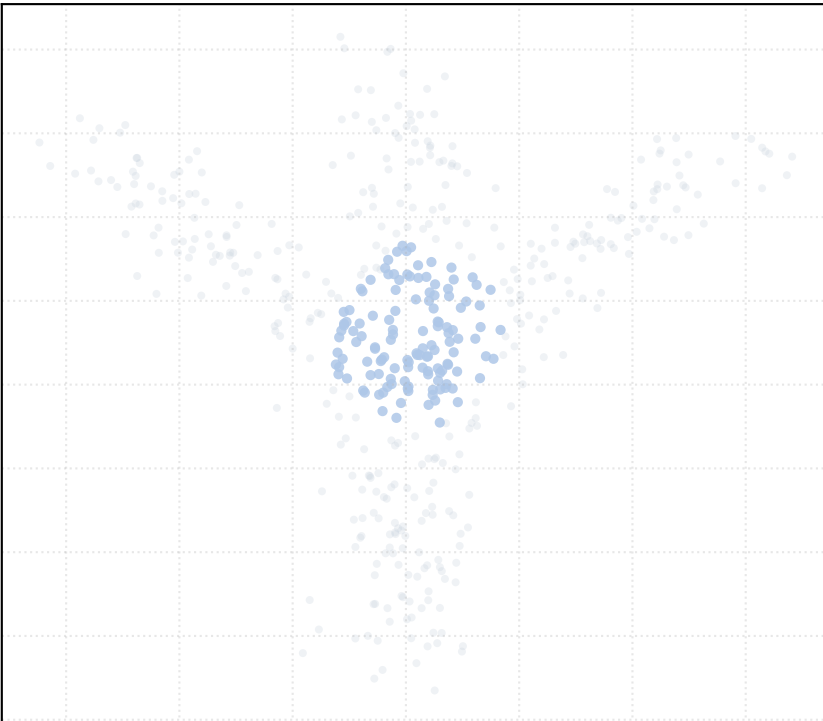


min_intersection = 20 [12 Clusters]

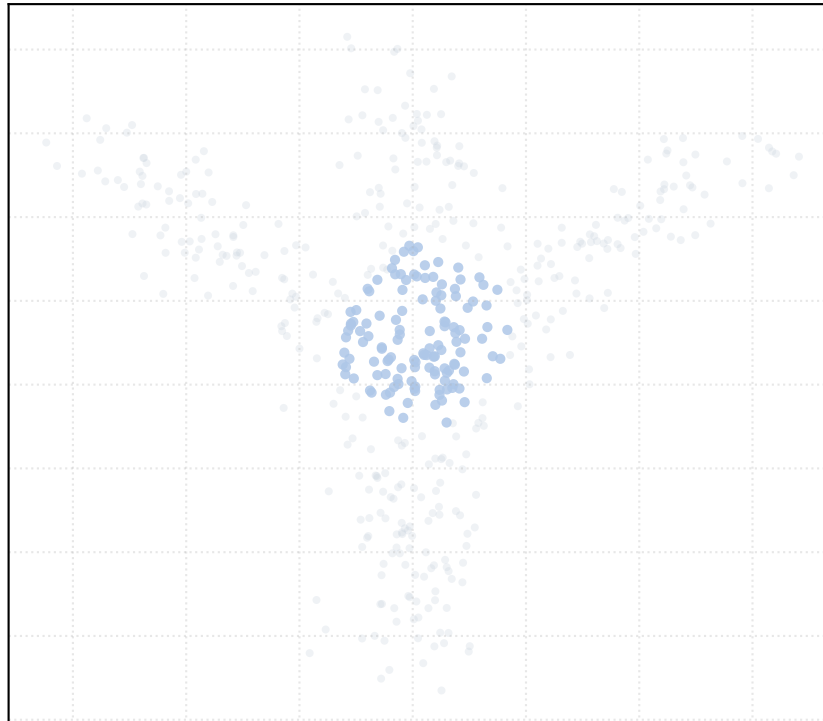


Dataset: bifurcation_3way_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

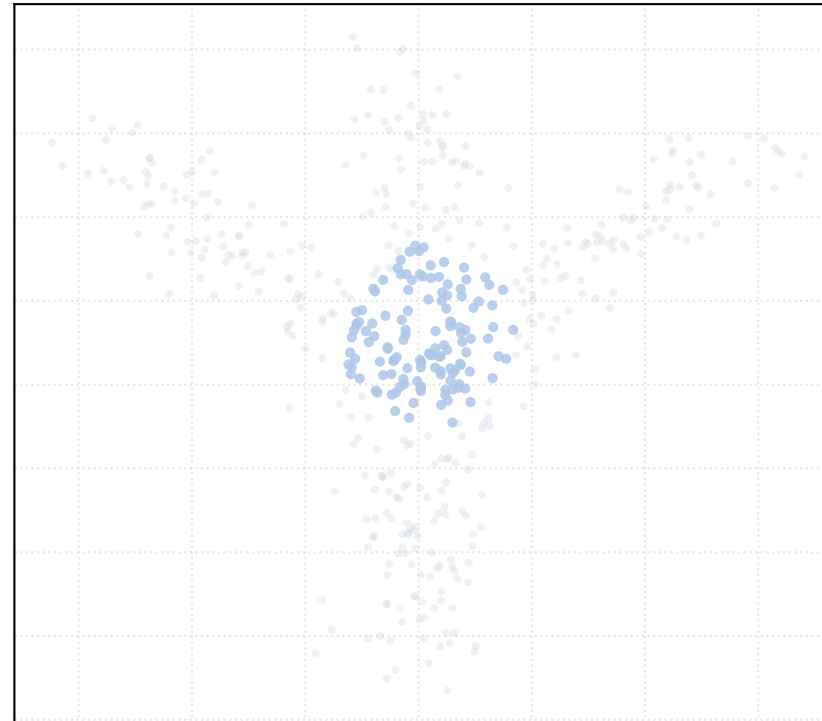
min_intersection = 1 (Default) [1 Clusters]



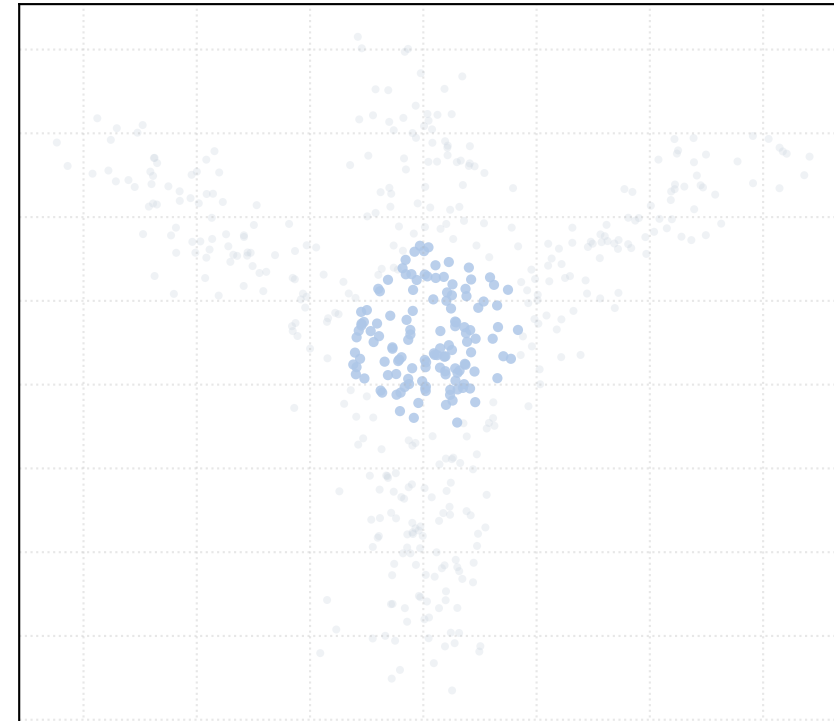
min_intersection = 2 [1 Clusters]



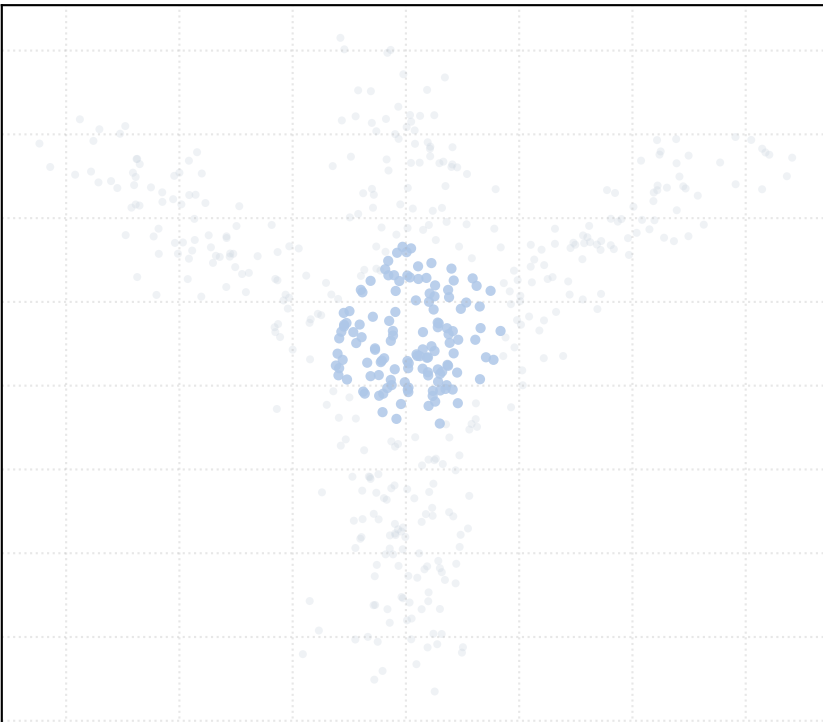
min_intersection = 3 [1 Clusters]



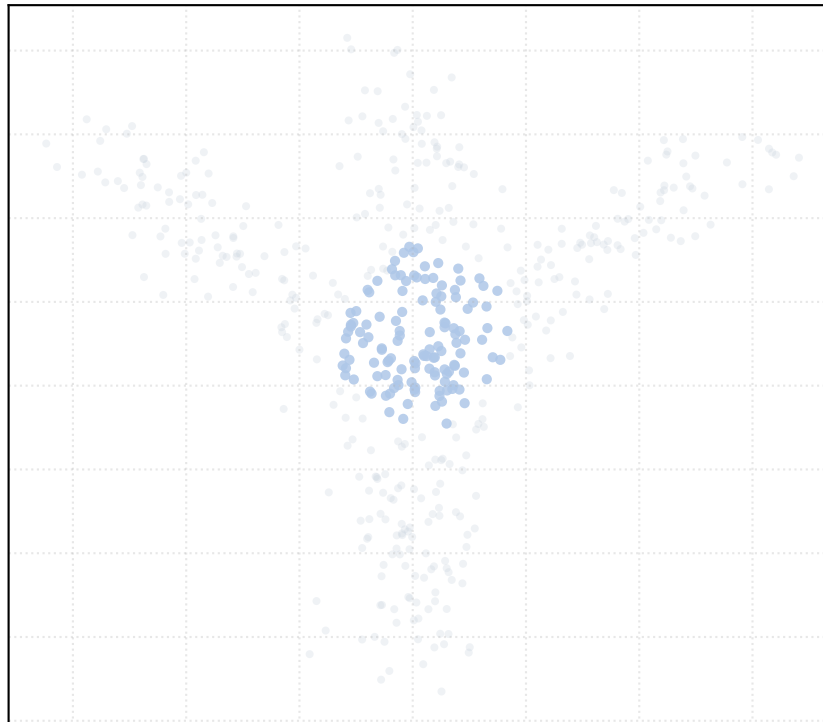
min_intersection = 5 [1 Clusters]



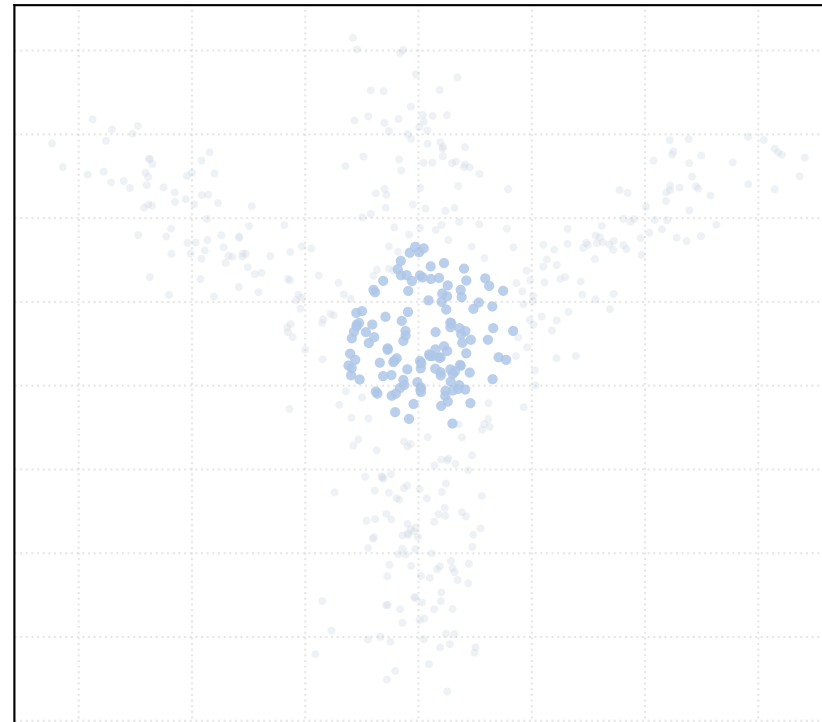
min_intersection = 8 [1 Clusters]



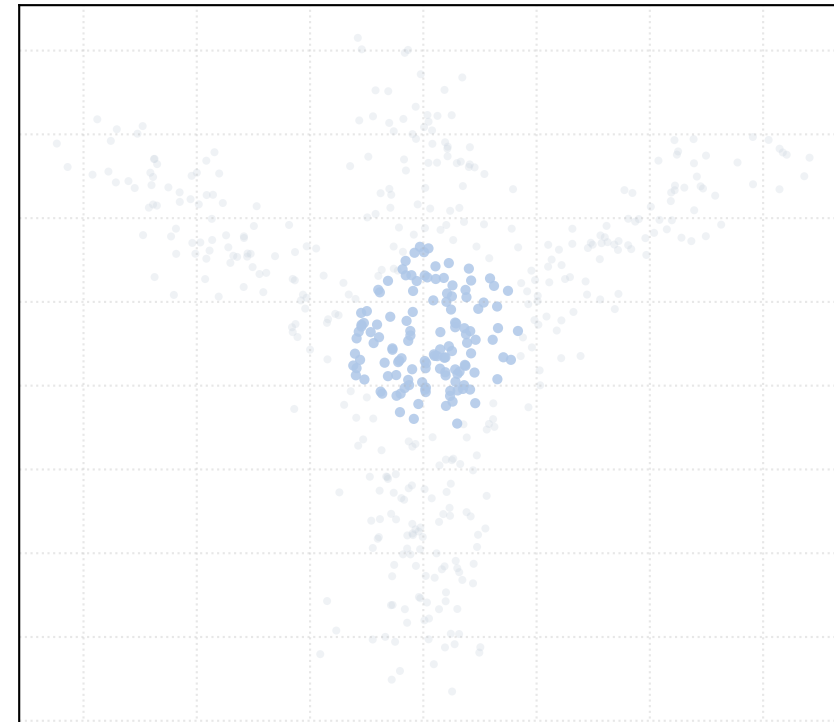
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

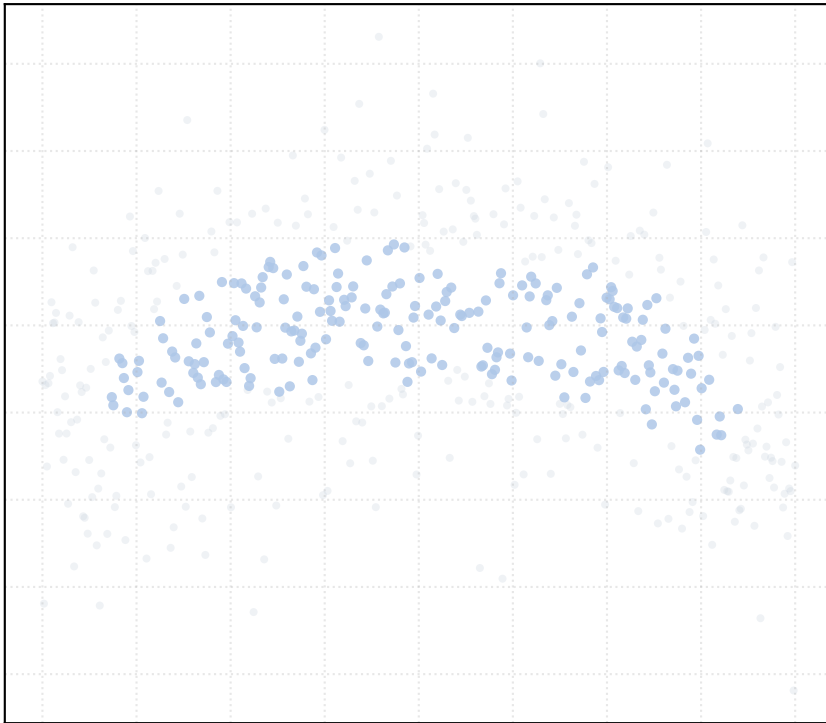


min_intersection = 20 [1 Clusters]

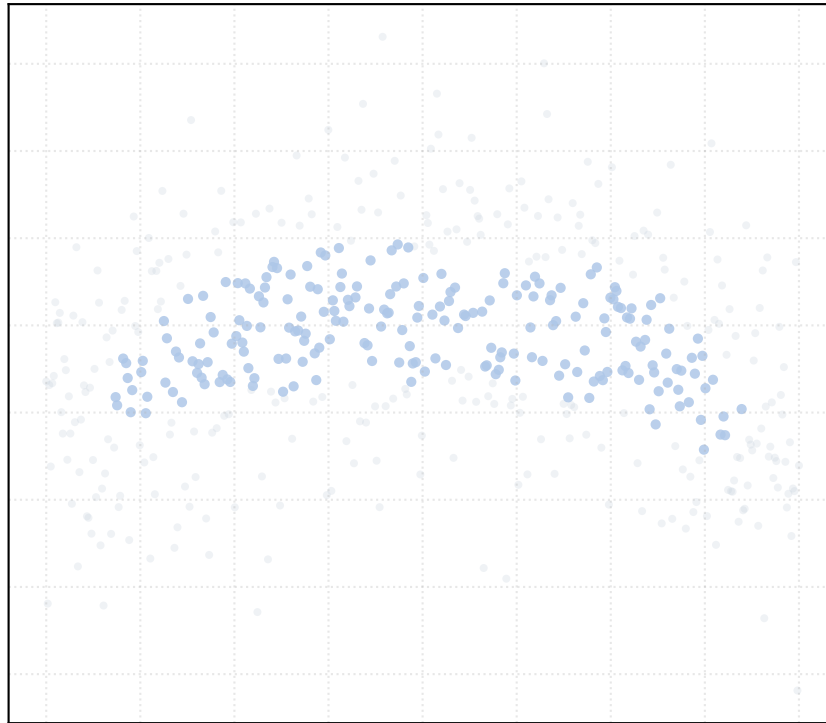


Dataset: circular_arc_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

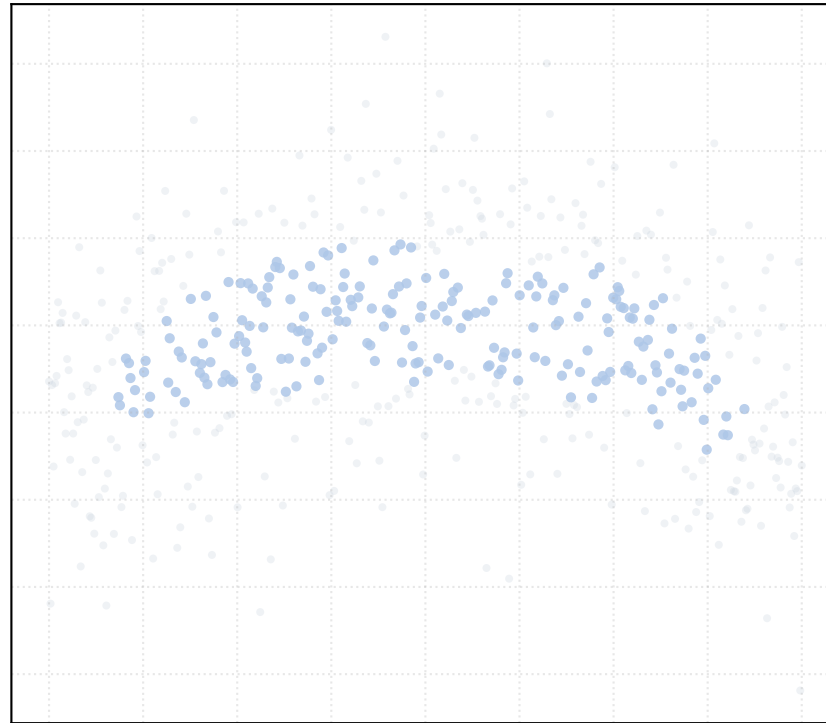
min_intersection = 1 (Default) [1 Clusters]



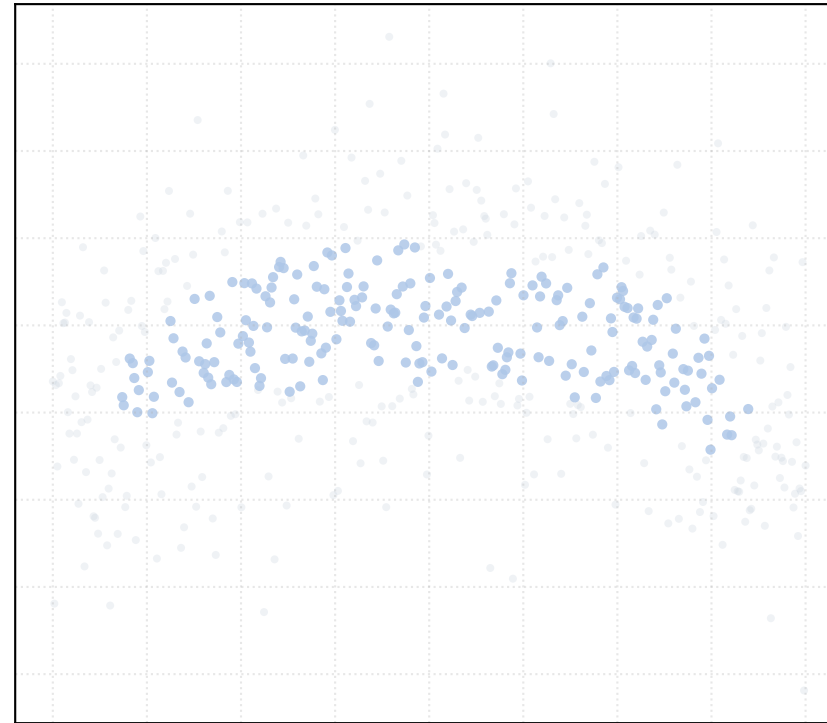
min_intersection = 2 [1 Clusters]



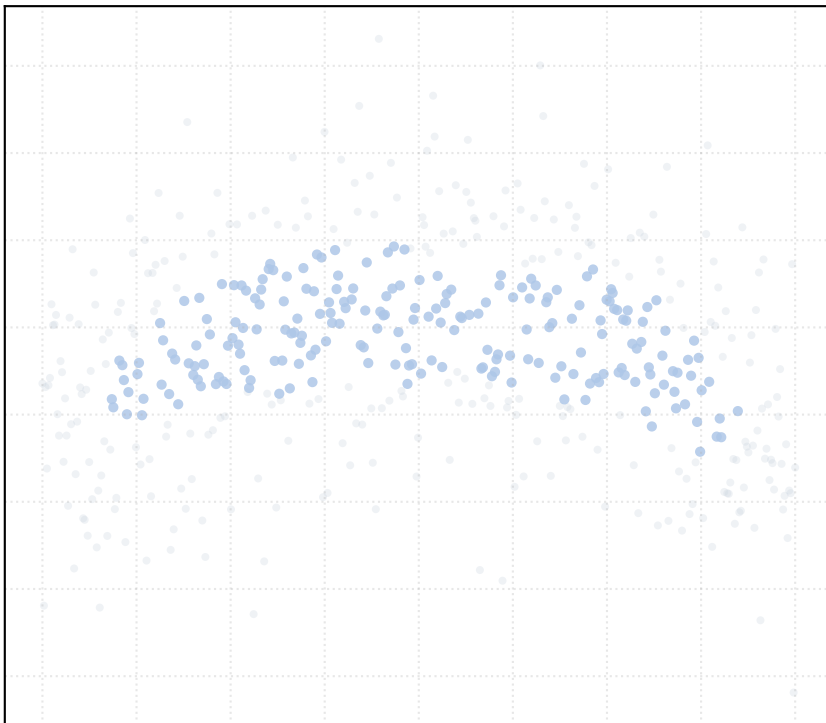
min_intersection = 3 [1 Clusters]



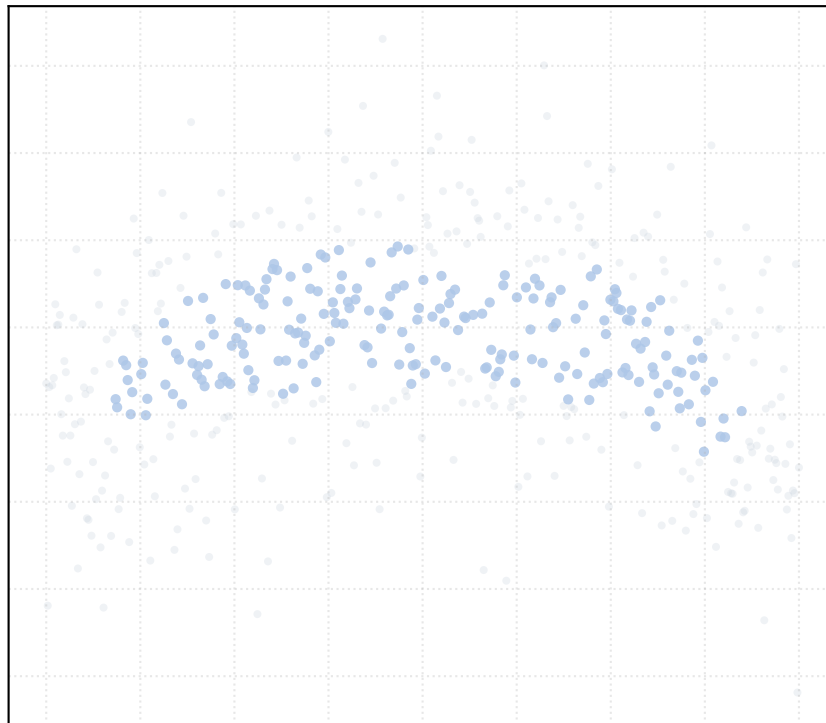
min_intersection = 5 [1 Clusters]



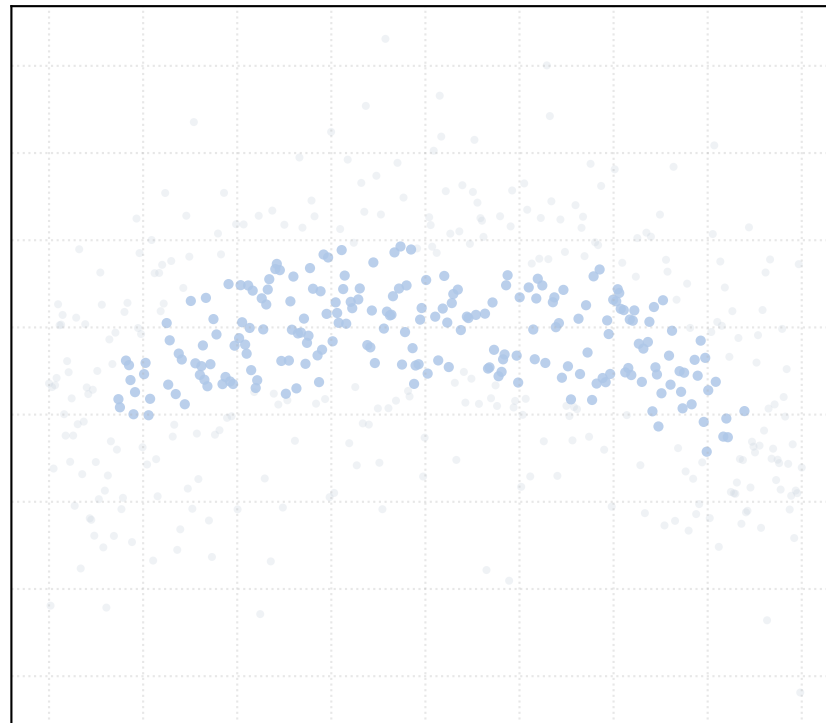
min_intersection = 8 [1 Clusters]



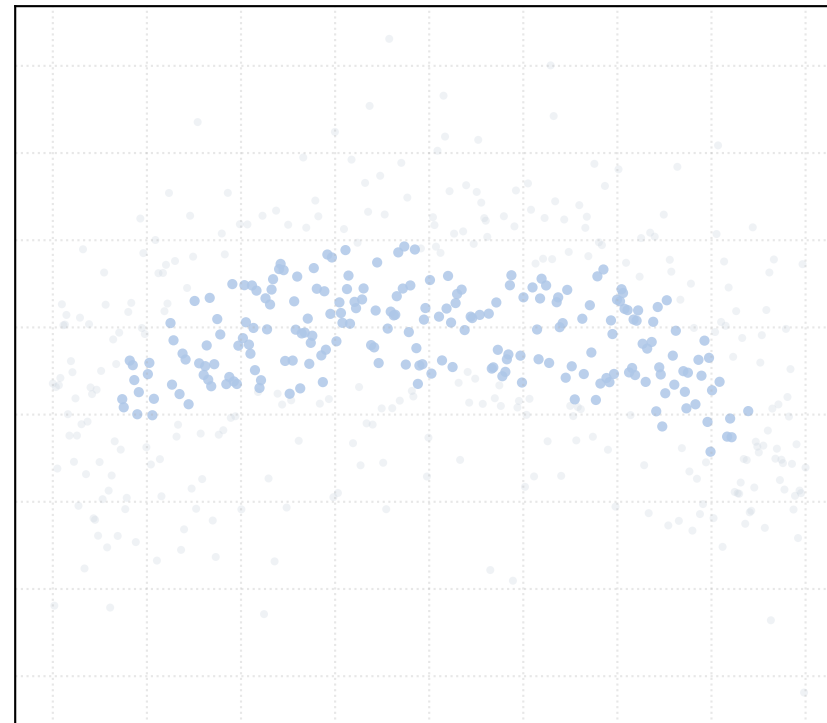
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

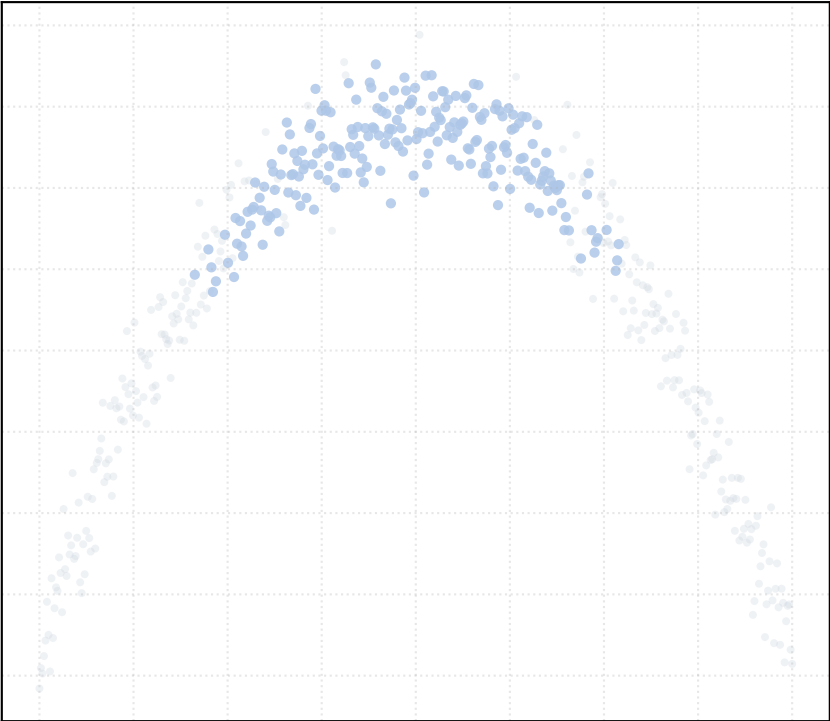


min_intersection = 20 [1 Clusters]

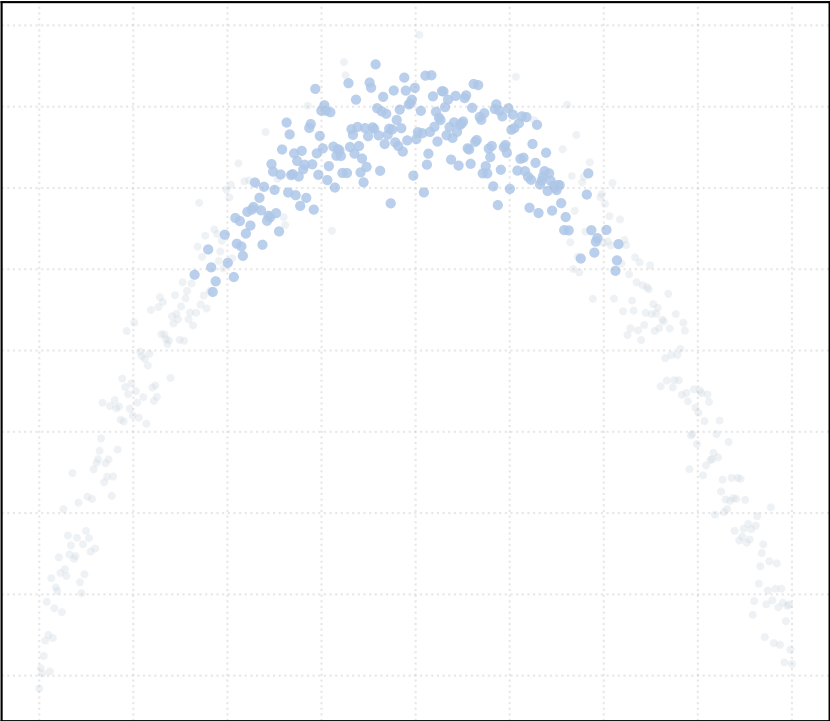


Dataset: circular_arc_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

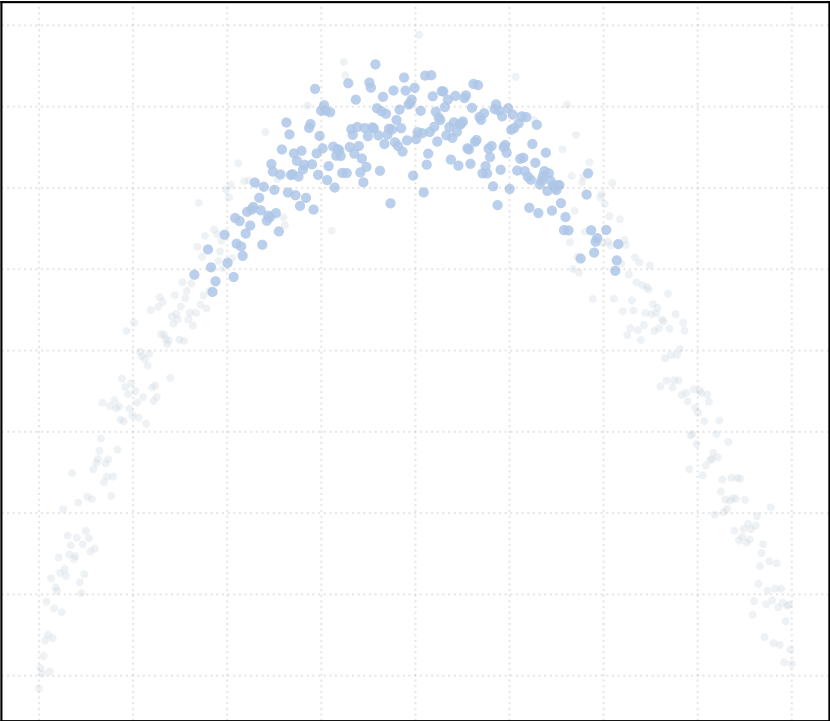
min_intersection = 1 (Default) [1 Clusters]



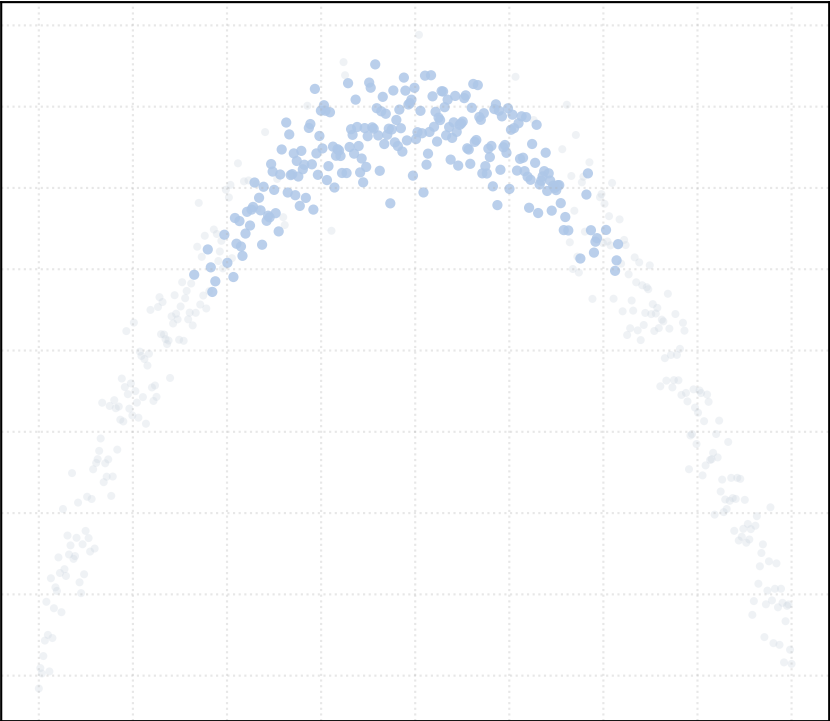
min_intersection = 2 [1 Clusters]



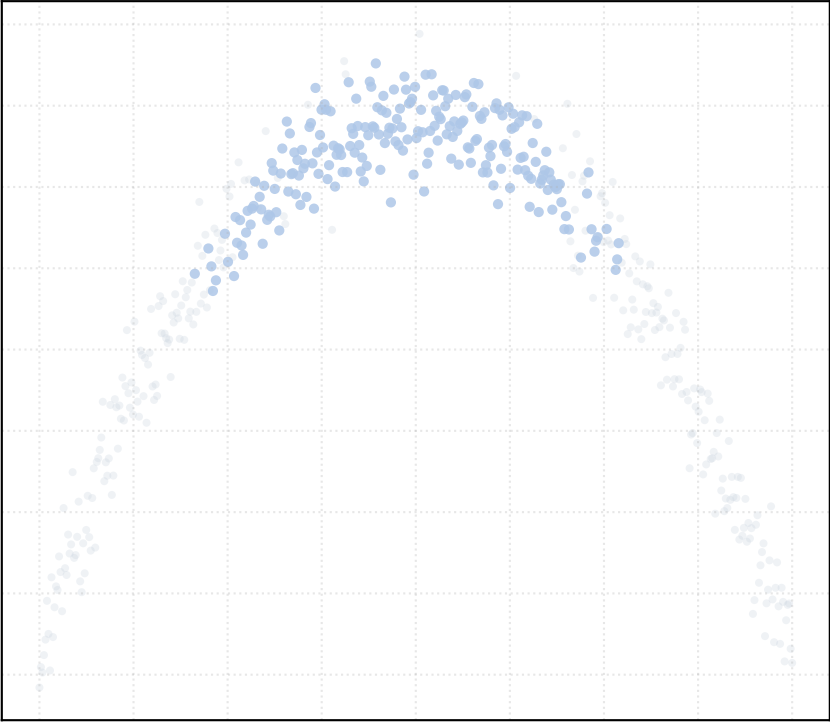
min_intersection = 3 [1 Clusters]



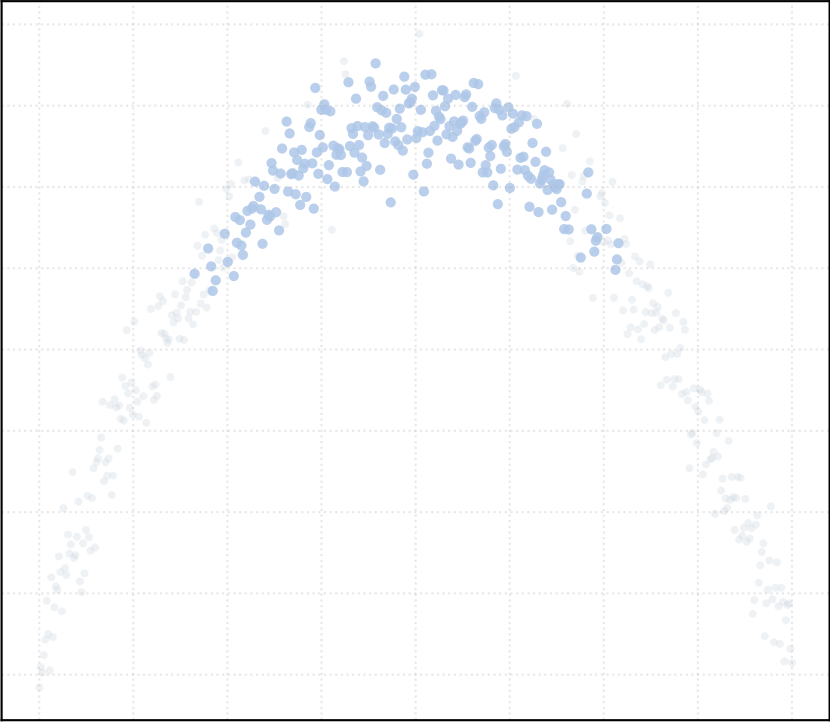
min_intersection = 5 [1 Clusters]



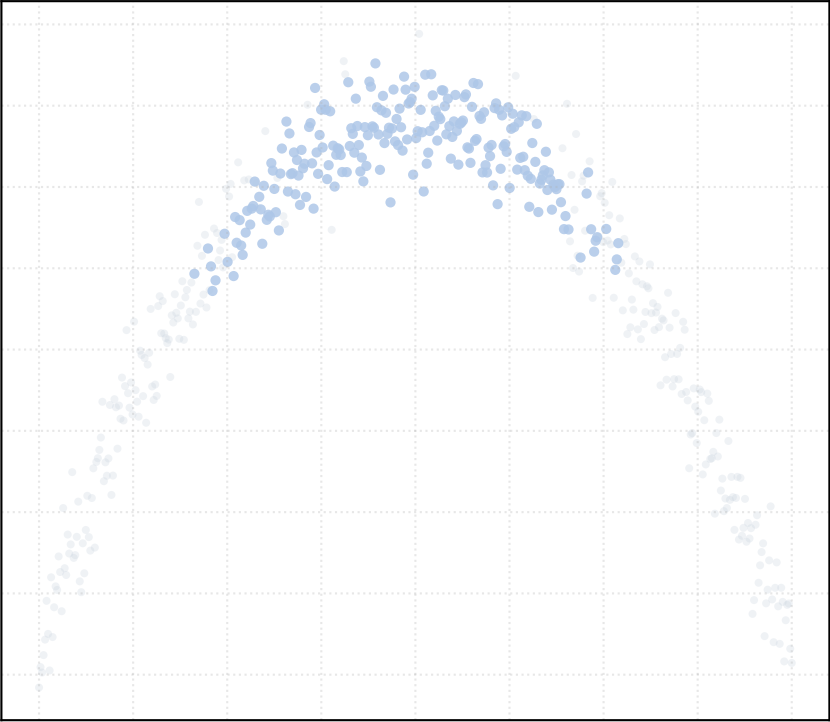
min_intersection = 8 [1 Clusters]



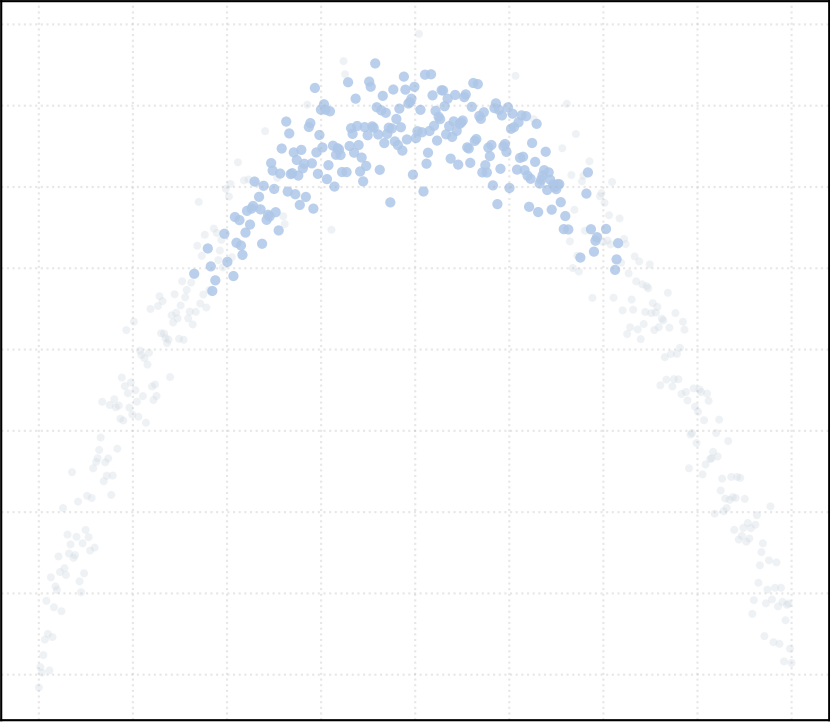
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

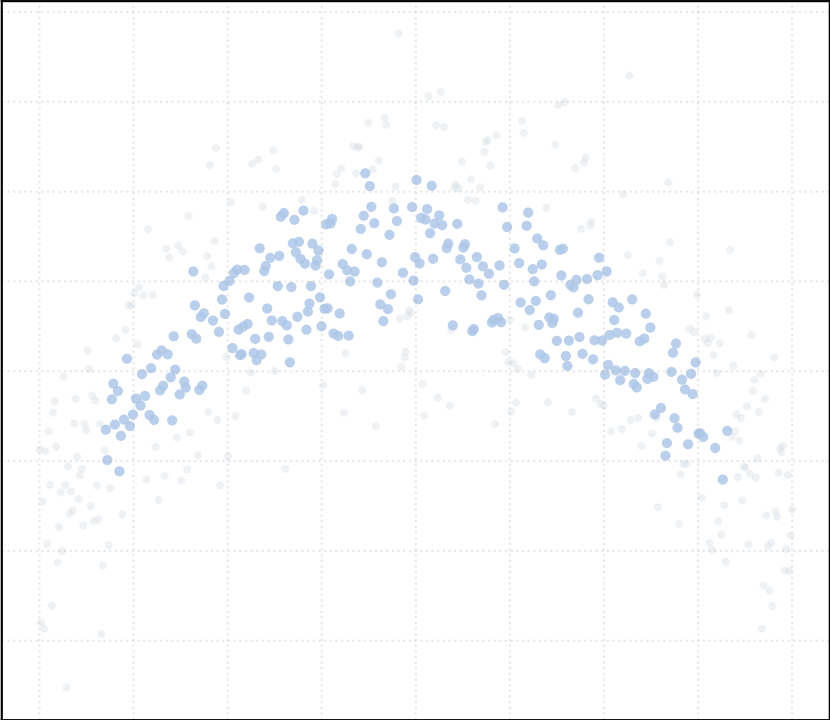


min_intersection = 20 [1 Clusters]

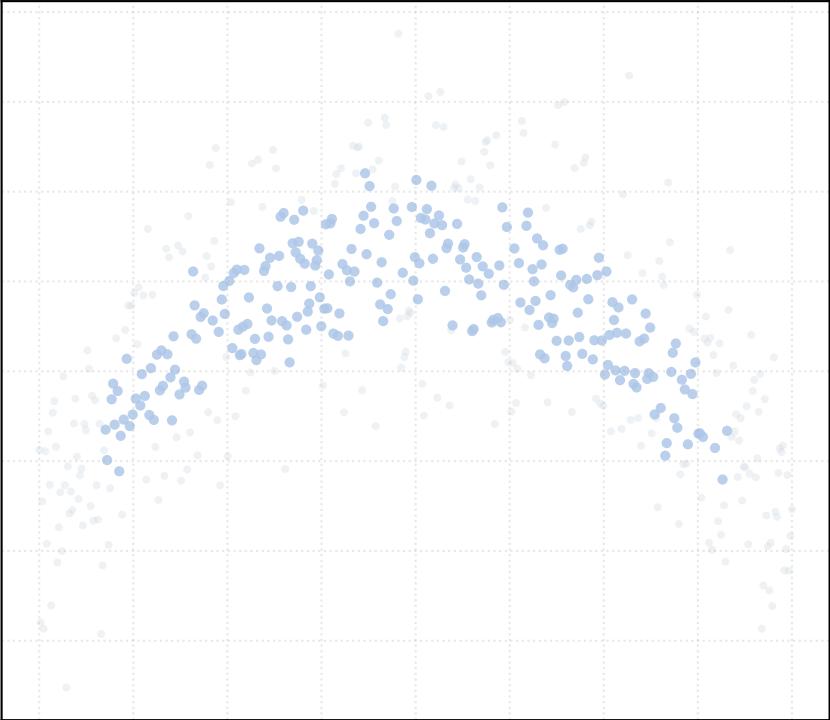


Dataset: circular_arc_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

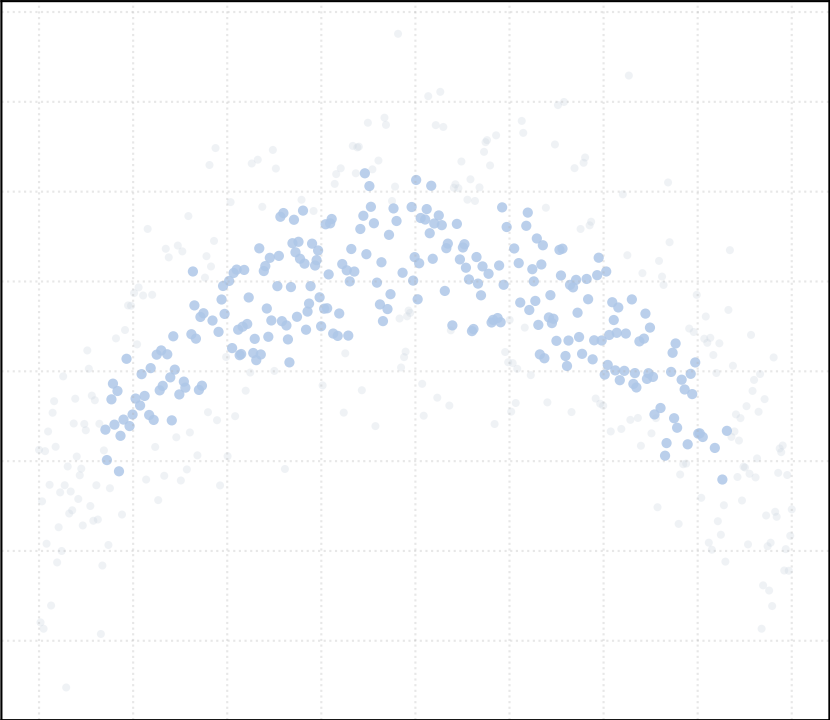
min_intersection = 1 (Default) [1 Clusters]



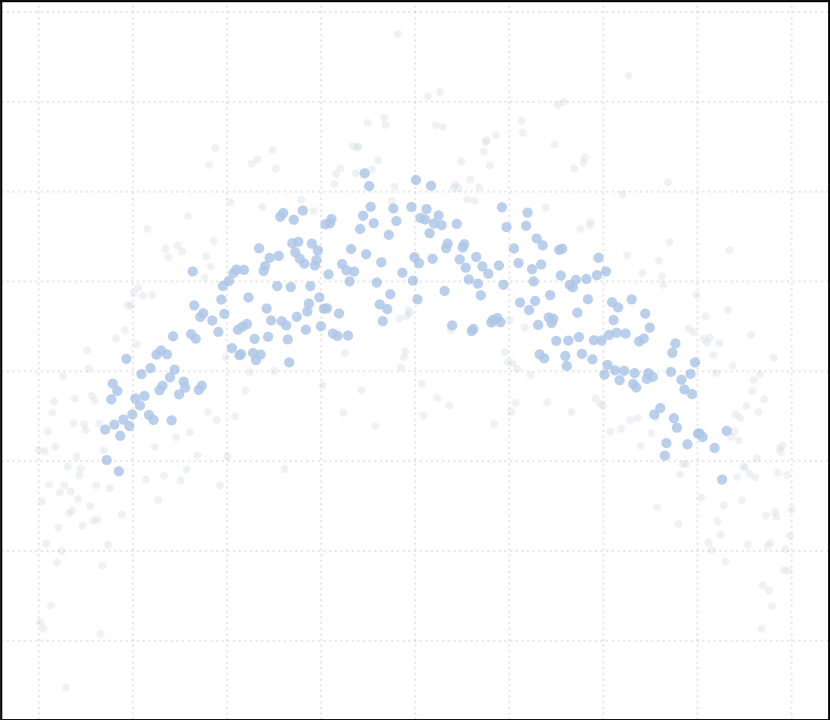
min_intersection = 2 [1 Clusters]



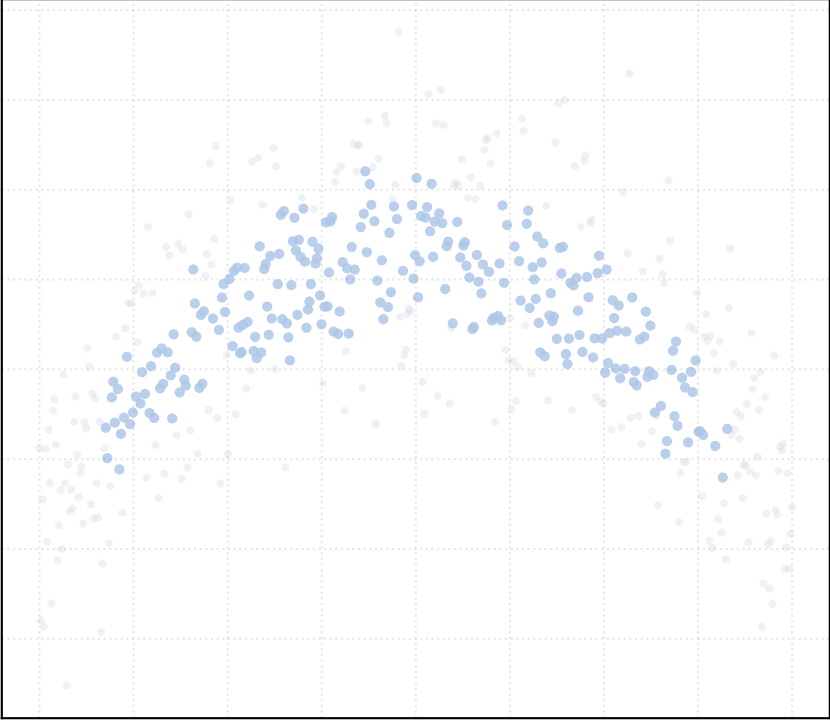
min_intersection = 3 [1 Clusters]



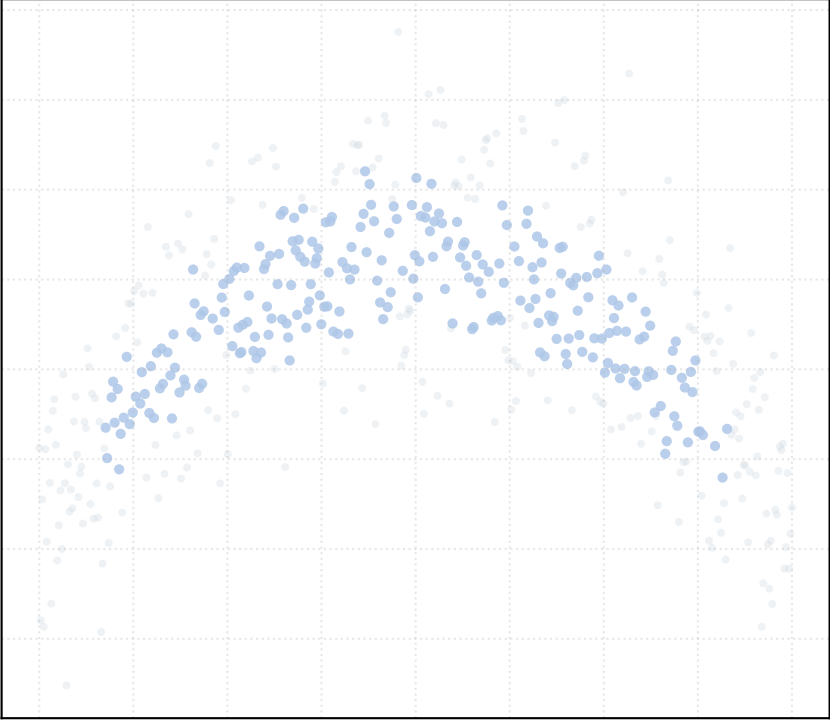
min_intersection = 5 [1 Clusters]



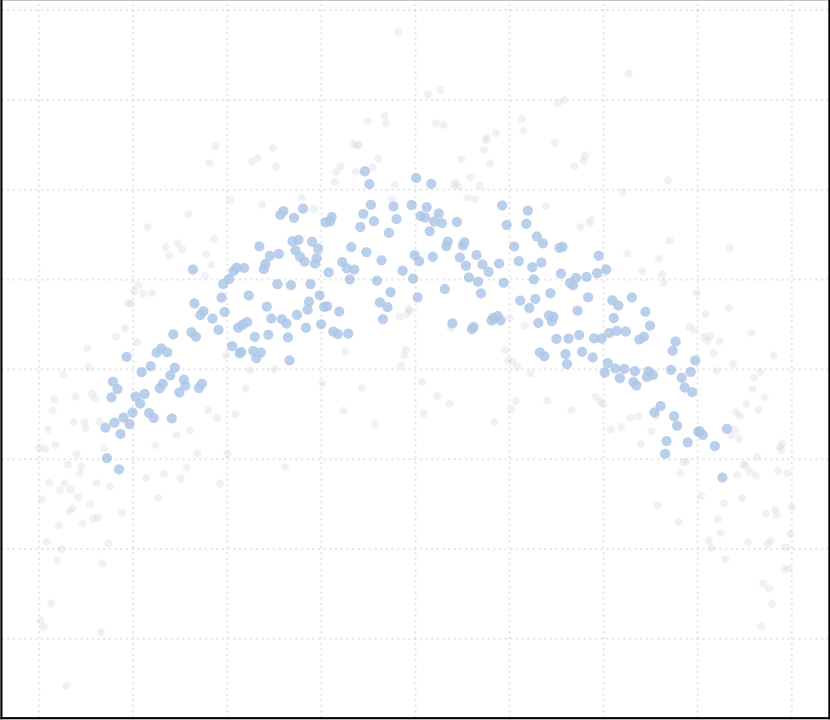
min_intersection = 8 [1 Clusters]



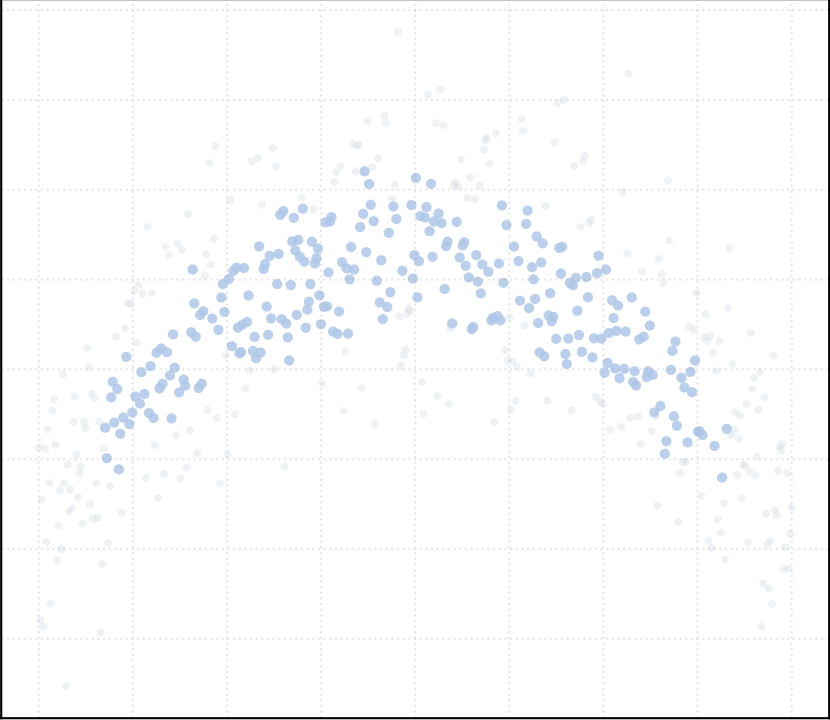
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

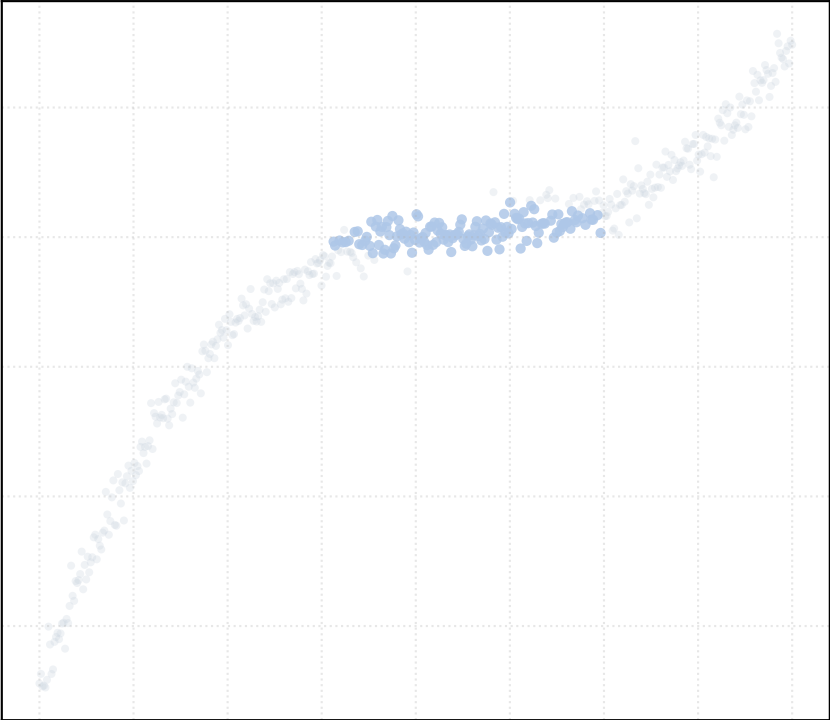


min_intersection = 20 [1 Clusters]

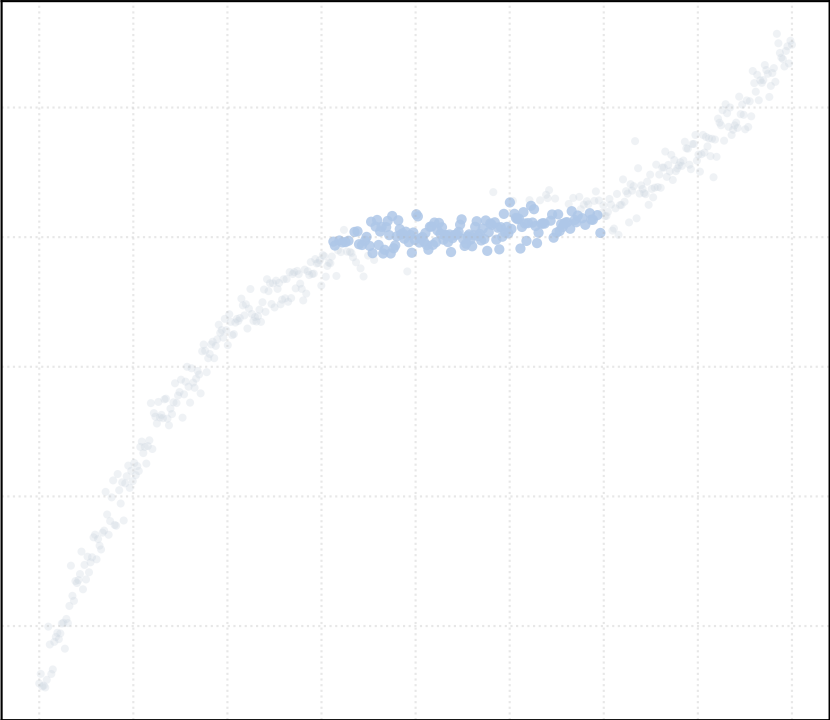


Dataset: cubic_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

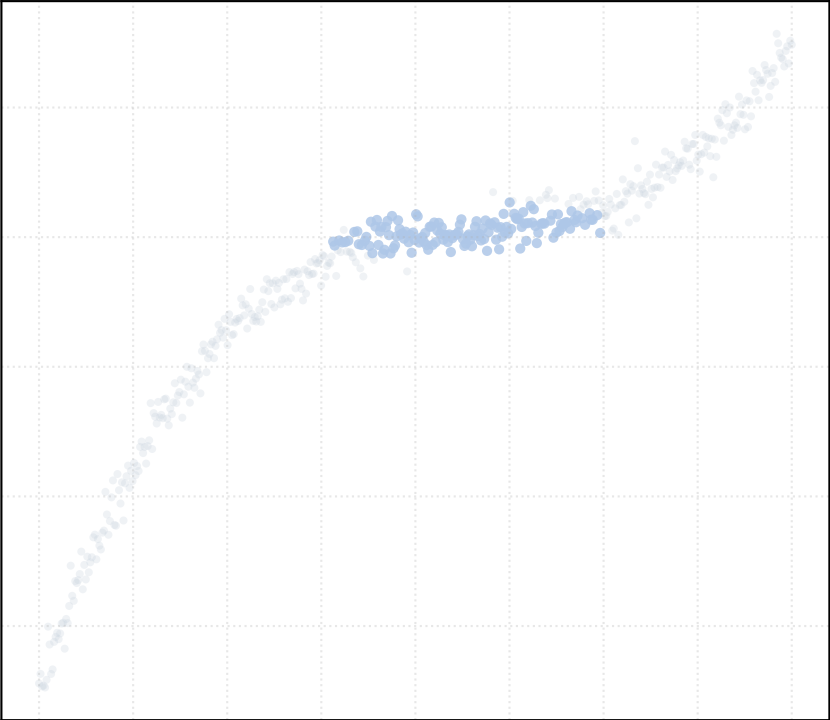
min_intersection = 1 (Default) [1 Clusters]



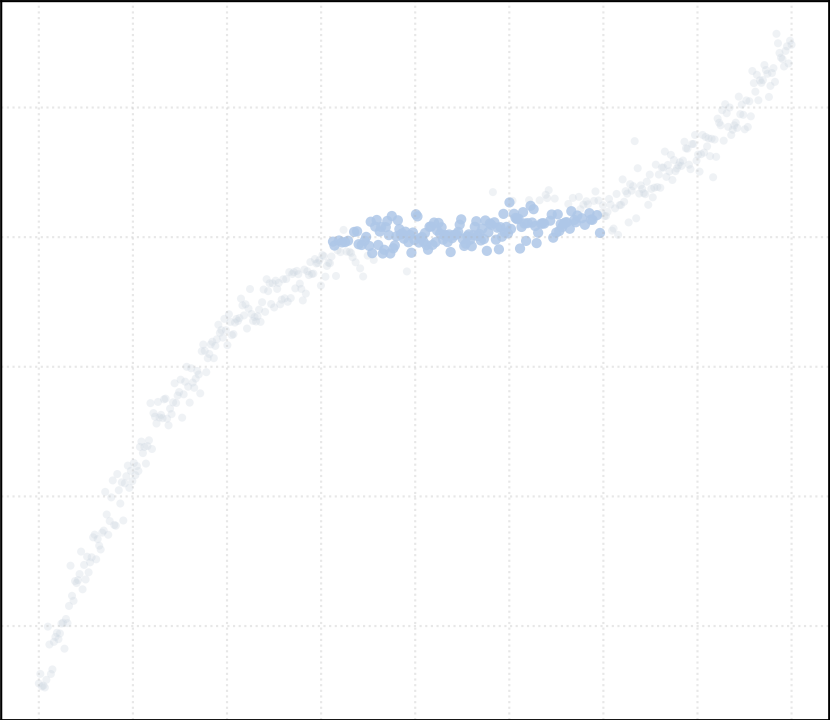
min_intersection = 2 [1 Clusters]



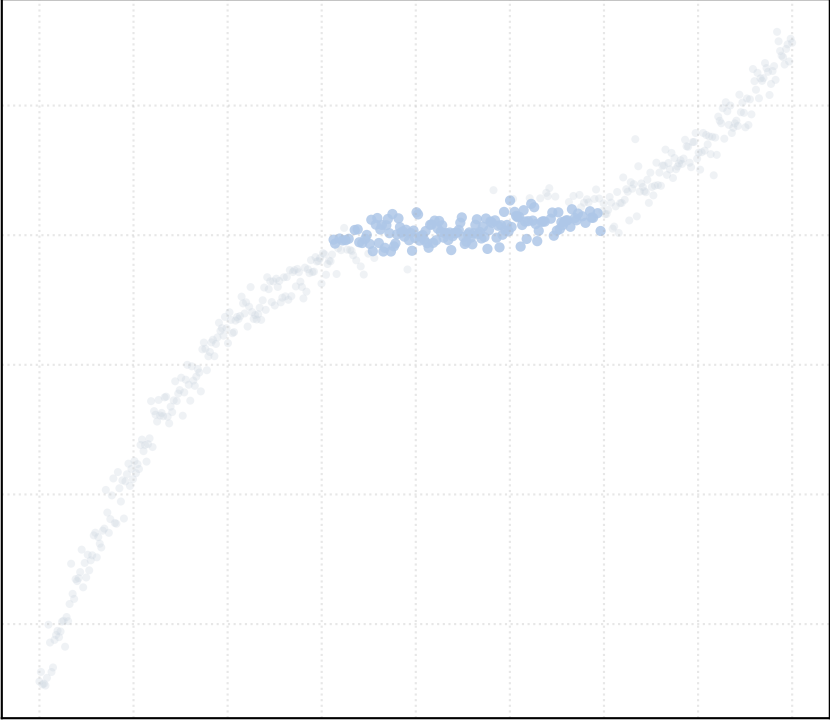
min_intersection = 3 [1 Clusters]



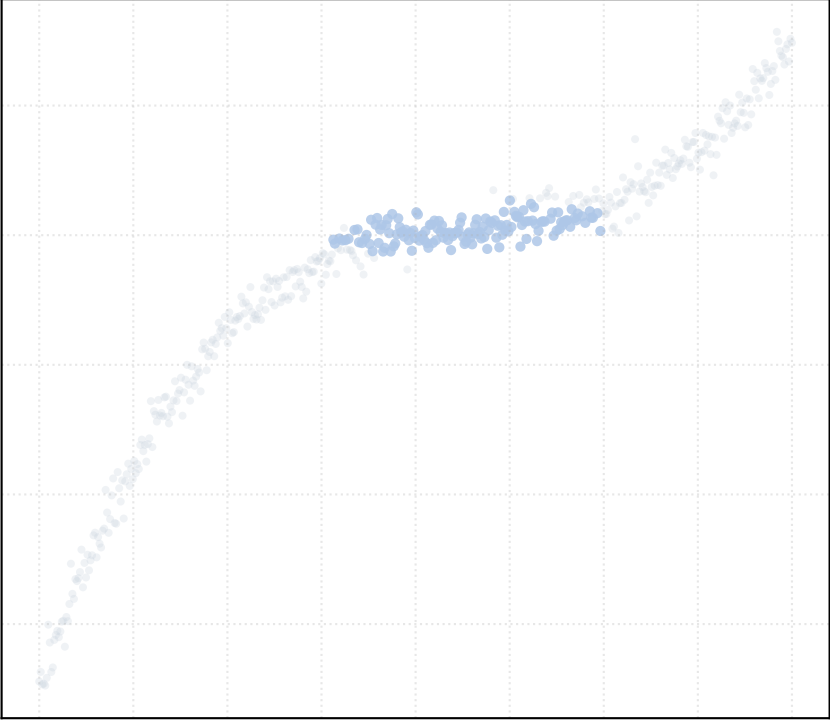
min_intersection = 5 [1 Clusters]



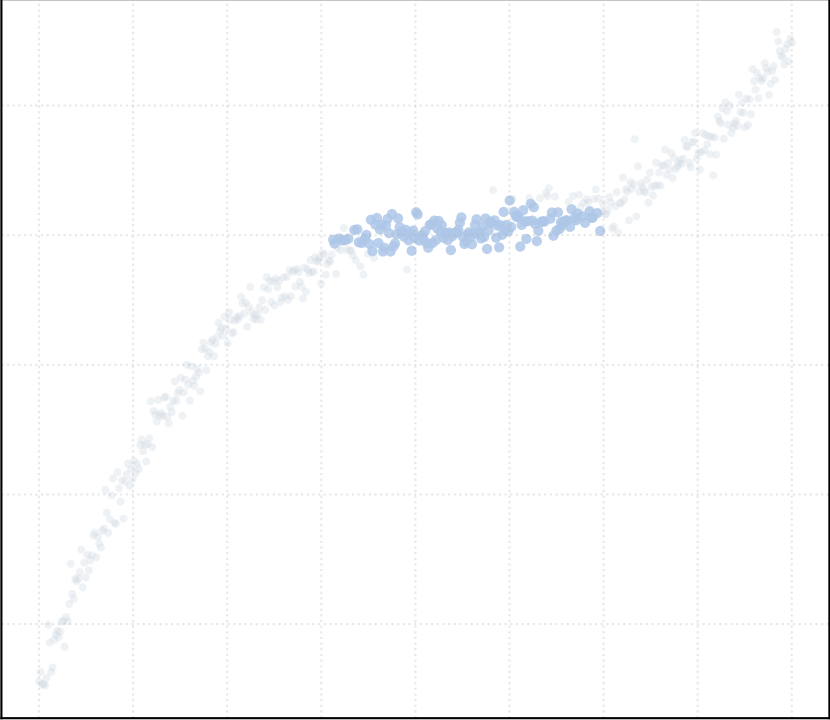
min_intersection = 8 [1 Clusters]



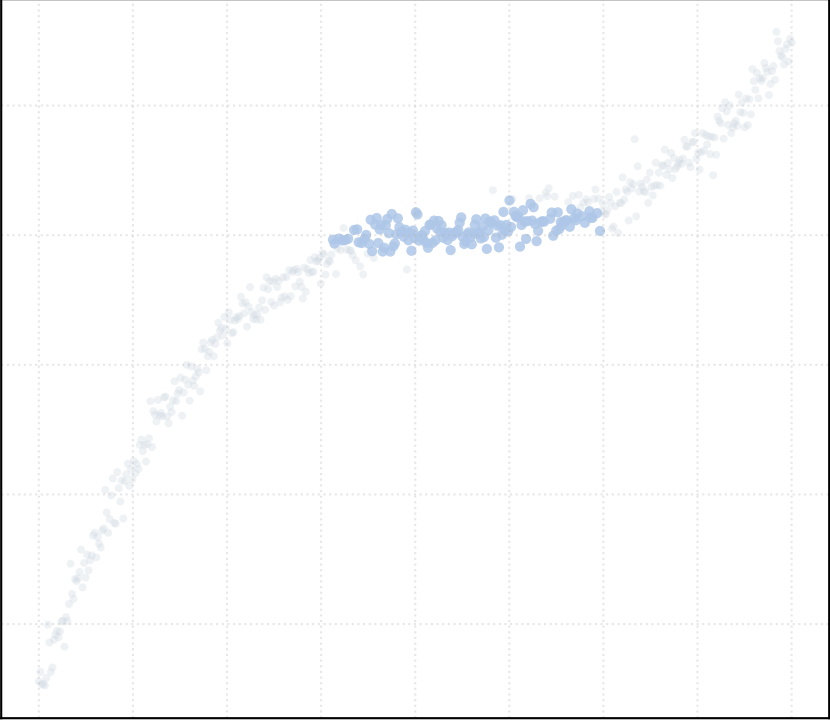
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

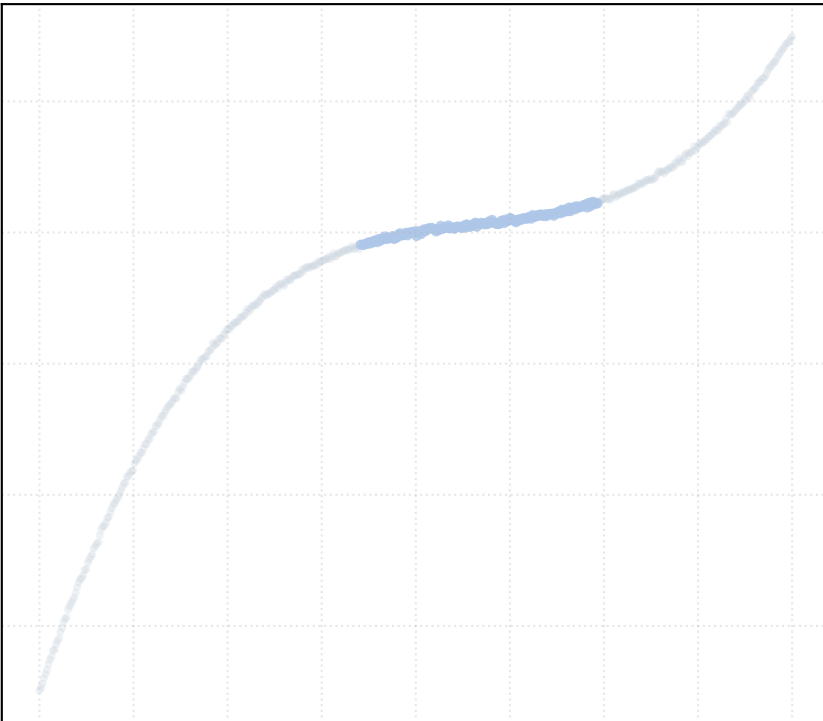


min_intersection = 20 [1 Clusters]

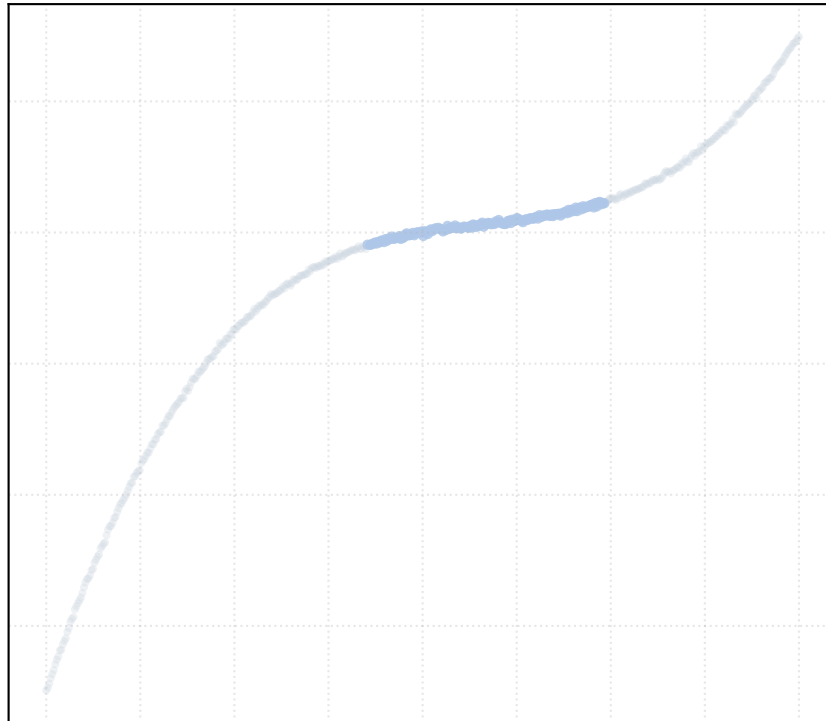


Dataset: cubic_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

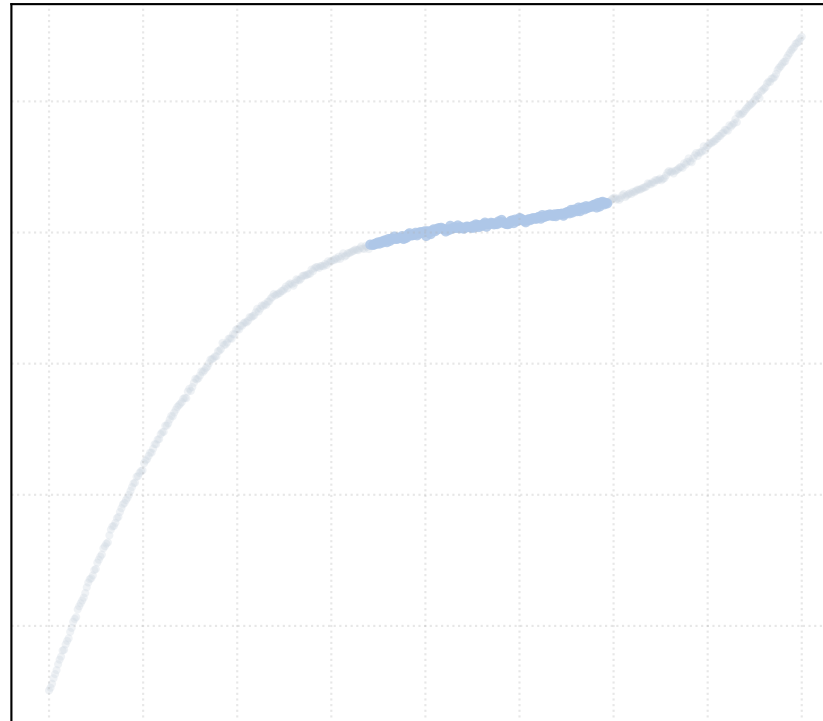
min_intersection = 1 (Default) [1 Clusters]



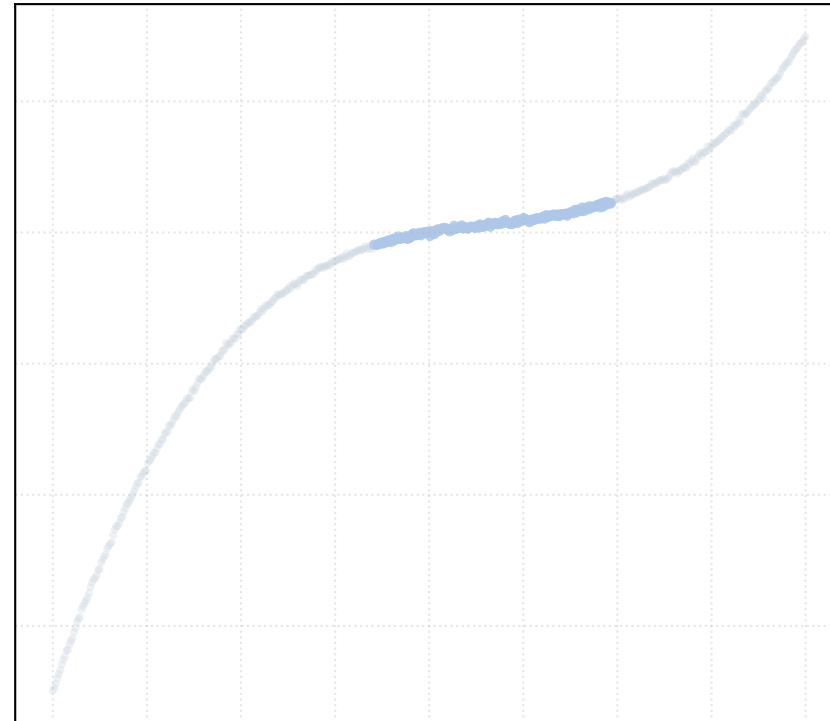
min_intersection = 2 [1 Clusters]



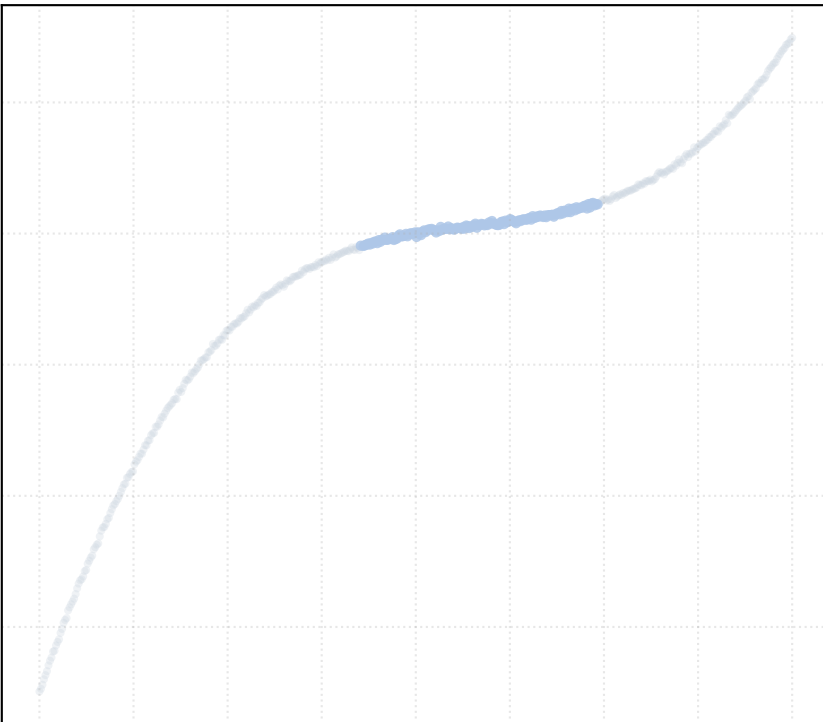
min_intersection = 3 [1 Clusters]



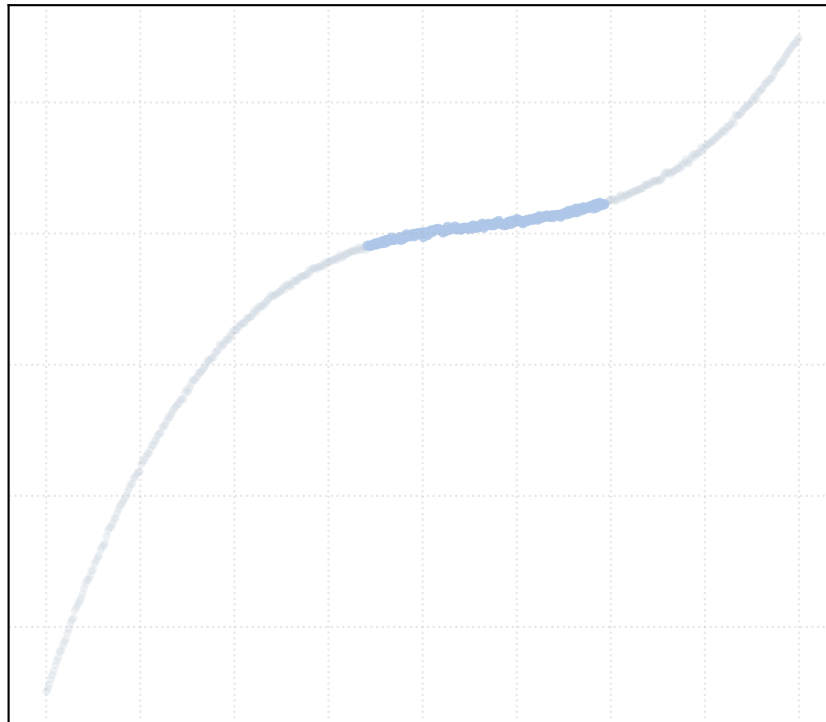
min_intersection = 5 [1 Clusters]



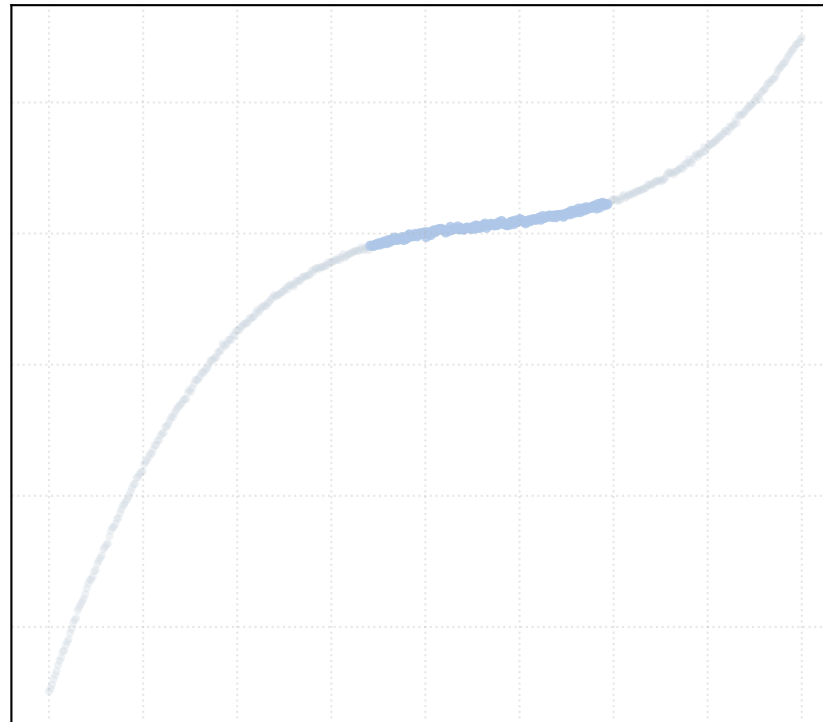
min_intersection = 8 [1 Clusters]



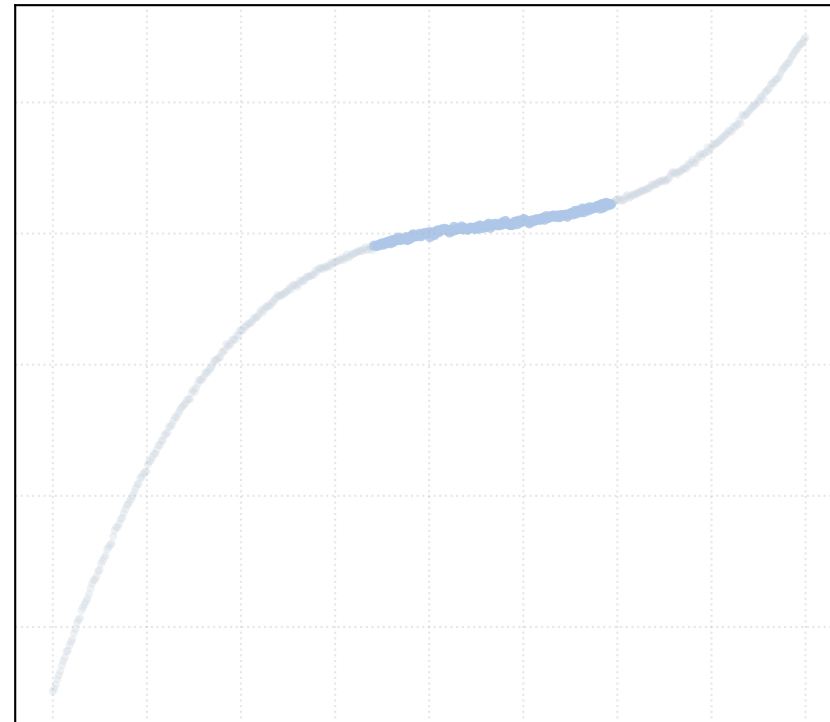
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

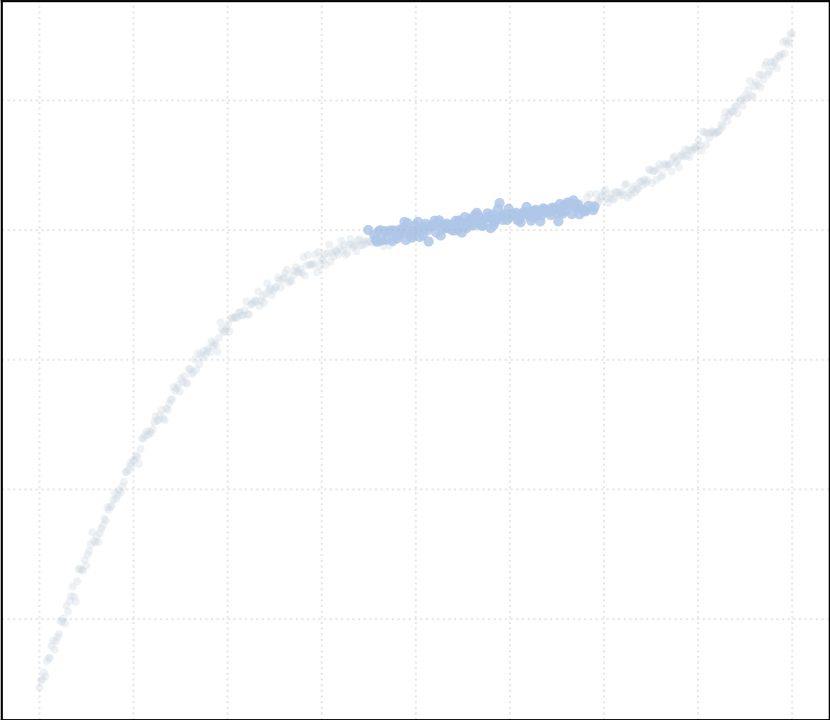


min_intersection = 20 [1 Clusters]

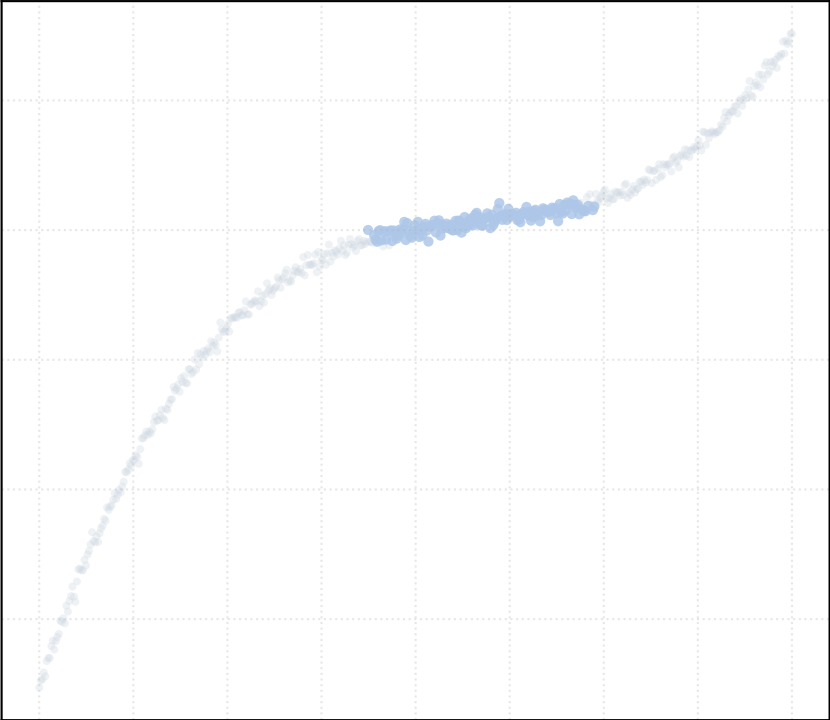


Dataset: cubic_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

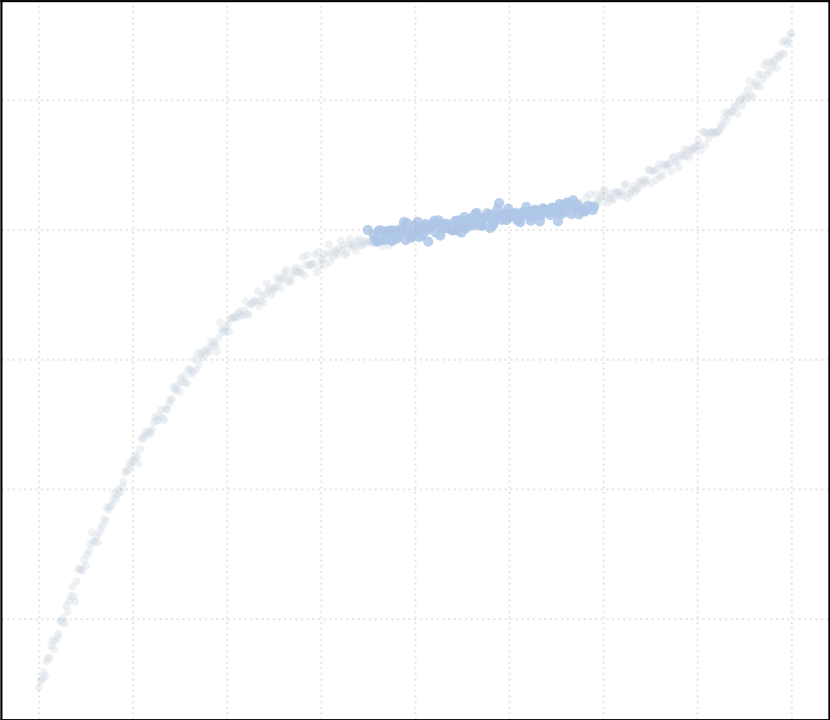
min_intersection = 1 (Default) [1 Clusters]



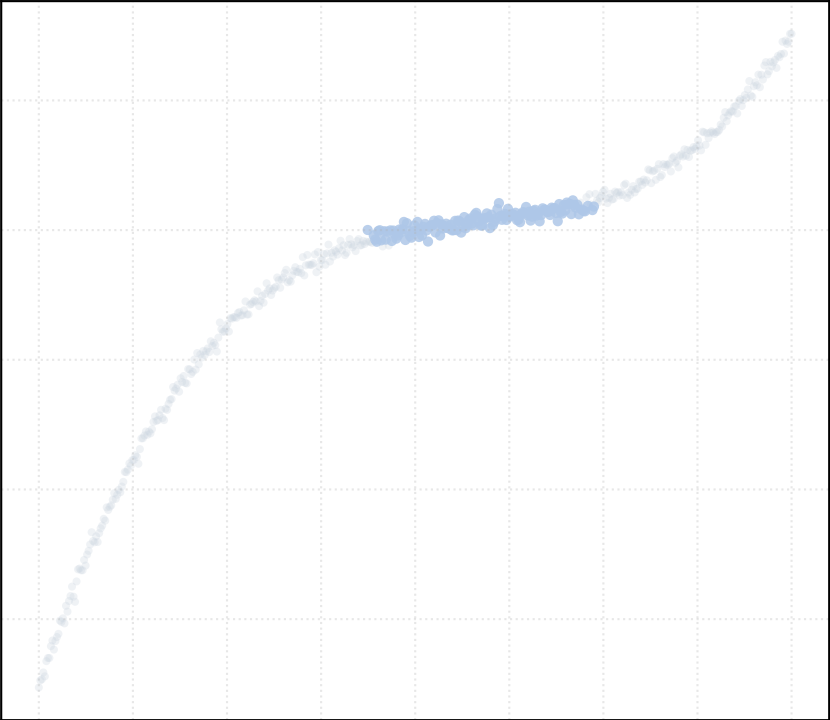
min_intersection = 2 [1 Clusters]



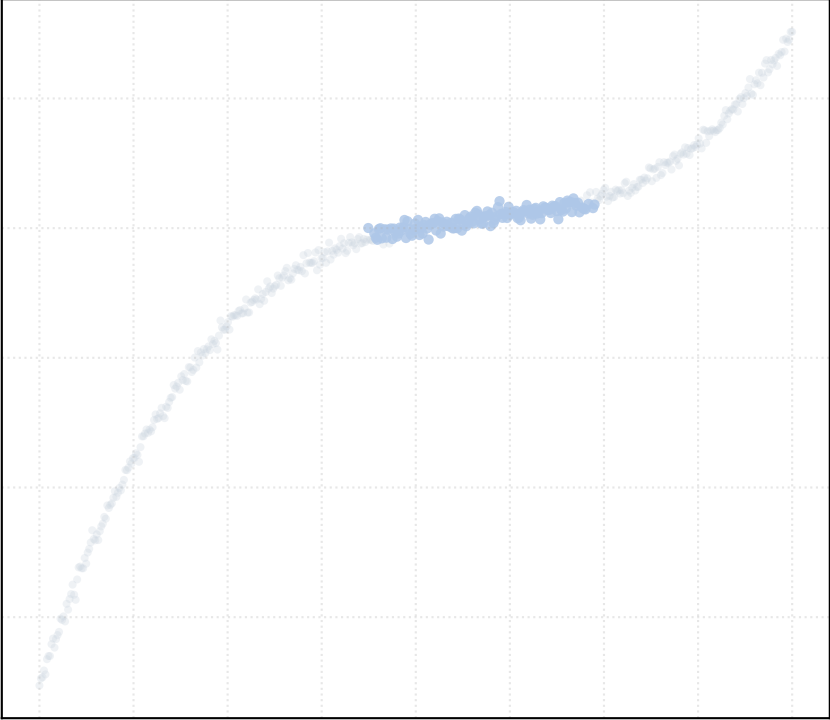
min_intersection = 3 [1 Clusters]



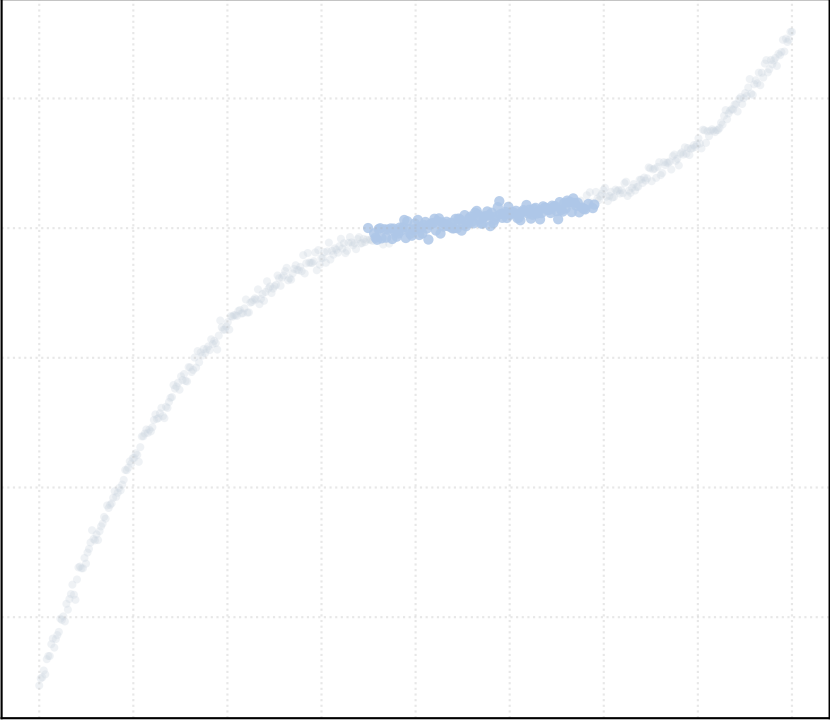
min_intersection = 5 [1 Clusters]



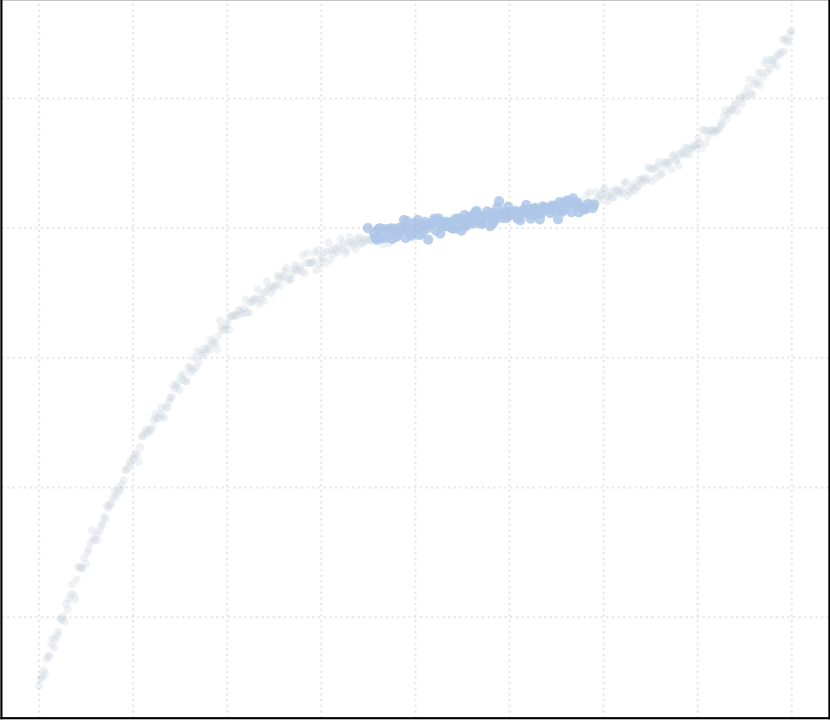
min_intersection = 8 [1 Clusters]



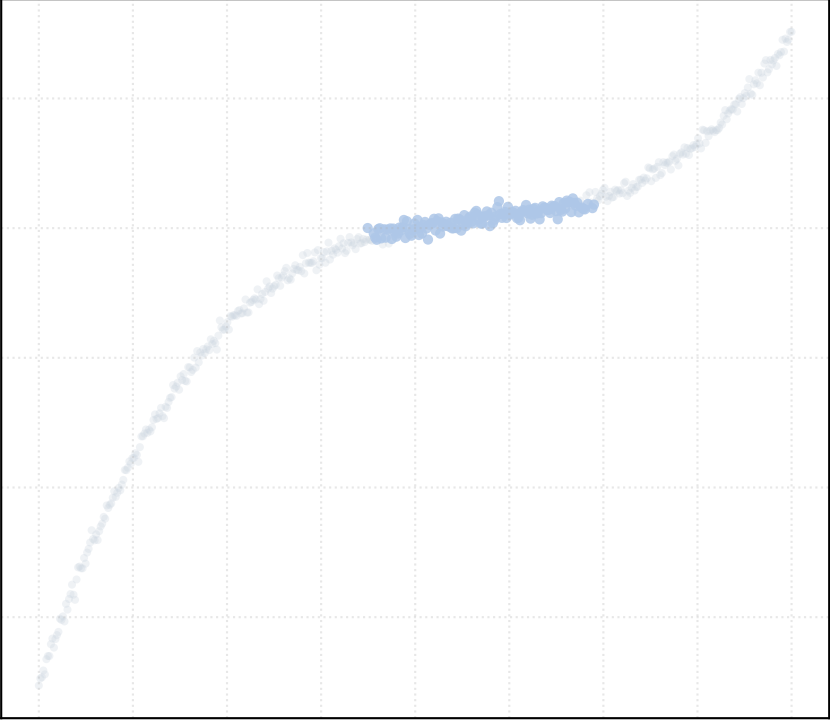
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

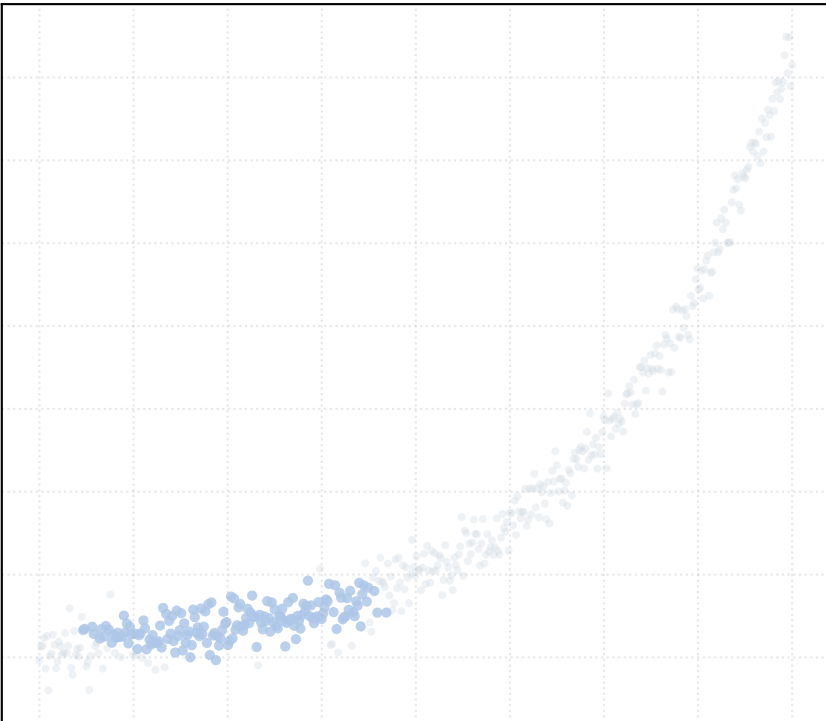


min_intersection = 20 [1 Clusters]

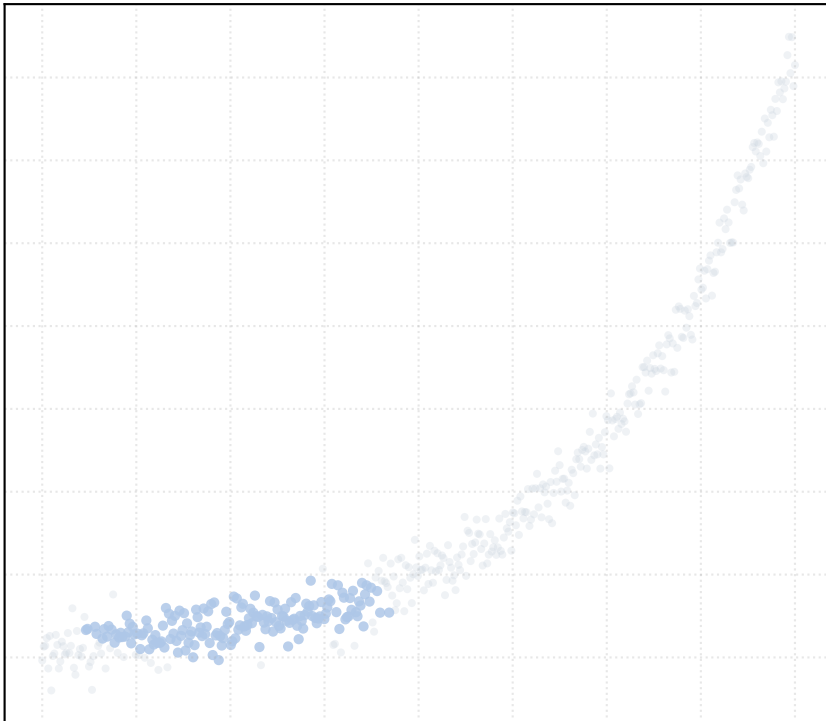


Dataset: exponential_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

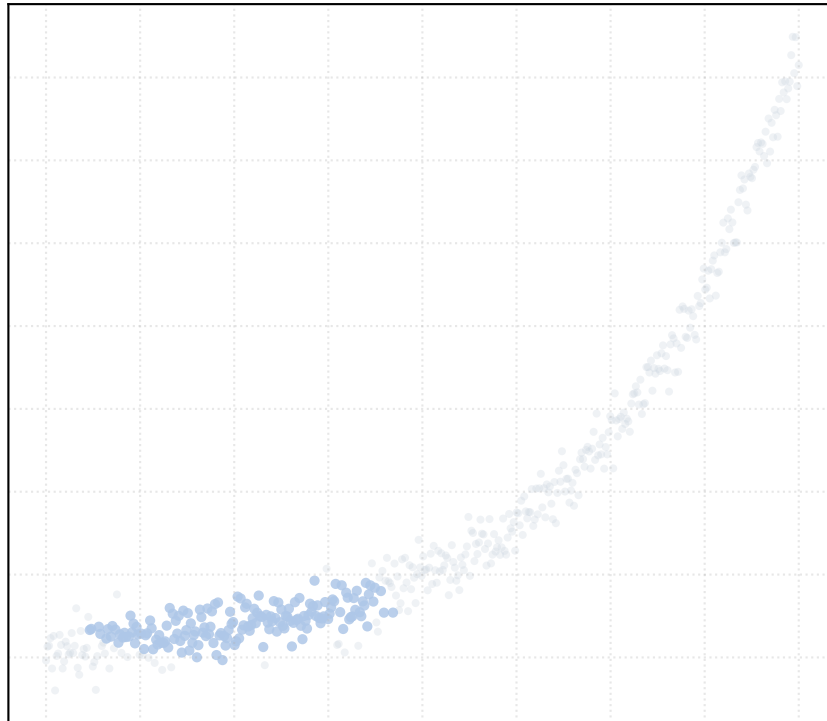
min_intersection = 1 (Default) [1 Clusters]



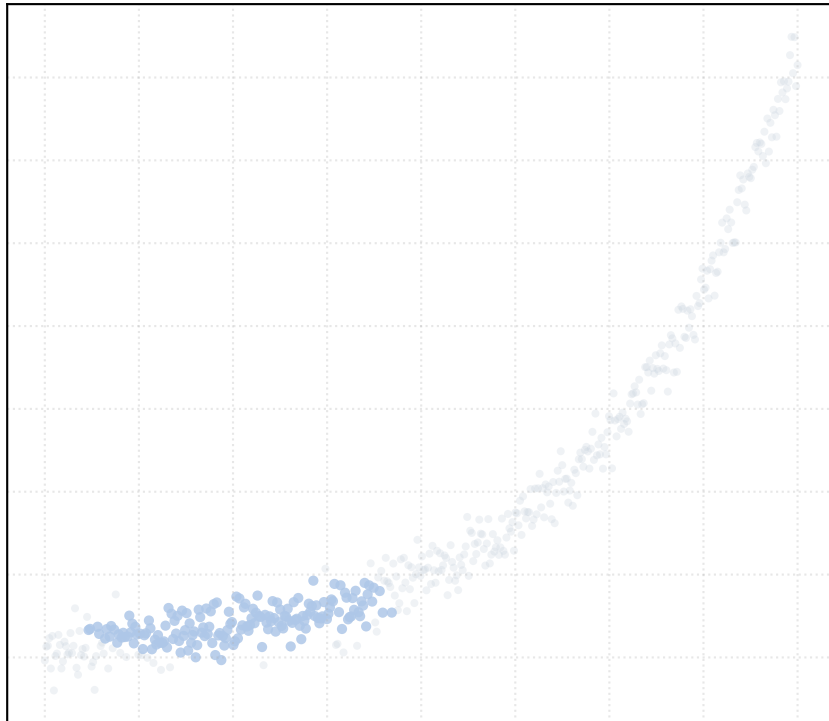
min_intersection = 2 [1 Clusters]



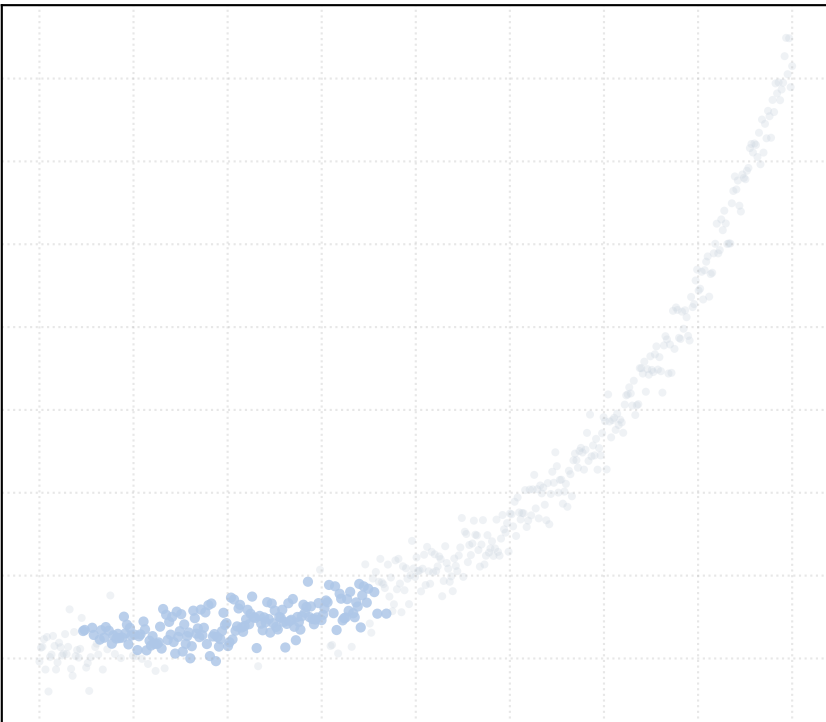
min_intersection = 3 [1 Clusters]



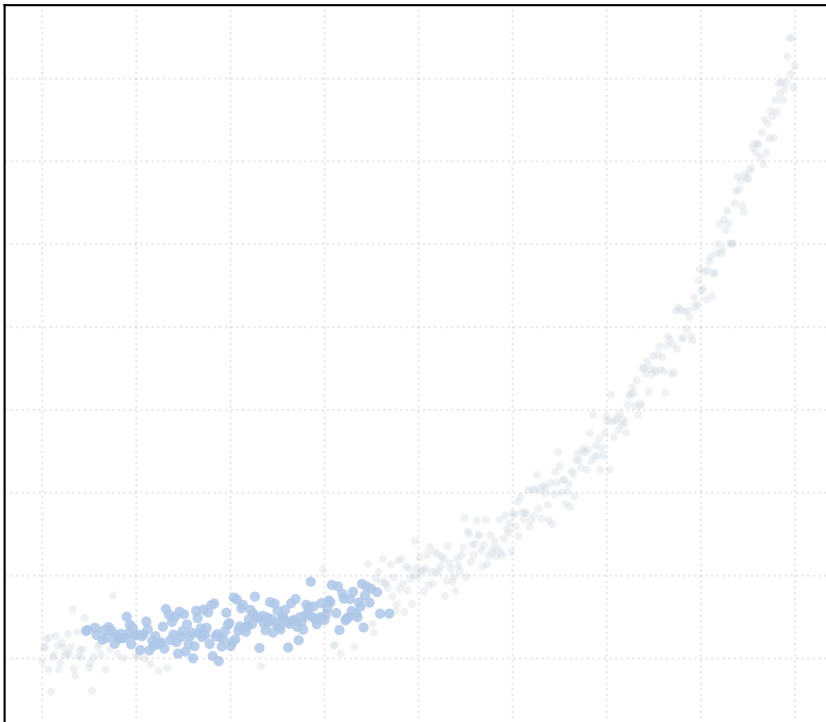
min_intersection = 5 [1 Clusters]



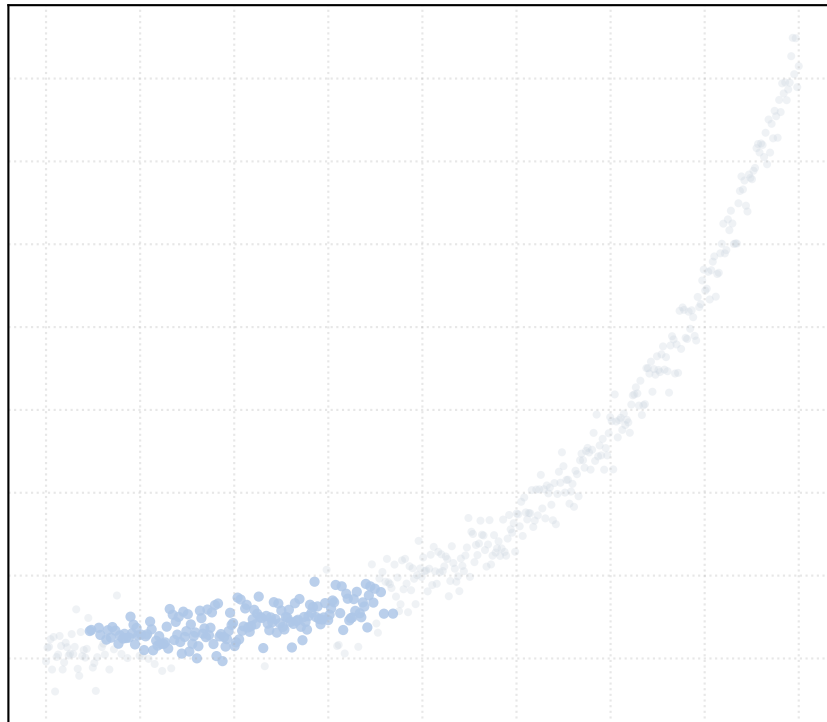
min_intersection = 8 [1 Clusters]



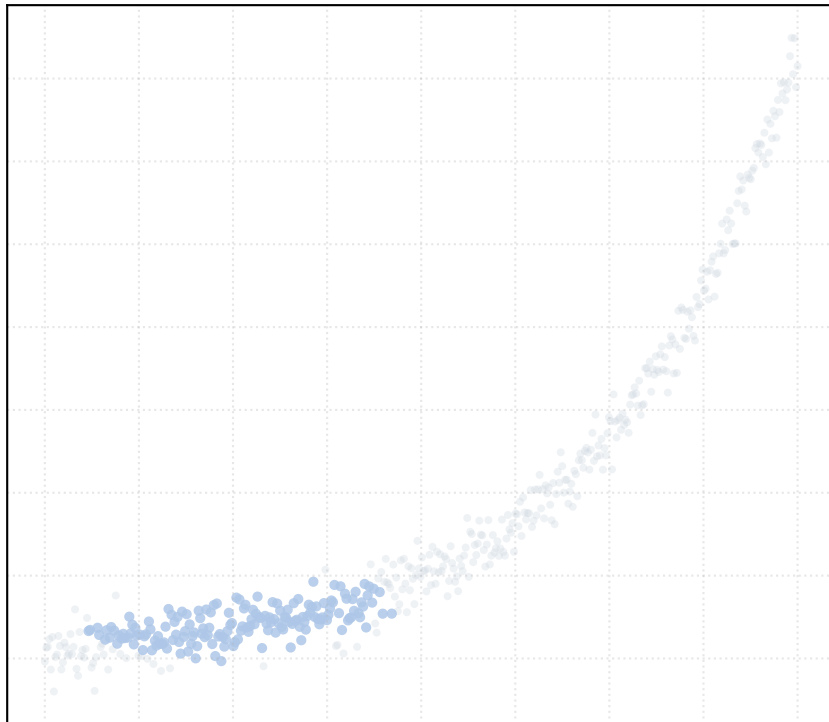
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

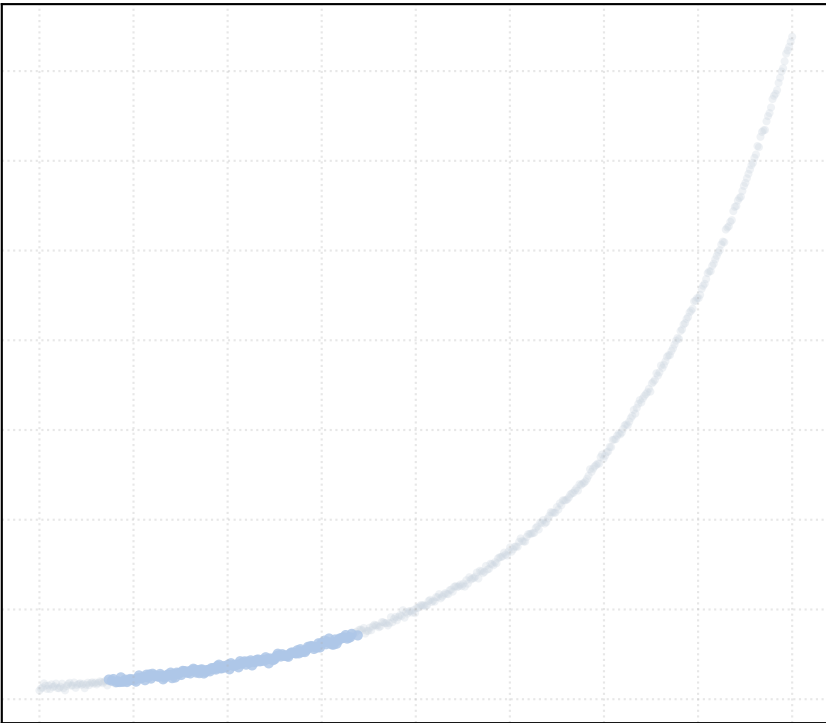


min_intersection = 20 [1 Clusters]

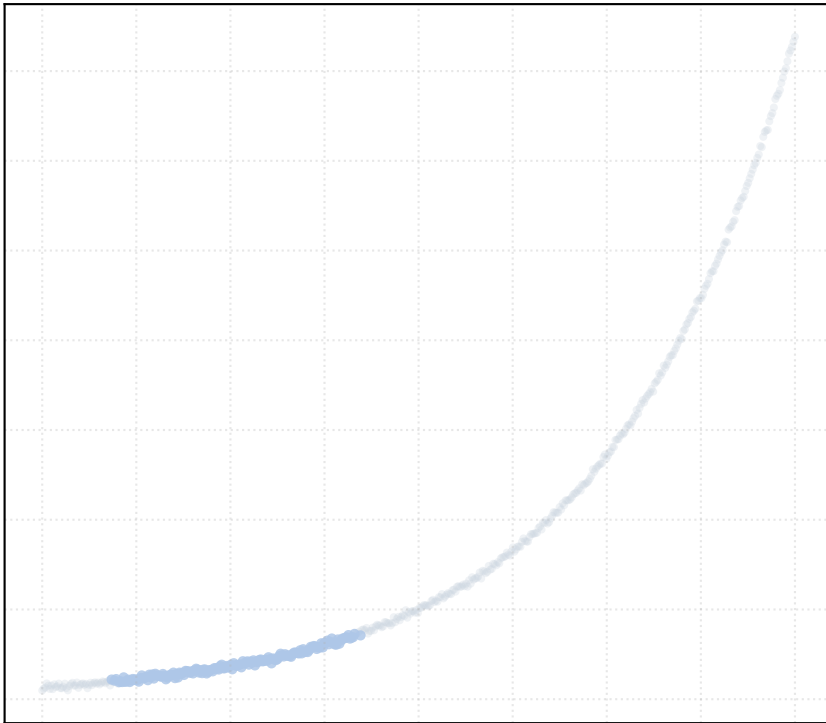


Dataset: exponential_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

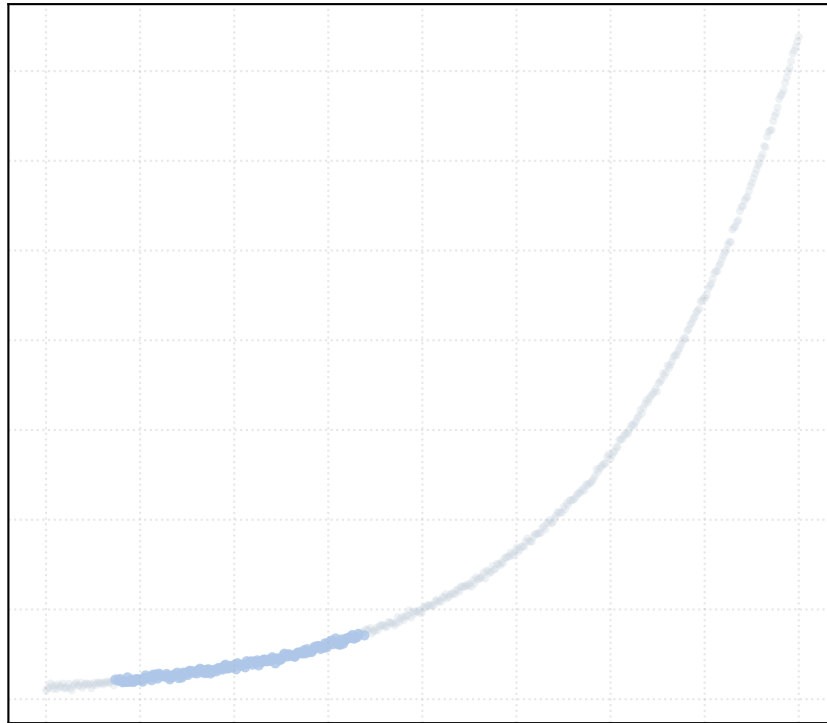
min_intersection = 1 (Default) [1 Clusters]



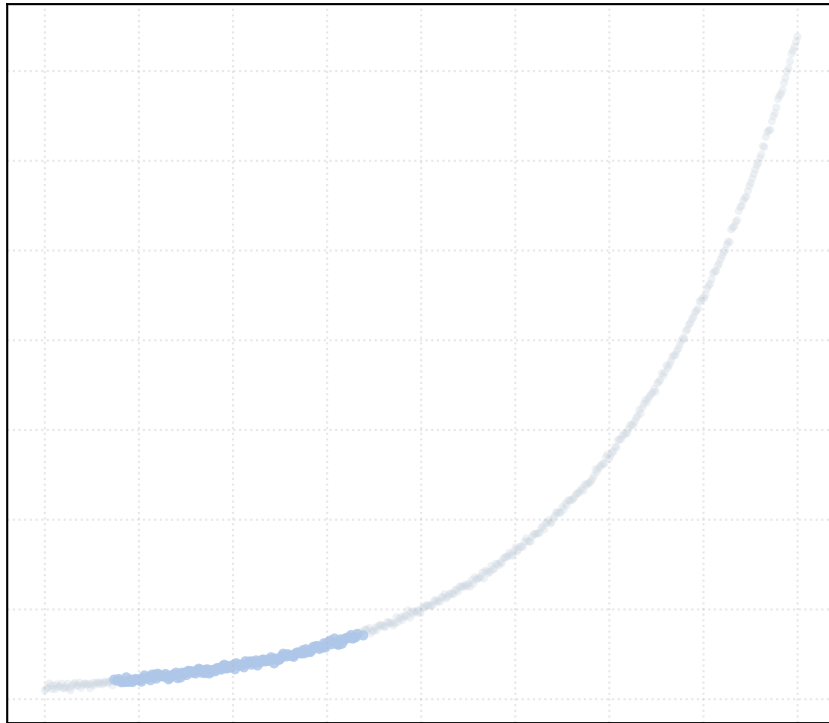
min_intersection = 2 [1 Clusters]



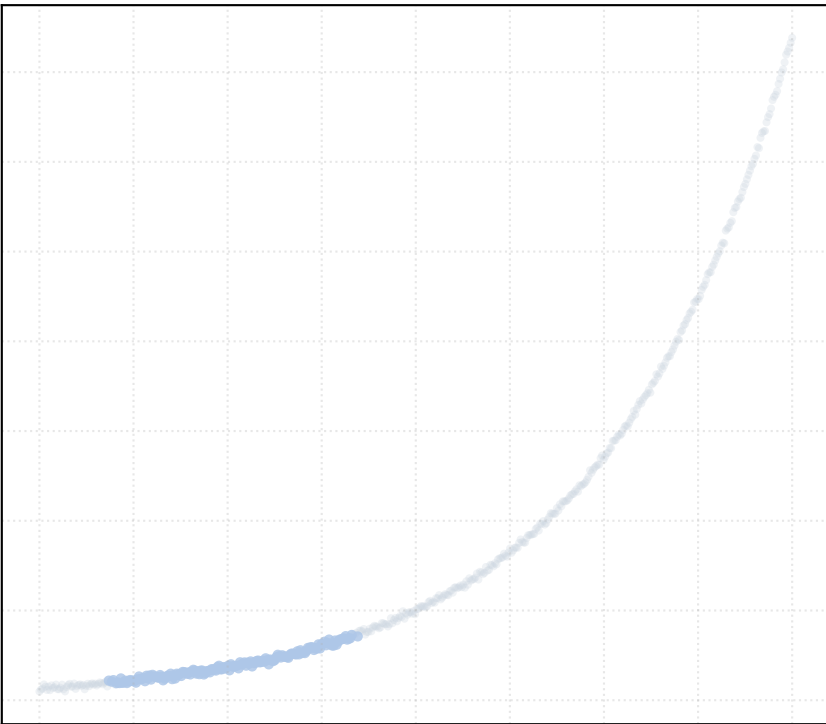
min_intersection = 3 [1 Clusters]



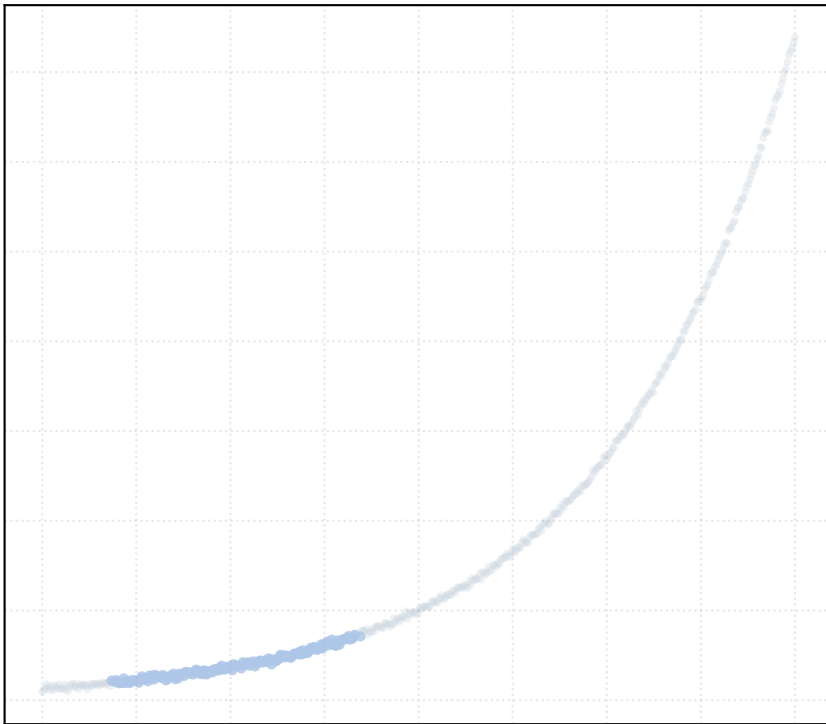
min_intersection = 5 [1 Clusters]



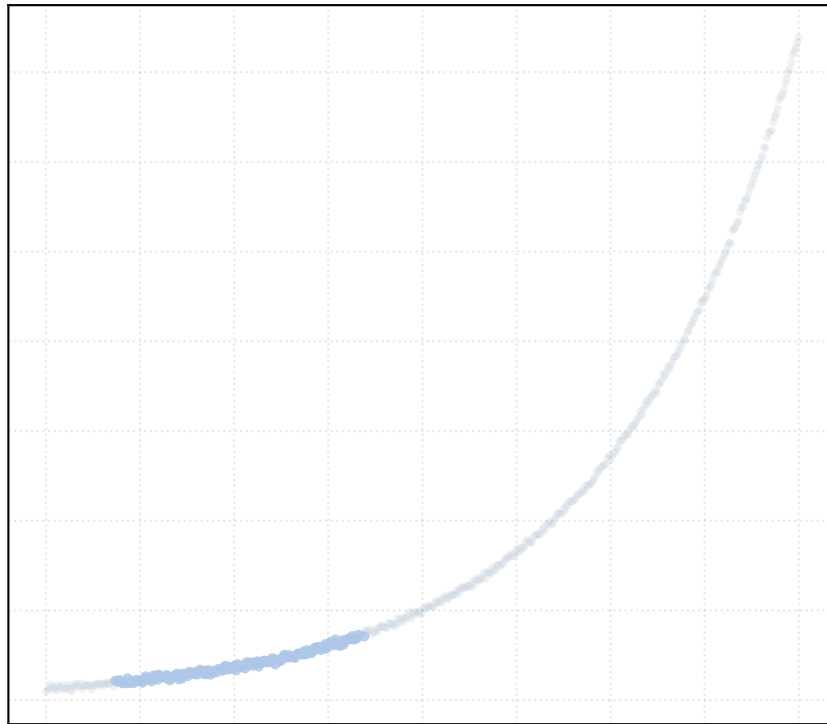
min_intersection = 8 [1 Clusters]



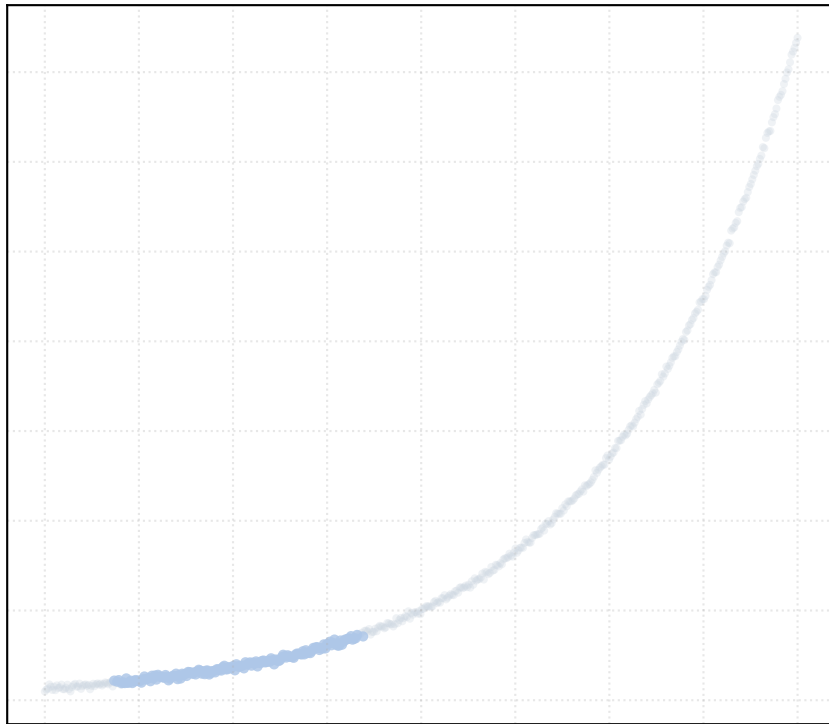
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

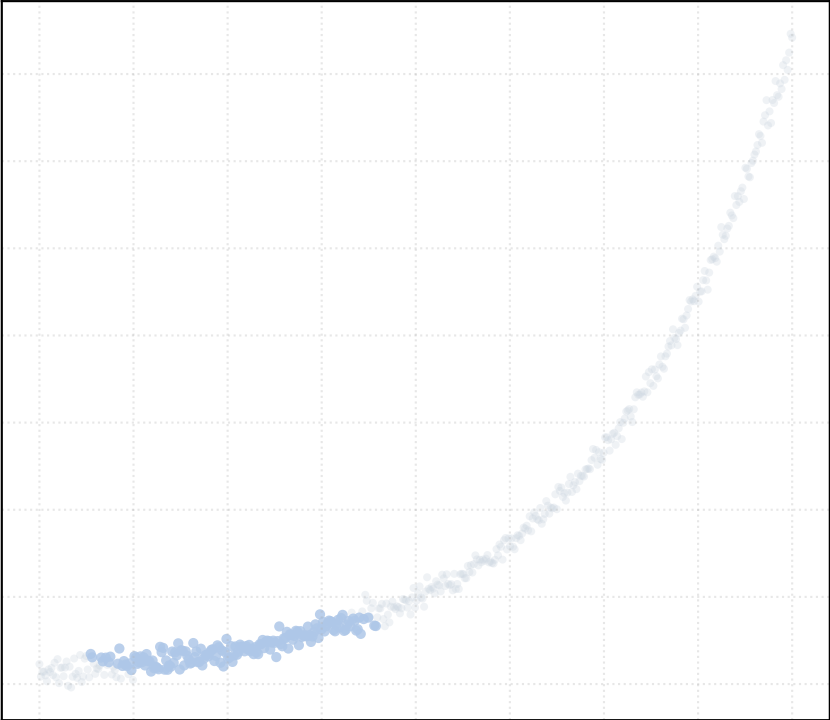


min_intersection = 20 [1 Clusters]

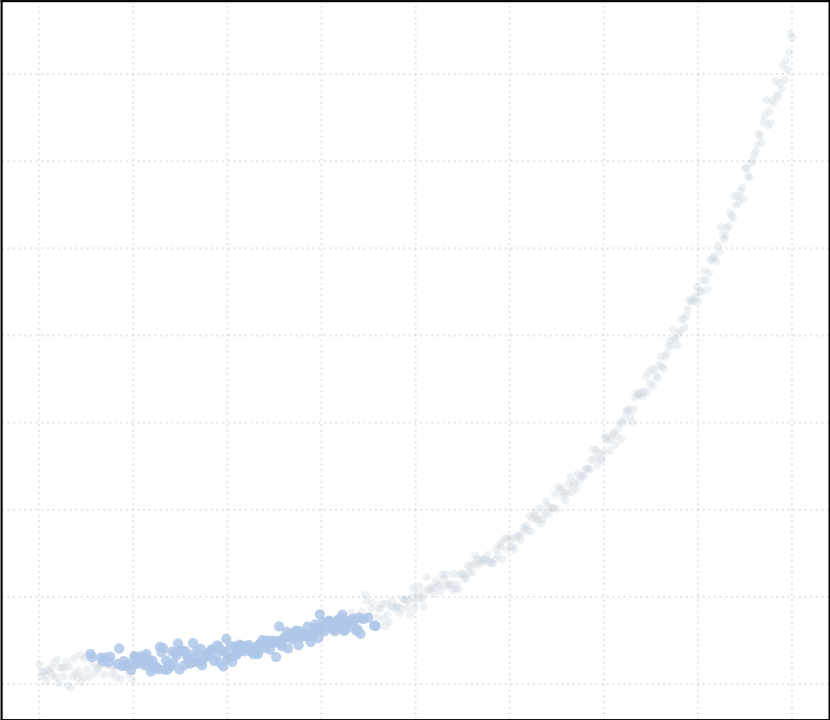


Dataset: exponential_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

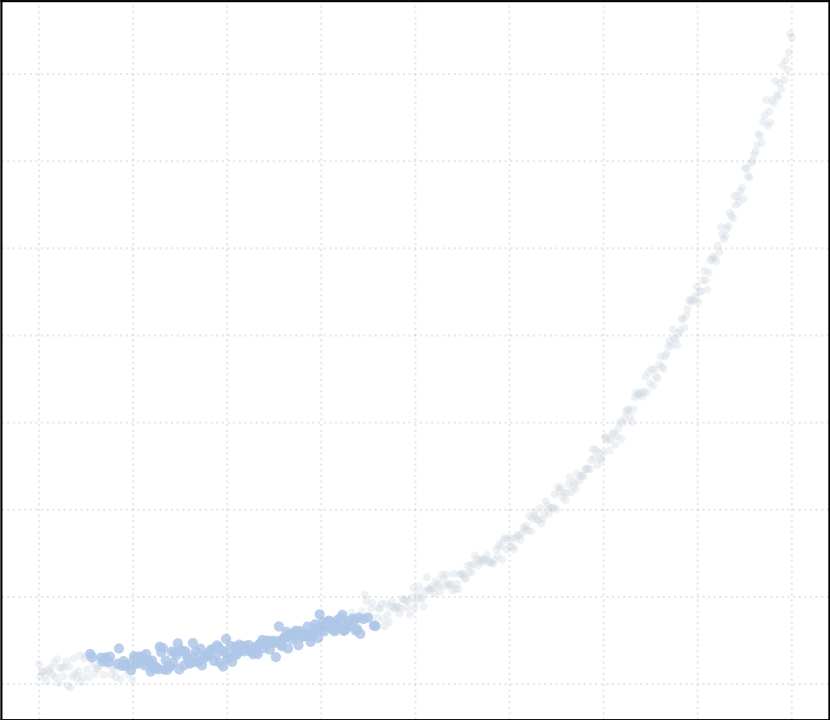
min_intersection = 1 (Default) [1 Clusters]



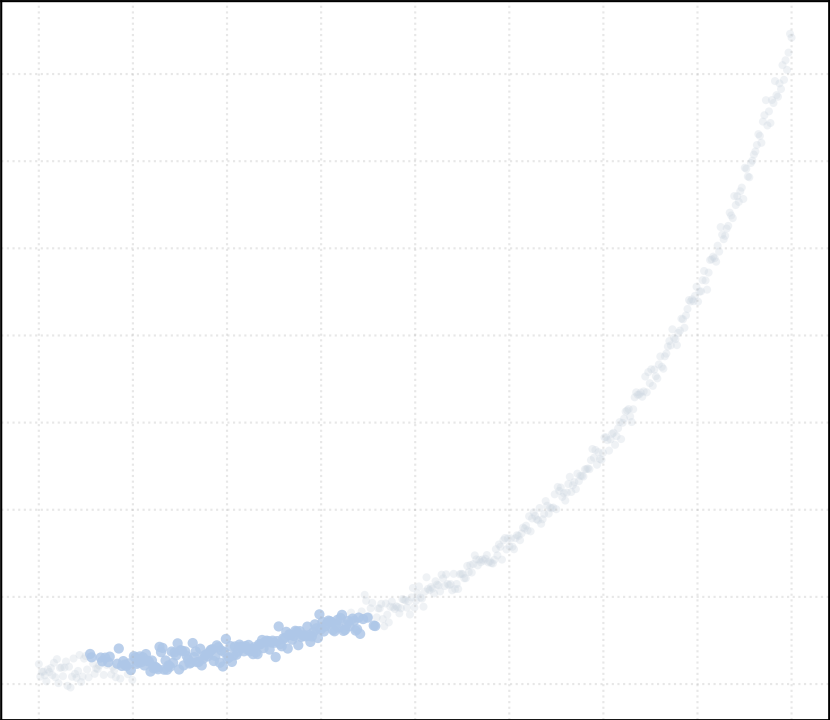
min_intersection = 2 [1 Clusters]



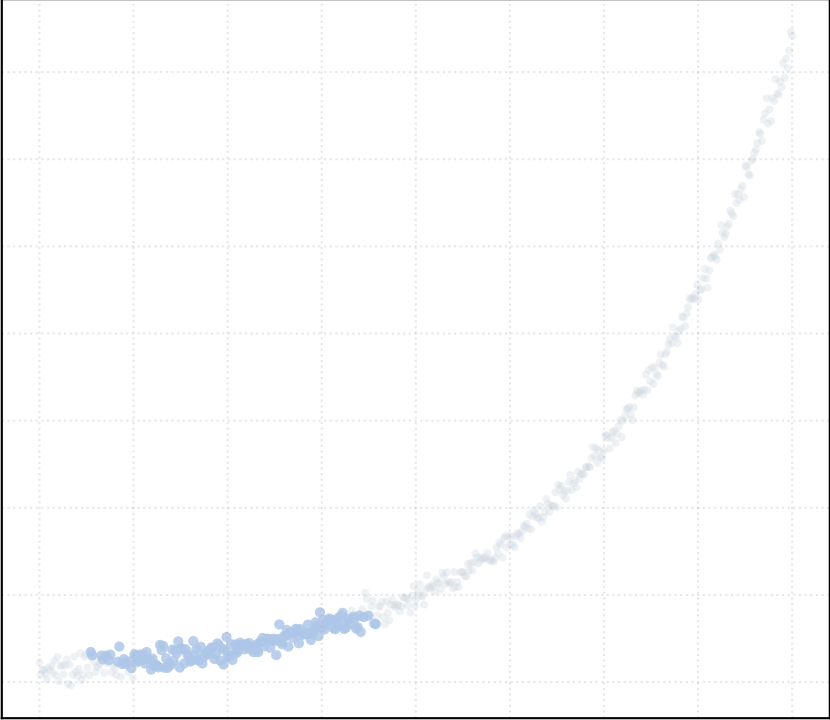
min_intersection = 3 [1 Clusters]



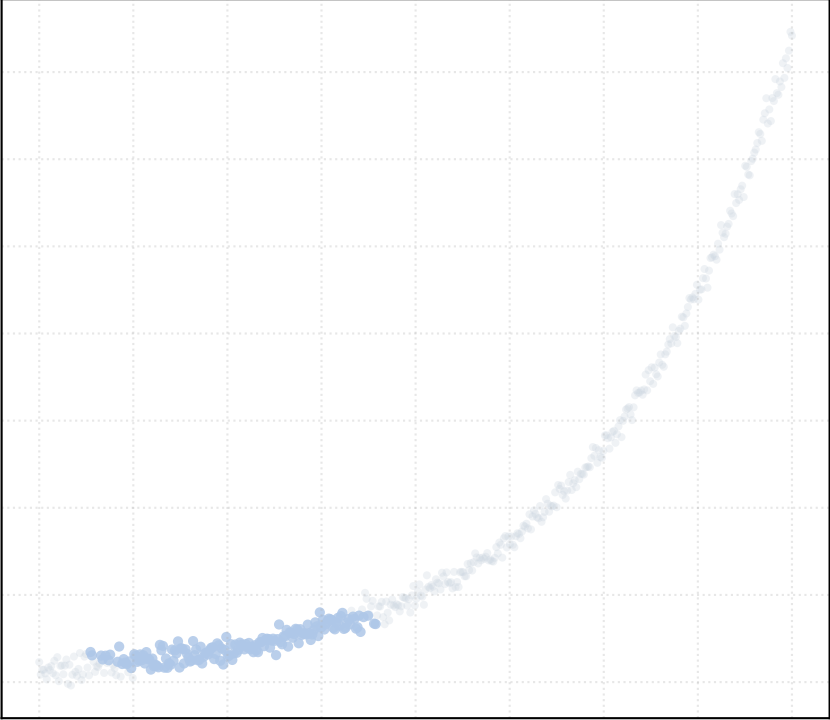
min_intersection = 5 [1 Clusters]



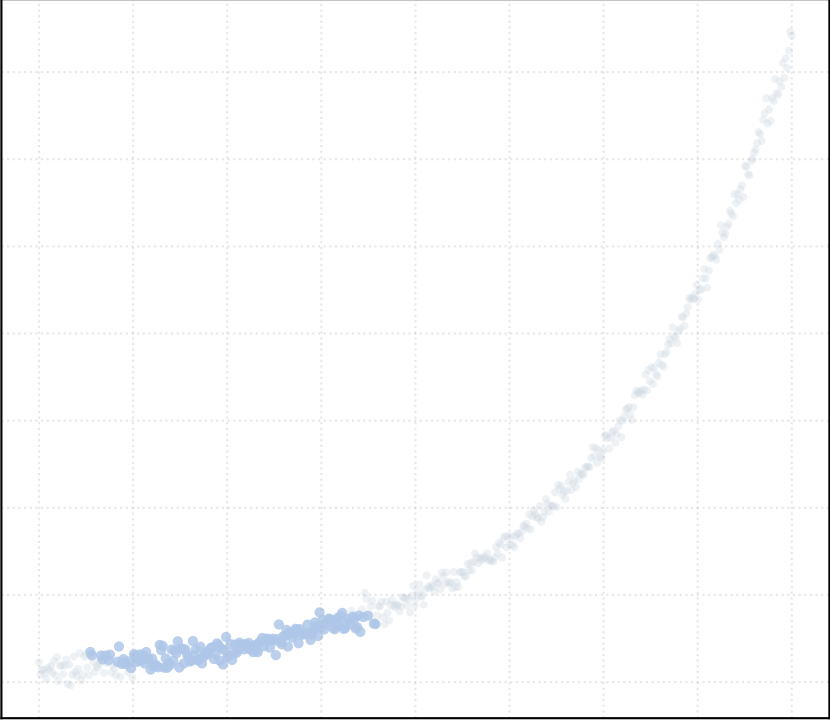
min_intersection = 8 [1 Clusters]



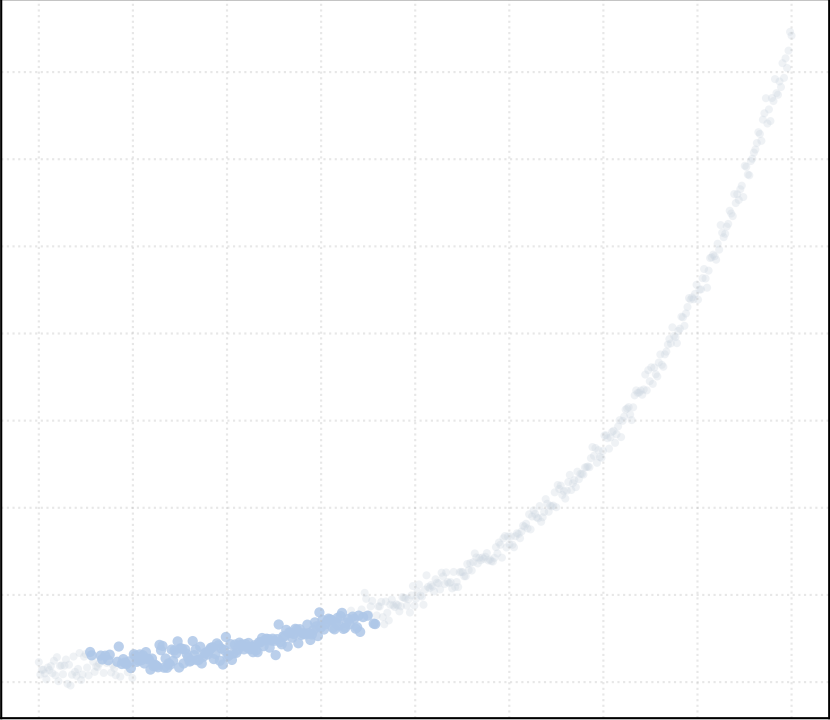
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

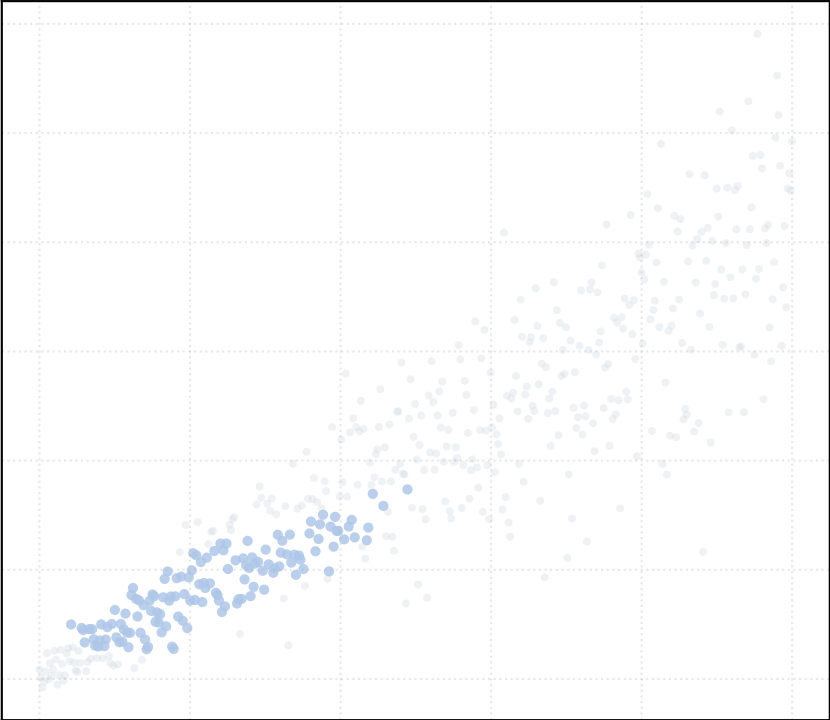


min_intersection = 20 [1 Clusters]

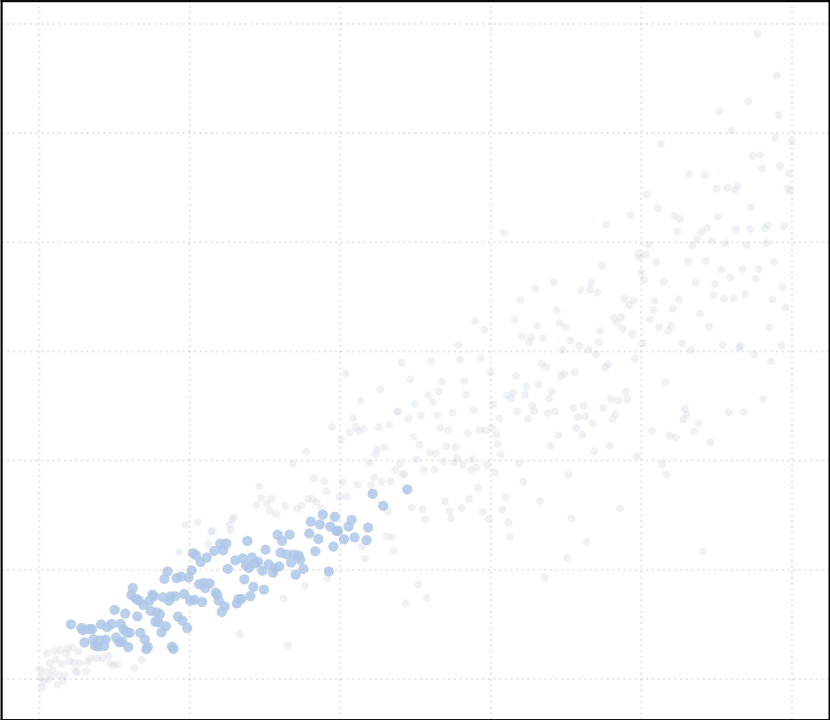


Dataset: heteroscedastic_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

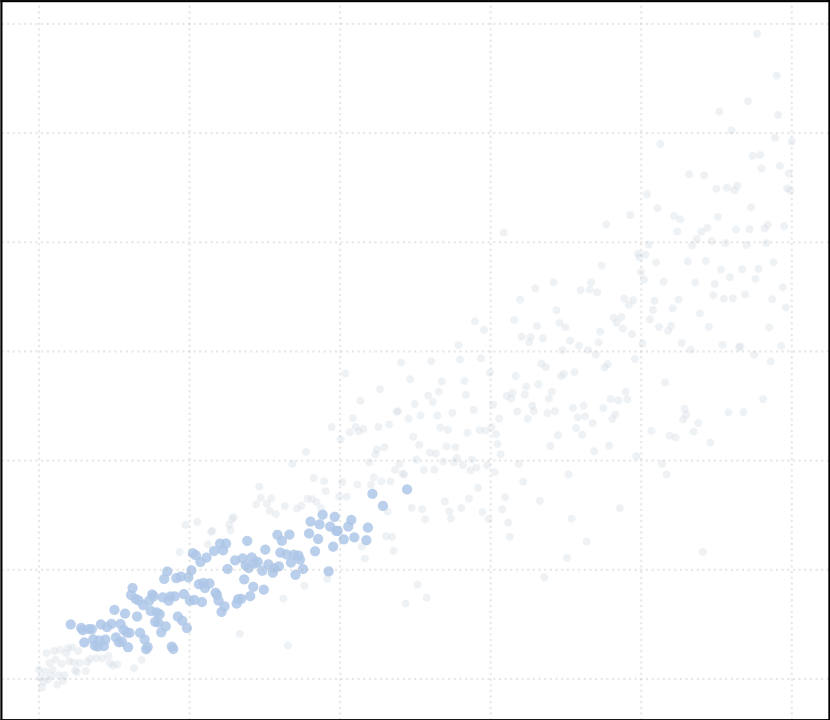
min_intersection = 1 (Default) [1 Clusters]



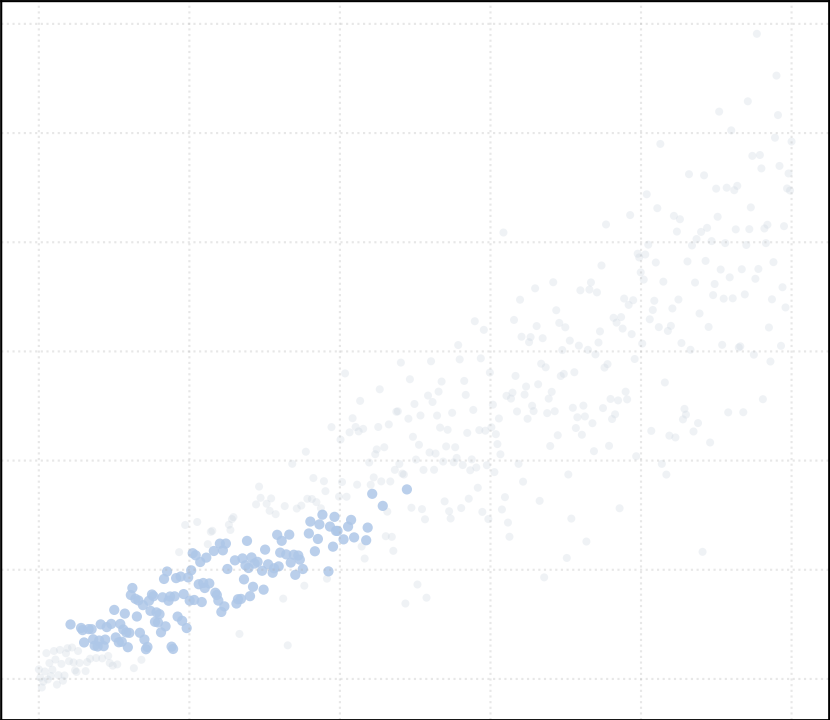
min_intersection = 2 [1 Clusters]



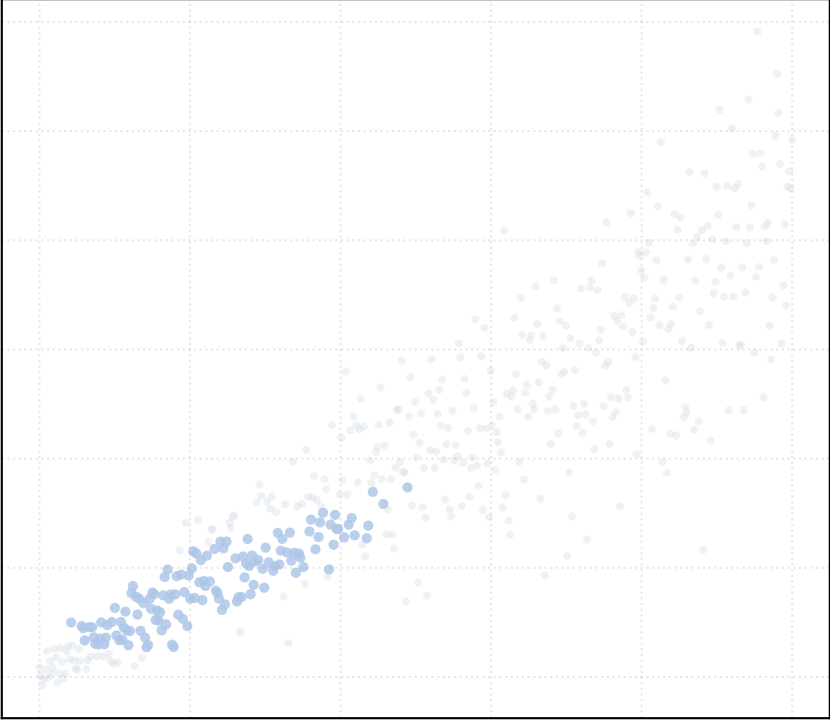
min_intersection = 3 [1 Clusters]



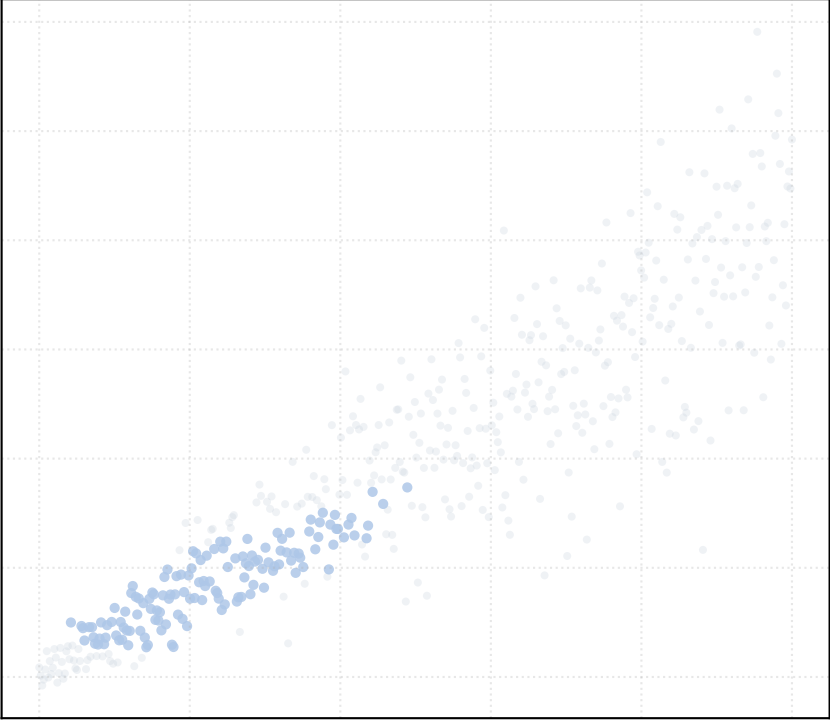
min_intersection = 5 [1 Clusters]



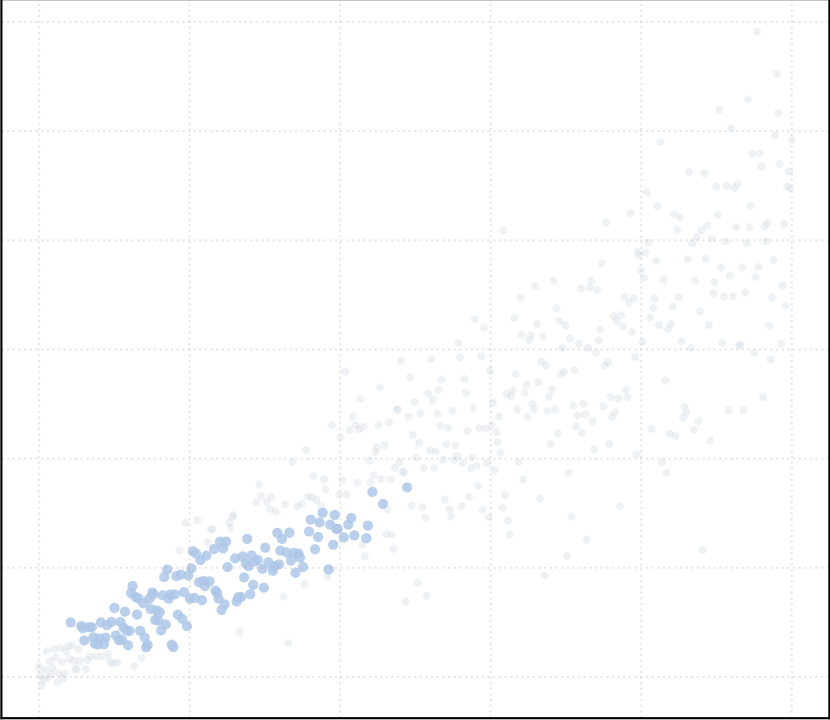
min_intersection = 8 [1 Clusters]



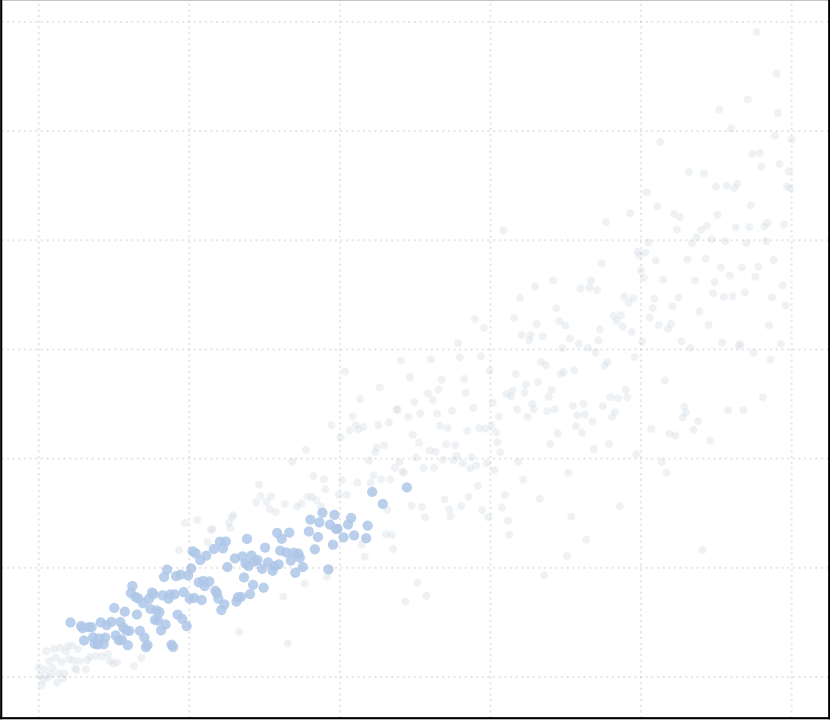
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

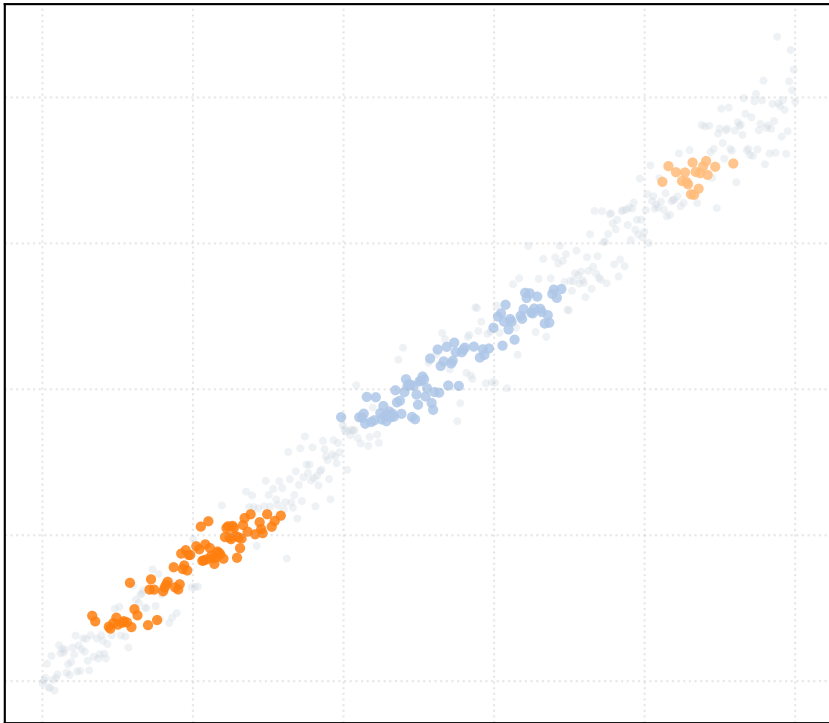


min_intersection = 20 [1 Clusters]

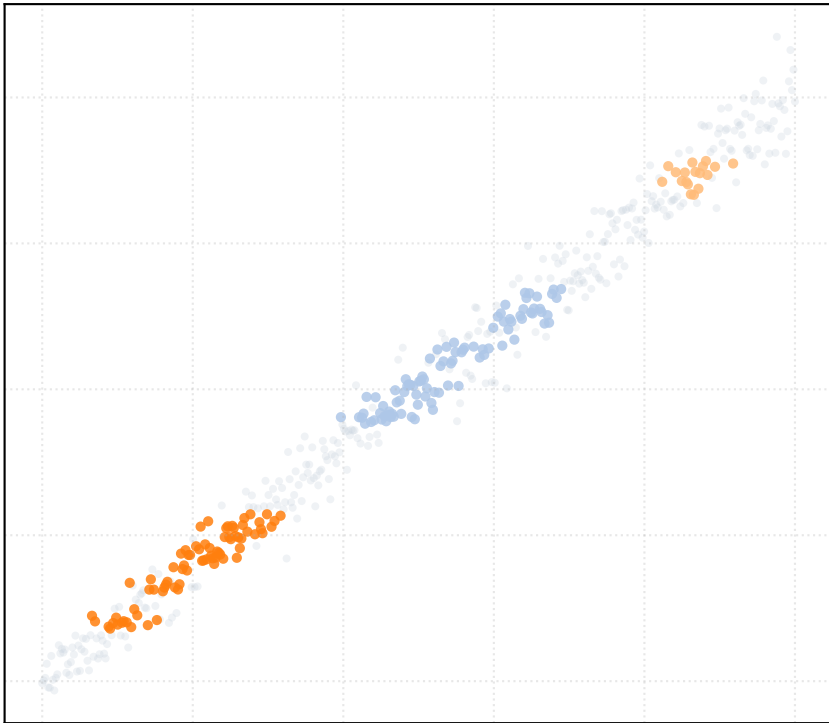


Dataset: heteroscedastic_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

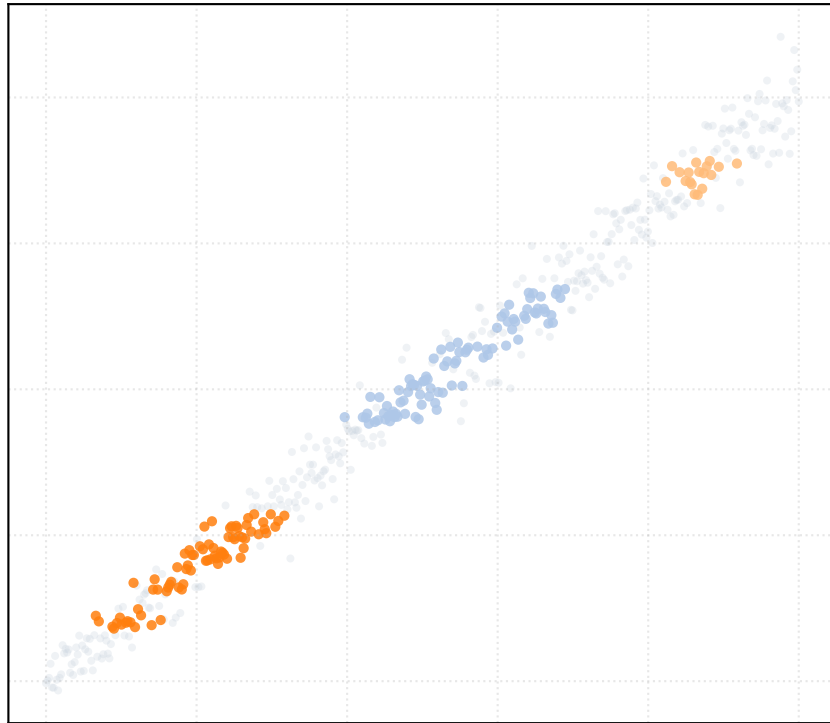
min_intersection = 1 (Default) [3 Clusters]



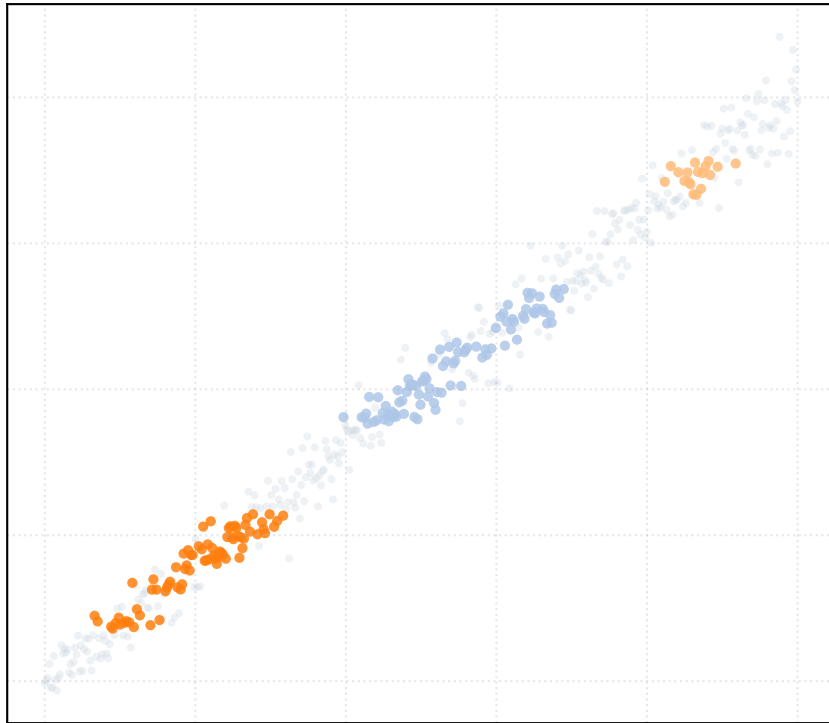
min_intersection = 2 [3 Clusters]



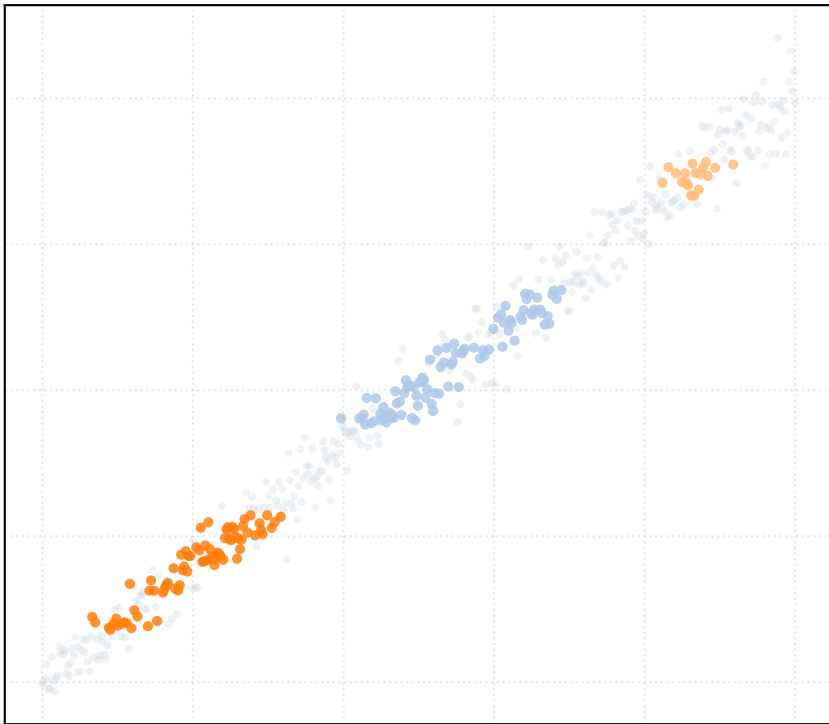
min_intersection = 3 [3 Clusters]



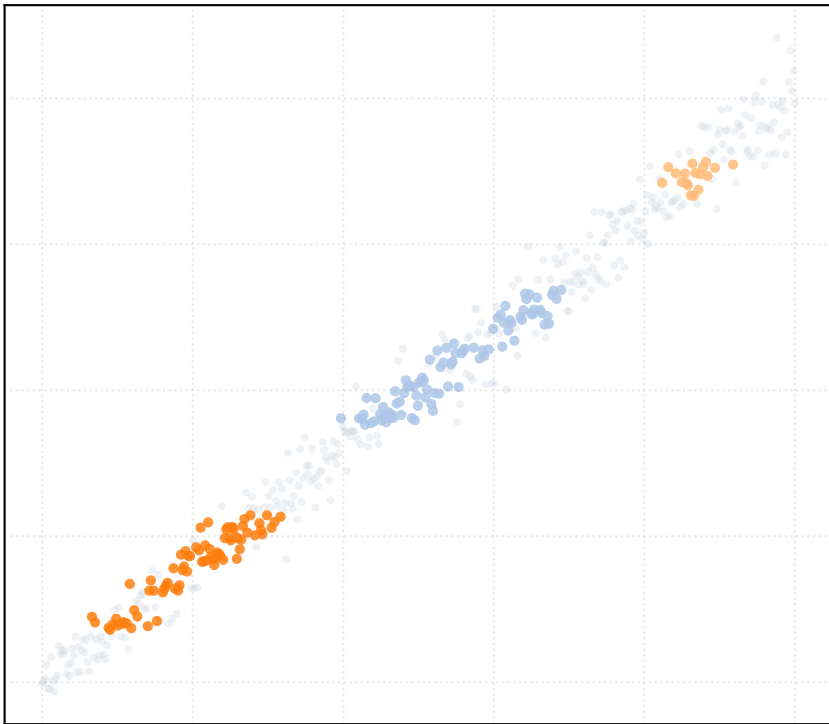
min_intersection = 5 [3 Clusters]



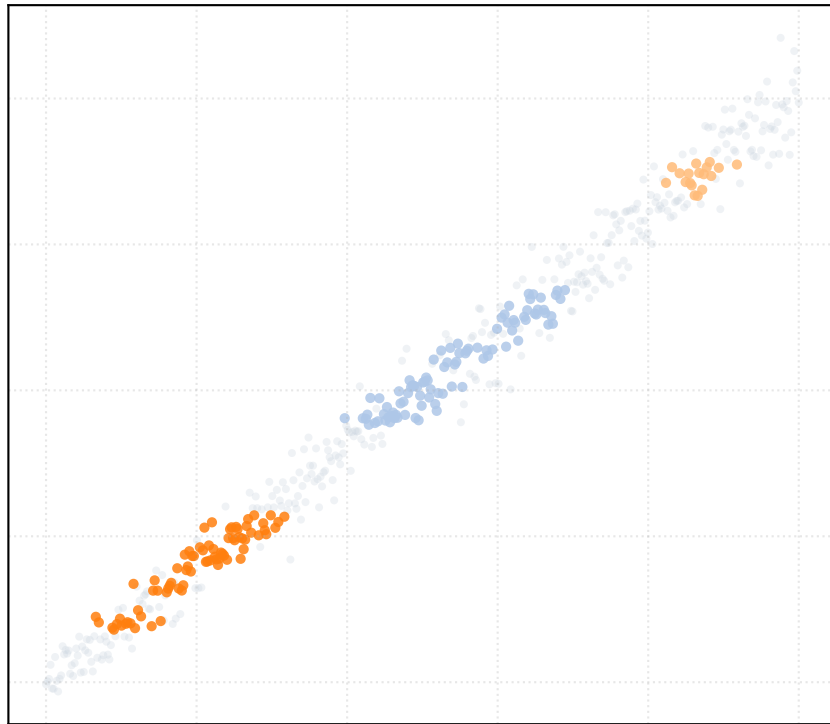
min_intersection = 8 [3 Clusters]



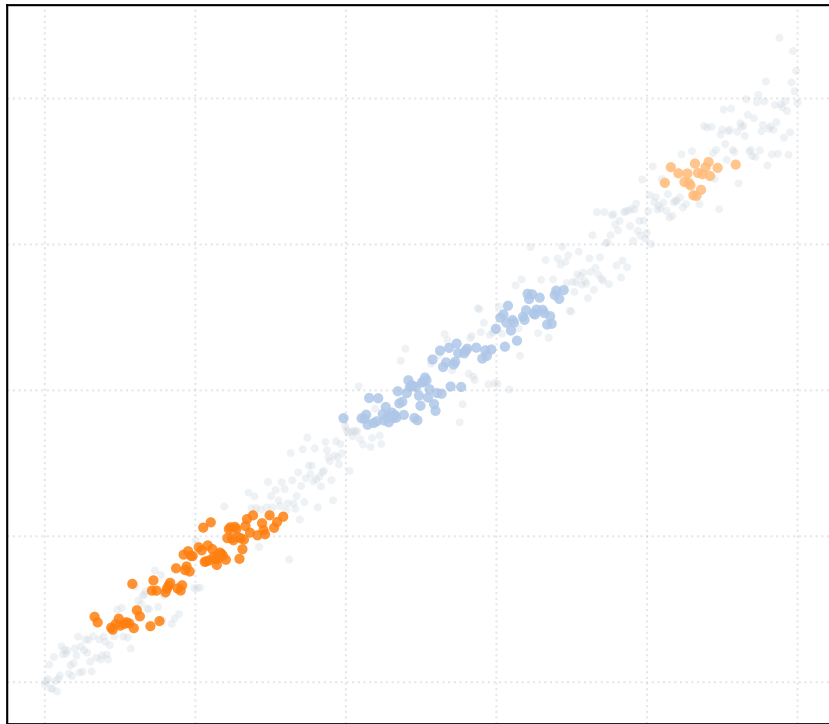
min_intersection = 10 [3 Clusters]



min_intersection = 15 [3 Clusters]

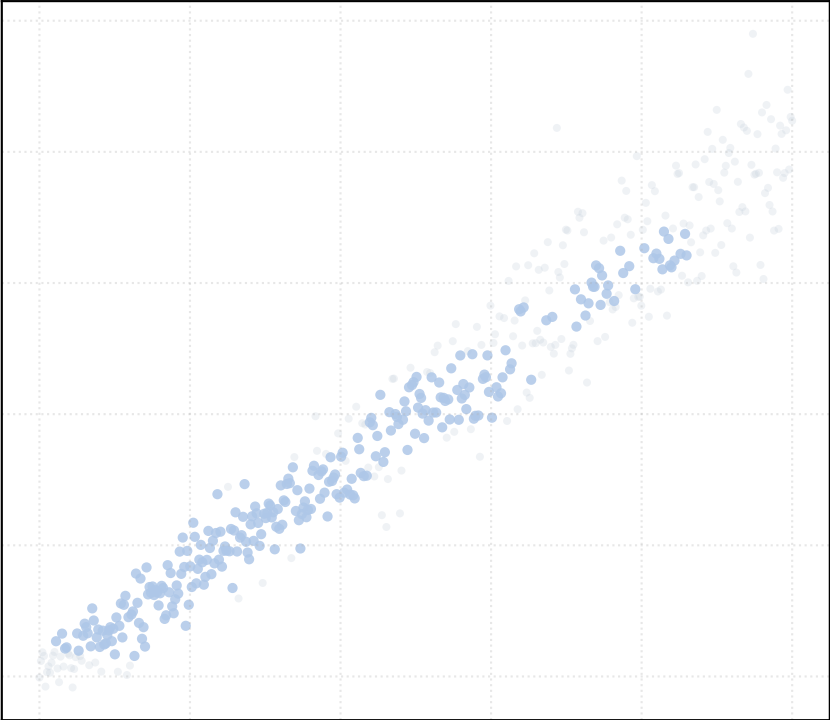


min_intersection = 20 [3 Clusters]

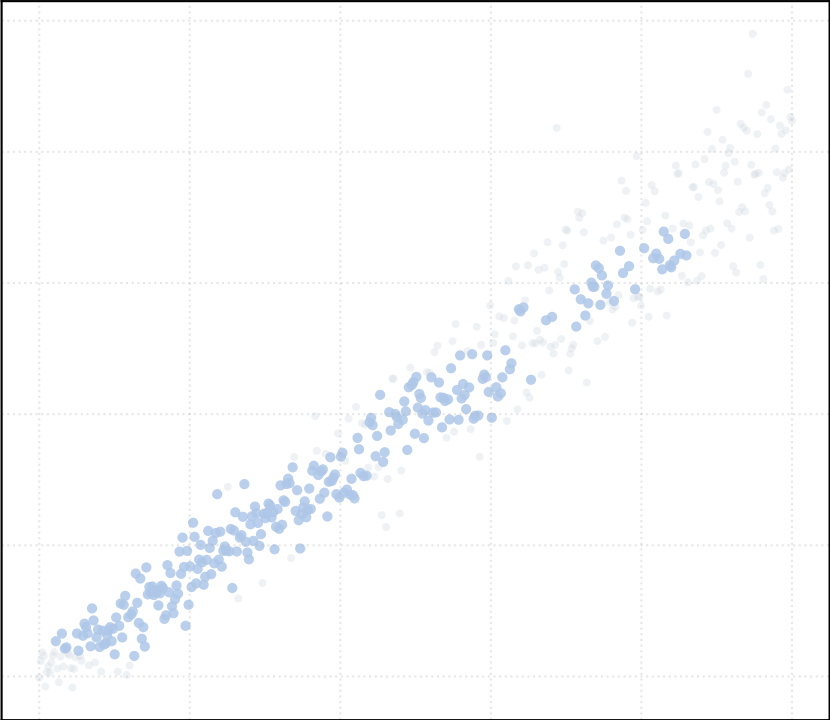


Dataset: heteroscedastic_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

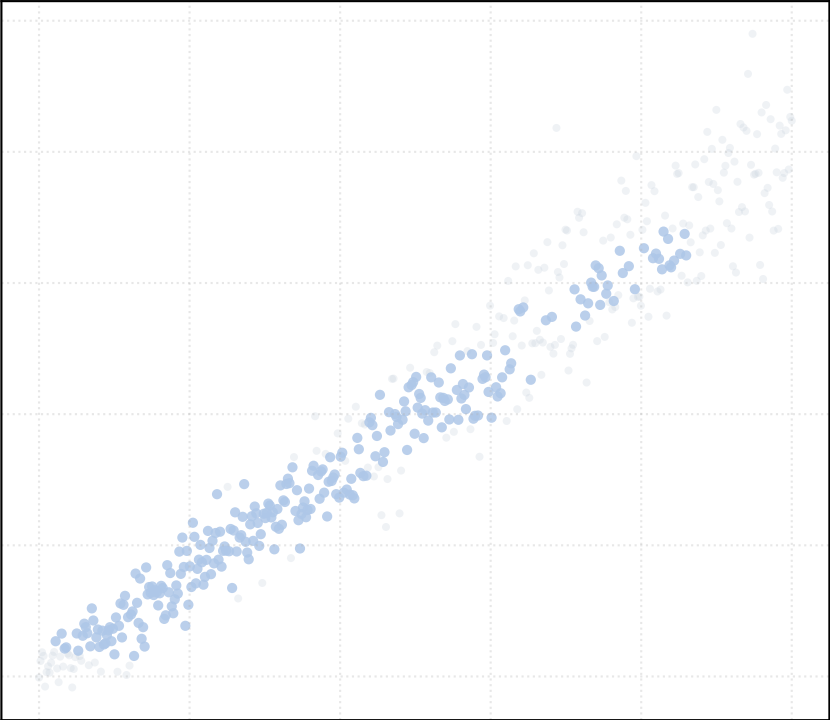
min_intersection = 1 (Default) [1 Clusters]



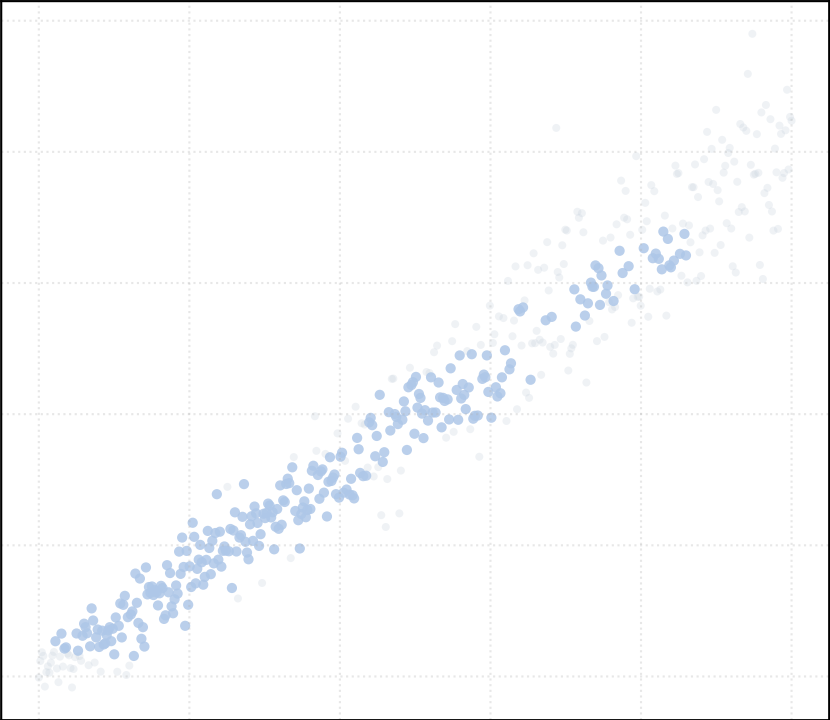
min_intersection = 2 [1 Clusters]



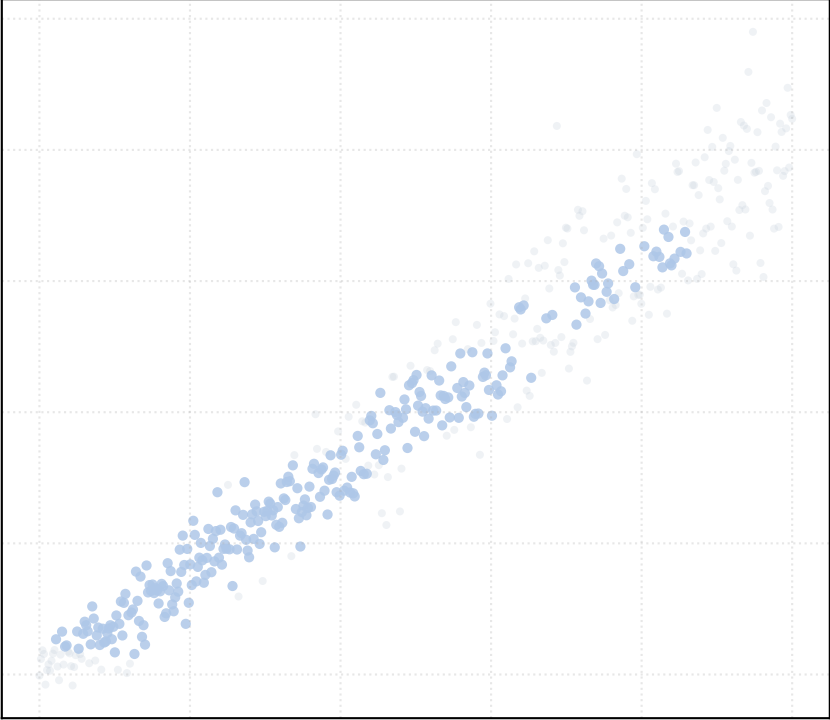
min_intersection = 3 [1 Clusters]



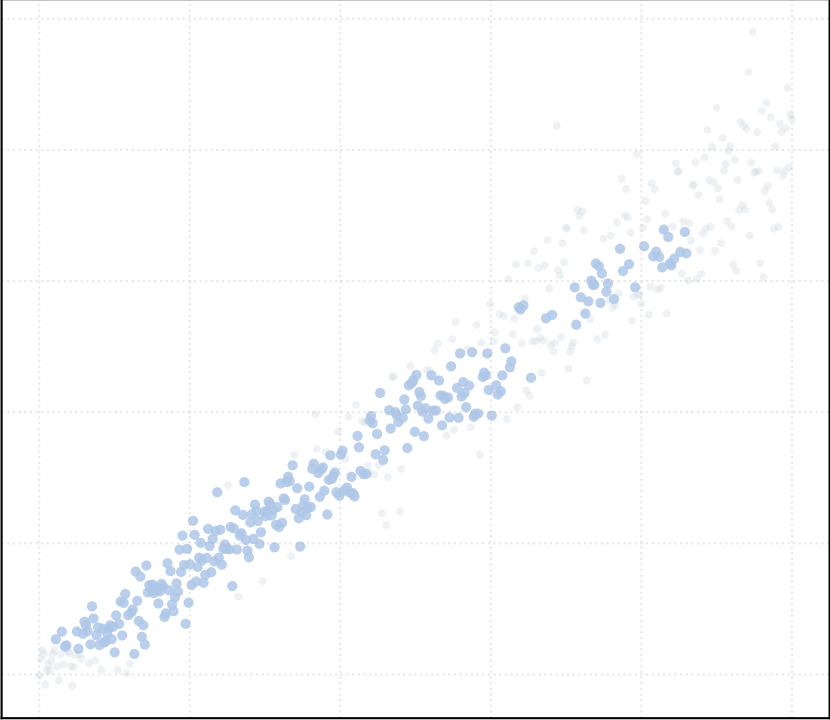
min_intersection = 5 [1 Clusters]



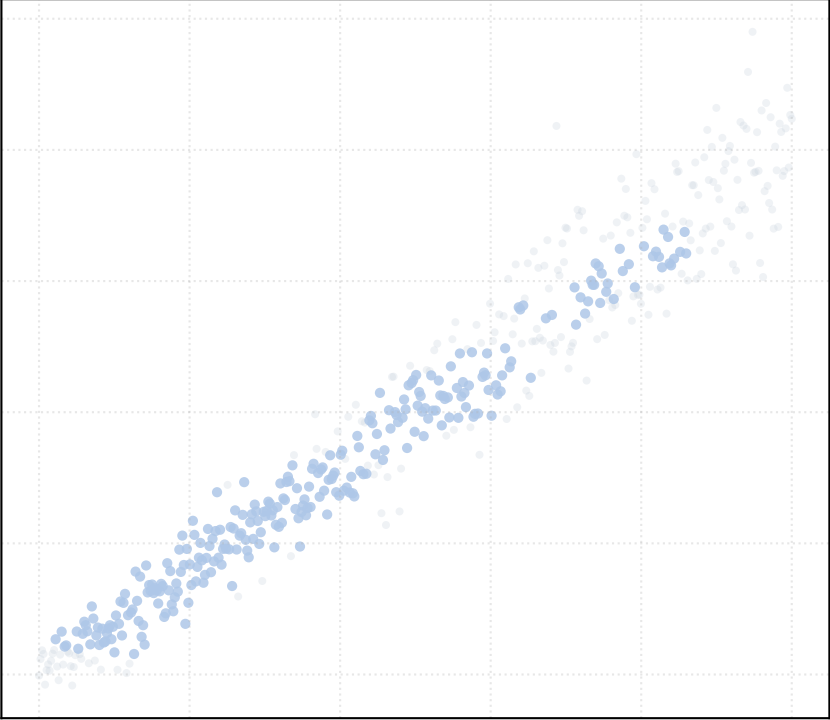
min_intersection = 8 [1 Clusters]



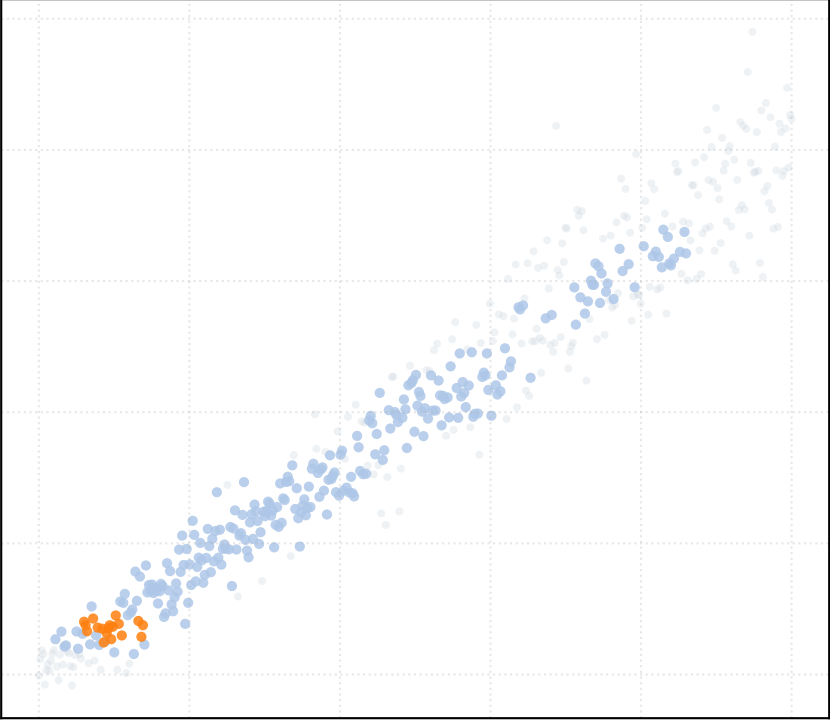
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

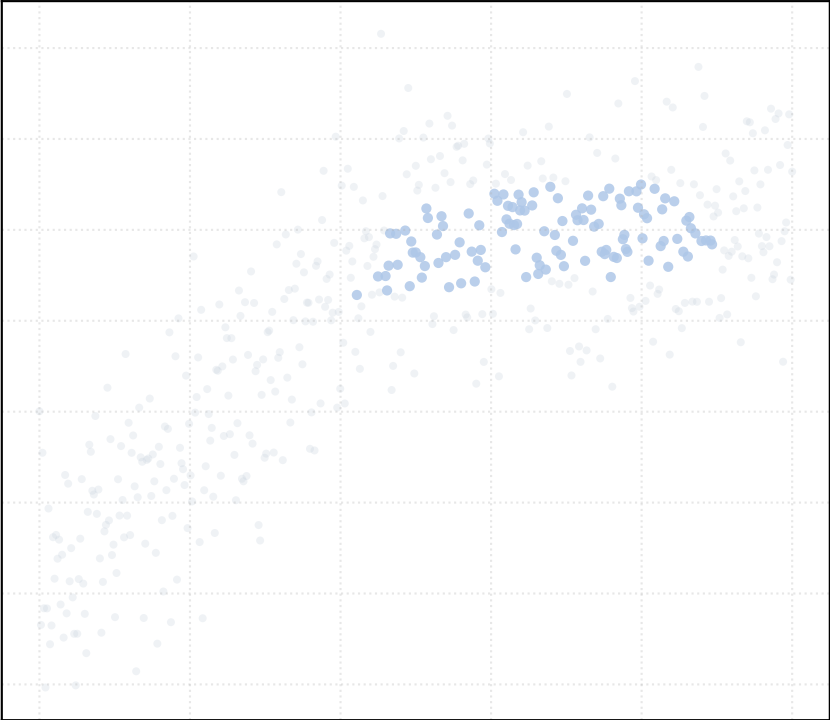


min_intersection = 20 [2 Clusters]

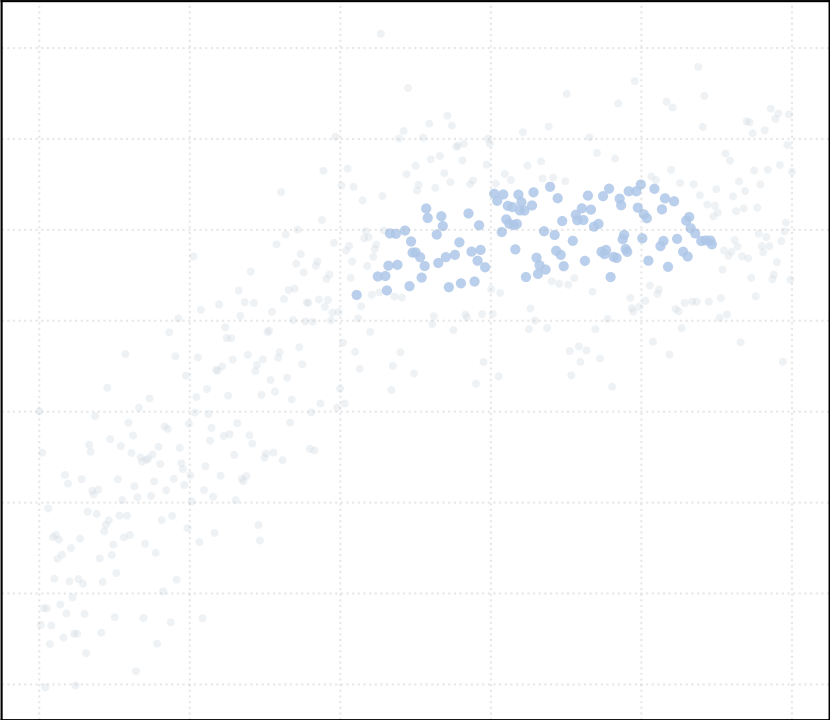


Dataset: kinked_curve_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

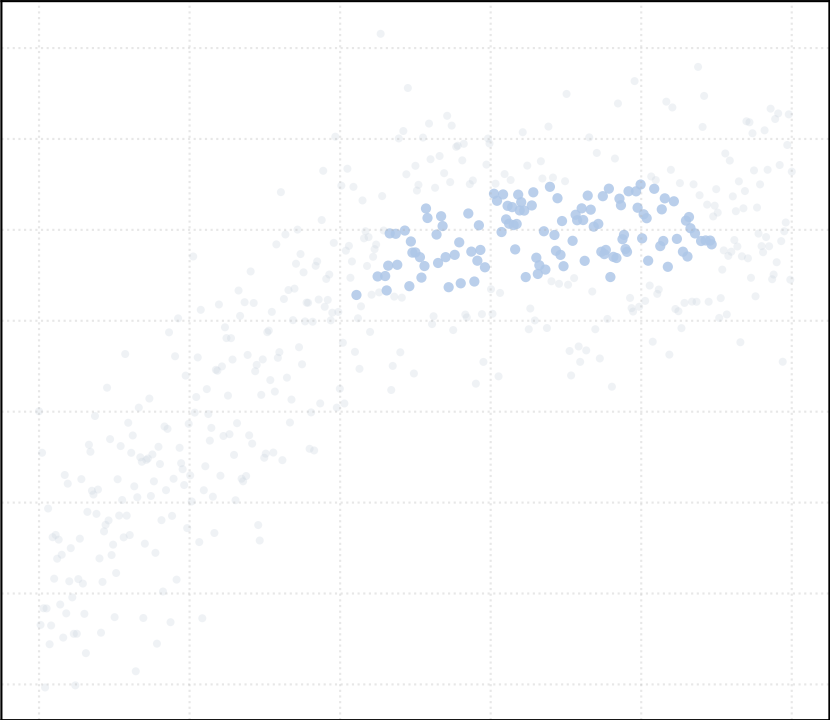
min_intersection = 1 (Default) [1 Clusters]



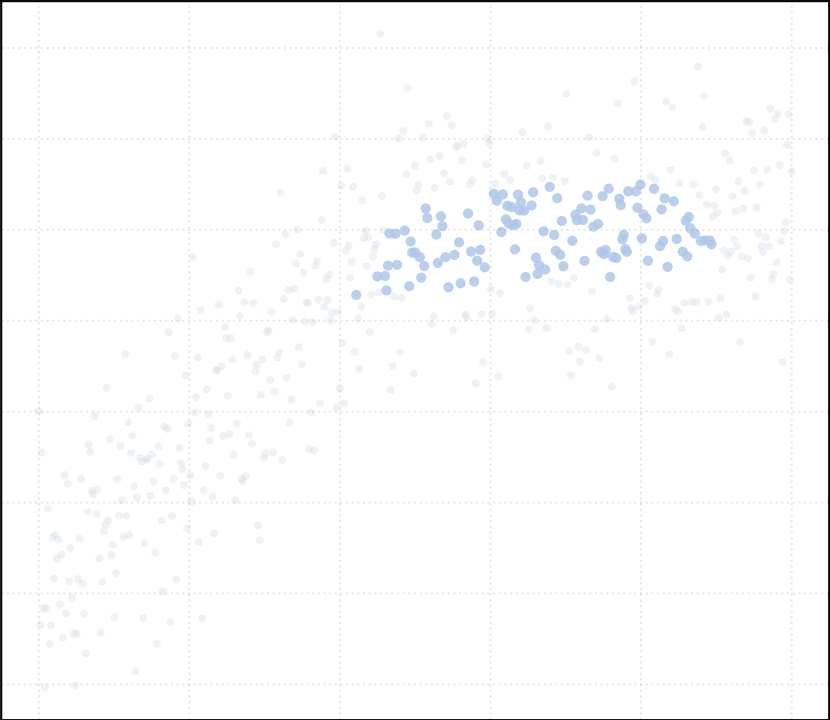
min_intersection = 2 [1 Clusters]



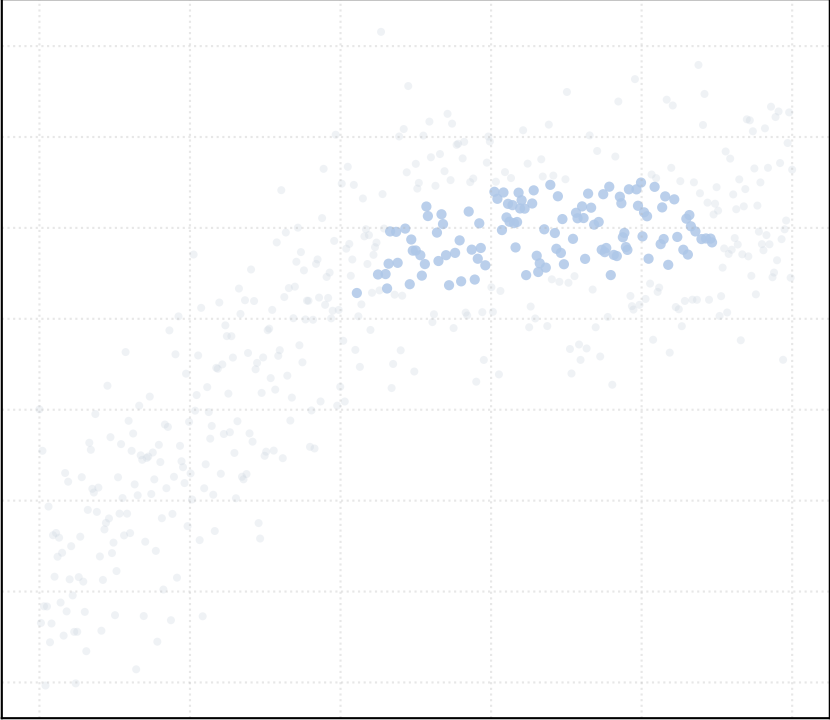
min_intersection = 3 [1 Clusters]



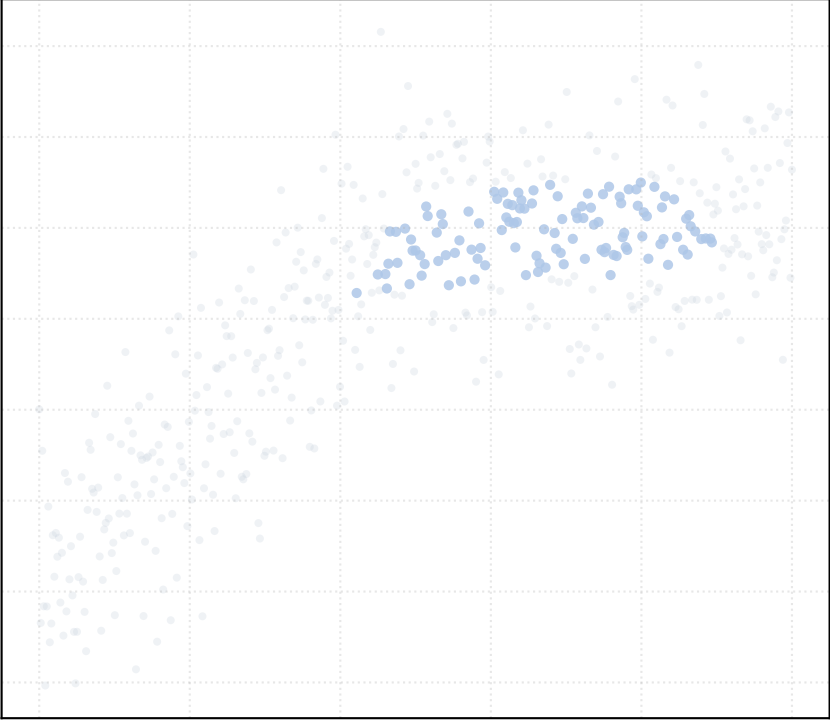
min_intersection = 5 [1 Clusters]



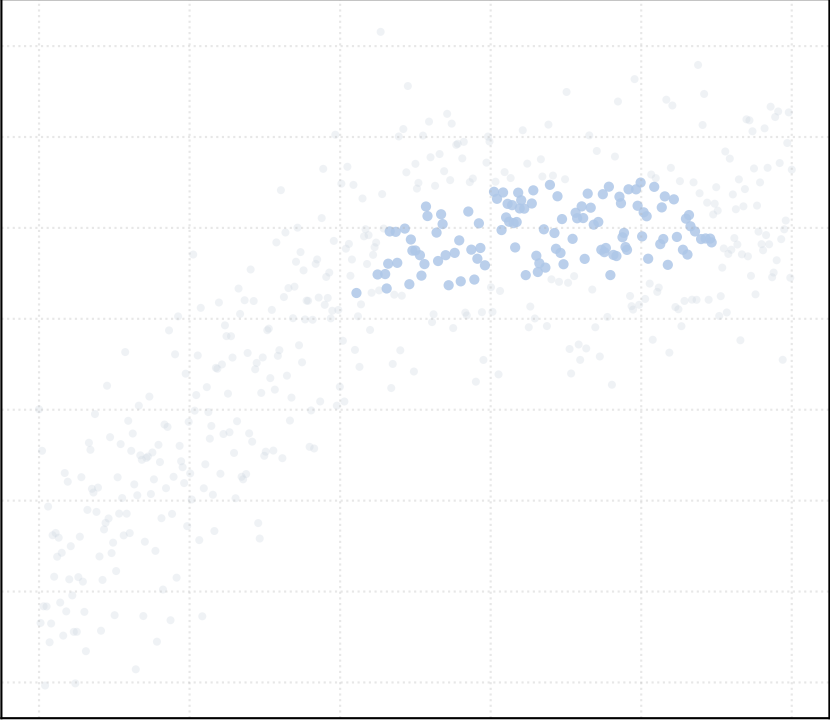
min_intersection = 8 [1 Clusters]



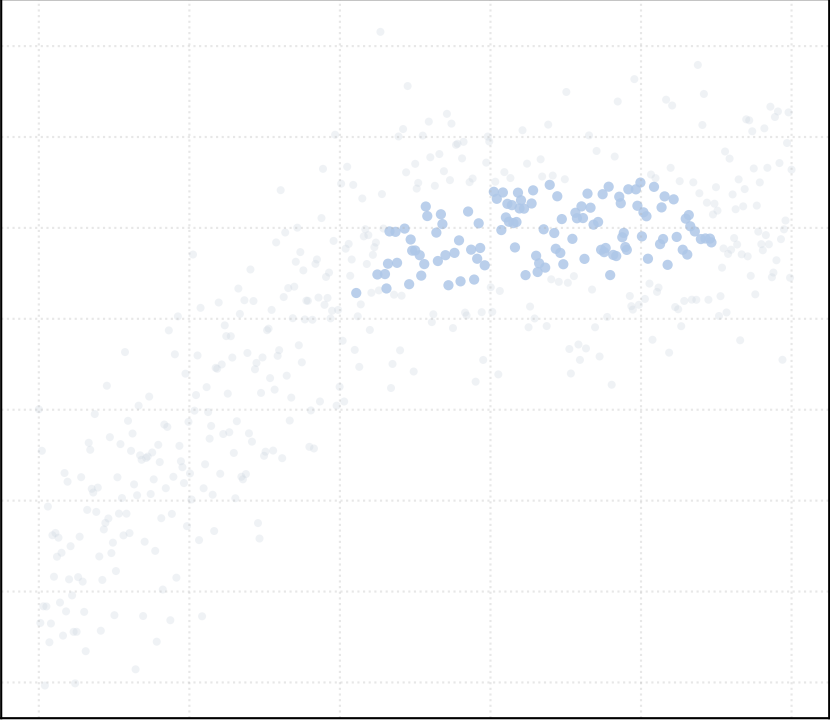
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

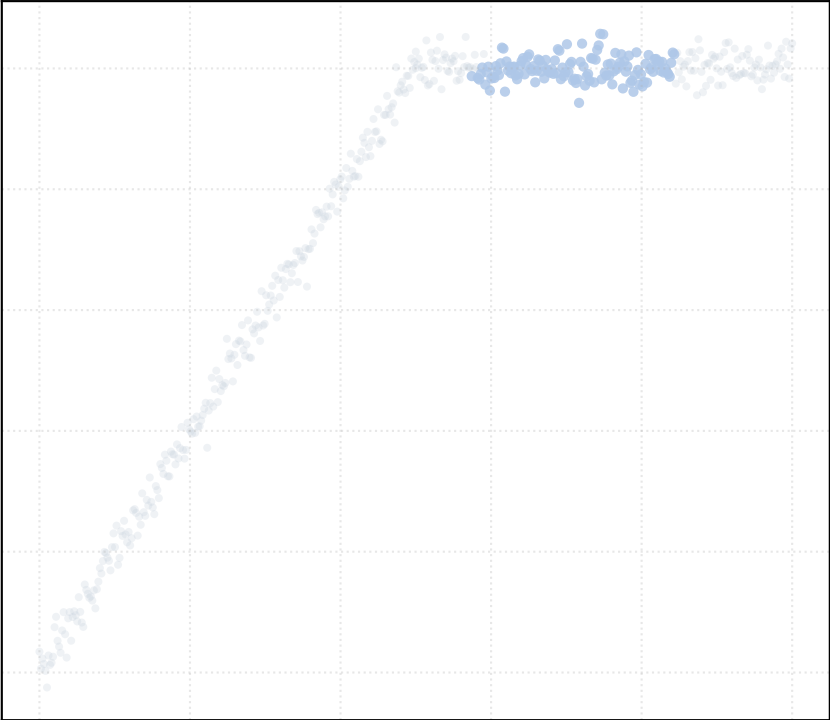


min_intersection = 20 [1 Clusters]

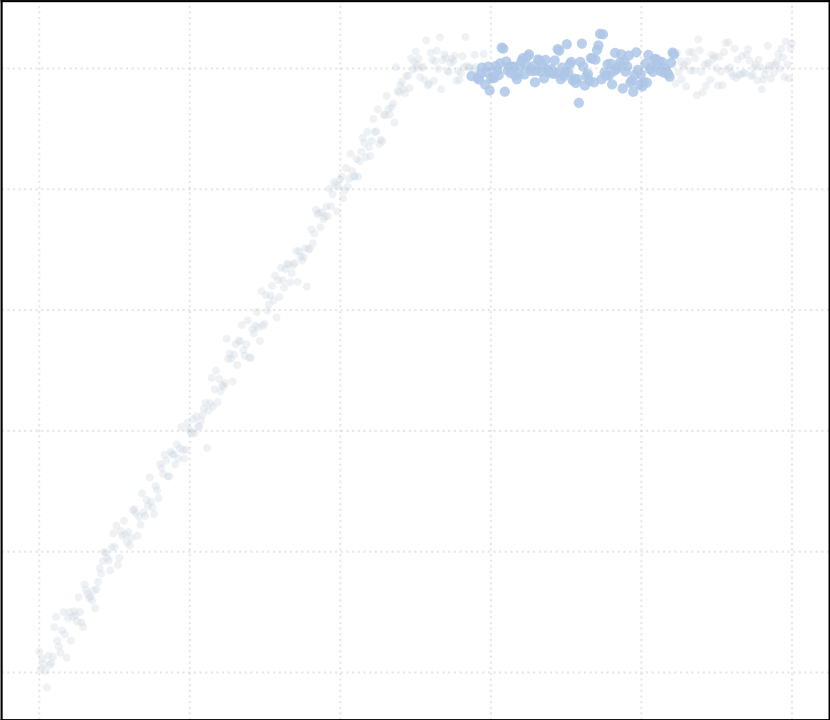


Dataset: kinked_curve_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

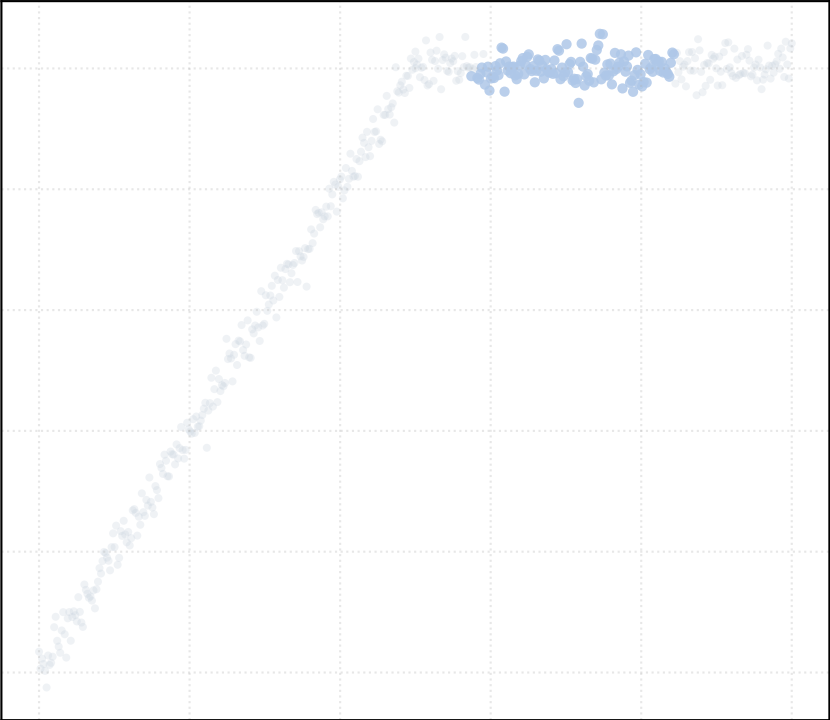
min_intersection = 1 (Default) [1 Clusters]



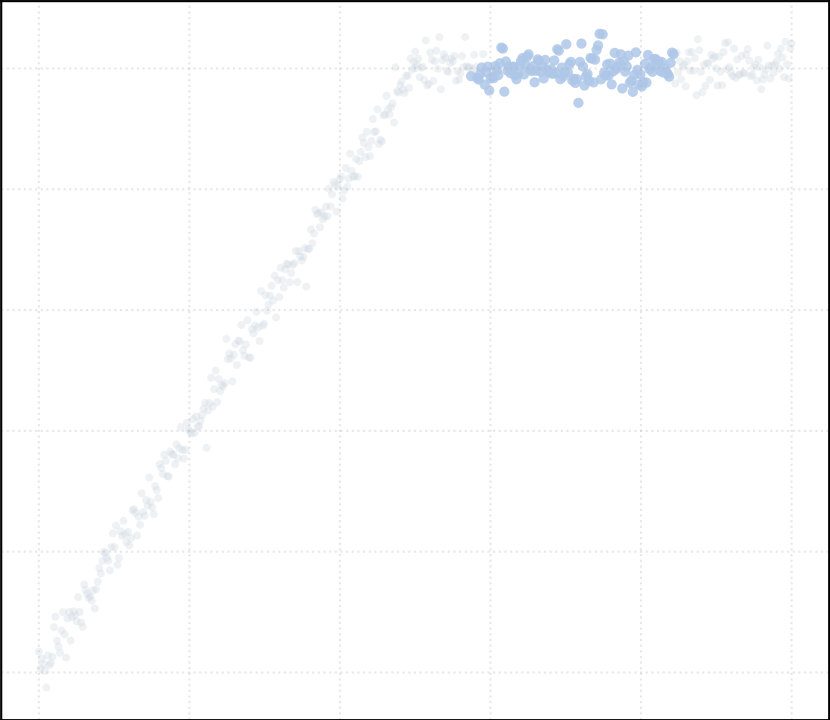
min_intersection = 2 [1 Clusters]



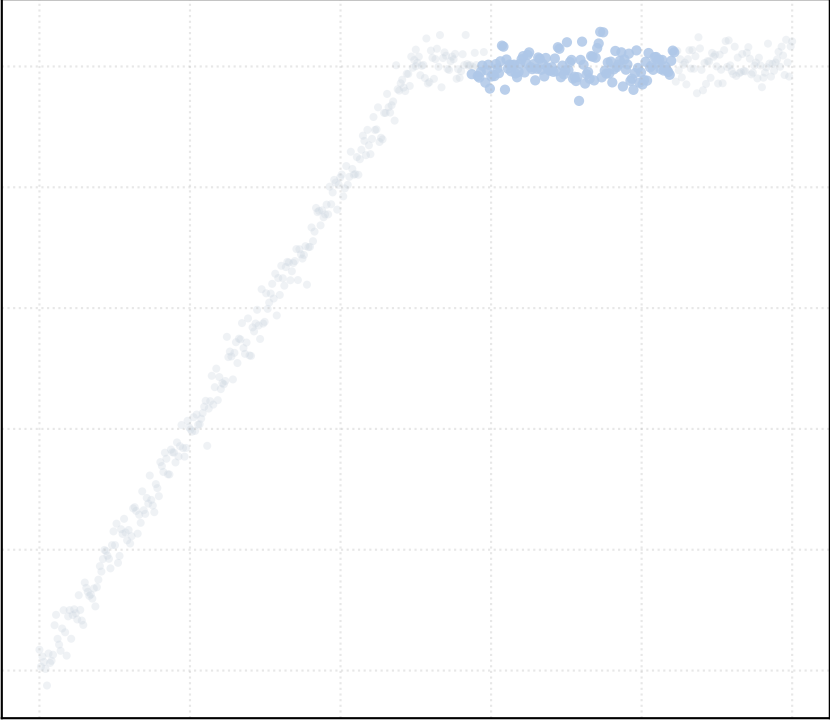
min_intersection = 3 [1 Clusters]



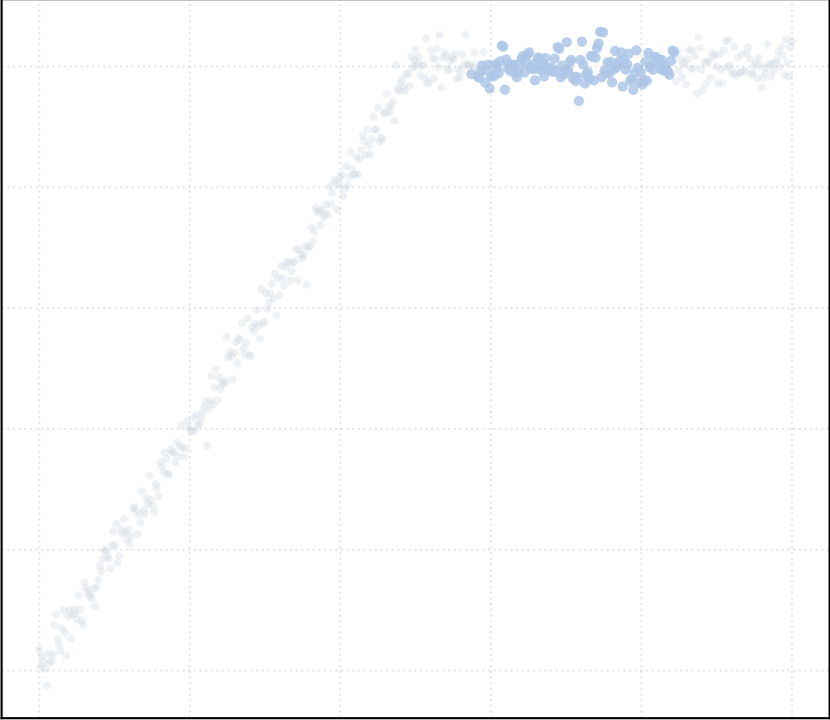
min_intersection = 5 [1 Clusters]



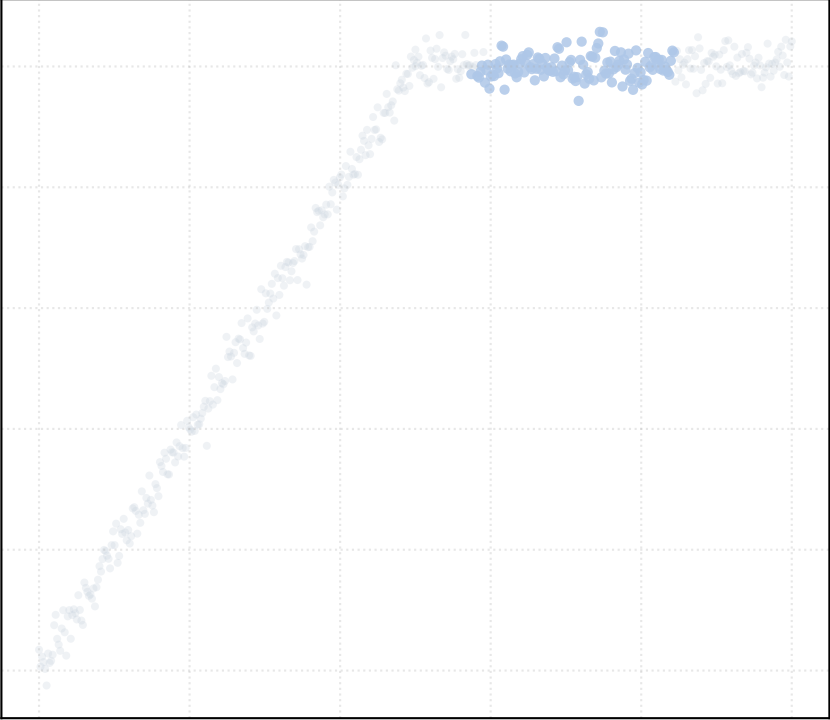
min_intersection = 8 [1 Clusters]



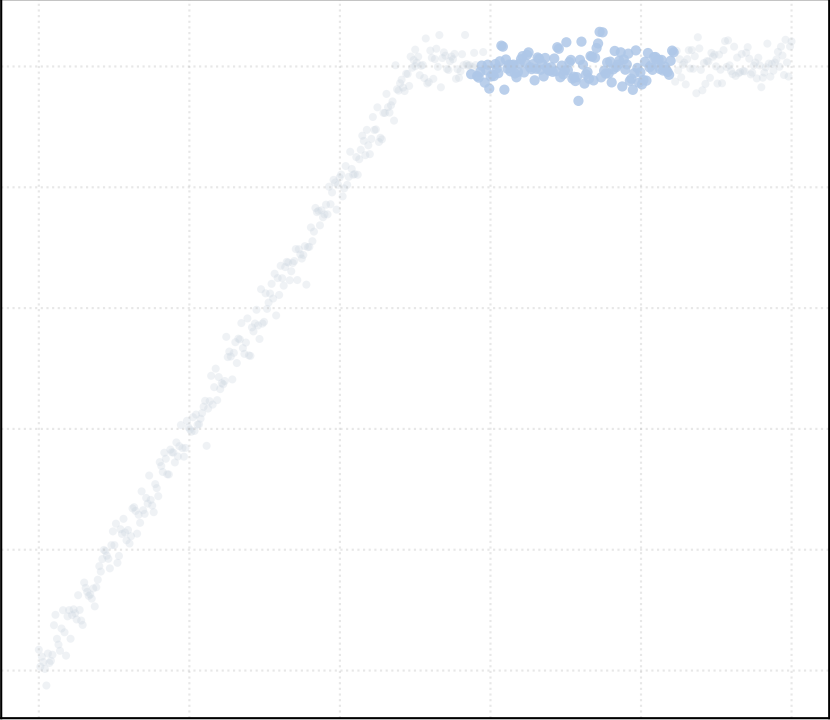
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

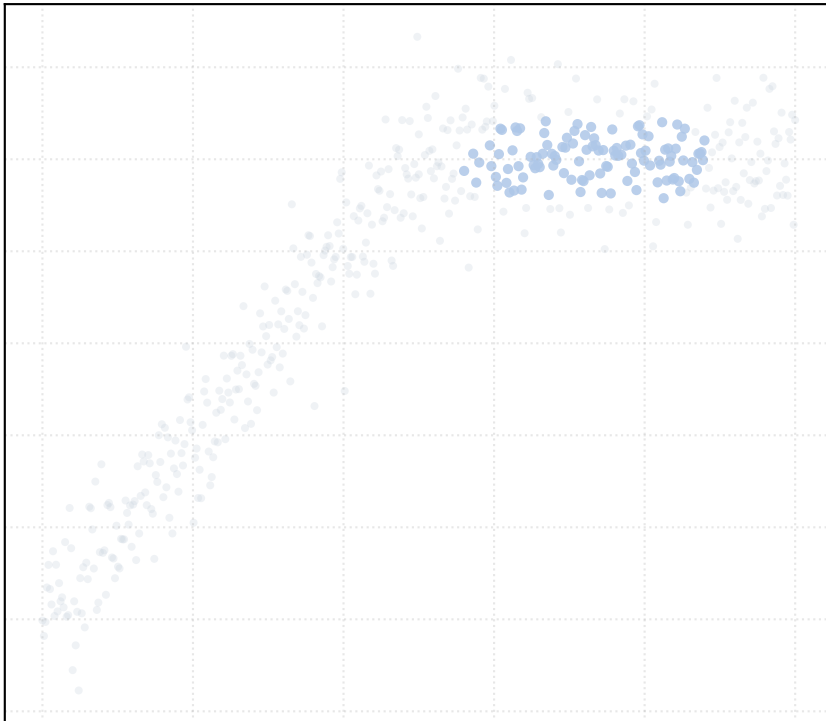


min_intersection = 20 [1 Clusters]

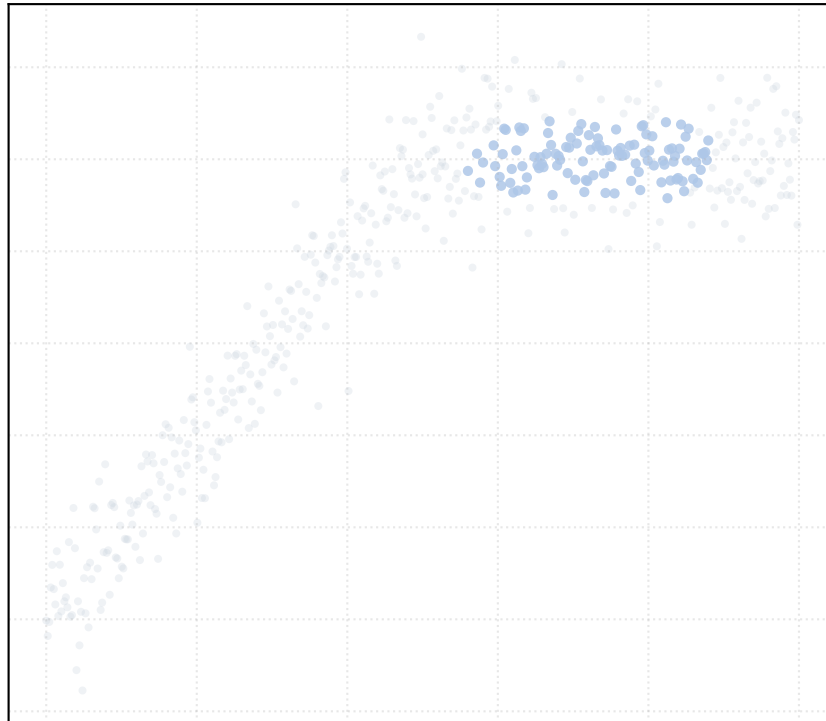


Dataset: kinked_curve_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

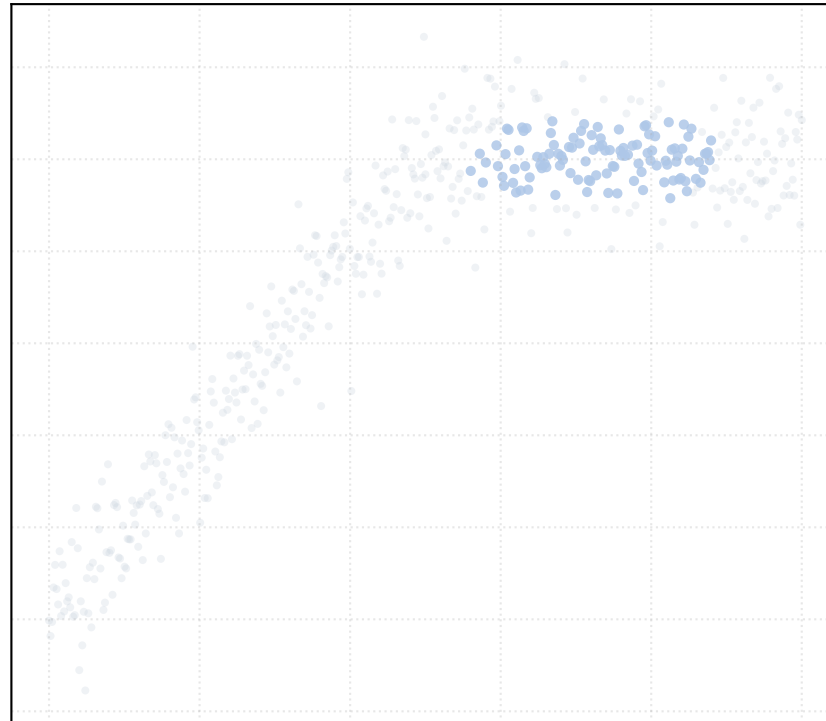
min_intersection = 1 (Default) [1 Clusters]



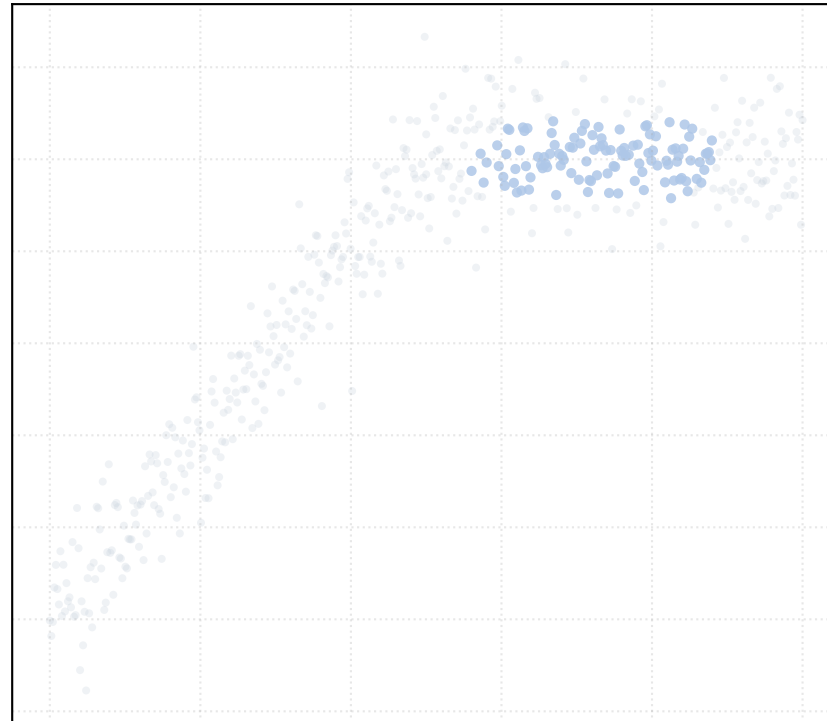
min_intersection = 2 [1 Clusters]



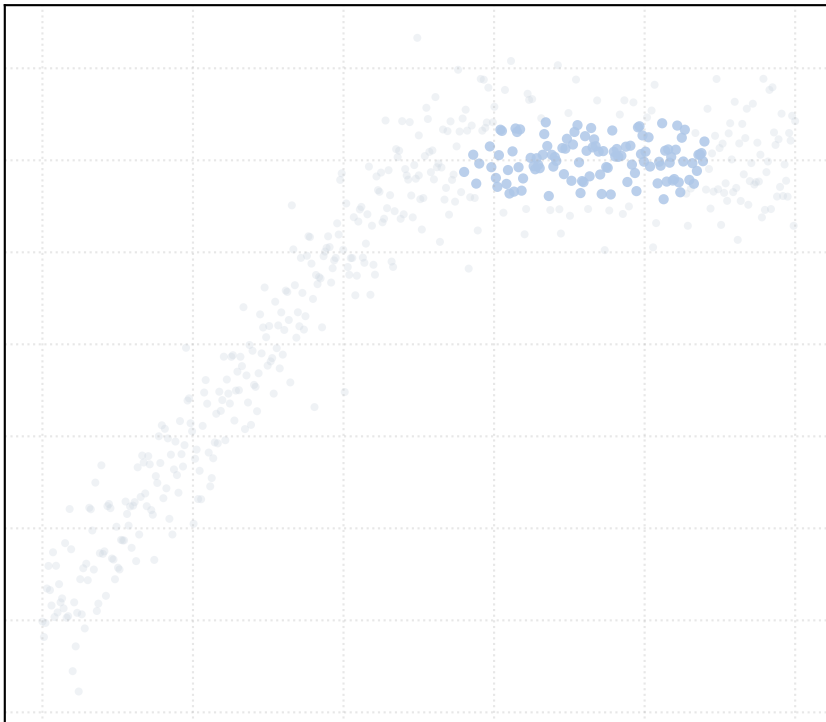
min_intersection = 3 [1 Clusters]



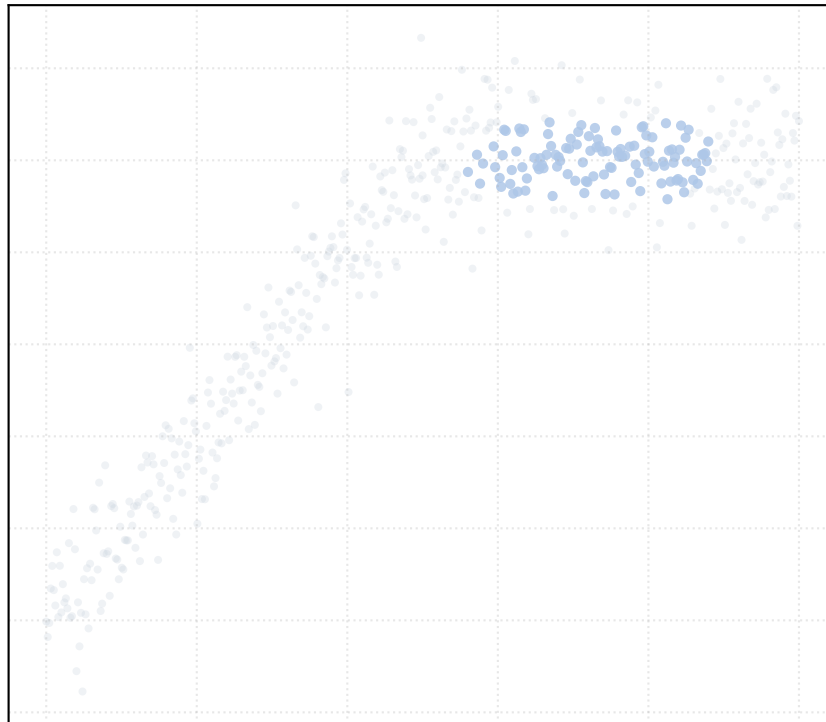
min_intersection = 5 [1 Clusters]



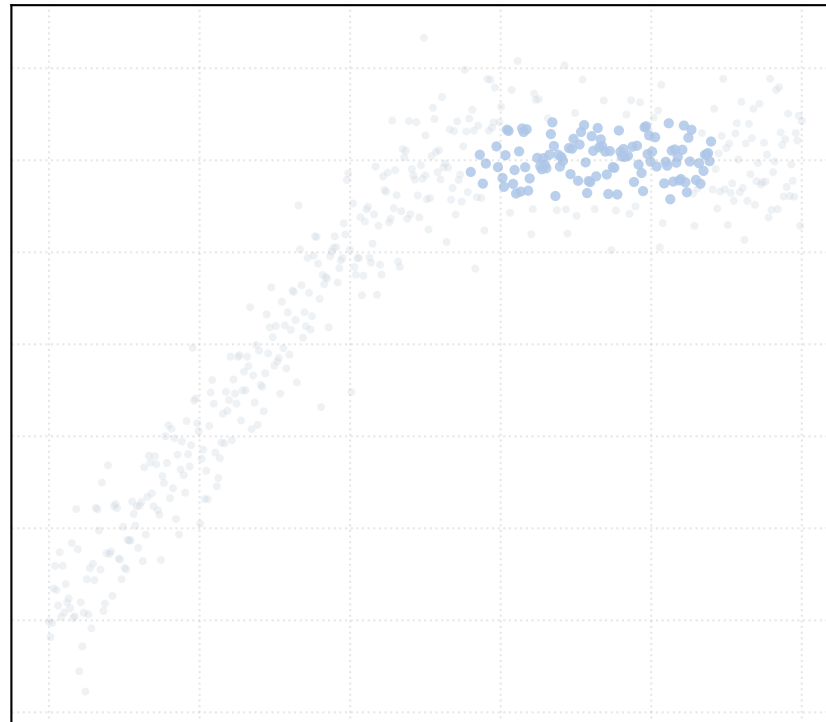
min_intersection = 8 [1 Clusters]



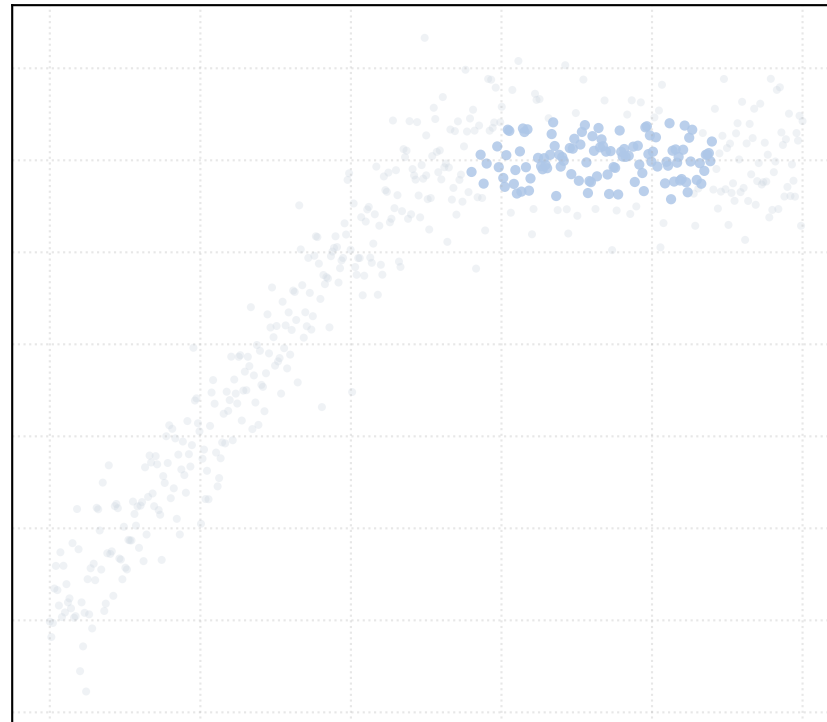
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

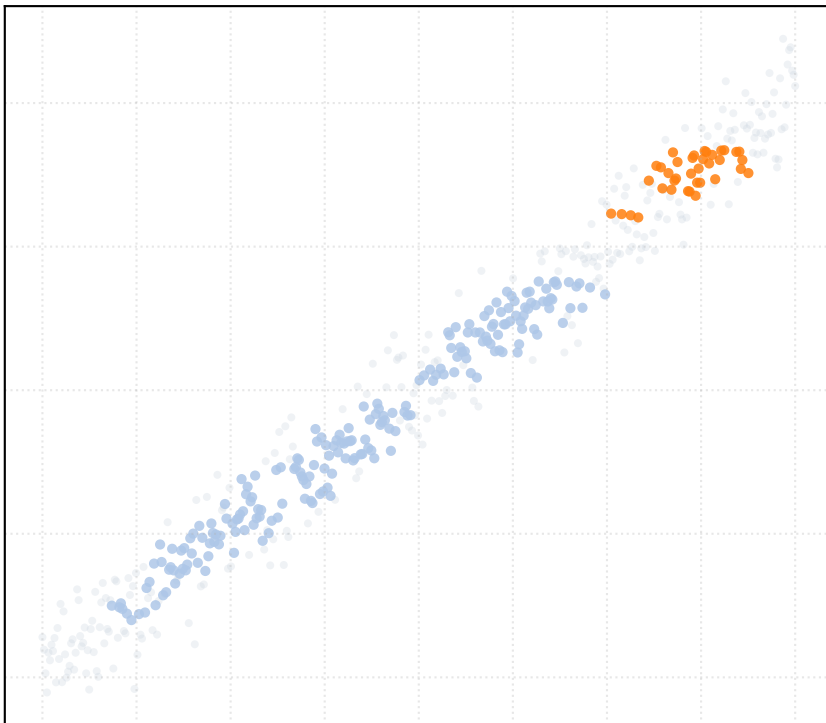


min_intersection = 20 [1 Clusters]

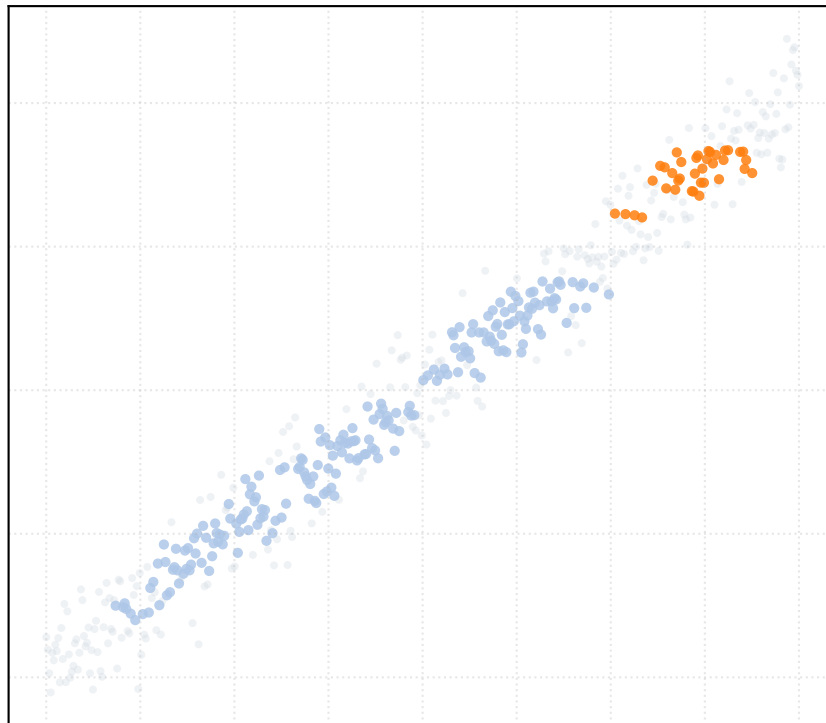


Dataset: linear_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

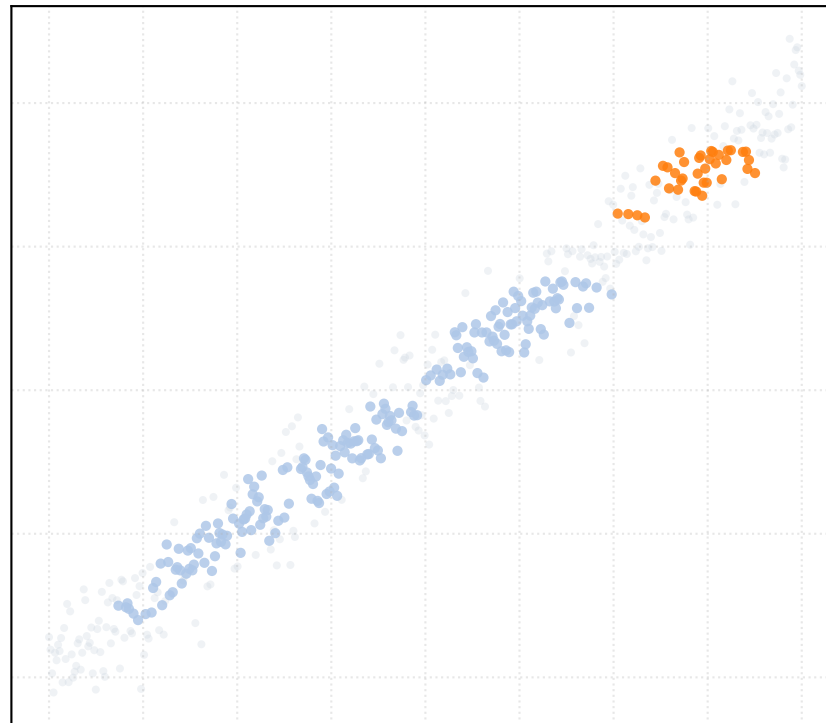
min_intersection = 1 (Default) [2 Clusters]



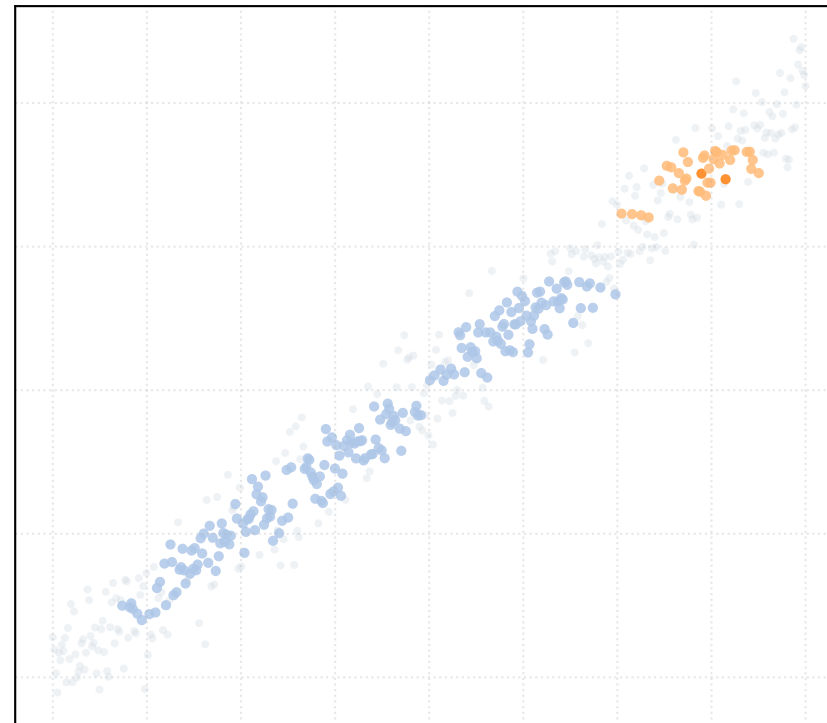
min_intersection = 2 [2 Clusters]



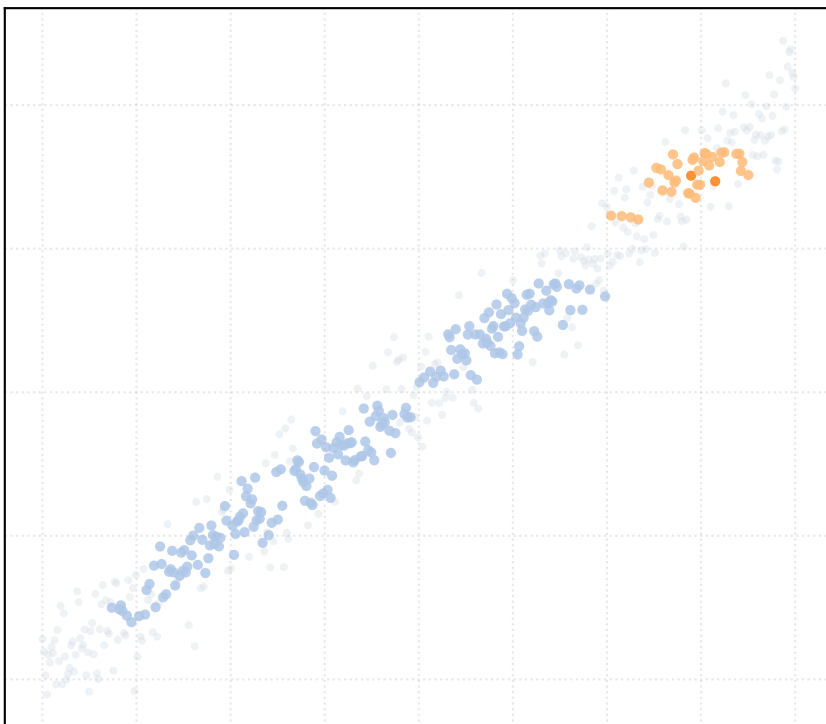
min_intersection = 3 [2 Clusters]



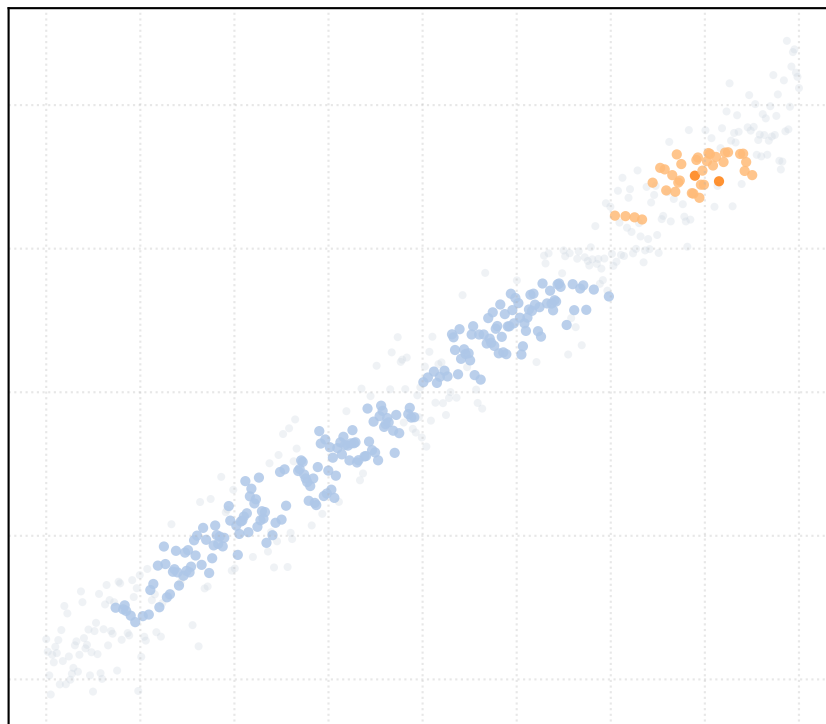
min_intersection = 5 [3 Clusters]



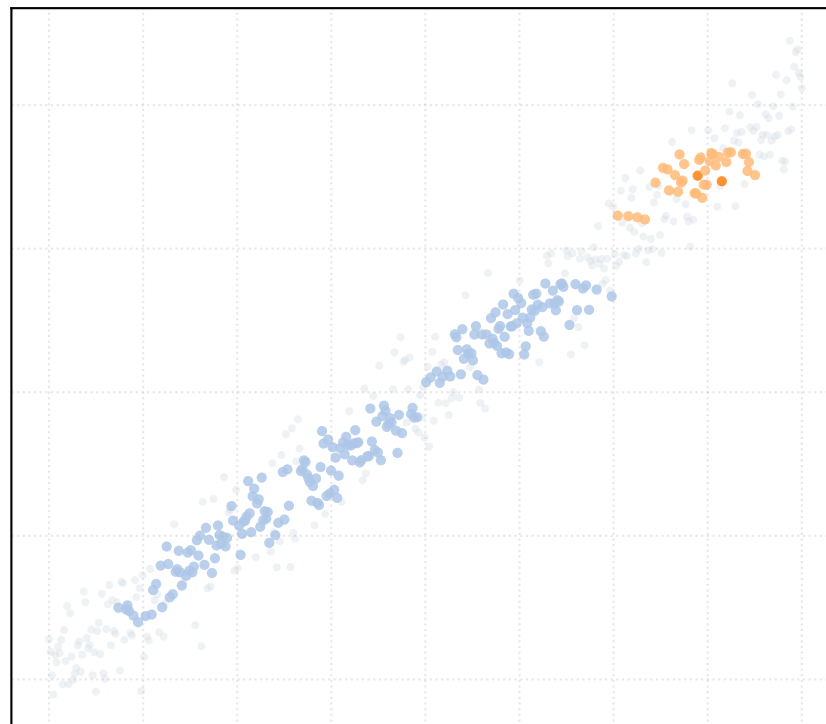
min_intersection = 8 [3 Clusters]



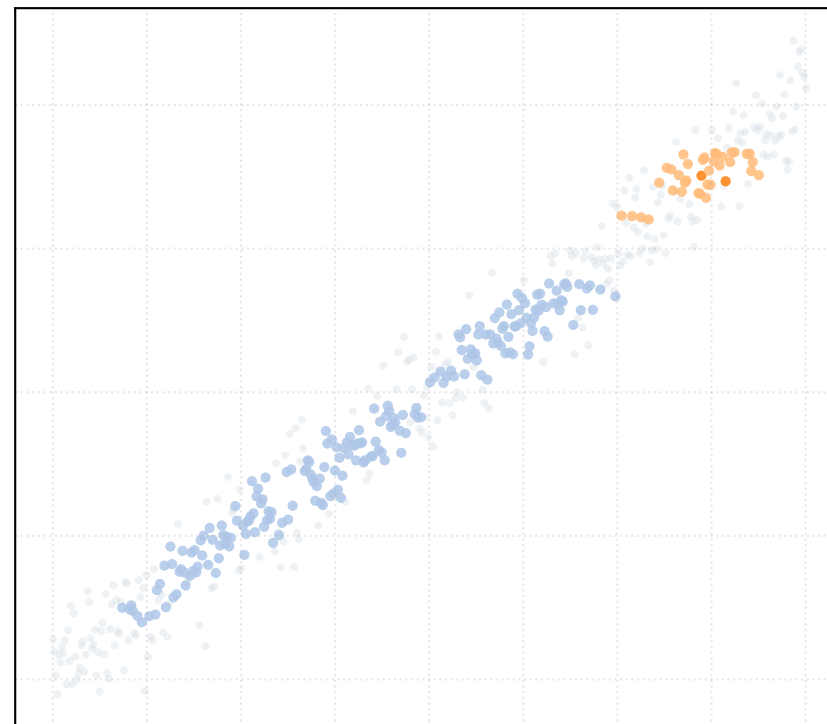
min_intersection = 10 [3 Clusters]



min_intersection = 15 [3 Clusters]

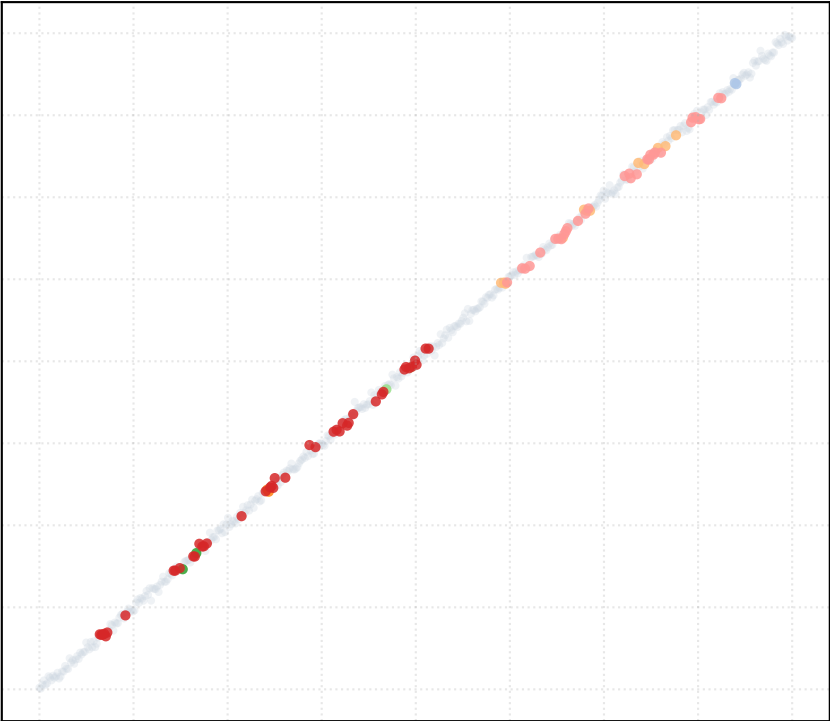


min_intersection = 20 [3 Clusters]

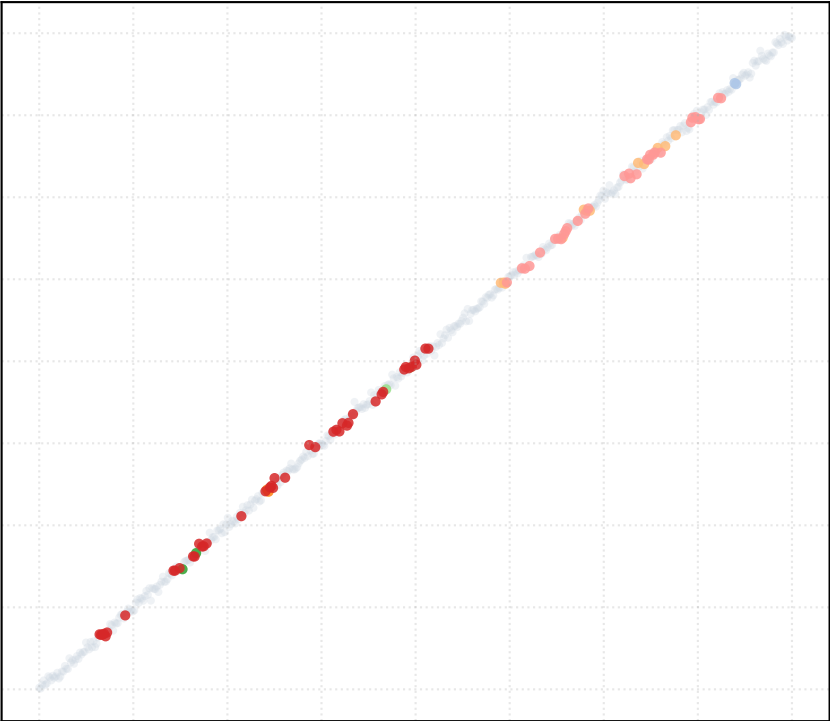


Dataset: linear_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

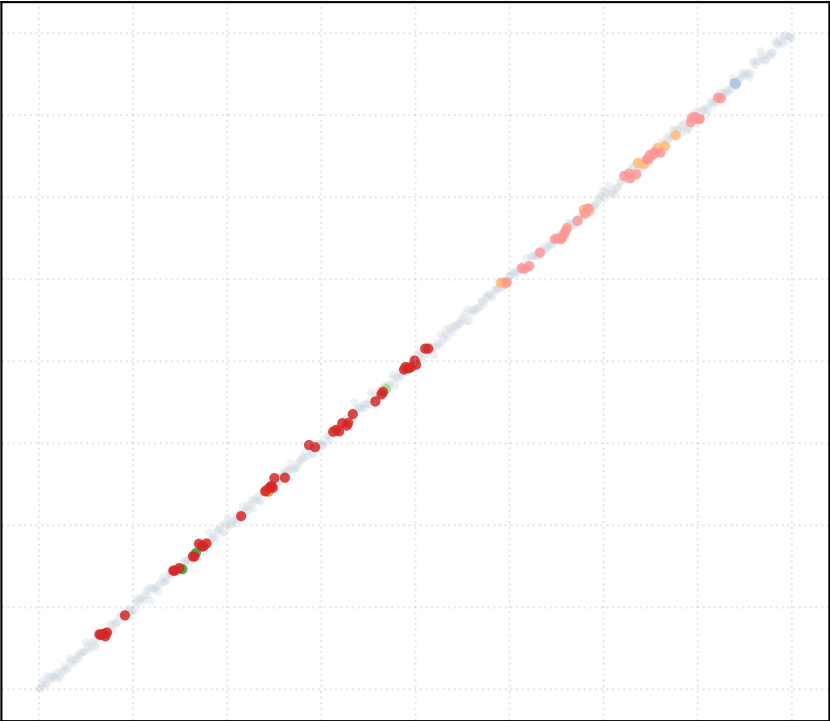
min_intersection = 1 (Default) [7 Clusters]



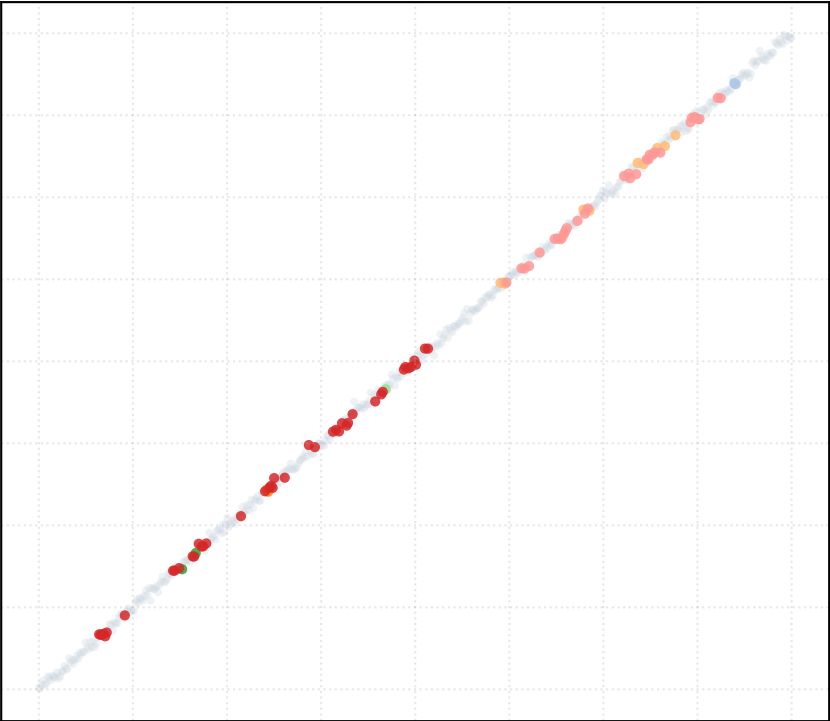
min_intersection = 2 [7 Clusters]



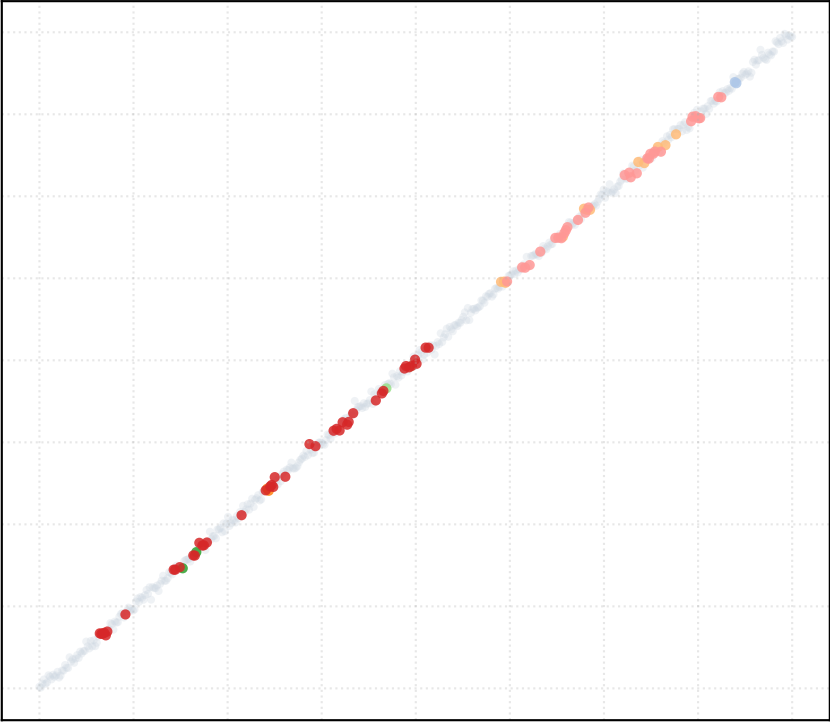
min_intersection = 3 [7 Clusters]



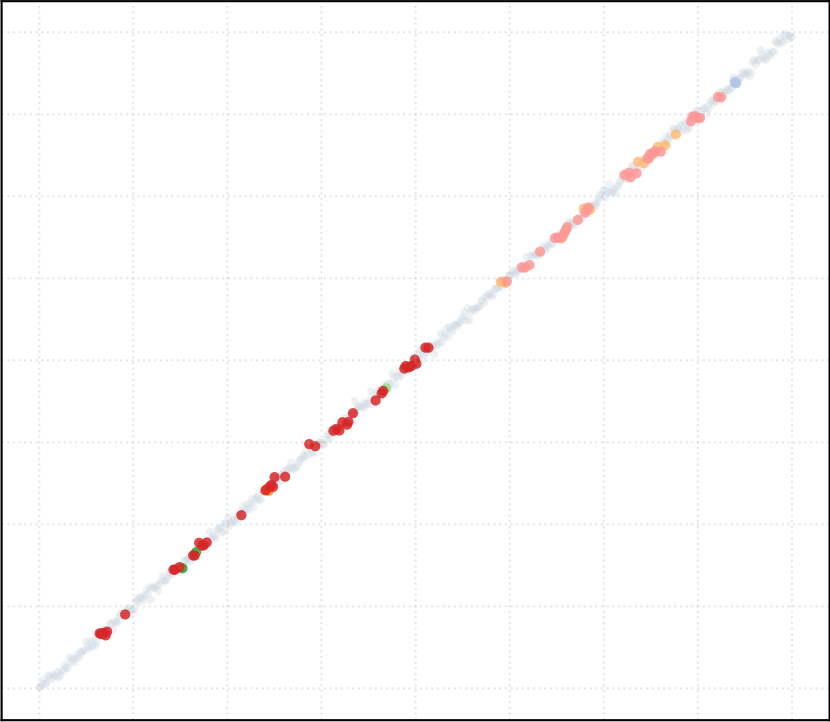
min_intersection = 5 [7 Clusters]



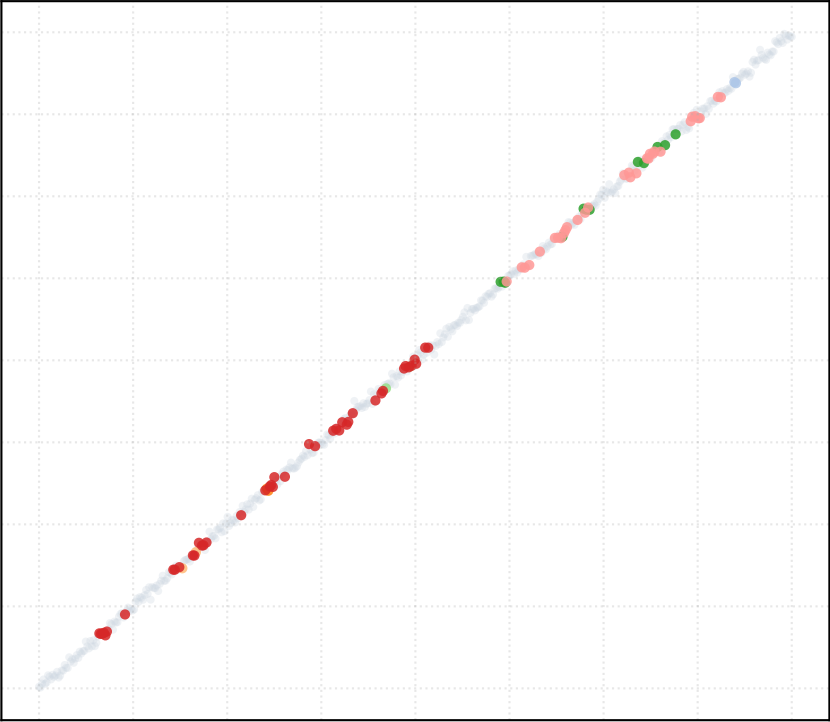
min_intersection = 8 [7 Clusters]



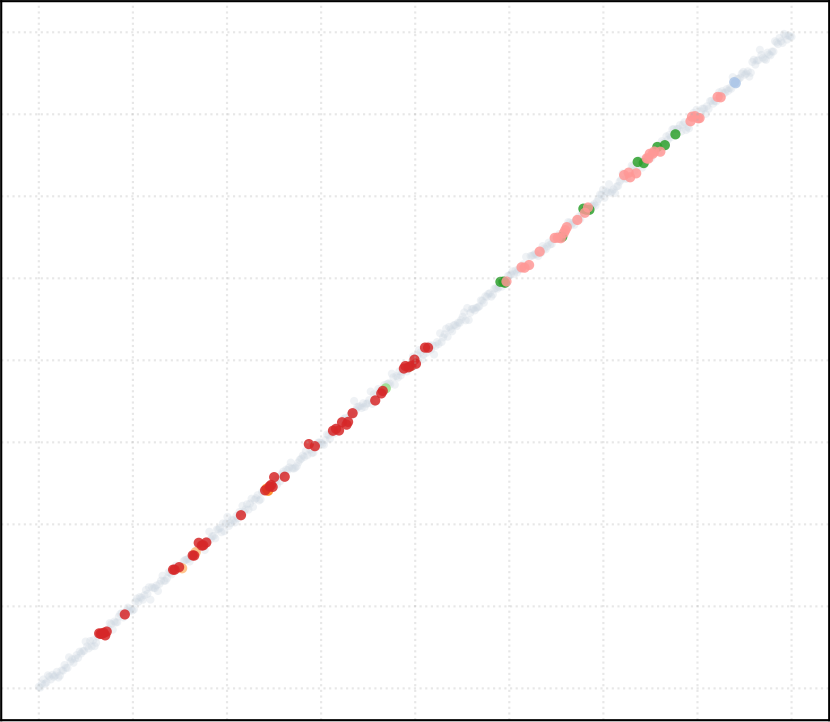
min_intersection = 10 [7 Clusters]



min_intersection = 15 [7 Clusters]

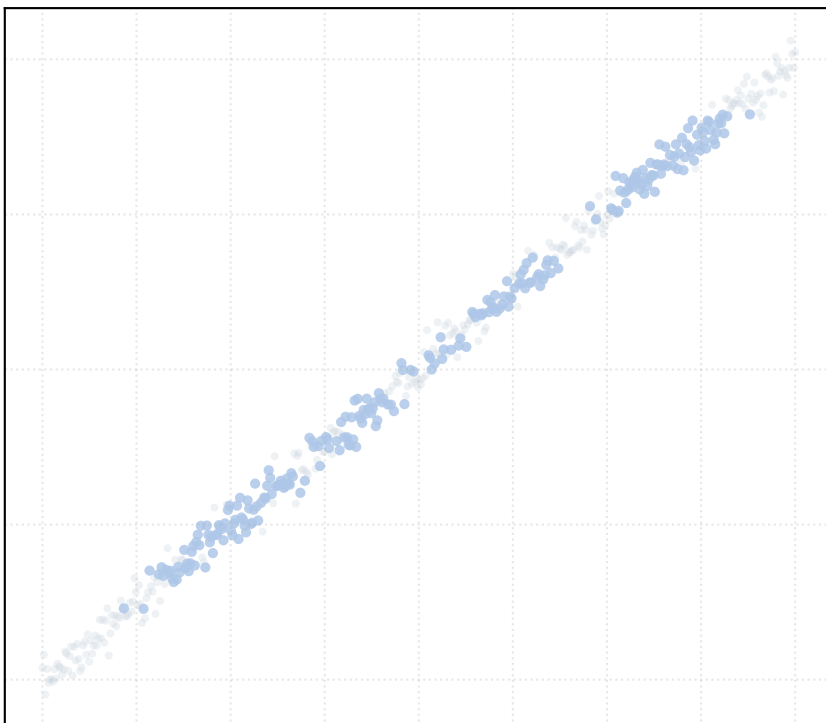


min_intersection = 20 [7 Clusters]

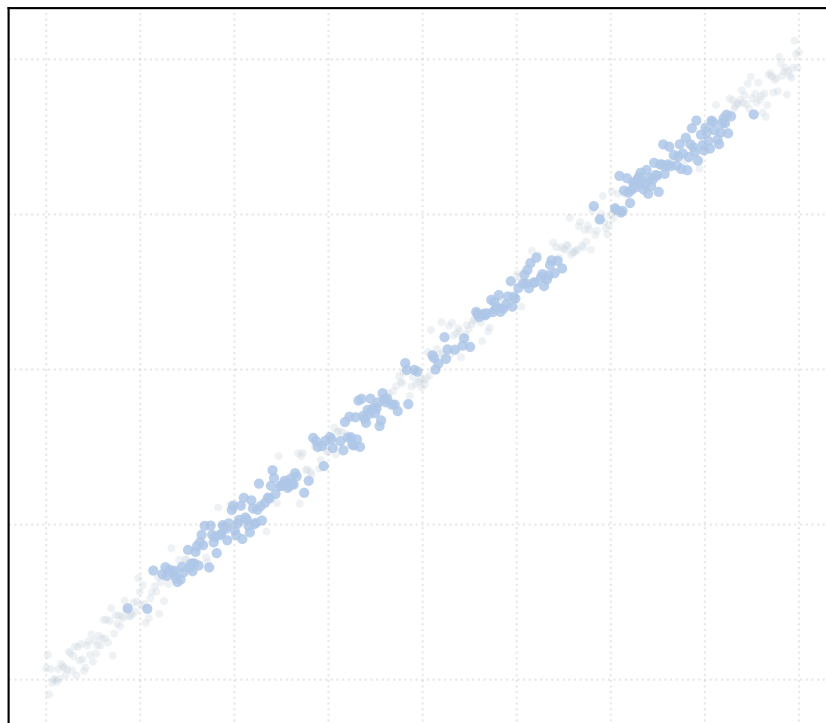


Dataset: linear_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

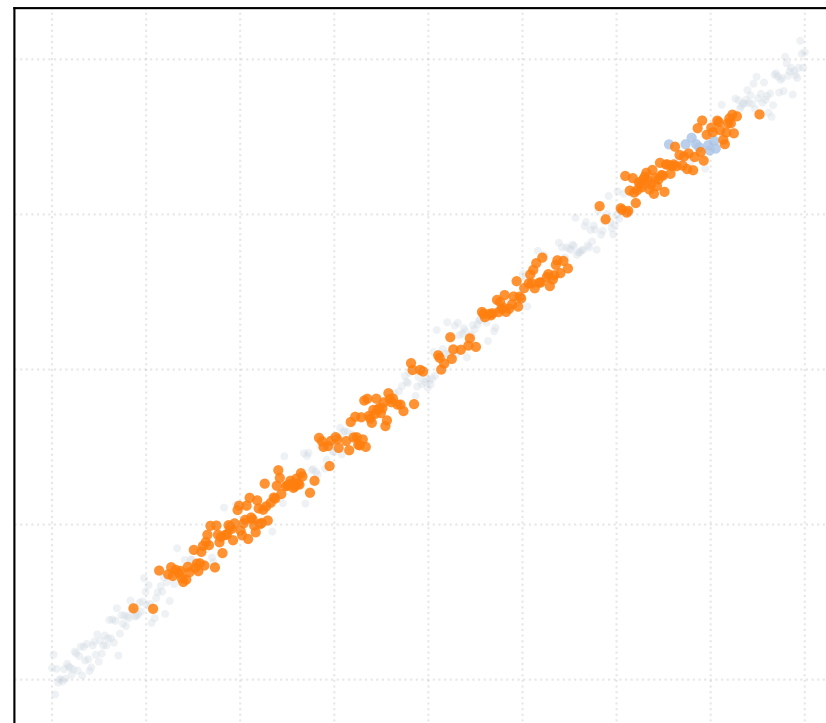
min_intersection = 1 (Default) [1 Clusters]



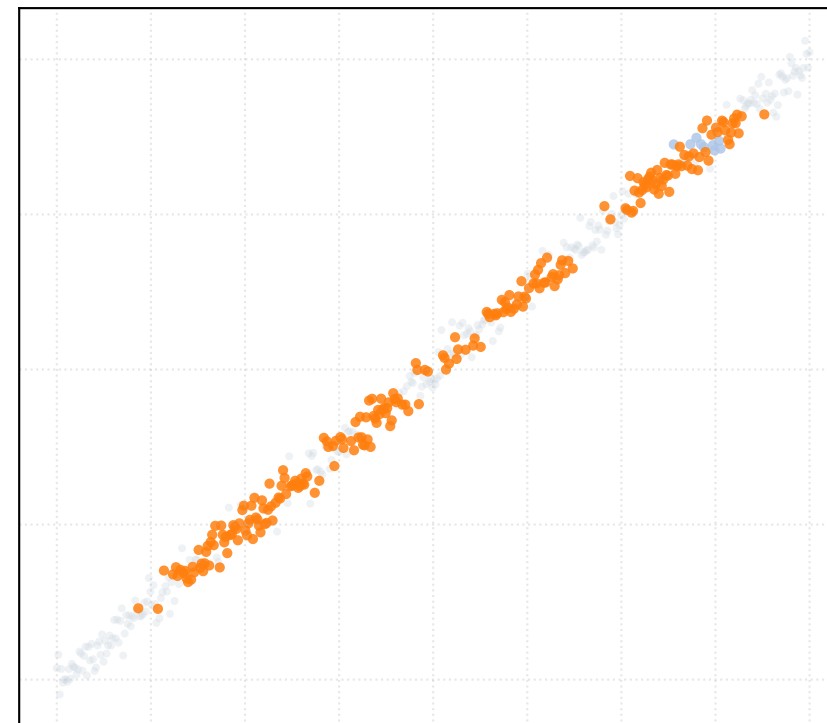
min_intersection = 2 [1 Clusters]



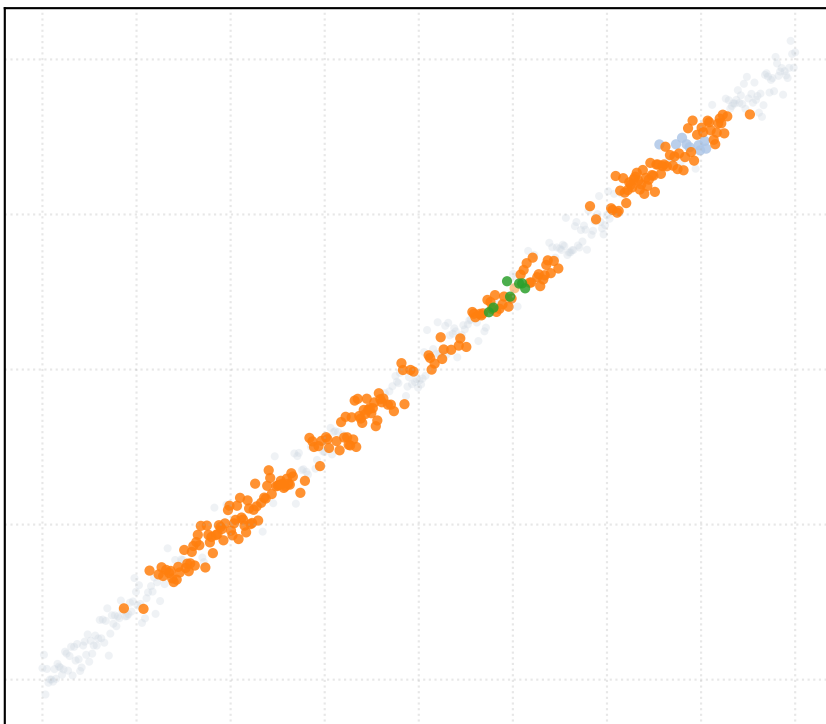
min_intersection = 3 [2 Clusters]



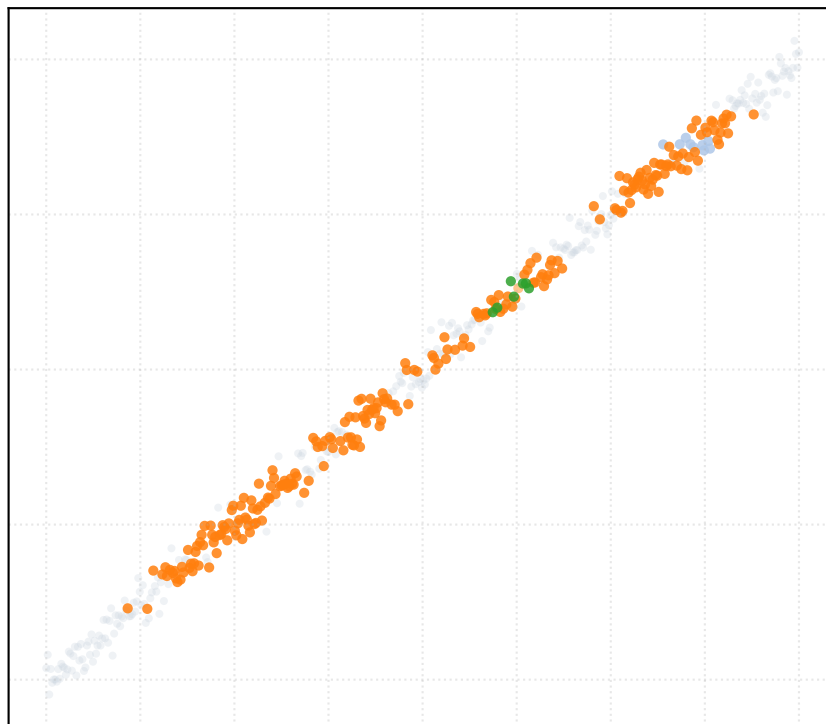
min_intersection = 5 [2 Clusters]



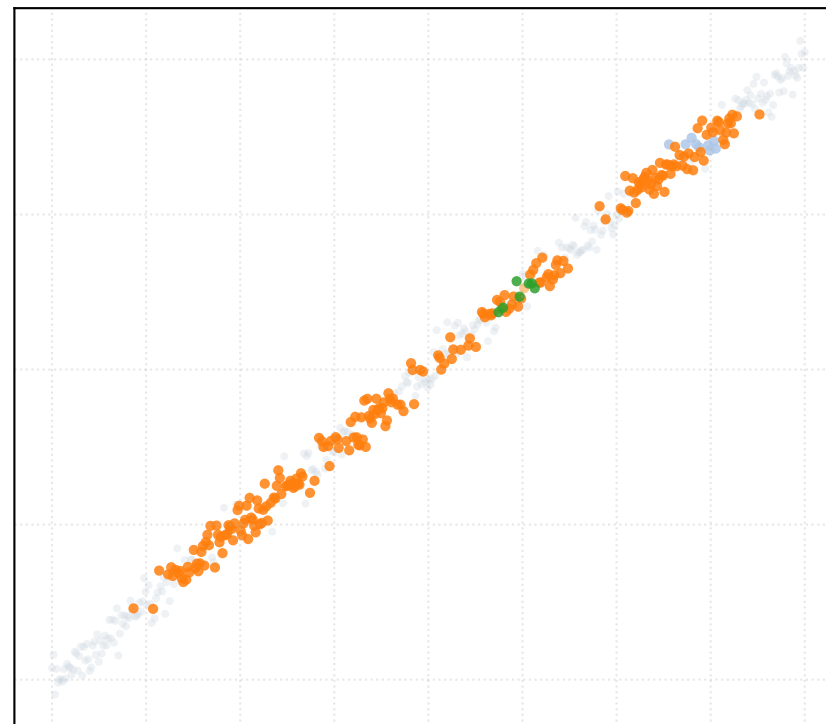
min_intersection = 8 [4 Clusters]



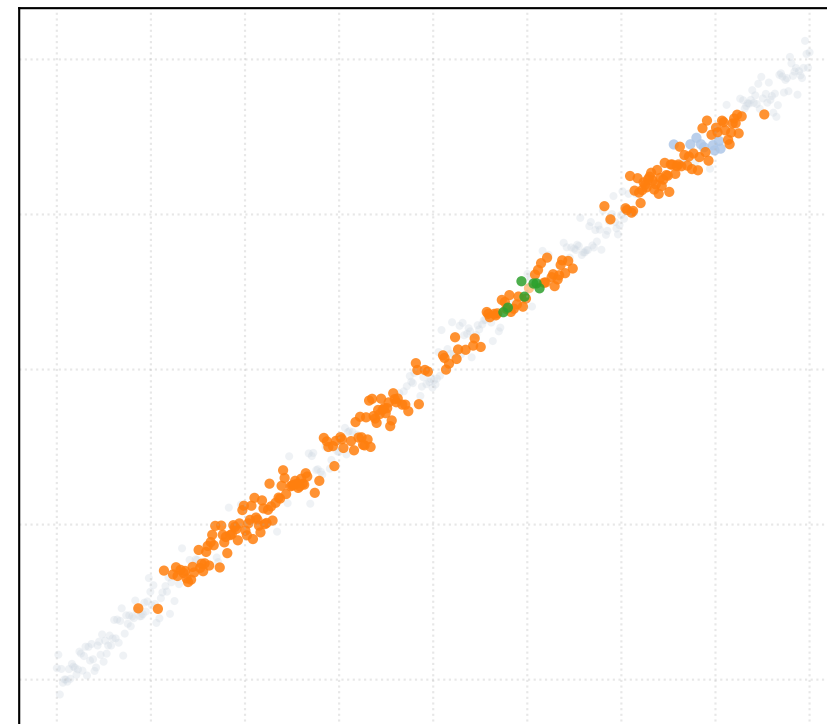
min_intersection = 10 [4 Clusters]



min_intersection = 15 [4 Clusters]

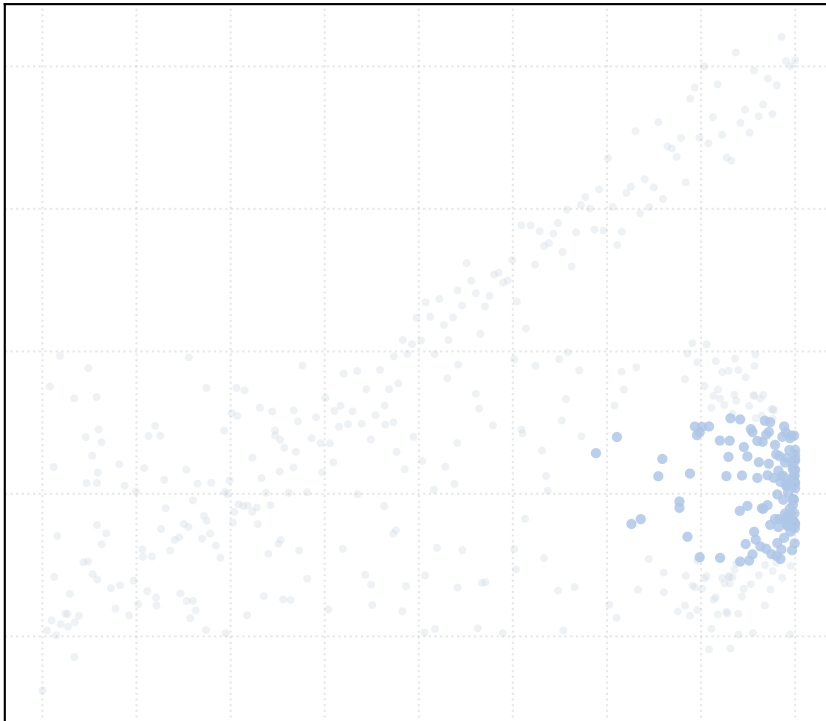


min_intersection = 20 [4 Clusters]

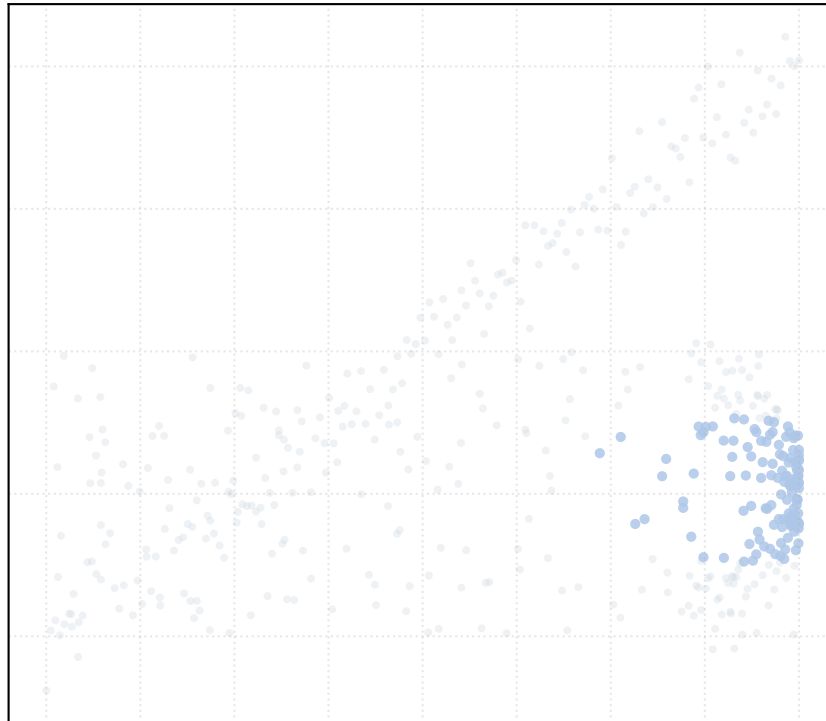


Dataset: mixed_generators_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

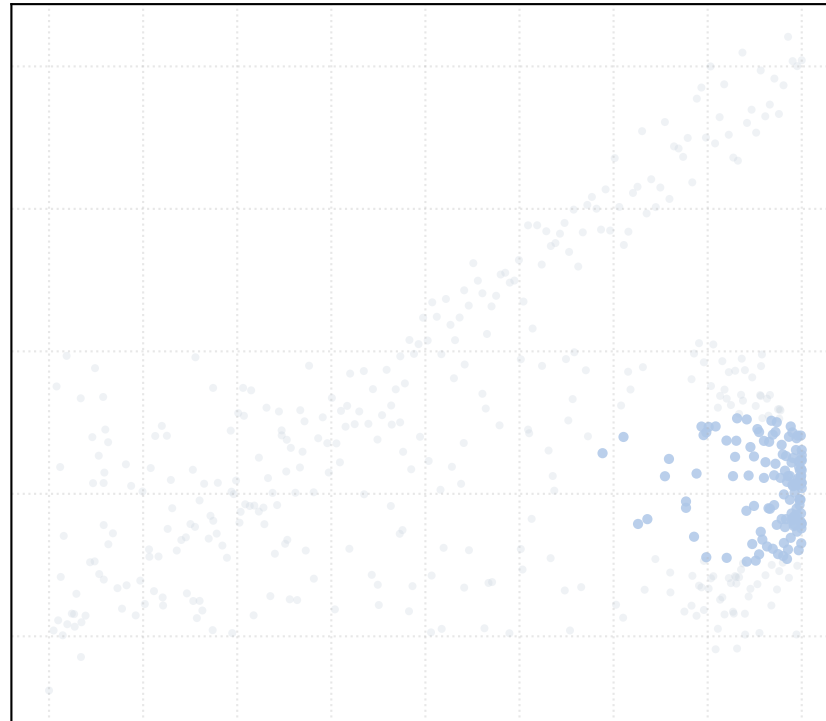
min_intersection = 1 (Default) [1 Clusters]



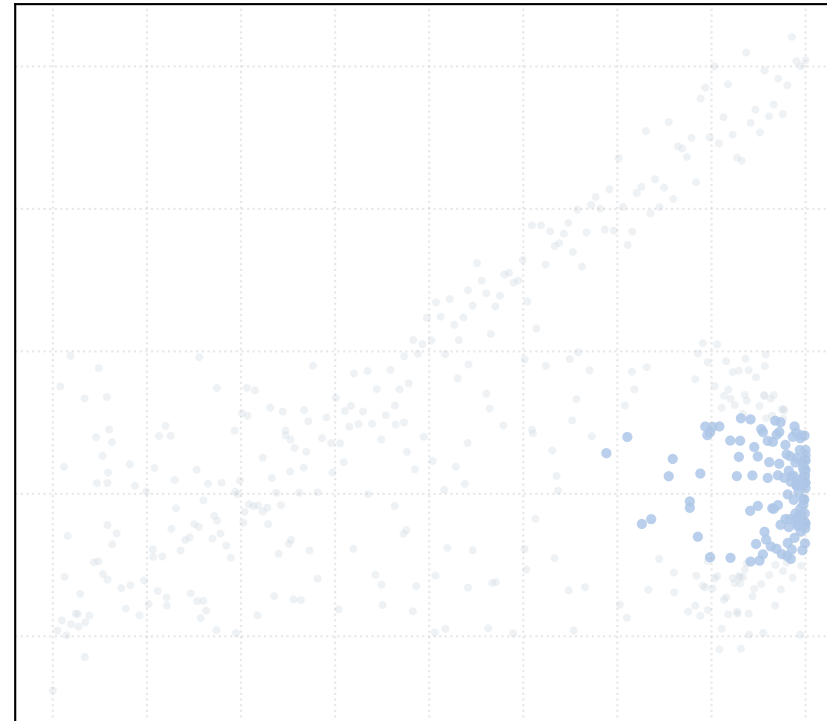
min_intersection = 2 [1 Clusters]



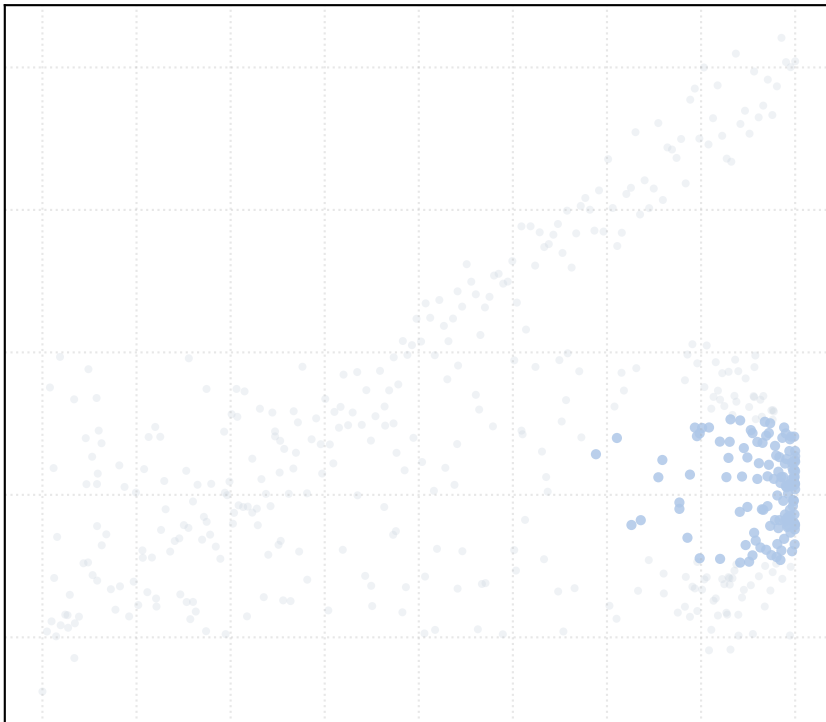
min_intersection = 3 [1 Clusters]



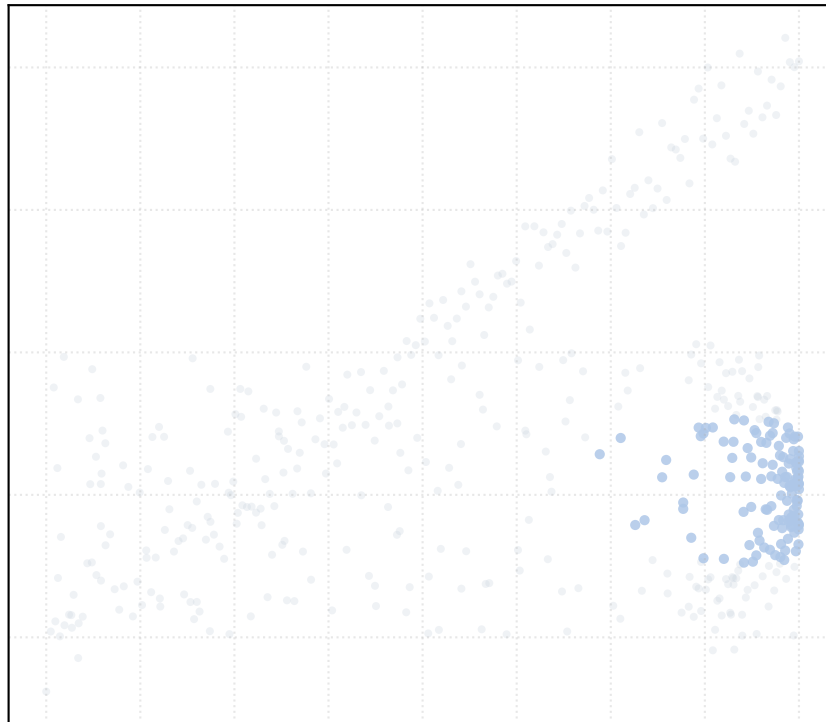
min_intersection = 5 [1 Clusters]



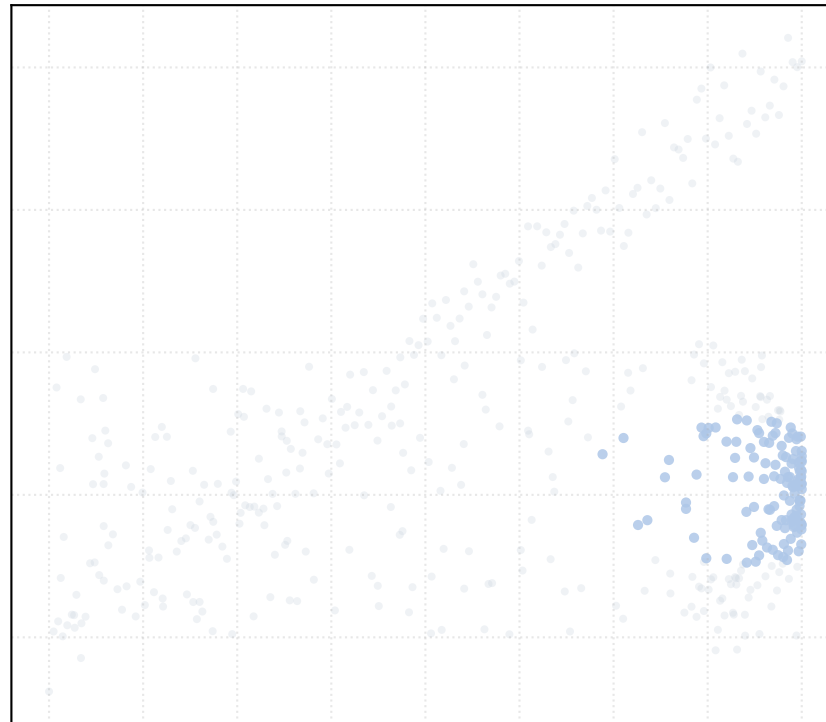
min_intersection = 8 [1 Clusters]



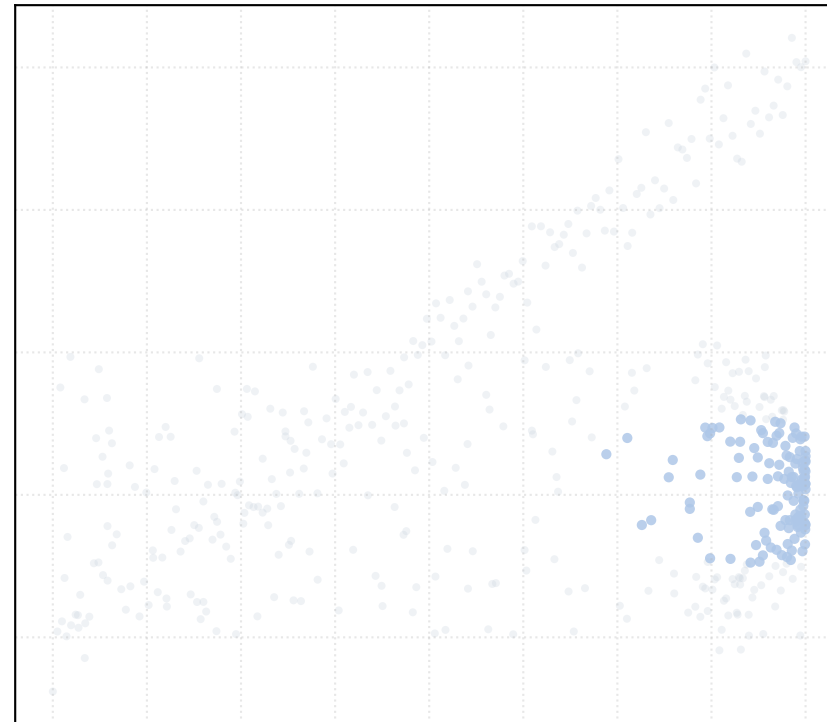
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

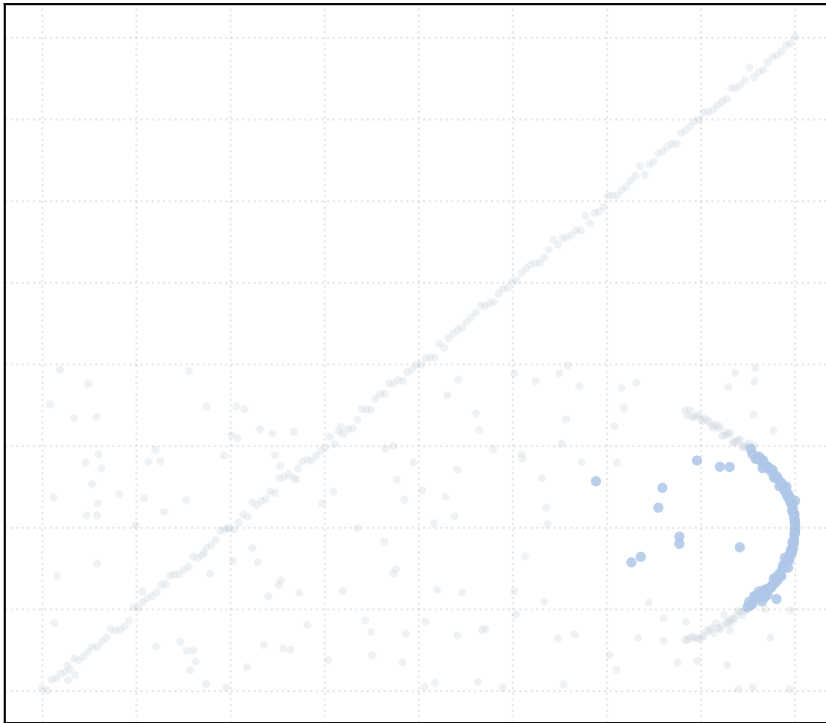


min_intersection = 20 [1 Clusters]

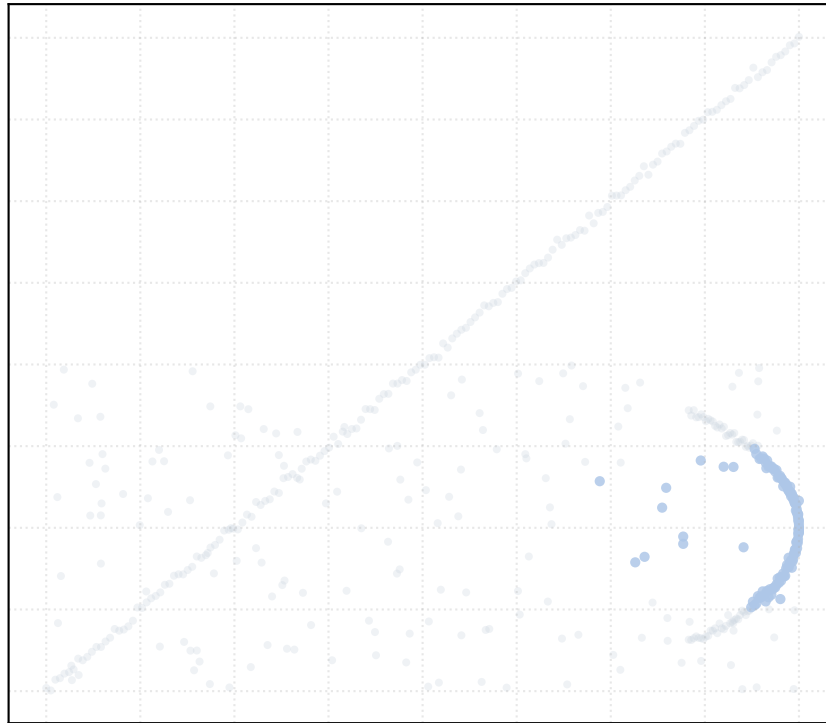


Dataset: mixed_generators_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

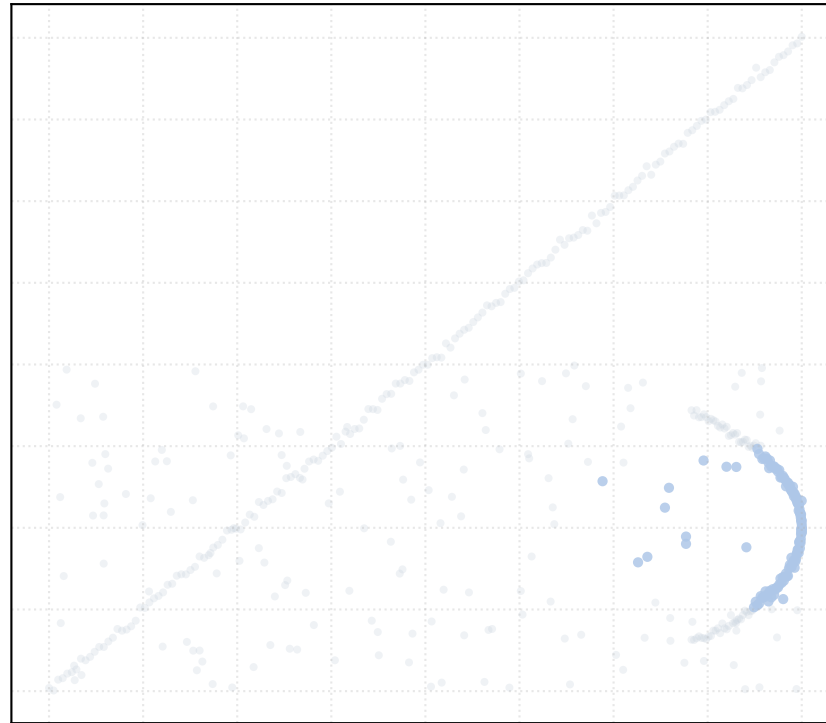
min_intersection = 1 (Default) [1 Clusters]



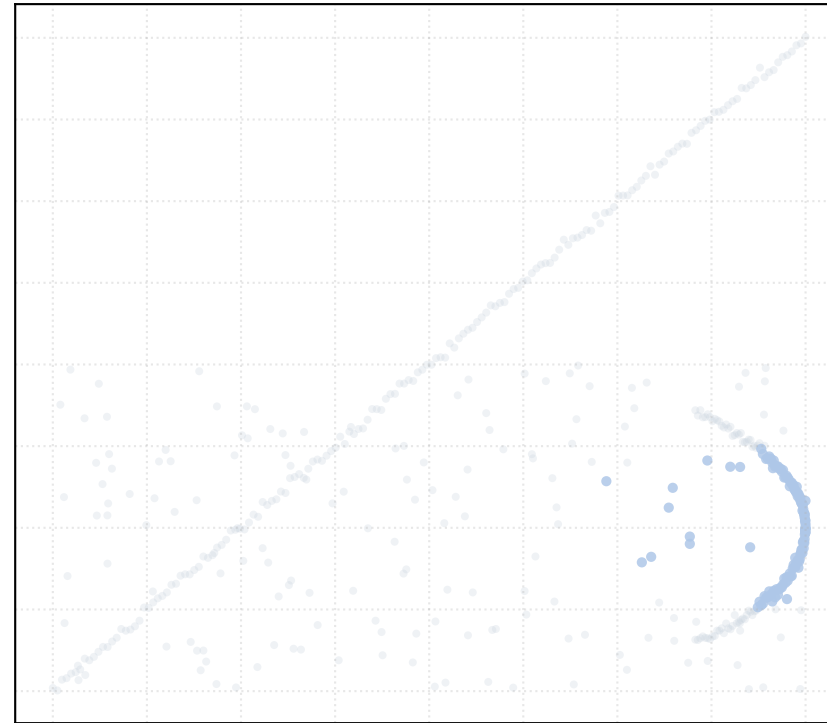
min_intersection = 2 [1 Clusters]



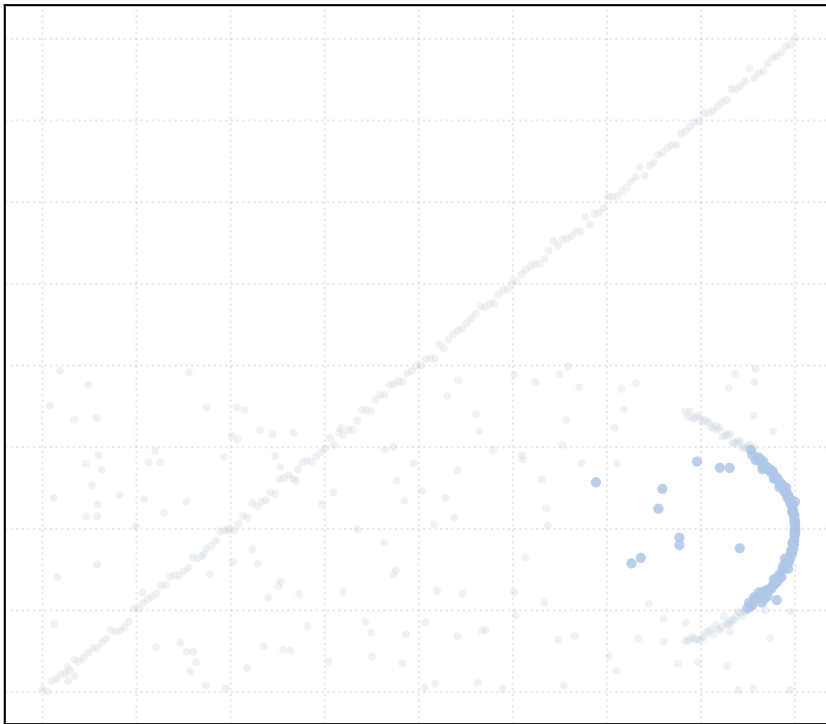
min_intersection = 3 [1 Clusters]



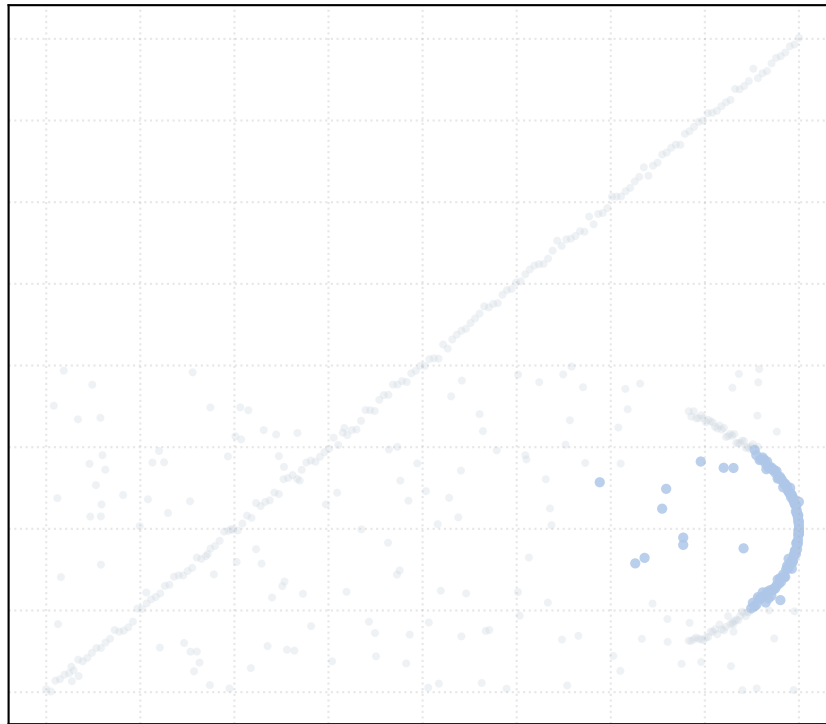
min_intersection = 5 [1 Clusters]



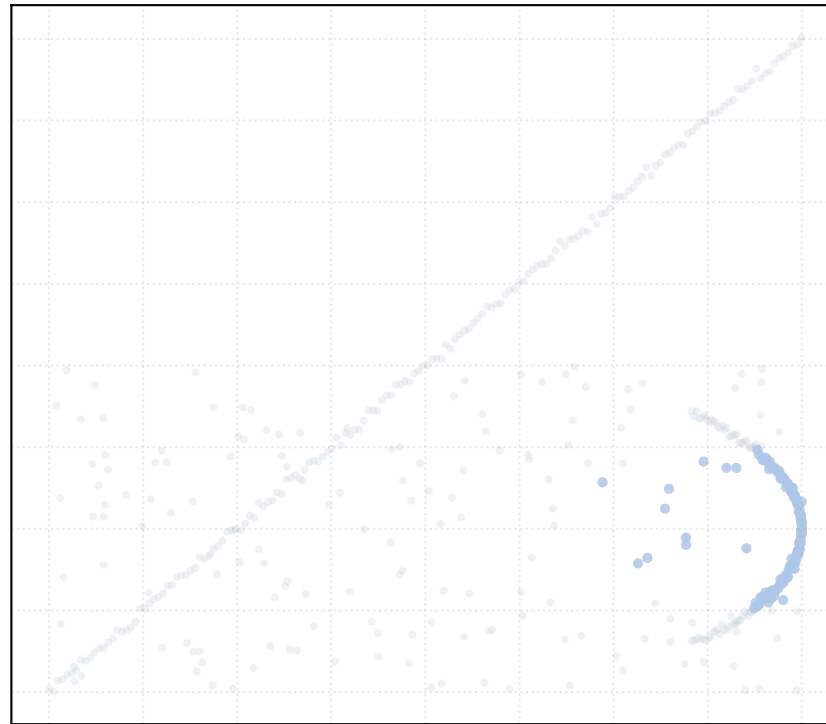
min_intersection = 8 [1 Clusters]



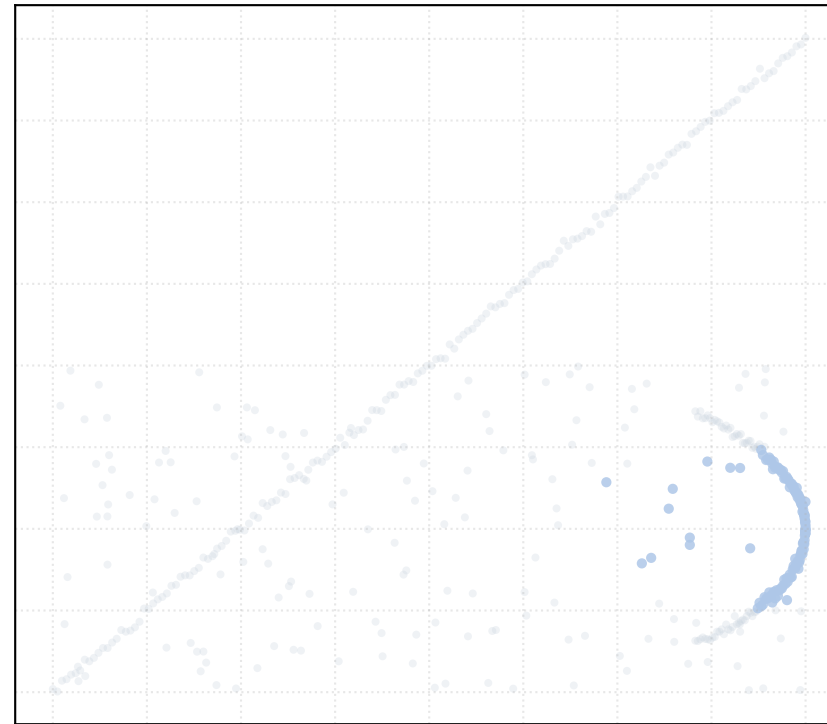
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

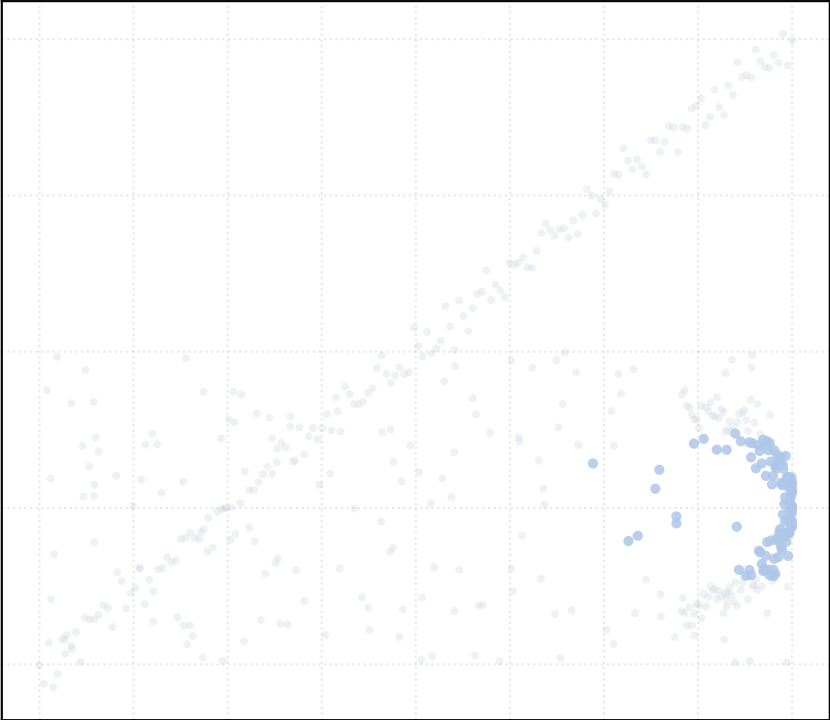


min_intersection = 20 [1 Clusters]

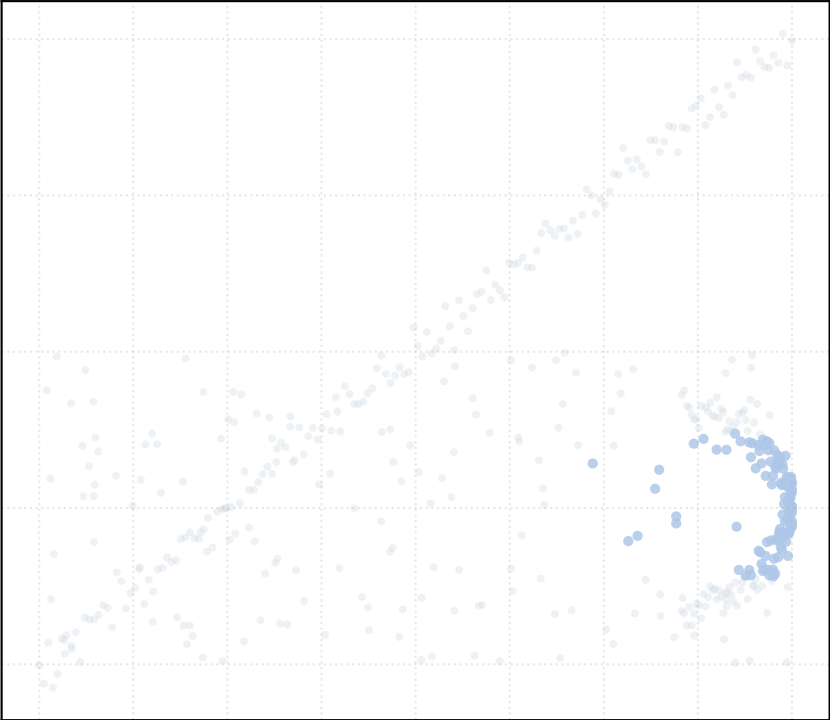


Dataset: mixed_generators_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

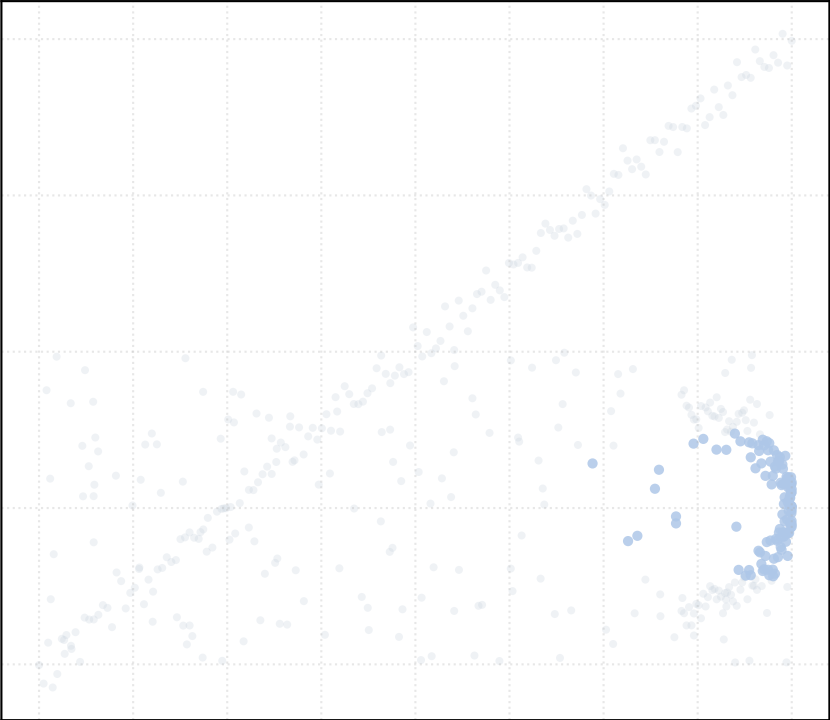
min_intersection = 1 (Default) [1 Clusters]



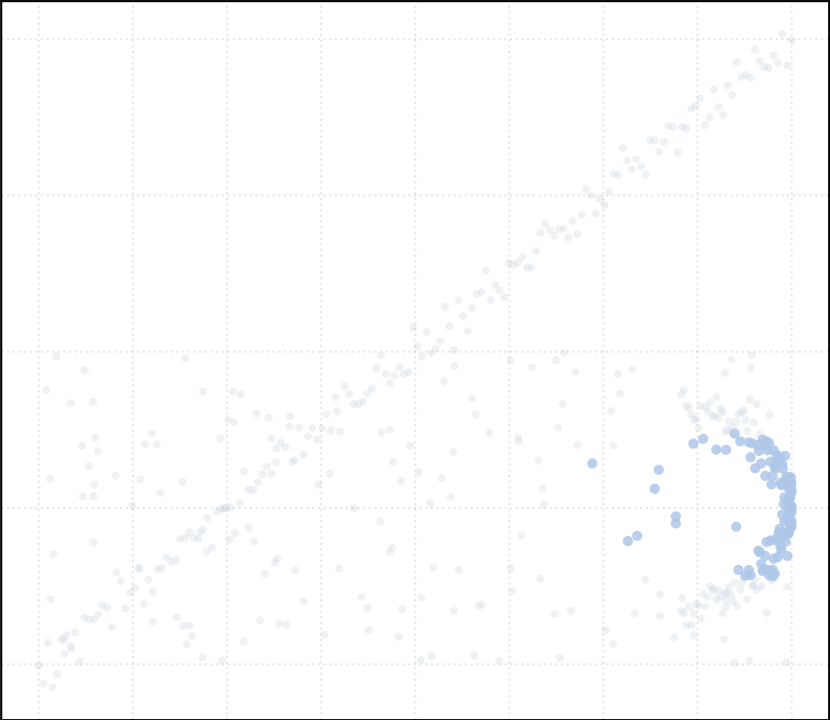
min_intersection = 2 [1 Clusters]



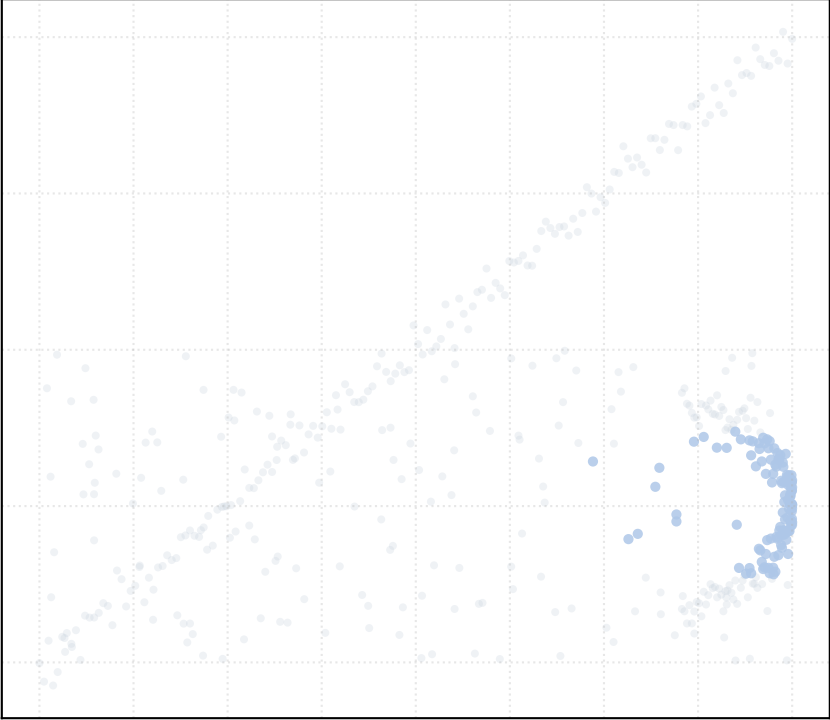
min_intersection = 3 [1 Clusters]



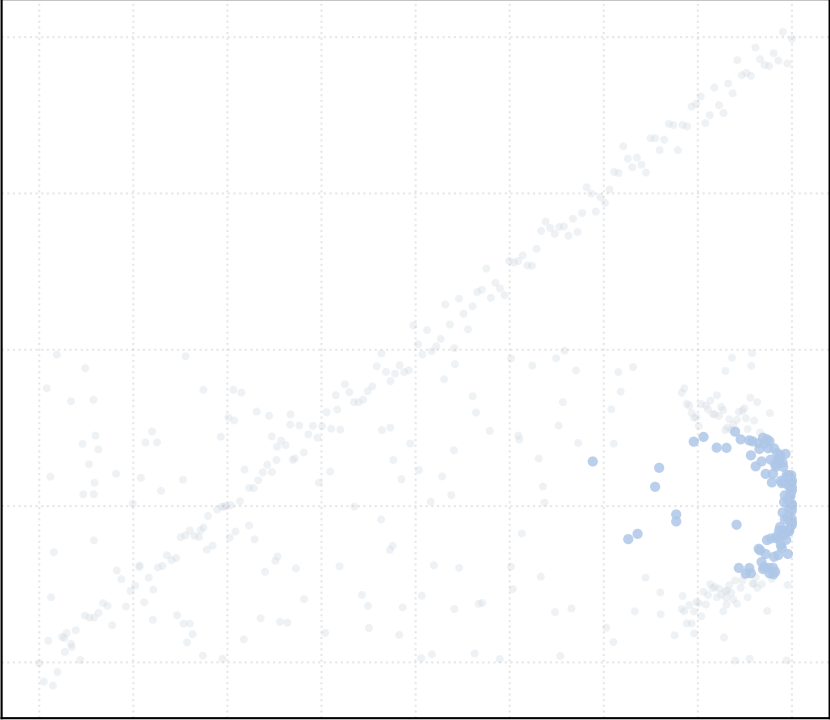
min_intersection = 5 [1 Clusters]



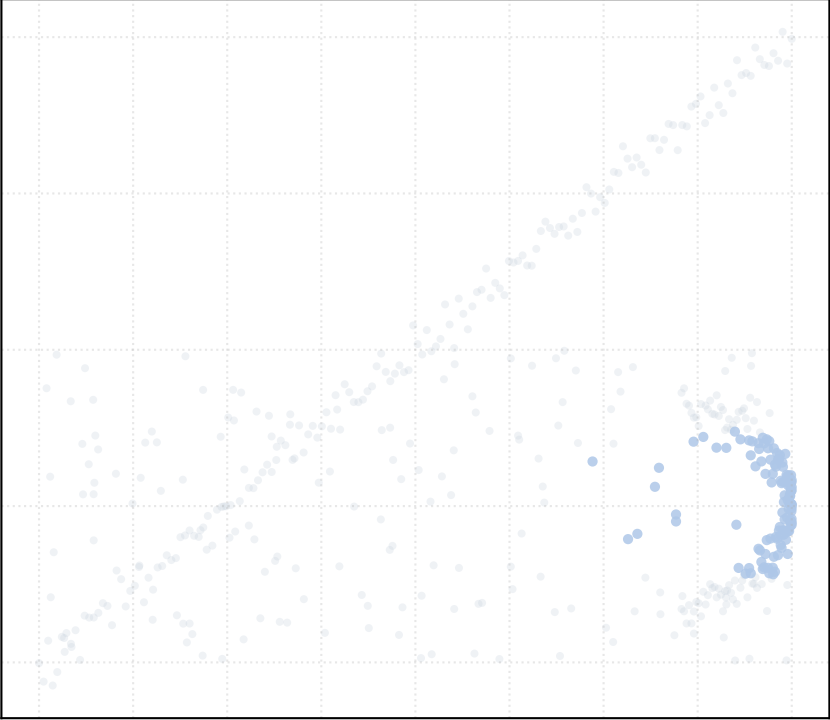
min_intersection = 8 [1 Clusters]



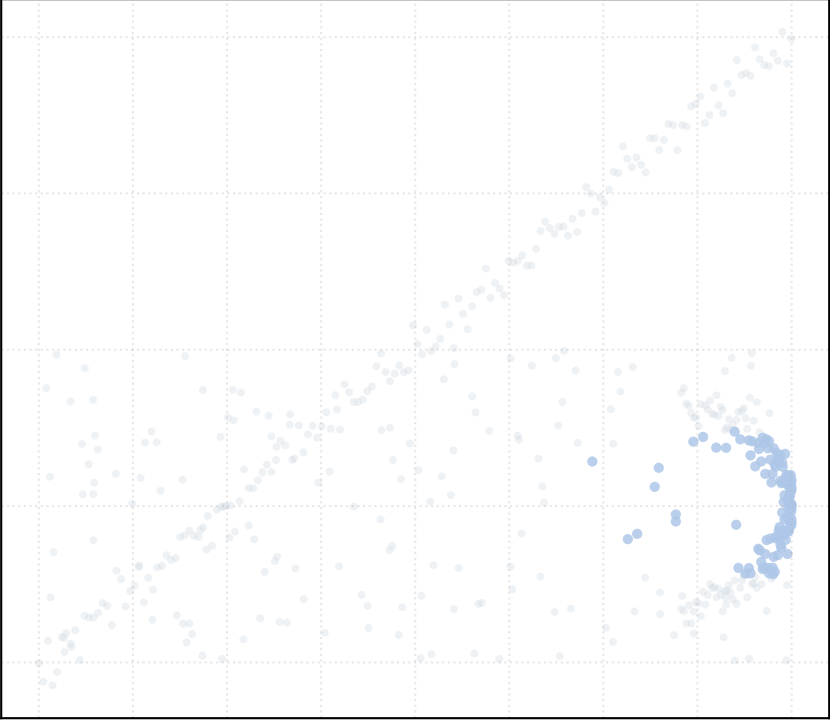
min_intersection = 10 [1 Clusters]



min_intersection = 15 [1 Clusters]

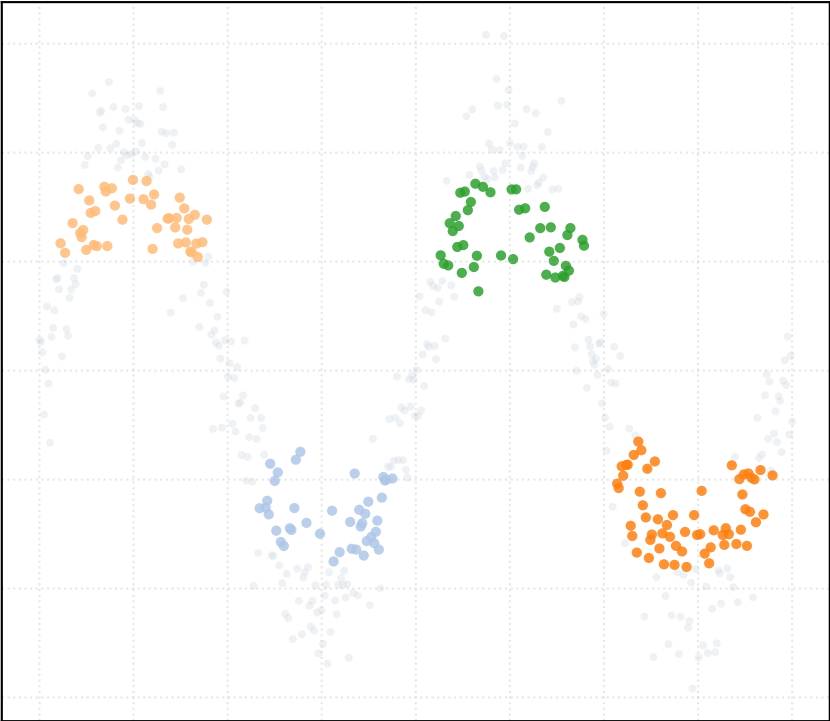


min_intersection = 20 [1 Clusters]

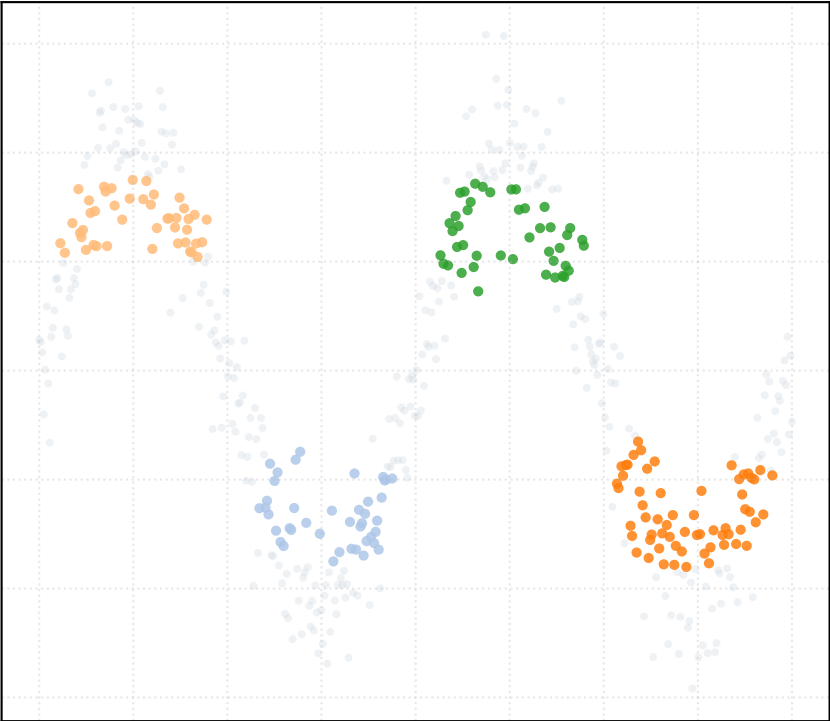


Dataset: s_curve_noise_high | Ensemble Merge Overlap Sweep (min_intersection)

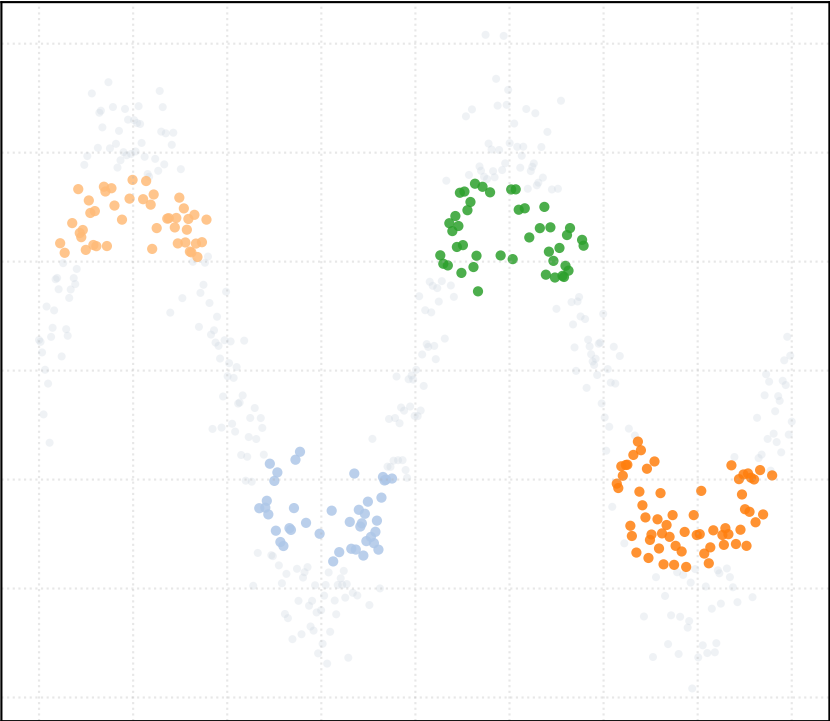
min_intersection = 1 (Default) [4 Clusters]



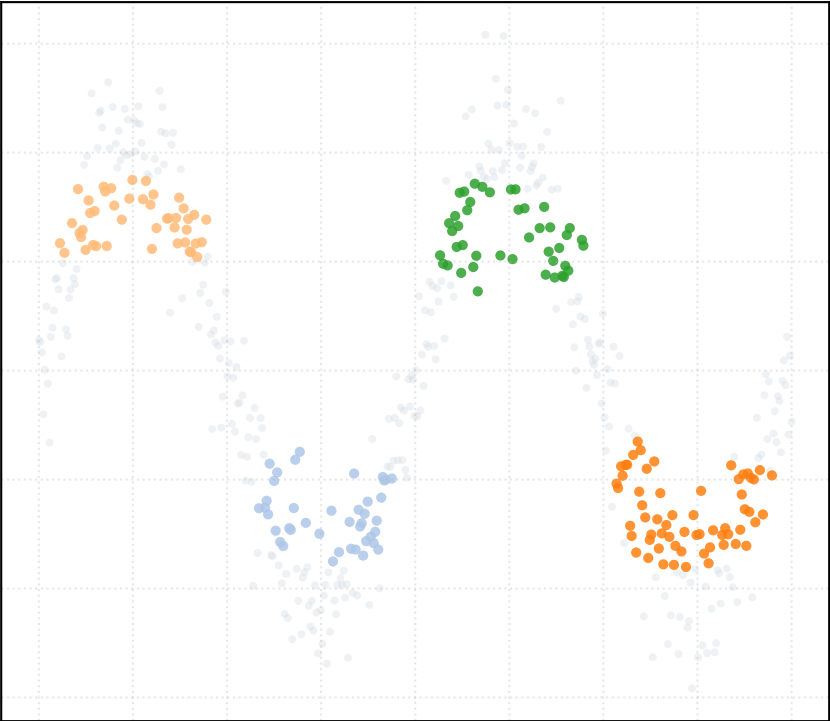
min_intersection = 2 [4 Clusters]



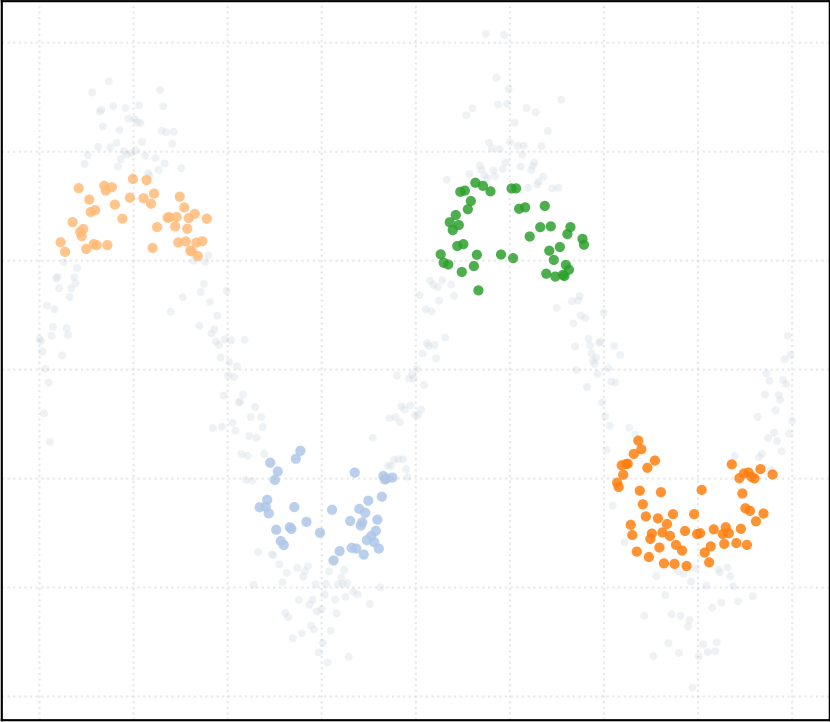
min_intersection = 3 [4 Clusters]



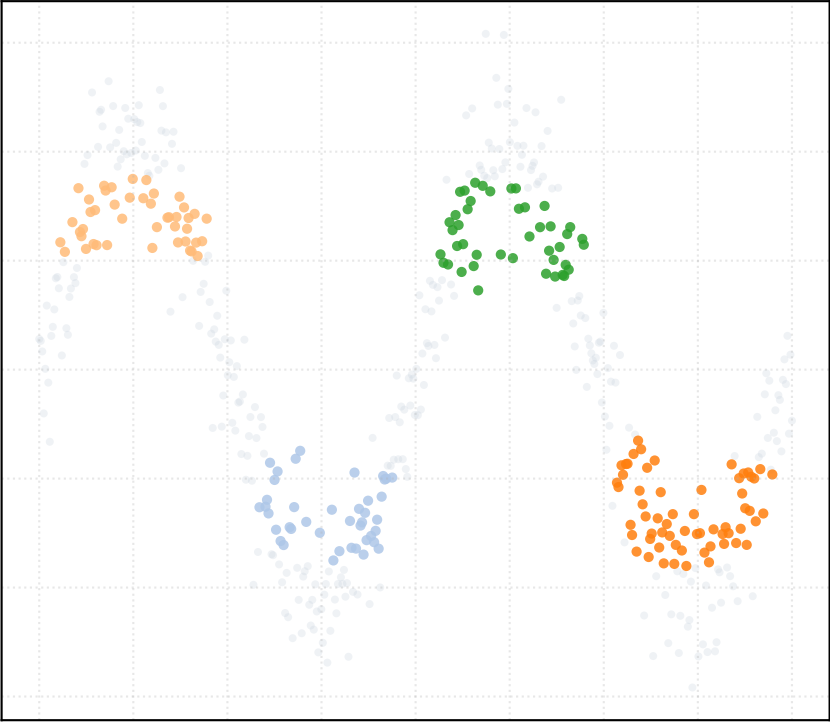
min_intersection = 5 [4 Clusters]



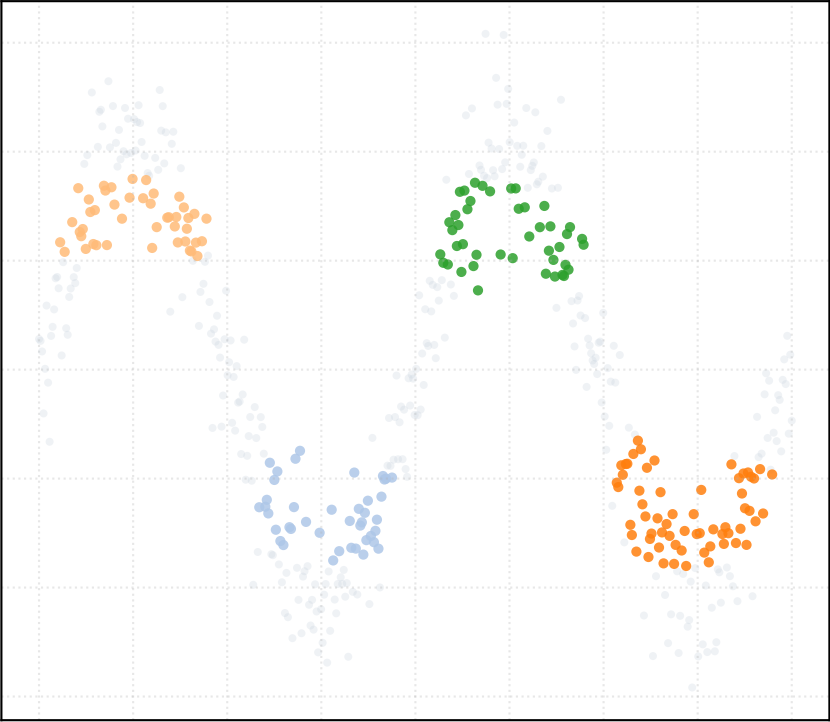
min_intersection = 8 [4 Clusters]



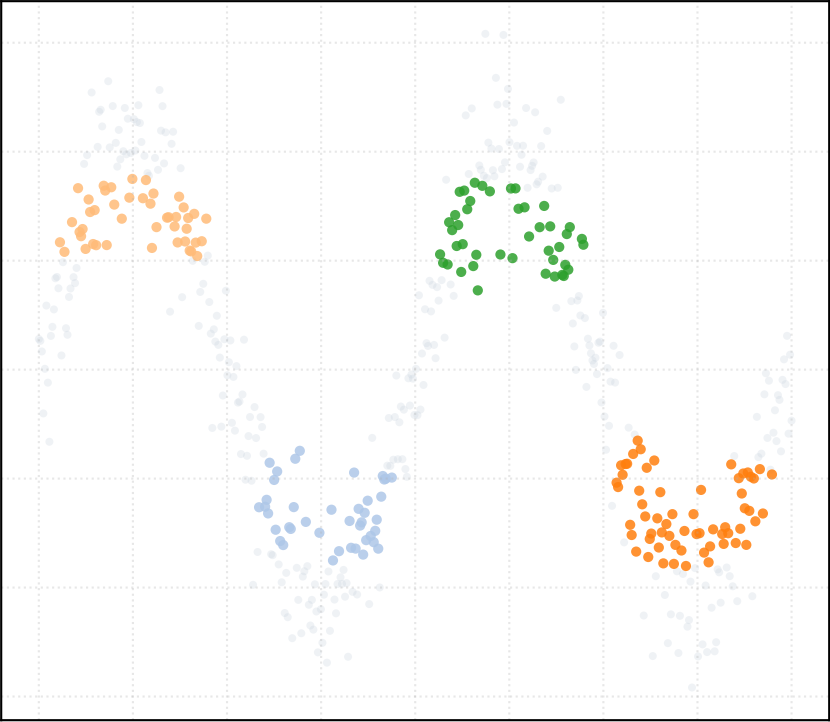
min_intersection = 10 [4 Clusters]



min_intersection = 15 [4 Clusters]

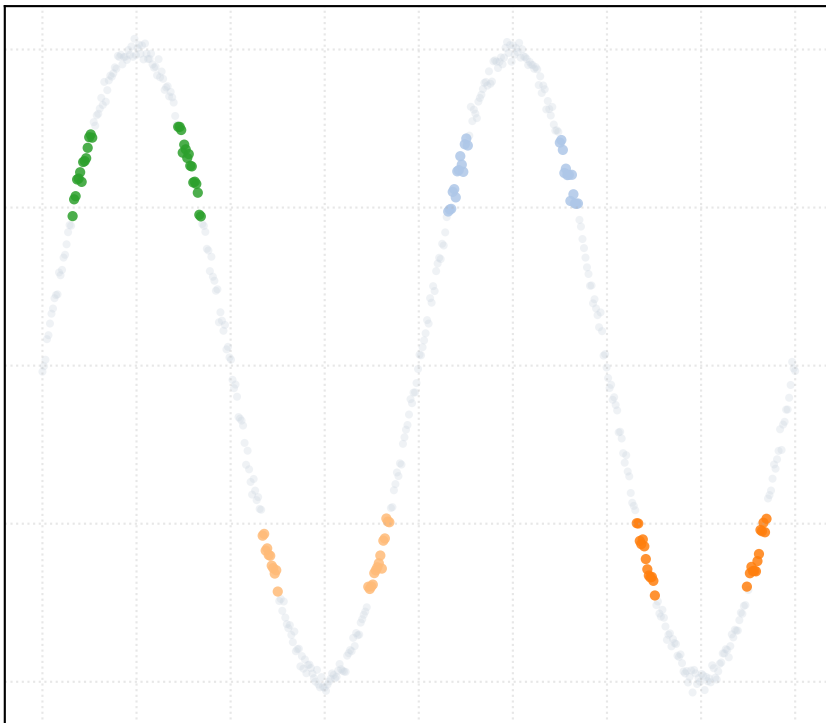


min_intersection = 20 [4 Clusters]

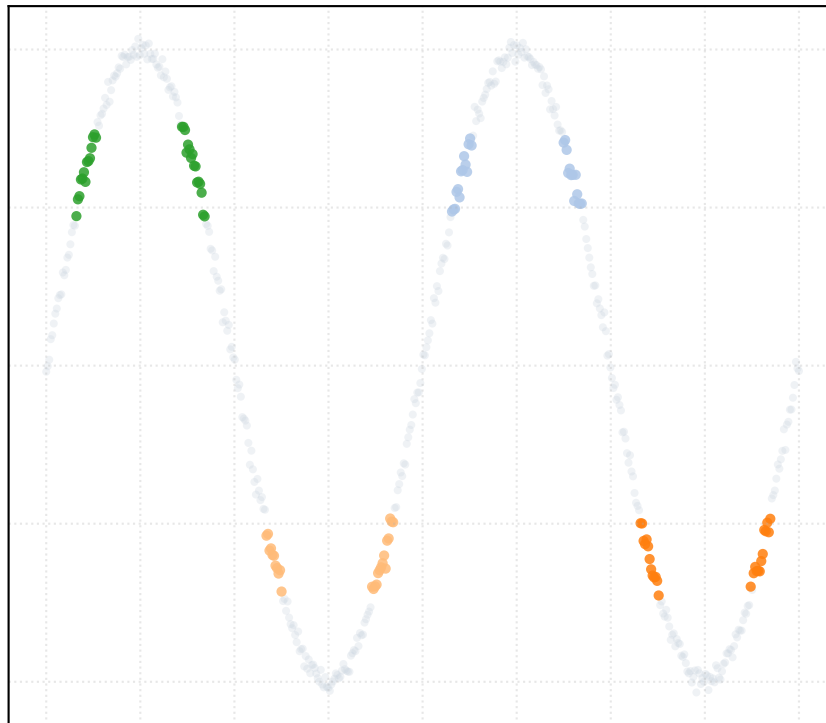


Dataset: s_curve_noise_low | Ensemble Merge Overlap Sweep (min_intersection)

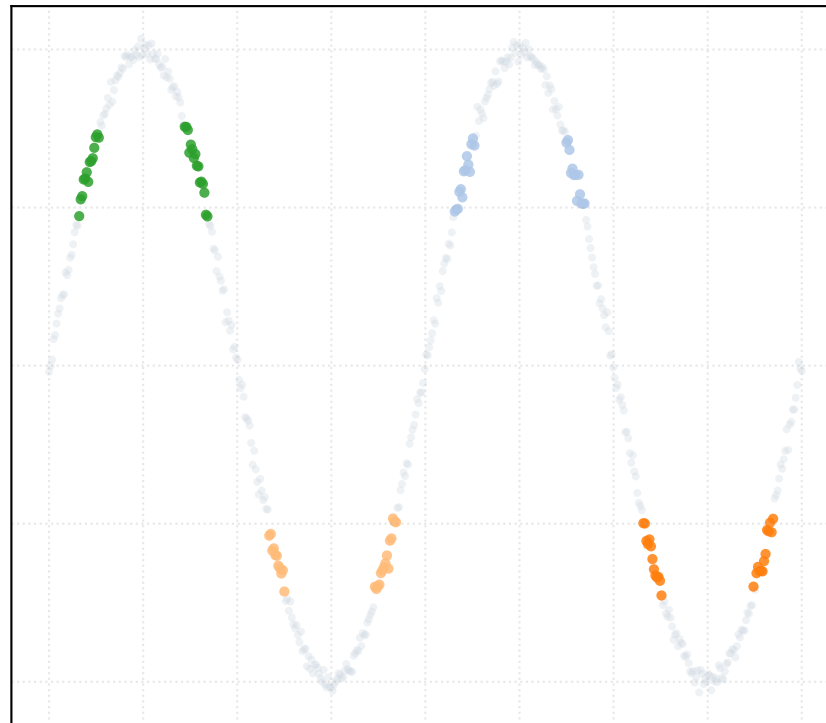
min_intersection = 1 (Default) [4 Clusters]



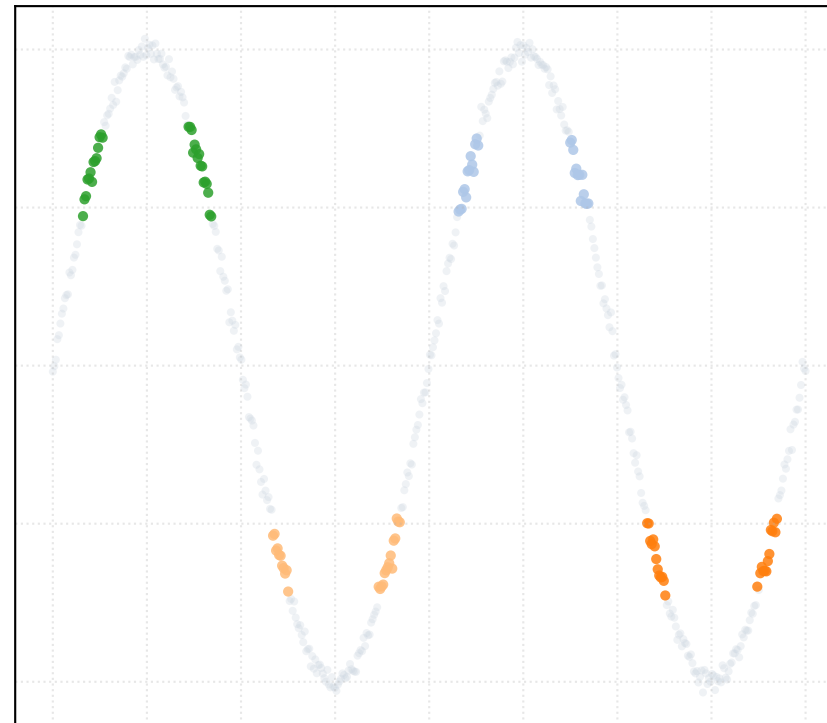
min_intersection = 2 [4 Clusters]



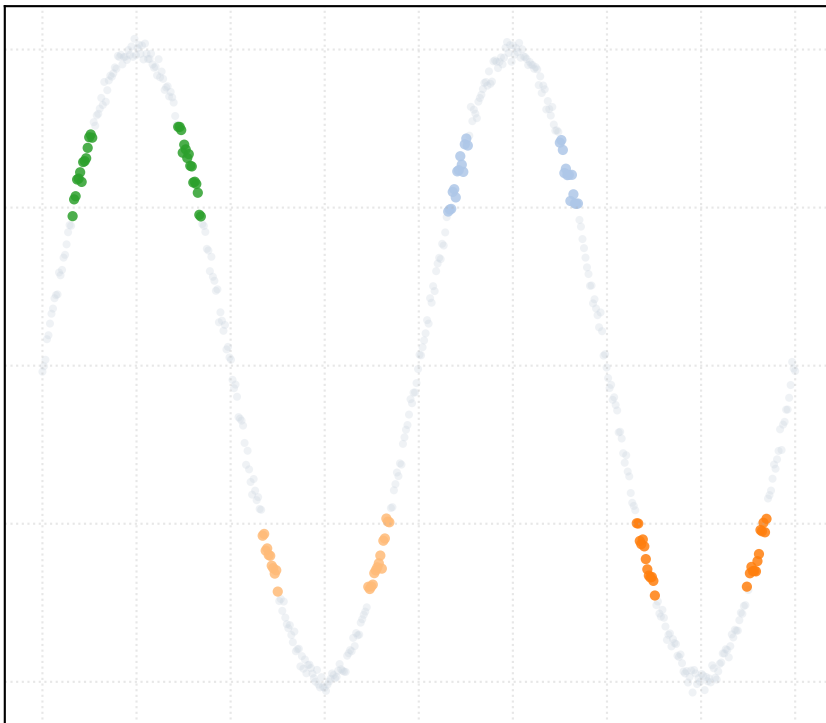
min_intersection = 3 [4 Clusters]



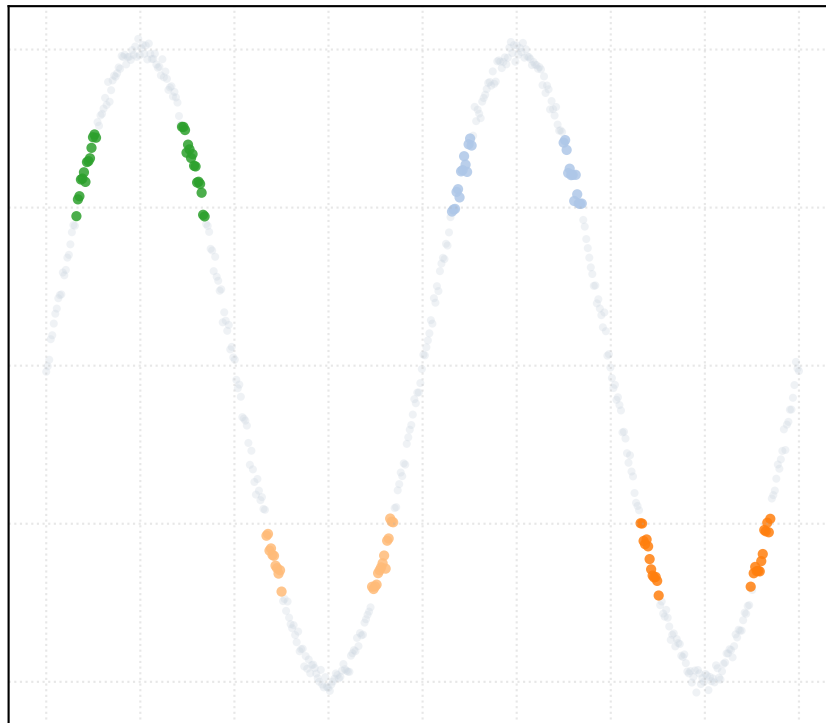
min_intersection = 5 [4 Clusters]



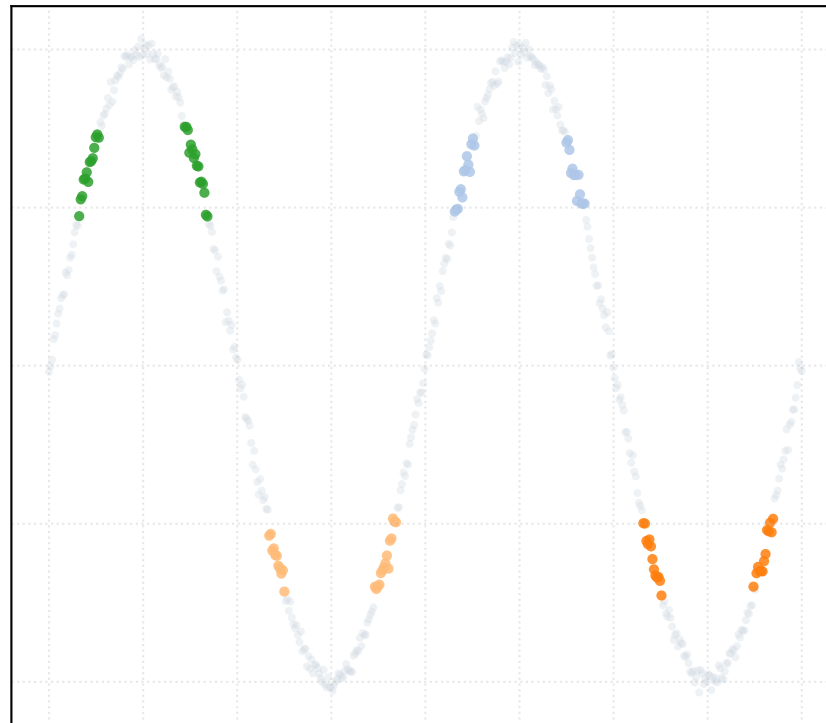
min_intersection = 8 [4 Clusters]



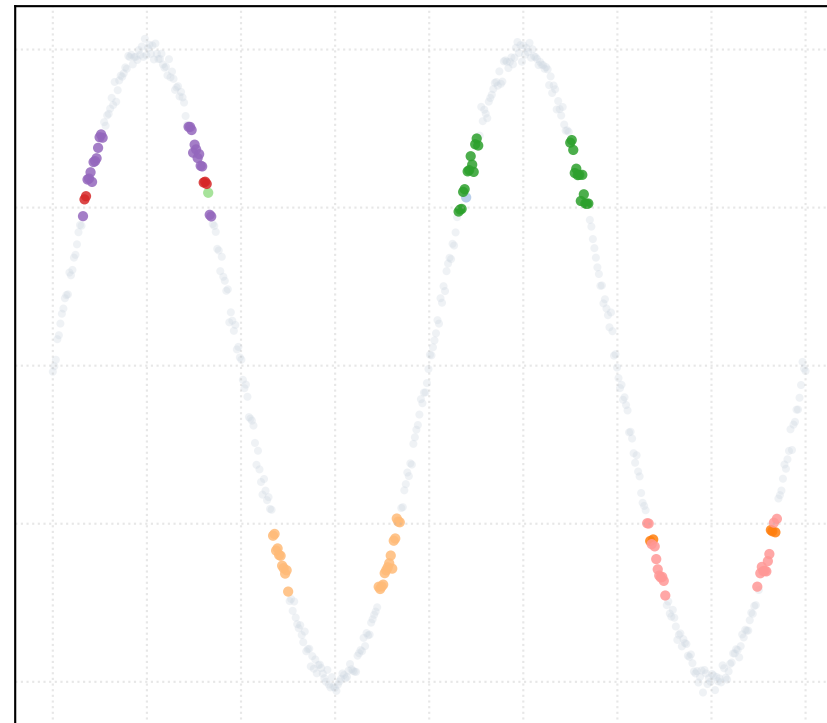
min_intersection = 10 [4 Clusters]



min_intersection = 15 [4 Clusters]

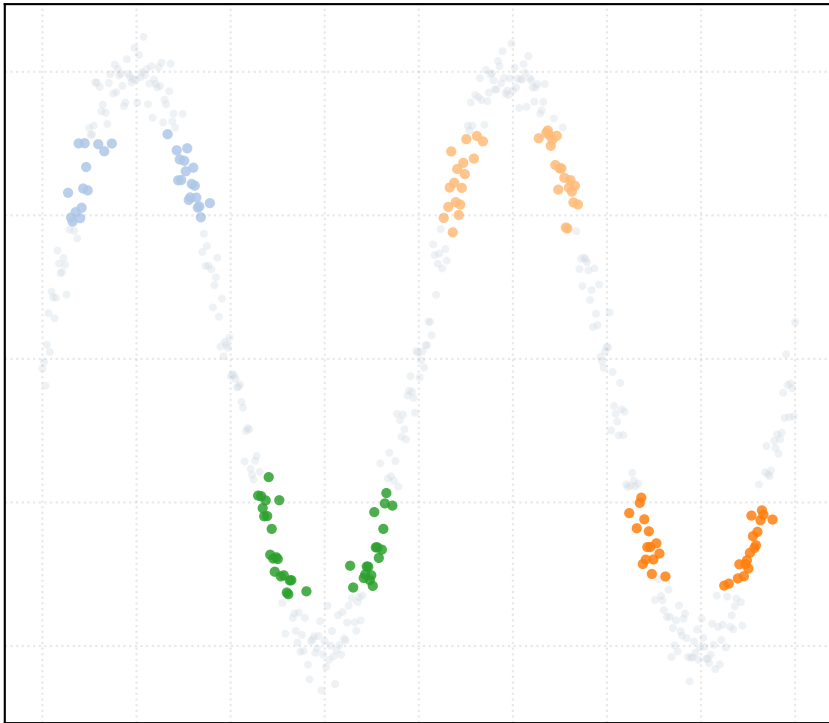


min_intersection = 20 [8 Clusters]

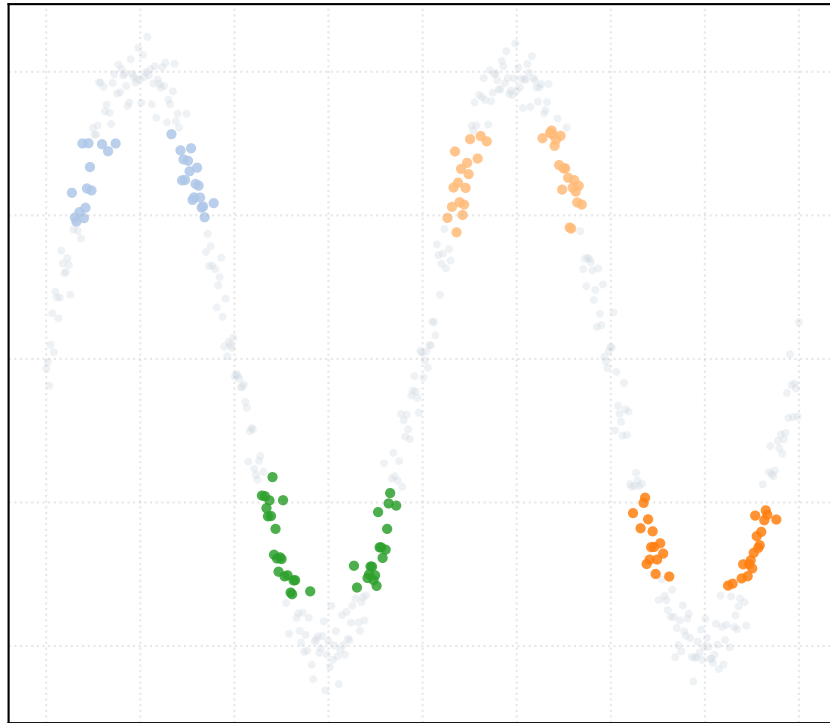


Dataset: s_curve_noise_medium | Ensemble Merge Overlap Sweep (min_intersection)

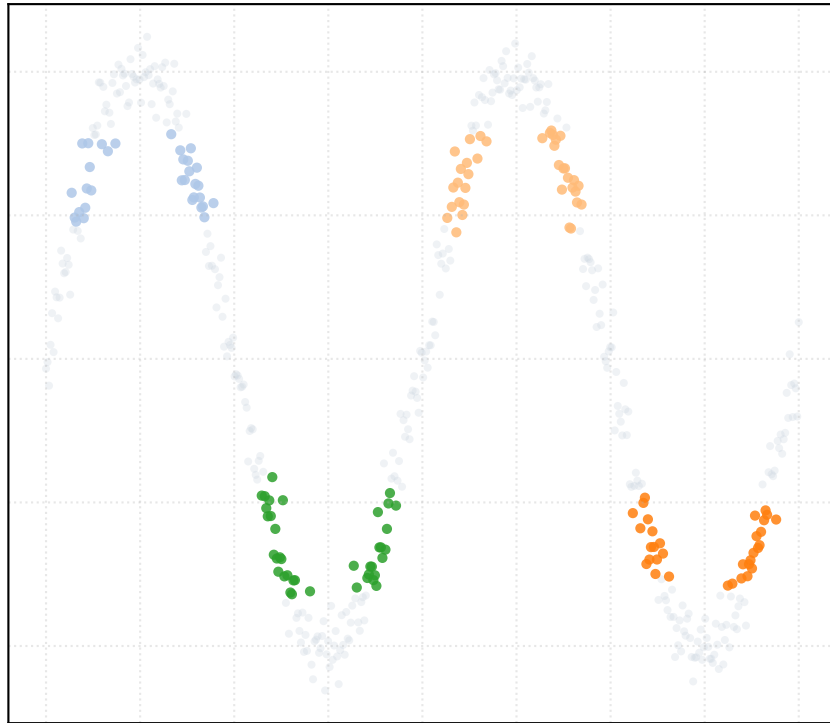
min_intersection = 1 (Default) [4 Clusters]



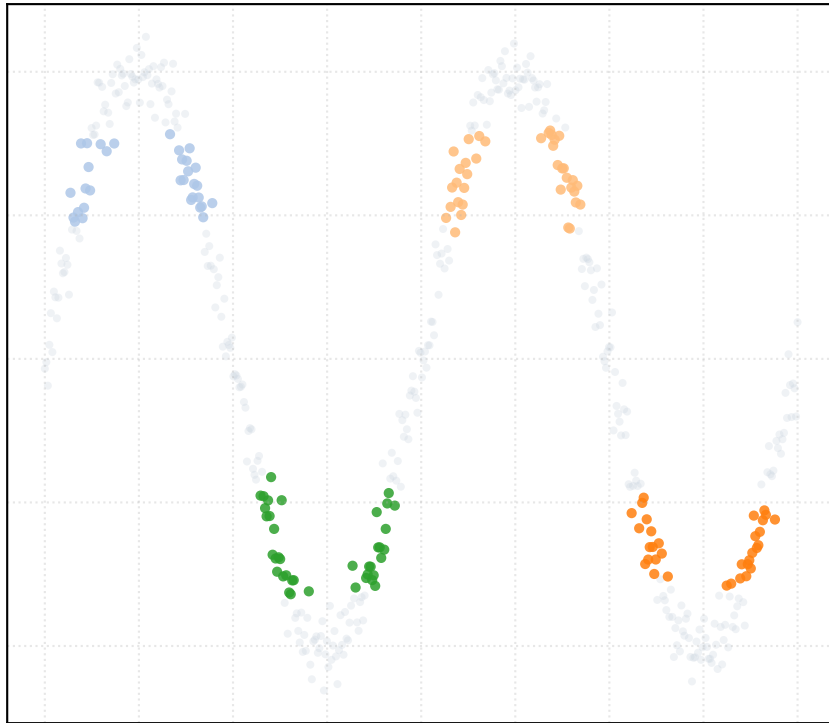
min_intersection = 2 [4 Clusters]



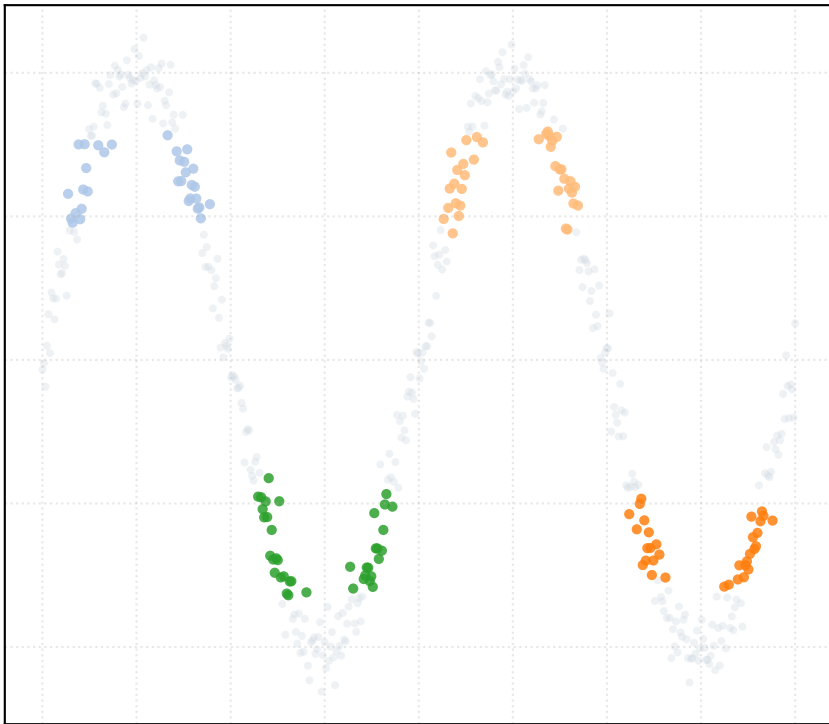
min_intersection = 3 [4 Clusters]



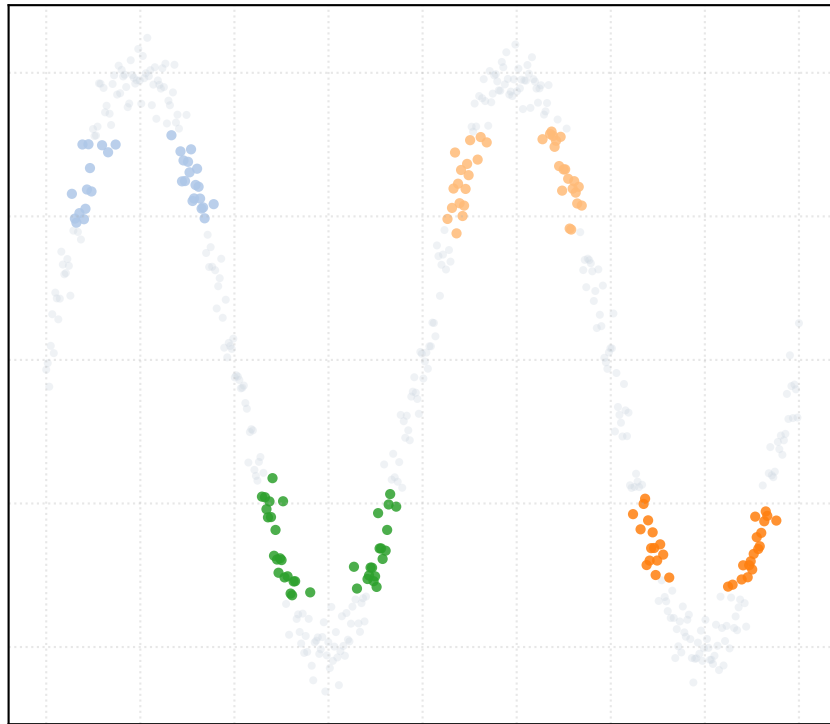
min_intersection = 5 [4 Clusters]



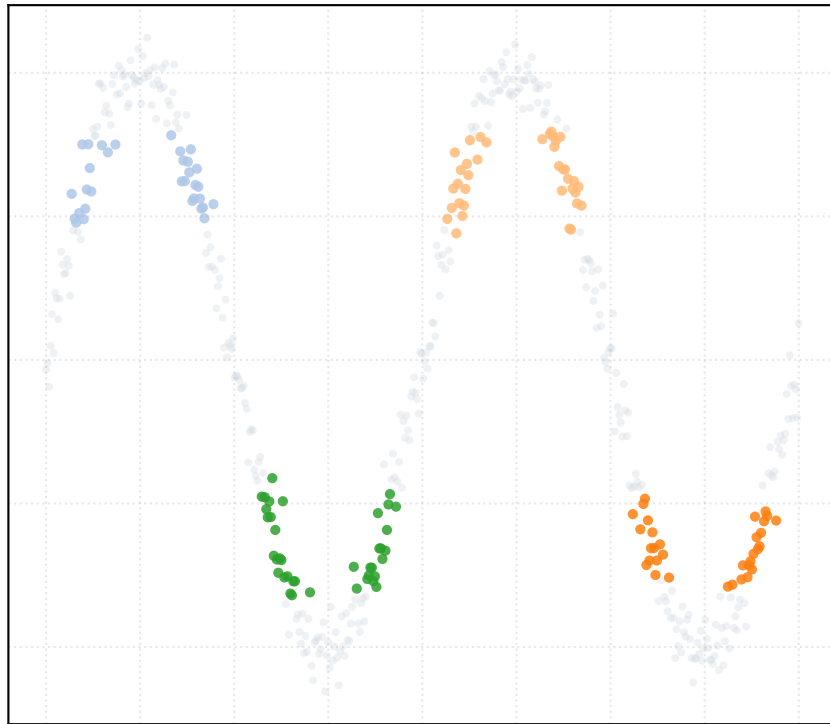
min_intersection = 8 [4 Clusters]



min_intersection = 10 [4 Clusters]



min_intersection = 15 [4 Clusters]



min_intersection = 20 [4 Clusters]

